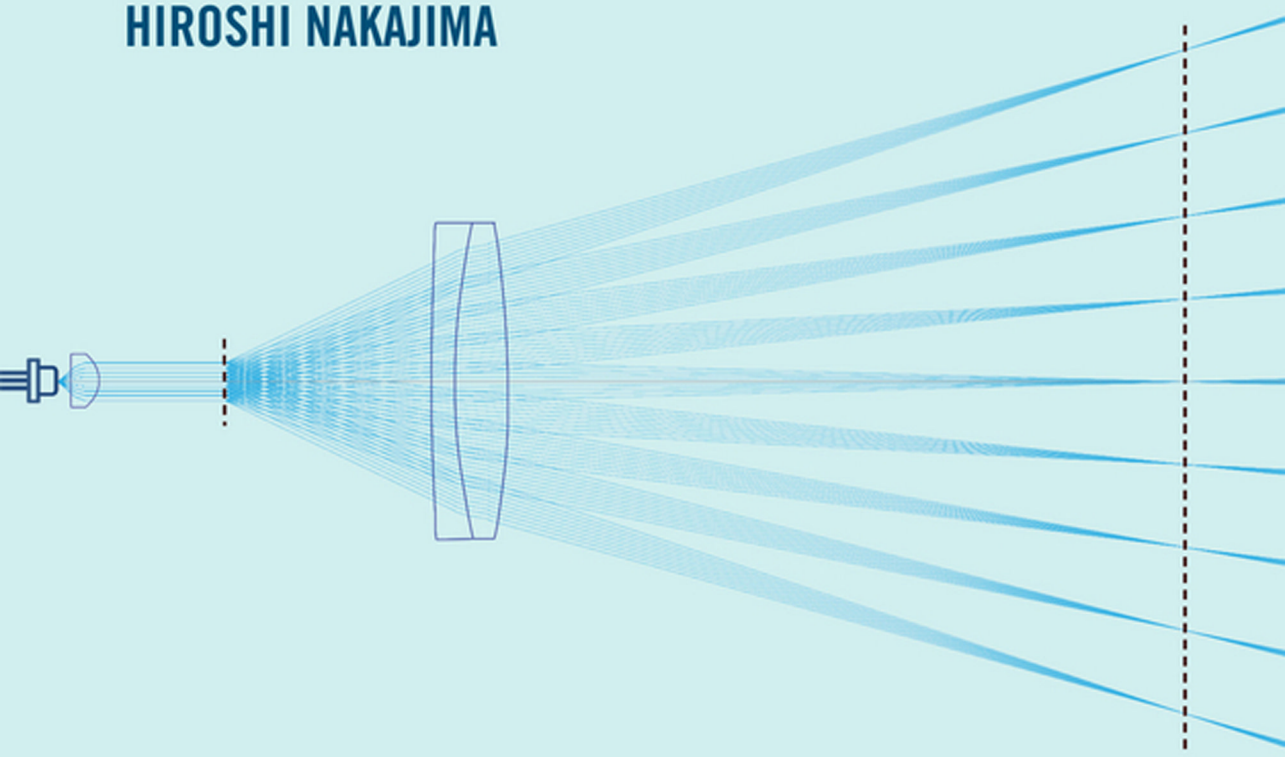
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Practical Calculations for Laser Optical System

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PRACTICAL CALCULATIONS FOR LASER OPTICAL SYSTEMS

Hiroshi Nakajima

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About the Author

Hiroshi Nakajima graduated from Doshisha University (Kyoto, Japan) with a degree in electronics. From 1967 to 2002, he worked for Hitachi Electronics Engineering Co., Ltd, as an engineer. He was responsible for developing automated manufacturing systems for a variety of electrical products, including the following: automatic adjustment systems for color picture tubes (CPTs), optical inspection systems for the silicon wafer surfaces of integrated circuits (ICs), and optical inspection systems for hard disk (HD) surfaces. From 2003 to 2012, he worked for Hitachi Electronics Engineering Co. and Hitachi High-Technologies Corporation as a technical consultant. During this time, he was responsible for developing optical inspection systems for HD surfaces. The author's specialty is developing machines which are capable of detecting microscopic surface defects (less than 0.1 μm in diameter) using laser optical systems. This book is largely based on his practical experience of working with laser optical systems.

Preface

Allow me to introduce myself. My name is Hiroshi Nakajima, and I am an electronic and optical engineer in Japan. During my job training, I had to study optics. It was tougher than I had anticipated. It took me a great deal of effort to acquire an understanding of optics and the various technologies that it employs. While I was studying optics, I always wondered if there was an easier way to master the subject. That is what motivated me to write this book.

There are lots of excellent books about theoretical and applied optics. However, there are very few books on the market that explain the practical calculation methods used for designing optical systems, with the aid of fully worked-out examples. This book is intended for the non-specialist reader, whose grasp of the mathematical complexities of optical theory is somewhat limited. For ease of comprehension, the technical explanations have been made as clear and simple as possible. The book provides a detailed explanation of the design of optical systems, starting from the basic theoretical principles and then moving on to practical, real-life examples, with supporting calculations included. By the time the reader finishes this book on optics, he/she should have a fairly complete understanding of the subject.

One of the distinguishing features of this book is that it contains a detailed treatment of optical laser systems, which generate a laser-focused beam and then use it to irradiate an object, thereby obtaining images through processes such as reflection, refraction, and diffraction. From a theoretical standpoint, light emitted from a laser is the simplest kind of light. Starting with lasers is an ideal way of acquiring an understanding of general optics, with a minimum of effort.

This book employs various calculation methods to describe the focusing of a laser beam on a spot. After exploring the basic principles of geometrical optics and simple optical systems, the reader will be provided with multiple worked examples of two- and three-dimensional ray tracing techniques. The more complex phenomena of interference and diffraction require a mathematical grasp of diffraction theory, which treats light as a wave. Next, we shall derive the equations for Gaussian beam focusing, a very simple and practical method for performing laser optical calculations. Finally, the reader will be introduced to some basic techniques that make use of diffraction calculations in order to solve applied problems relating to the wave motion of light.

Some of the calculations in this book (ray trace calculations and Gaussian beam focusing calculations) are performed directly on Excel spreadsheets, which makes them easy to understand, while others (ray trace calculations and diffraction calculations) employ VBA programming, where users can perform complex calculations by inputting simple settings. Readers can either choose to perform the ray trace calculations directly on Excel spreadsheets, or use VBA programs instead. Additionally, all of the calculations have been recorded on the Excel files available on the book's companion web site. I sincerely hope that my book will help readers obtain an understanding of what is needed to design optical systems – especially laser systems.

Lastly, I would like to thank Dr Vincent J. Torley for patiently proofreading my English.

Hiroshi Nakajima

OUTLINE OF CONTENTS

Chapter 1 – Geometrical Optics

In this chapter, the reader is introduced to the basic principles of geometrical optics. The fundamental characteristics of light are discussed, including Fermat's principle. Starting from these basic principles, we can derive the principles of paraxial theory, the five Seidel aberrations and the sine condition, which together form the basis of geometrical optics.

Chapter 2 – Examples of Simple Optical Design Using Paraxial Theory

The basic principles for designing optical systems are discussed in this chapter, for various types of lenses. Examples of simple optical systems are also provided. Additionally, various technical considerations relating to the design of laser optical systems are discussed.

Chapter 3 – Ray Tracing Applications of Paraxial Theory

In this brief chapter, we examine a method for performing ray tracings, using paraxial theory. Problems with practical applications of this method are also included.

Chapter 4 – Two-Dimensional Ray Tracing

Here, we discuss a method for performing ray tracing calculations in two dimensions, using the laws of refraction. We show how ray tracing calculations can be performed for rays traveling from the object to the image, for a variety of optical systems. We can calculate the ray aberration occurring on the image plane, as well as the wave-front aberration of an optical system. As this book is intended to give the reader an understanding of the practical applications of optical systems, this chapter includes fully worked solutions to real-life problems relating to several different types of lenses, including biconvex lenses, plano-convex (concave) lenses, cylindrical lenses, meniscus lenses, achromatic lenses, and aspheric lenses. We can also use aberration-free lenses in ray tracing, as a substitute for lenses whose specifications we do not know in detail.

Chapter 5 – Three-Dimensional Ray Tracing

Using the method of ray tracing discussed in Chapter 4, we can expand the number of dimensions from two to three, and thereby obtain a method for performing three-dimensional ray tracing calculations. Using this method, we can then generate spot diagrams on the image plane. Two- and three-dimensional ray tracing calculations can be performed on Excel cells directly, or on VBA programs. Support is provided for the reader, whether he/she chooses Excel or VBA.

Chapter 6 – Mathematical Formulae for Describing Wave Motion

Chapters 1–5 relate to geometrical optics. However, geometrical optics alone cannot provide us with a complete analysis of the behavior of light in all situations. In particular, it does not tell us how to design optical systems relating to the phenomena of interference and diffraction. In these cases, we need a mathematical theory which enables us to deal with light as a wave. In this chapter, we study the fundamental characteristics of light, the wave equations, and the mathematical expressions used to describe the amplitude and phase of waves, including Argand diagrams. (For the benefit of those readers who may be interested in further study, a detailed derivation of the equations for plane waves, spherical waves, and Gaussian beams is contained in the Appendices.)

Chapter 7 – Calculations for Focusing Gaussian Beams

By combining the lens formula in Chapter 1 and the Gaussian beam equations in this chapter, we can derive focusing equations for a Gaussian beam. In this chapter, we discuss the characteristics of Gaussian beams and derive the focusing equations for a Gaussian beam. The chapter also contains a treatment of the M-squared factor, which can be used to calculate the focused spot size of an actual laser beam whose beam quality is a little worse than the ideal Gaussian beam because of being superimposed with higher order lateral modes.

Chapter 8 – Diffraction: Theory and Calculations

In this chapter, we examine the theory and calculation methods of diffraction for various apertures, including a slit, a rectangular aperture, and a circular aperture.

Chapter 9 – Calculations for Gaussian Beam Diffraction

Here, we discuss methods for calculating diffraction irradiance, especially for Gaussian beams, as they are the laser beams which are generally used in real-life situations. In this book, the diffraction calculations are performed by numerical integration methods using the VBA program. The VBA program includes one-dimensional diffraction calculations (for the case of diffraction from a slit), two-dimensional R-θ diffraction calculations (diffraction from a circular aperture) and two-dimensional x-y diffraction calculations (diffraction from an aperture with an arbitrary shape). The numerous examples at the end of the chapter provide the reader with solutions to practical problems with real-life applications, which the reader is likely to encounter in an everyday context.

Appendices

During the preparation of this book, some of the more detailed theoretical explanations were edited from the text, for the sake of ease of comprehension. However, some of these detailed explanations have been included in the Appendices for the benefit of those seeking a deeper understanding of optics, and for anyone who may be interested in undertaking further studies in the future.

1

Geometrical Optics

1.1 Characteristics of Lasers

This book is about optical calculation methods and the principles for applying these methods to actual optical devices. Most of these devices use lasers, so we will begin by briefly examining the characteristics of lasers. The following exposition will be especially beneficial for readers whose understanding of lasers is rather limited.

The term LASER is an acronym for "Light Amplification by Stimulated Emission of Radiation." Lasers have special characteristics that distinguish them from most light-emitting devices: a narrow, low-divergence beam (sharp *directivity*), and a very narrow wavelength spectrum (*monochromaticity*). These features of lasers make them ideally suited for the generation of high intensity beams.

The history of lasers goes back about half a century to 1960, when the first working laser was demonstrated. Lasers are now widely used in a variety of fields, including optical storage (e.g., CD drives and DVD drives), fiber-optic communication, manufacturing (especially for cutting, bending, welding, and marking materials), scientific measurement, and medicine. This diversity of application is due to the following four properties of lasers, which give them a commercial and scientific edge over other light-emitting devices.

1. *Monochromaticity:* Radiation that has a very narrow frequency band (or wavelength band) is said to possess the property of *monochromaticity.* Because the band is so narrow, the radiation can be regarded as having a single frequency (or alternatively, a single wavelength). Laser light typically has a very narrow frequency band (or wavelength band), which makes it an ideal example of the property of monochromaticity. Sunlight, by contrast, has a very broad band spectrum, with multiple frequencies and wavelengths.
2. *Beam directivity:* The term *directivity*, when used in relation to lasers, refers to the directional properties of the electromagnetic radiation they emit. A laser beam exhibits a very sharp (narrow) directivity, allowing it to propagate in a straight line with almost no expansion. By contrast, normal light sources such as flashlights and car headlights have broader directivity than lasers, so their beams cannot travel as far.
3. *Coherence:* If split laser beams which were emitted from the same source are superimposed, a fringe pattern will appear. This fringe pattern is never observed in an isolated beam. We refer to this phenomenon as *interference* caused by the wave characteristics of light. If these two beams traveling onto the same plane are superimposed while they are *in phase* (with their crests and troughs lined up), the resultant beam will appear brighter, but if these waves are superimposed with a 180° phase difference

Optical Design Using Excel®: Practical Calculations for Laser Optical Systems, First Edition. Hiroshi Nakajima.
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(i.e., if crests and troughs are superimposed), the beam will appear dark and the resultant amplitude will be zero. A laser can easily generate interference patterns because of its *coherence* (uniformity of phase). Sunlight cannot readily generate interference patterns, due to its incoherence: because its coherence time and coherence length are very short, and the phases of the superimposed waves will not come into phase very easily. Sunlight will only exhibit coherence over a very short interval both of time and space.

4. *High concentration of energy (high intensity):* A sheet of paper can be burnt simply by focusing sunlight on it, using a convex lens. A laser is much more concentrated: it can even weld two pieces of steel together. This is not merely due to the high power of the laser, but also because of the extremely high *intensity* of the laser beam, where the light energy is narrowly focused. It is relatively easy to concentrate a laser beam on a small target. A high intensity beam can be generated very easily using a laser.

All of these characteristics of lasers can best be summed up by the word "coherent." A laser is both *temporally coherent* and *spatially coherent.* What this means is that a laser has a uniform phase over time at an arbitrary point in space, and it also has a uniform phase in space at an arbitrary point in time. Thus a laser has a uniform phase both in time and in space. A laser radiates a single-wavelength (more precisely, a very narrow wavelength spectrum) beam with a constant phase and it can propagate in a specific direction. An electric light, by contrast, radiates a multitude of different wavelengths at various phases and in all directions. Thus it is both spatially and temporally incoherent.

1.2 The Three Fundamental Characteristics of Light Which Form the Basis of Geometrical Optics

To understand geometrical optics, which is the most basic form of optics, we need to first study the fundamental characteristics of light – including light emitted by lasers. The following properties of light are confirmed by everyday experience:

1. Light rays travel in straight lines in a uniform medium.
2. Light rays are independent of one another.
3. Light rays can be reflected and refracted: they change their direction at a boundary between different media, in accordance with the laws of reflection and refraction.

The whole science of geometrical optics can be derived from these three characteristics of light.

1.2.1 Light Rays Travel in Straight Lines

Many common optical phenomena attest to the fact that light rays travel in straight lines. For instance, when the sun is shining outdoors, a tree casts a shadow whose shape is identical with its own. Without using any lenses, we can construct a pinhole camera that can capture the image of an object, simply by making a pinhole in one of the walls of a black box as shown in Figure 1.1.

1.2.2 Light Rays Act Independently of One Another

Figure 1.2 illustrates the independent action of light rays using the example of three spotlights whose light is of different colors: red, blue, and green. When we irradiate the same area on a white sheet of paper with

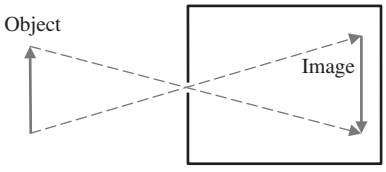


Figure 1.1 Pinhole camera

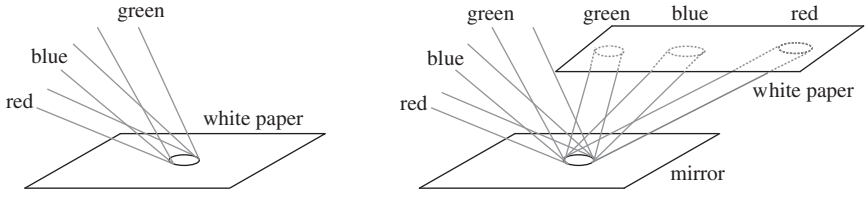


Figure 1.2 Independent action of light rays

these three spotlights, we perceive the light as "white." However, if we replace the paper with a mirror, and then project the reflected light onto another sheet of paper, the three separate beams of light reappear in their original colors of red, blue, and green. The fact that these three beams reappear in their original colors demonstrates that light rays coming from different sources act independently of one another after being reflected by the mirror. Likewise, the fact that the light from the three spotlights appears white when they are all focused on the same area of paper can be explained in terms of the constituent light wavelengths (red, blue, and green) reaching our eyes and acting on our retinas independently. It is the superposition of waves which causes us to perceive them as white.

1.2.3 Reflection of Light Rays

As shown in Figure 1.3a, when a light ray is reflected by a mirror at the point of incidence O, we can define the normal as an imaginary line through point O perpendicular to the mirror. The reflected ray will lie in the same plane as the incident ray and the normal to the mirror surface, and the angle of reflection will be the same as the angle of incidence. We are all familiar with this phenomenon from everyday experience. It can be described by the following equation:

$$\text{Angle of incidence } \theta_i = \text{angle of reflection } \theta_r \tag{1.1}$$

1.2.4 Refraction of Light Rays

An object lying in a tub of water appears to be at a shallower depth than it actually is. This phenomenon can be explained by the refraction of light at the interface between the water and the air. As shown in Figure 1.3b, when a ray is refracted at a boundary plane between different media, the relationship between the angle of incidence and angle of refraction can be described by the following equation (Snell's law):

$$n_1 \sin\theta_i = n_2 \sin\theta_t \tag{1.2}$$

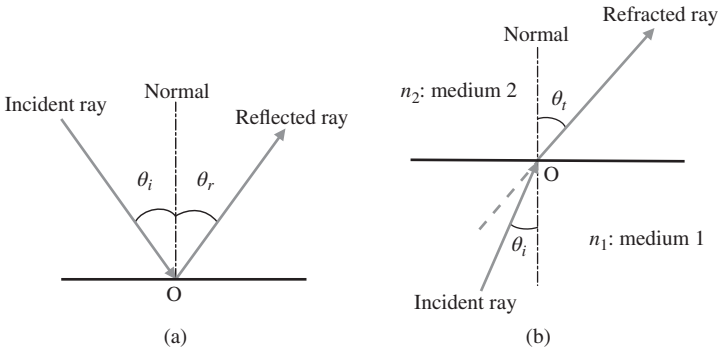


Figure 1.3 (a) Reflection of a light ray and (b) refraction of a light ray

where

n_1 = Refractive index in medium 1
n_2 = Refractive index in medium 2
θ_i = Angle of incidence
θ_t = Angle of refraction.

1.3 Fermat's Principle

The law of rectilinear propagation and the law of reflection and refraction of light rays can both be derived from Fermat's principle, as explained below [1].

(i) The velocity of light is inversely proportional to the refractive index of the medium in which light propagates. (In other words, light travels more slowly in a medium having a higher refractive index.)
The velocity of light in a vacuum is a constant, *c:*

$$c = 2.99792458 \times 10^8 \text{ m/s} \tag{1.3}$$

The velocity of light v in a medium having refractive index n is:

$$v = c/n \tag{1.4}$$

In optical calculations, the optical path length L is defined as follows:

$$L = nL_0 \tag{1.5}$$

where L_0 = physical length traversed by light in the medium. (L_0 is defined as the optical path length in a vacuum, and the refractive index in a vacuum, $n_0 = 1$.)

(ii) The path taken by a ray of light between any two points in a system is always the path that can be traversed in the least time (or the path that has the minimum optical path length) (Fermat's Principle)
A more accurate and complete statement of Fermat's principle is that a ray of light traveling from one point to another point must traverse an optical path length which corresponds to a stationary point, which means that it can either be a minimum, a maximum, or a point of inflection.

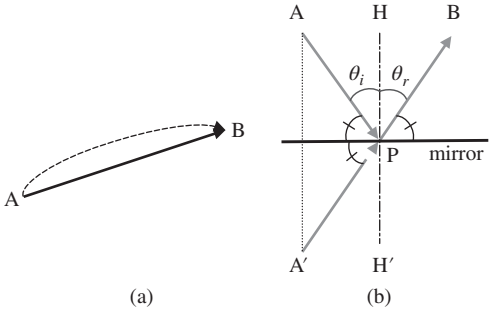


Figure 1.4 (a) Rectilinear propagation and (b) law of reflection

1.3.1 Rectilinear Propagation

Fermat's principle entails that a ray of light must travel along a rectilinear path in a uniform medium.

As shown in Figure 1.4a, a ray of light traveling from A to B in a uniform medium takes the path that can be traversed in the least time, that is, the straight line AB.

1.3.2 Reflection

When a ray of light is reflected, it will be reflected at an angle that minimizes its optical path length.

Figure 1.4b illustrates a ray of light which proceeds from point A, is reflected by the mirror at P, and reaches B. Now let us assume that the starting point is A′ instead of A. (A′ is the "reflection" of A in the mirror.) Clearly, the length of A′PB is equal to APB. It is also obvious that the optical path length A′PB is minimized when A′PB is rectilinear. When this is the case, the angle of reflection θ_r will equal the angle of incidence θ_i.

$$\text{Angle of incidence } \theta_i = \text{angle of reflection } \theta_r \tag{1.1}$$

1.3.3 Refraction

A ray of light traveling from one medium to another will be refracted at an angle that minimizes the optical path length.

Suppose that a ray of light proceeds from point A(0,a) inside a medium with a refractive index of n_1, then passes through the boundary plane at point B(x,0), and finally reaches point C(d,c), inside a medium with a refractive index n_2, as shown in Figure 1.5a.

The optical path length ABC will be:

$$\begin{aligned} L &= n_1 \mathrm{AB} + n_2 \mathrm{BC} \\ &= n_1 \sqrt{x^2 + a^2} + n_2 \sqrt{(d-x)^2 + c^2} \end{aligned} \tag{1.6}$$

The derivative of L with respect to x will be:

$$\frac{dL}{dx} = \frac{n_1 x}{\sqrt{x^2 + a^2}} - \frac{n_2 (d-x)}{\sqrt{(d-x)^2 + c^2}} \tag{1.7}$$

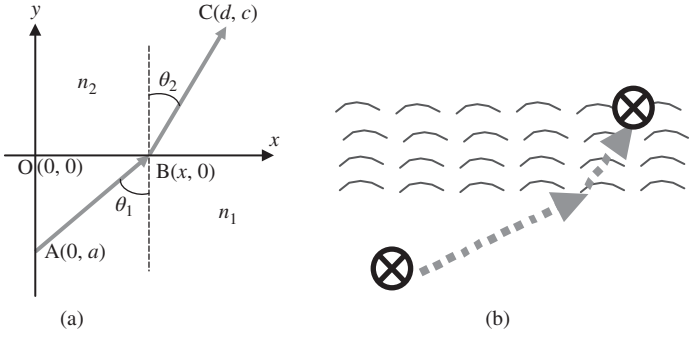


Figure 1.5 (a) Angle of refraction and (b) emergency path of a life saver

When $dL/dx = 0$, the optical path length L is minimized. That is,

$$\frac{dL}{dx} = \frac{n_1 x}{\sqrt{x^2 + a^2}} - \frac{n_2(d-x)}{\sqrt{(d-x)^2 + c^2}} = 0 \tag{1.8}$$

From Figure 1.5a, the following equations can be derived:

$$\sin\theta_1 = \frac{x}{\sqrt{x^2 + a^2}} \tag{1.9}$$

$$\sin\theta_2 = \frac{d-x}{\sqrt{(d-x)^2 + c^2}} \tag{1.10}$$

Combining Equation (1.8), Equation (1.9) and Equation (1.10) yields:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2 \tag{1.2}$$

Equation (1.2) is the conditional equation of refraction.

Path Swum by a Lifesaver in An Emergency Rescue

A lifesaver at a beach will swim in a path that minimizes the time to reach the victim in an emergency rescue (Figure 1.5b).

This is identical to the behavior of a ray of light which is described by Fermat's principle.

1.3.4 An Ideal Imaging System Using Lenses

Let us imagine an ideal imaging system using multiple lenses. A ray of light proceeds from the point object A, passes through the lens system, and reaches imaging point B, as in Figure 1.6. If the system is an ideal imaging system, these three routes must have the same optical path length (Fermat's principle).

In an ideal imaging system, the optical path length from the object to the image will have a constant value of ΣL:

$$\sum L = n_1 L_1 + n_2 L_2 + n_3 L_3 + n_4 L_4 + n_5 L_5 = \text{constant} \tag{1.11}$$

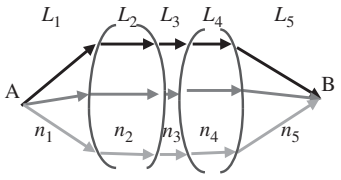


Figure 1.6 Optical path length in an ideal imaging system

Note

The refractive index of a medium is not always the same: it varies slightly, depending on the wavelength of the light traveling through it. Thus lenses will exhibit some degree of chromatic aberration. Spherical lenses also exhibit various monochromatic aberrations, which will be discussed in Section 1.6. In actual imaging systems, multiple lens systems are generally used in order to minimize or eliminate these aberrations.

1.4 Principle of Reversibility

Since Fermat's principle only relates to the path and not the direction traveled, all of the foregoing equations describing light rays apply equally to light traveling along the same path in a backwards direction.

If there is a ray of light which starts from point A, passes through an optical system, and reaches point B, it follows that the ray starting at B and reaching A must travel along exactly the same path. This insight, which has immense practical value, is referred to as the "Principle of reversibility."

1.5 Paraxial Theory Using Thin Lenses

We will now examine several methods of imaging using lenses [1]. First, using the concept of a minimal optical path length, we will discuss how rays can create an image of an object.

1.5.1 Equation of a Spherical Lens Surface

The equation of a spherical lens surface S whose center of curvature lies on the z-axis and whose surface touches the origin as in Figure 1.7, can be expressed by:

$$x^2 + y^2 + (z - r)^2 = r^2 \tag{1.12}$$

where r is the curvature radius of the lens surface.

Equation (1.12) can be rewritten as:

$$z = \frac{1}{2r}(x^2 + y^2 + z^2) \tag{1.13}$$

When we use a thin lens, we can neglect z^2 because $z^2 \ll x^2 + y^2$ in thin lenses. Hence

$$z = \frac{1}{2r}(x^2 + y^2) \tag{1.14}$$

$$z = \frac{1}{2}c\,(x^2 + y^2) \tag{1.15a}$$

$$c \equiv \frac{1}{r}\ (\text{curvature}) \tag{1.15b}$$

In Equation (1.15a), since the surface of the lens bulges toward the incident rays coming from the light source, it is convex and has a positive curvature. In a two-dimensional space, Equation (1.15a)

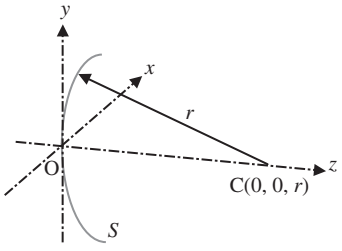


Figure 1.7 Spherical lens surface

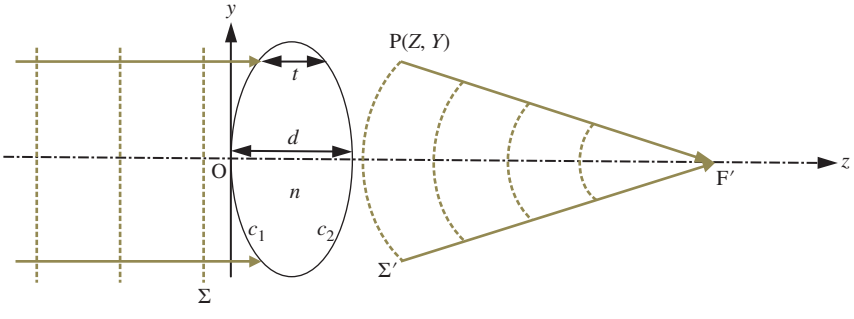


Figure 1.8 Refraction of a light ray by a convex lens

simplifies to:

$$z = \frac{1}{2}cy^2 \tag{1.16}$$

Equation (1.16) describes a convex lens with a curvature of c, whose thickness decreases as the vertical height of the lens increases.

1.5.2 Wave Front Radii of Incident Rays versus Rays Refracted by a Convex Lens

A collimated light beam (whose rays are parallel) is incident onto a convex lens having a refractive index of n, a thickness of d in the center, and front and back surfaces whose curvatures are c_1 ($\equiv 1/r_1$) and c_2 ($\equiv 1/r_2$), respectively. Suppose that the front surface of the lens touches the origin, as shown in Figure 1.8. The wave front surface (defined as a set of points whose optical path lengths from a point light source all have the same value) of the incident rays will lie on a plane Σ, as the light beam is collimated. Using the concept of optical path length, we can illustrate how the shape of the wave front changes, after passing through the lens.

Suppose the ray at a height Y travels a distance Z parallel to the z-axis, while the ray going along the z-axis travels through the center of a lens whose thickness is d. After passing through the lens, the wave front will be a surface consisting of points having the same optical path length from the light source.

From Equation (1.16), the thickness t of the lens at height Y will be:

$$t = d - \frac{1}{2}(c_1 - c_2)\, Y^2 \tag{1.17}$$

The optical path length of the ray L_0 which travels through the center of the lens will be:

$$L_0 = nd \tag{1.18}$$

The optical path length of the ray L_Y which travels through the lens at a vertical height of Y will be:

$$L_Y = nt + Z - t \tag{1.19}$$

The condition for both rays having the same optical path length is that $L_0 = L_Y$. Thus

$$nd = n\left\{ d - \frac{1}{2}\left(c_1 - c_2\right)\, Y^2 \right\} + Z - d + \frac{1}{2}(c_1 - c_2)\, Y^2 \tag{1.20}$$

In other words,

$$Z = d + \frac{1}{2}(n-1)\,(c_1 - c_2)Y^2 \tag{1.21}$$

Equation (1.21) represents a sphere with a curvature of $(n-1)(c_1 - c_2)$, which includes the rear surface of the lens on the axis. From the preceding discussion, it follows that the surface of the wave front, after passing through the lens, will be transformed into a sphere whose equation is represented by Equation (1.21).

Note

The foregoing equations all presuppose that we can apply the thin lens approximation: light rays maintain the same vertical height while passing through a lens.

1.5.3 The Refractive Power of a Lens and the Thin Lens Equations

Let us now apply Equation (1.21) to incident rays which are not parallel but diverging. From Equation (1.21), the ray of light at a vertical height of Y will travel a distance of $1/2(n-1)(c_1 - c_2)Y^2$ further than the ray along the z-axis, which travels through the lens at the point where it is thickest (Figure 1.9). Putting it another way, after passing through a convex lens, the wave front curvature will increase by a factor of $(n-1)(c_1 - c_2)$. We refer to this value as the refractive power of the lens, K.

$$K = (n-1)\,(c_1 - c_2) \tag{1.22}$$

As a result, a convex lens will refract the incident rays inwards. Figure 1.9 shows that a convex lens will refract a diverging beam of incident light (which has a negative wave front curvature) inwards, generating a converging beam of light (with a positive wave front curvature). The refractive power K will be the difference in wave front curvatures between the light rays behind the lens and those in front of the lens.

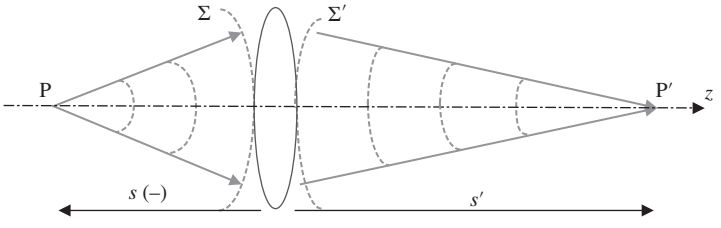


Figure 1.9 Refraction by a lens

We can define K as the (refractive) power of a lens. The point light source P situated at a distance $s(-)$ in front of the lens generates a wave front Σ (a sphere with its center at P) at the lens front. After passing through the lens, the wave front Σ' is transformed into a sphere centered at P′ which is situated at a distance $s'(+)$ to the right of the lens along the z-axis. The wave front curvatures of Σ and Σ' will be $1/s$ and $1/s'$. The curvature after passing through the lens $1/s'$ should be the value added by the (refractive) power K to the initial curvature of the wave front before passing through the lens $1/s$. That is,

$$\frac{1}{s'} = \frac{1}{s} + K \tag{1.23}$$

More generally, we can replace K with the inverse of the focal length $1/f$. This gives us:

$$\frac{1}{s'} = \frac{1}{s} + \frac{1}{f} \tag{1.24}$$

where s is the distance of the object from the lens, and s' is the distance of the image from the lens. These values will be positive when they are situated to the right of the lens and negative when they are situated to the left of the lens. f is the focal length of the lens.

Equation (1.24) is the famous Gaussian lens formula. Using this formula, we can easily calculate the distance of the image. If $s > 0$, the object is situated behind the lens, so the object will be a virtual object. If $s' < 0$, the image is situated in front of the lens, so the image will be a virtual image.

From Equation (1.22), the focal length of a thin lens can be expressed as follows:

$$\frac{1}{f} = (n-1)(c_1 - c_2) = (n-1)\left(\frac{1}{r_1} - \frac{1}{r_2}\right) \tag{1.25}$$

If the lens is plano-convex (or plano-concave), $c_2 = 0$ ($r_2 = \infty$), and we can then write:

$$\frac{1}{f} = \frac{(n-1)}{r} \tag{1.26}$$

1.5.4 Imaging Equations for a Lens

1.5.4.1 A Method of Drawing Rays Before and After Passing through the Lens, in Order to Obtain Their Image Point

Figure 1.10 illustrates the rays of light before and after going through the lens, where

F = First focal point
F′ = Second focal point
H = Primary principal point (H_1HH_2 = primary principal plane)
H′ = Secondary principal point ($H_1'H'H_2'$ = secondary principal plane)
f = First focal length (−)
f' = Second focal length (+)
z = Distance of object from the first focal point (−)
s = Distance of object from the primary principal point (−)
z' = Distance of image from the second focal point (+)
s' = Distance of image from the secondary principal point (+)
A = Object
AB = Normal to the axis (B is situated on the axis)
A′ = Image
A′B′ = Normal to the axis (B′ is situated on the axis).

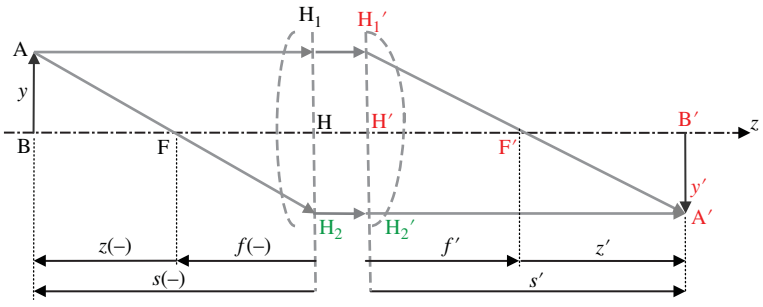


Figure 1.10 A method of drawing rays in order to obtain their image point

In Figure 1.10,

① A ray leaves object A, travels parallel to the axis, is refracted at the secondary principal plane H_1' and then passes through the second focal point F′.
② Another ray leaves A and passes through the first focal point F, is refracted at the primary principal plane H_2, and then travels parallel to the axis.

Rays ① and ② are illustrated in Figure 1.10. Rays ① and ② meet again at A′ after passing through the lens. This point will be the image point A′, that is, the lens will make an image A′B′ of the object AB.

1.5.4.2 Lateral Magnification β

The lateral magnification β is defined as the value of the height of the image divided by the height of the object. In Figure 1.10,

$$\Delta FAB \propto \Delta FH_2H, \qquad \Delta F'H_1'H' \propto \Delta F'A'B'$$

Hence,

$$\beta = \frac{y'}{y} = -\frac{f}{z} = -\frac{z'}{f'} \tag{1.27}$$

1.5.4.3 Newton's Formula

From Equation (1.27), the relation between z, z', f, and f' can be expressed as follows:

$$zz' = ff' \tag{1.28}$$

Equation (1.28) is referred to as Newton's formula. When the lens is surrounded by air, we have $f = -f'$. The lateral magnification β can then be re-expressed as:

$$\beta = -\frac{f}{z} = -\frac{z'}{f'} = \frac{z'+f'}{z+f} = \frac{s'}{s} \tag{1.27′}$$

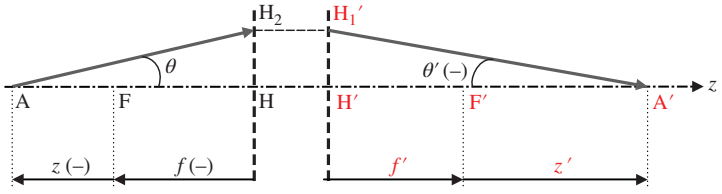


Figure 1.11 Angular magnification γ

1.5.4.4 Angular Magnification γ

As shown in Figure 1.11, when a lens forms an image A′ of an object A situated on the optical axis, then for any arbitrary ray proceeding from A and reaching A′, we have:

$$\tan\theta = \frac{HH_1}{HA} = \frac{HH_1}{(-z)+(-f)} \qquad \tan(-\theta') = \frac{H'H_1'}{H'A'} = \frac{H'H_1'}{f'+z'}$$

where θ and θ' are the angles of the incident ray and the imaging ray, respectively, with the optical axis. Since $HH_1 = H'H_1'$,

$$\gamma \equiv \frac{\tan\theta'}{\tan\theta} = \frac{f+z}{f'+z'} \tag{1.29}$$

From Newton's formula, Equation (1.28), the following equation can be derived:

$$\frac{z}{f'} = \frac{f}{z'} \tag{1.28′}$$

Combining Equation (1.29) and Equation (1.28′), we obtain:

$$\gamma = \frac{f+z}{f'+z'} = \frac{z}{f'} = \frac{f}{z'} \tag{1.30}$$

1.5.4.5 Longitudinal Magnification α

Longitudinal magnification denotes the factor by which an image increases in size, as measured along the optical axis. In Figure 1.12, the size of the object and the image along the optical axis are Δz and $\Delta z'$, respectively. Using Newton's formula, Equation (1.28), the longitudinal magnification α will be:

$$\alpha = \frac{\Delta z'}{\Delta z} = \frac{dz'}{dz} = -\frac{z'}{z} = -\frac{ff'}{z^2} \tag{1.31}$$

1.5.4.6 Relation between the Three Magnifications α, β, γ

Using Equation (1.27), Equation (1.30) and Equation (1.31), we can show that

$$\frac{\alpha}{\beta^2} = -\frac{f'}{f} \qquad \beta\gamma = -\frac{f}{f'}$$

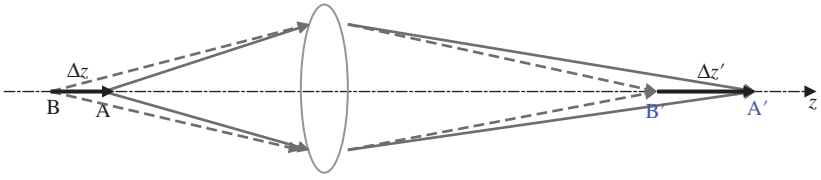


Figure 1.12 Longitudinal magnification α

When the lens is surrounded by air, we have $f = -f'$. The relation between α, β, and γ will then be:

$$\alpha = \beta^2 \tag{1.32a}$$

$$\gamma = \frac{1}{\beta} \tag{1.32b}$$

1.5.4.7 Paraxial Ray Equations with the Origin at the Principal Point

Suppose that the point object along the optical axis is O and that of its image is O′ (the conjugate point of O), and the lateral magnification is β_0. From Equation (1.27), the coordinates of O and O′ will be:

$$-f/\beta_0 \text{ for O, and } -\beta_0 f' \text{ for O}'.$$

Let us postulate two more conjugate points A and A′, with a lateral magnification of β. The coordinates of A and A′ will then be:

$$-f/\beta \text{ for A, and } -\beta f' \text{ for A}'.$$

Now let us define s and s' as the coordinates of A and A′, relative to their respective origins O and O′. If we define $s = \text{OA}$ and $s' = \text{O}'\text{A}'$, s and s' will then be:

$$s = -\frac{f}{\beta} - \left(-\frac{f}{\beta_0}\right) = -\frac{f}{\beta} + \frac{f}{\beta_0} \qquad s' = -\beta f' - (-\beta_0 f') = -\beta f' + \beta_0 f'$$

β can be eliminated from the two preceding equations, as shown below:

$$\frac{f}{\beta_0 s} + \frac{\beta_0 f'}{s'} = 1$$

If we let $\beta_0 = 1$, then the above equation becomes:

$$\frac{f}{s} + \frac{f'}{s'} = 1 \tag{1.33}$$

In the special case where $\beta_0 = 1$, the objective point O and image point O′ then coincide with the principal points H and H′, respectively.

If the lens is surrounded by air, then $f = -f'$. Equation (1.33) can then be rewritten as Equation (1.24′):

$$\frac{1}{s'} = \frac{1}{s} + \frac{1}{f'} \tag{1.24′}$$

This equation is the same as the one which we previously obtained. If we rewrite f' as f,

$$\frac{1}{s'} = \frac{1}{s} + \frac{1}{f} \tag{1.24}$$

Once again, we have obtained the Gaussian lens formula. This time we have introduced principal points in order to allow us to treat actual lenses, which have finite thicknesses.

Figure 1.13 illustrates how the rays should be drawn, in accordance with Equation (1.24).

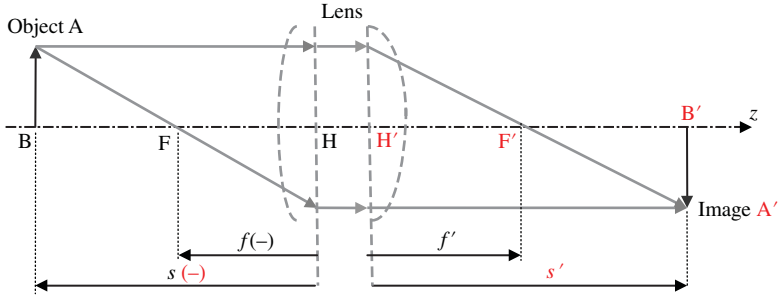


Figure 1.13 Illustration of rays, showing the lens, the object, and its image

where

AB = Object
A′B′ = Image
F = First focal point
F′ = Second focal point
H = Primary principal point
H′ = Secondary principal point
f = Focal length of the lens
s = Distance HB of object from lens, which has a − sign when s is to the left of the lens.
s' = Distance H′B′ of image from lens, which has a + sign when s' is to the right of the lens.

As shown in Figure 1.13, both the distance of the object s and of its image s' will have a positive sign when they are to the right of the lens, and a negative sign when they are to the left of the lens. Likewise, the sign of the focal length f will be positive when the lens is convex and negative when it is concave.

1.5.5 Simple Lenses

For a simple lens having a refractive index of n, surface radii of r_1 and r_2, and a central thickness of t, we can calculate the first and second focal lengths f and f' and the positions of primary and secondary principal points p and p', using the following equations (Figure 1.14) (see Appendix A for supporting details).

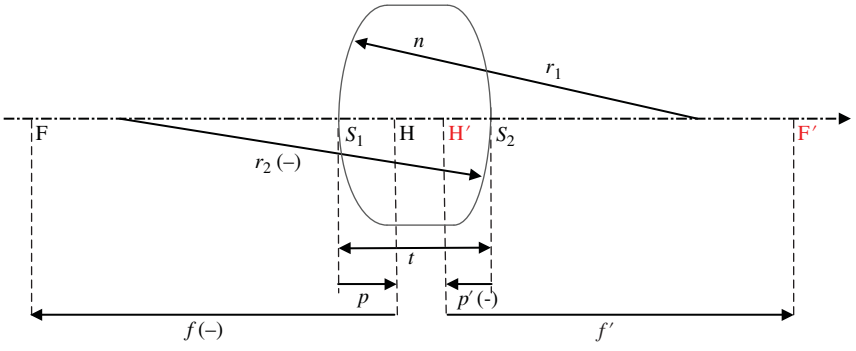


Figure 1.14 Focal length and principal point of a simple lens

$$\frac{1}{f'} = -\frac{1}{f} = (n-1)\left\{\left(\frac{1}{r_1} - \frac{1}{r_2}\right) + \frac{(n-1)t}{nr_1r_2}\right\} \tag{1.34}$$

$$p = -\frac{r_1 t}{n(r_2 - r_1) + (n-1)t} \tag{1.35}$$

$$p' = -\frac{r_2 t}{n(r_2 - r_1) + (n-1)t} \tag{1.36}$$

where the sign of the range is positive (negative) when it is from left (right) to right (left), and $f = -f'$. These equations can be derived by treating the two parts of the lens, that is, the front surface and the back surface, as a combination.

If the curvatures of the front and back surface have the same value, we can substitute $r = r_1 = -r_2$ in the preceding equations. This yields the following result:

$$\frac{1}{f'} = -\frac{1}{f} = (n-1)\left\{\frac{2}{r} - \frac{(n-1)\,t}{nr^2}\right\} \tag{1.37}$$

Given *f,* we can calculate the value of *r,* using Equation (1.38).

$$r = \frac{n(n-1)f' + \sqrt{n^2(n-1)^2 f'^2 - n(n-1)^2 f' t}}{n} \tag{1.38}$$

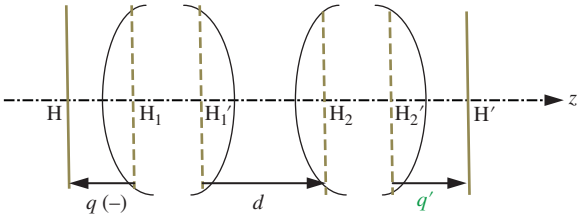


Figure 1.15 Combination of two simple lenses

1.5.6 The Focal Lengths and Principal Points of a Two-Lens Combination

Let us now consider a combination of two simple lenses. In Figure 1.15, let f_1 and f_1' denote the first and the second focal lengths of the first lens and let H_1 and H_1' denote the primary and secondary principal points, respectively. Let f_2, f_2', H_2 and H_2' denote the corresponding values for the second lens. The first and second focal lengths f and f' and the primary and the secondary principal points H and H′ of the combination of two lenses can be calculated by applying the following equations. The results for a combination of three or more lenses can easily be obtained by iterating this process (see Appendix A for further details).

$$f = \frac{f_1 f_2}{d - f_1' + f_2} \tag{1.39}$$

$$f' = -\frac{f_1' f_2'}{d - f_1' + f_2} \tag{1.40}$$

$$q = H_1H = \frac{f_1 d}{d - f_1' + f_2} \tag{1.41}$$

$$q' = H_2'H' = \frac{f_2' d}{d - f_1' + f_2} \tag{1.42}$$

where d is the distance between the principal points of the first and the second lenses.

$$d = H_1'H_2$$

We can calculate the focal lengths and the principal points of any arbitrary lens system by using the equations for a simple lens and for a combination of two lenses [Equation (1.34), Equation (1.35), Equation (1.36), Equation (1.39), Equation (1.40), Equation (1.41) and Equation (1.42)]. All of these equations can be derived from paraxial theory (see Appendix A).

1.6 The Five Seidel Aberrations

Paraxial theory, which we examined in the previous section, can yield accurate results for *paraxial* rays (which lie close to the optical axis and make a small angle to the optical axis). However, in real life, rays emitted from a point object travel outwards in *all* directions. For this reason, no lens can make

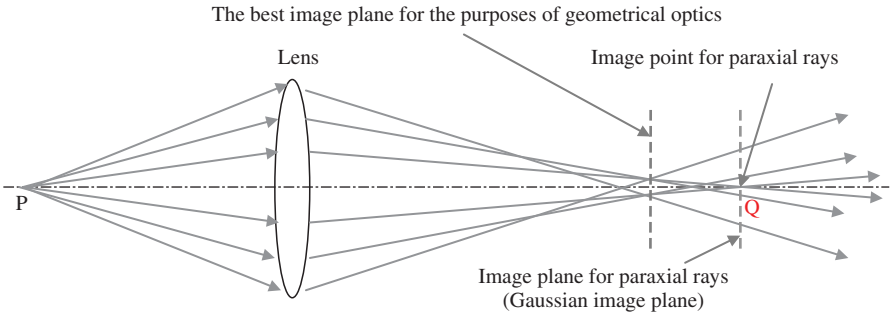


Figure 1.16 Spherical aberration

rays emitted from a point object converge exactly at a single point. *Aberration* can thus be defined as an imperfection in image formation by an optical system, caused by the inherent shortcomings of the lens. Aberrations fall into two main classes: chromatic aberrations and monochromatic aberrations. Here, we will confine our discussion of monochromatic aberrations to their significance in laser applications [2].

1.6.1 Monochromatic Aberration: A Brief Outline

Monochromatic aberrations include spherical aberration, coma aberration, astigmatism, field curvature, and distortion. These aberrations are referred to as the *five Seidel aberrations*, named after Philip Ludwig von Seidel, who first identified them in 1857.

1.6.1.1 Spherical Aberration

Spherical aberration is the only one of Seidel's five aberrations which occurs for an object lying on the optical axis. As shown in Figure 1.16, if rays from the point object P on the axis make a small angle with the optical axis, then after passing through the lens, they will converge on a point Q which also lies on the optical axis, and whose distance from the lens is equal to the distance calculated using paraxial theory, which we discussed in the previous section. However, rays passing through those parts of the lens which lie well above (or below) the optical axis will be refracted slightly further inwards than the paraxial, "ideal" rays which converge to form an image at point Q. For this reason, the image of a point object on the image plane which includes Q will not be a point, but a circle. This aberration is referred to as *spherical aberration*. The image plane on point Q is referred to as the Gaussian image plane, and the best image plane from the point of view of geometrical optics will be one which is located slightly closer to the lens than the Gaussian image plane.

1.6.1.2 Coma Aberration

Coma is an aberration which occurs for an object lying *off-axis.* In Figure 1.17, the ray from the off-axis point source P passing through the center of the entrance pupil of the lens will form a point image P′ on the image plane. However, rays from P which pass through the large circle A on the entrance pupil will form a large circular image A′ on the image plane. Likewise, the rays from P that pass through the small circle B on the pupil will form a small circular image B′ on the image plane. Thus the image of the point source P will not be a point, but a small comet-shaped patch.

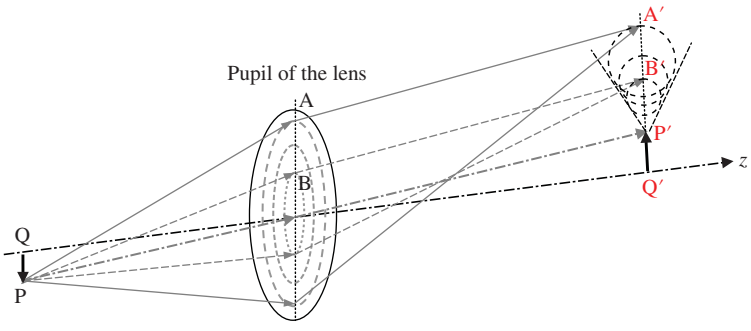


Figure 1.17 Coma aberration

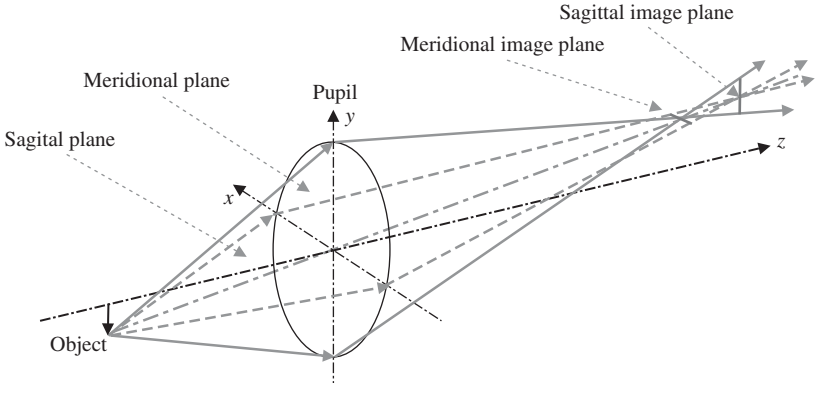


Figure 1.18 Astigmatism

1.6.1.3 Astigmatism

In Figure 1.18, we can define the meridional plane as the plane that includes the optical axis and the principal ray which passes through the center of the lens. The sagittal plane is the plane containing the principal ray which is also perpendicular to the meridional plane. Rays which proceed from a point which lies off the optical axis and then travel along the meridional plane will not converge on the same point as rays which proceed from the same off-axis point, but travel along the sagittal plane. The former will converge on the meridional image plane and the latter on the sagittal image plane. Because of this aberration, the image of the point object P will not be a point, but two images lying on different planes. This aberration is referred to as *astigmatism*.

1.6.1.4 Field Curvature

Due to aberrations in the optical system, a flat (planar) object cannot always form a perfectly flat image. Instead, the image will usually be on a curved surface. As shown in Figure 1.19, aberrations result when a flat screen is used to record the image formed by the curved image surface. This aberration is referred to as *field curvature*.

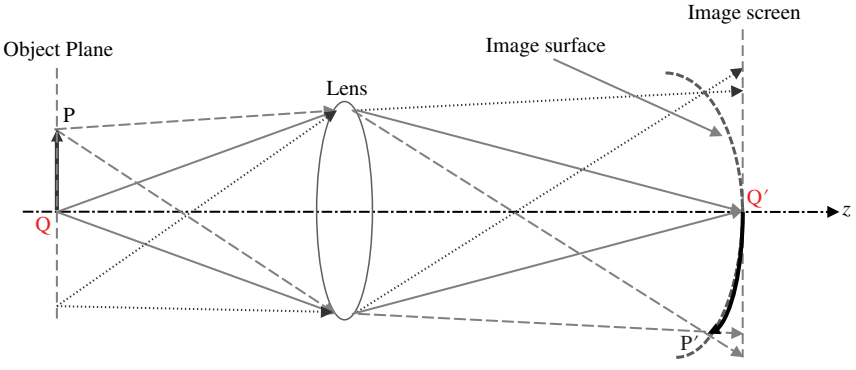


Figure 1.19 Aberration caused by field curvature

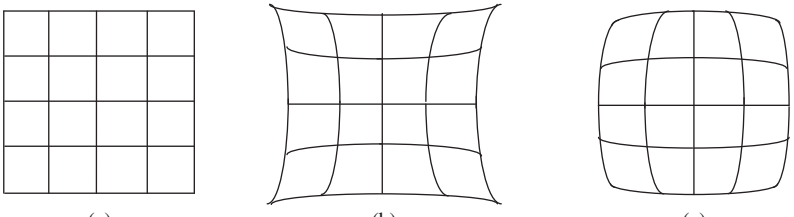


Figure 1.20 Distortion. (a) Original, (b) pin-cushion distortion, and (c) barrel distortion

1.6.1.5 Distortion

If we start with a picture such as the grid in Figure 1.20a as our object, the image formed may exhibit either pin-cushion distortion, as shown in Figure 1.20b, or barrel distortion, as in Figure 1.20c. These aberrations are collectively referred to as *distortion*.

1.6.2 Ray Aberration

Let us suppose that rays proceeding from a point object travel through an optical system and reach an image plane. In this process, (transverse) ray aberration denotes the distance between the rays on the ideal image plane between the ray passing through the actual optical system and the ideal optical system. In this section we will briefly discuss the magnitude of Seidel's five aberrations.

In Figure 1.21, a ray proceeds from an off-axis object and leaves the exit pupil at Q(ρ,φ). The x–y coordinates of Q can be expressed as follows:

$$x = \rho \sin \varphi \quad y = \rho \cos \varphi$$

P′(X,Y) denotes the coordinates of the ideal image point (for paraxial rays) on the ideal image plane.

Suppose that a ray starts at object P, leaves the exit pupil at Q, and creates an image at P″. Due to aberrations in the optical system, P″ will be located some distance from the point P′ where paraxial rays form their image. The distance between P′ and P″ along the optical axis is referred to as *longitudinal aberration*. The ray QP″ intercepts the paraxial image plane (the $\varepsilon_X - \varepsilon_Y$ plane

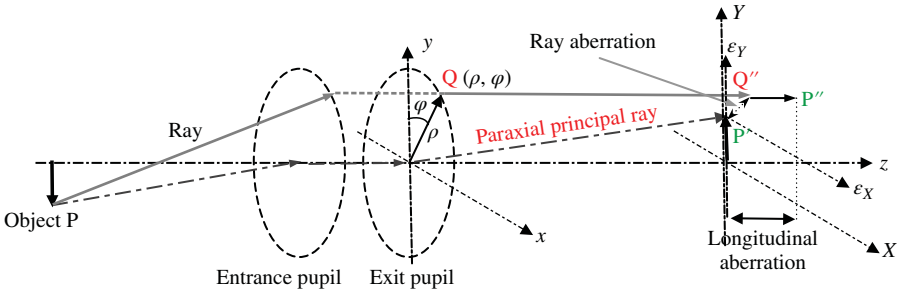


Figure 1.21 Ray aberration

in Figure 1.21) at Q″. The distance between Q″ and P′ along the $\varepsilon_X - \varepsilon_Y$ plane is referred to as *ray aberration*.

Ray aberration can be computed precisely, by performing a ray trace calculation, which applies the equations of refraction (Snell's law) to the boundaries of each lens surface. It is also possible to calculate ray aberration, using the following approximation:

$$\sin\theta \cong \theta - \frac{\theta^3}{3!} \quad \cos\theta \cong 1 - \frac{\theta^2}{2!}$$

We refer to the ray aberration using this approximation as a *third-order aberration.* The following are the third-order aberration results for ray aberration.

$$\varepsilon_X = S_1\rho^3 \sin\varphi + S_2 Y\rho^2 \sin 2\varphi + PY^2\rho\sin\varphi \tag{1.43a}$$

$$\varepsilon_Y = S_1\rho^3 \cos\varphi + S_2 Y\rho^2(2 + \cos 2\varphi) + (S_3 + P)Y^2\rho\cos\varphi + S_5 Y^3 \tag{1.43b}$$

We refer to these aberrations as *Seidel's five aberrations,* where

S_1 = Spherical aberration coefficient
S_2 = Coma coefficient
S_3 = Astigmatism coefficient
P = Petzval sum coefficient
S_5 = Distortion coefficient
ρ = Ray height at the exit pupil
φ = Ray position angle at the exit pupil
Y = Ideal image height on the image plane.

Equation (1.43a,b) describes the case where the image lies on the Y-axis. We have assumed this, in order to simplify the calculation results. Here, we are using lenses with spherical surfaces which are symmetrical with respect to the optical axis.

1.6.2.1 Spherical Aberration

In Equation (1.43a,b), if we set all of the coefficients to zero except for S_1, then we get:

$$\varepsilon_X{}^2 + \varepsilon_Y{}^2 = (S_1\rho^3)^2 \tag{1.44}$$

Equation (1.44) describes the equation of a circle. *Spherical aberration* distorts the image of a point object, so that it appears as a circle (Figure 1.16), whose radius is proportional to the cube of the ray height at the exit pupil. Since Equation (1.44) contains no term Y, the aberration observed will be independent of the image height. (In other words, the degree of aberration is uniform over the entire image plane.) The ($\pm$) sign appearing in front of the degree of spherical aberration will depend on whether the lens is convex or concave. The foregoing results imply that we can reduce spherical aberration by using a combination of lenses with convex and concave surfaces.

1.6.2.2 Coma Aberration

In Equation (1.43a,b), if we set all of the coefficients to zero except for S_2, then we get:

$$\varepsilon_X{}^2 + (\varepsilon_Y - 2S_2 Y\rho^2)^2 = (S_2 Y\rho^2)^2 \tag{1.45}$$

Equation (1.45) describes the equation of a circle centered at $(0, 2S_2 Y\rho^2)$ and having a radius of $S_2 Y\rho^2$. Hence the aberration spot will be shaped like a comet, as in Figure 1.17. When $Y = 0$, the radius equals zero. Hence there will not appear to be any *coma aberration* for objects on the optical axis. For off-axis objects, coma aberration increases in proportion to ρ^2 (the square of the height of the ray, on exiting the pupil). The condition for there being no coma for a small object near the axis will be:

$$\frac{n_1 \sin\theta_1}{n_2 \sin\theta_2} = \beta \tag{1.46}$$

Equation (1.46) was derived by Abbe, where

n_1, n_2 = Refractive indices of the media surrounding the object and the image, respectively
θ_1, θ_2 = Angles of the ray to the optical axis when it leaves the object and when it reaches the image plane, respectively
β = Lateral magnification.

Equation (1.46) is referred to as *Abbe's sine condition*. We will return to this equation in Section 1.7.

1.6.2.3 Astigmatism

In Equation (1.43a,b), if $S_3 \neq 0$ and $P = 0$, and if we set the other coefficients to zero, we get:

$$\varepsilon_X = 0 \tag{1.47a}$$

$$\varepsilon_Y = S_3 Y^2 \rho \cos\varphi \tag{1.47b}$$

Equation (1.47a,b) tells us that there is no aberration along the X-axis; rather there is an aberration of $\varepsilon_Y = S_3 Y^2 \rho$ in a line along the Y-axis. Originally, we assumed that the image lay on the Y-axis. However, the image may have both X- and Y-coordinates. By symmetry, the aberration could also be in a line along the X-axis, on a slightly different image plane. Thus, the beam from the off-axis object will converge on the two different image planes. We refer to this phenomenon as *astigmatism* (Figure 1.18).

1.6.2.4 Aberration Caused by (Petzval) Field Curvature

In Equation (1.43a,b), if $S_3 = 0$ and $P \neq 0$, and if we set the other coefficients to zero, we get:

$$\varepsilon_X = PY^2 \rho \sin\varphi \tag{1.48a}$$

$$\varepsilon_Y = PY^2 \cos\varphi \tag{1.48b}$$

From Equation (1.48a,b) we obtain:

$$\varepsilon_X{}^2 + \varepsilon_Y{}^2 = (PY^2\rho)^2 \tag{1.49}$$

Equation (1.49) describes the equation of a circle with a radius of $PY^2\rho$. The aberration radius will be proportional to ρ and Y^2 (Figure 1.19).

1.6.2.5 Distortion

In Equation (1.43a,b), if $S_5 \neq 0$, and if we set the other coefficients to zero, we get:

$$\varepsilon_X = 0 \tag{1.50a}$$

$$\varepsilon_Y = S_5 Y^3 \tag{1.50b}$$

Equation (1.50a,b) entails that this aberration will be proportional to the cube of the image height *Y*. In other words, this aberration will equal zero at the center of the image, but will increase rapidly as we approach the periphery. Distortion can be thought as an aberration caused by lateral magnification error, and depending on the image height (Figure 1.20).

1.7 The Sine Condition

1.7.1 The Abbe Sine Condition

When an optical system has no aberration for an on-axis object which emits wide angle (diverging) rays, the necessary condition for an object situated just a little way from the on-axis object in a perpendicular direction also having no aberration is referred to as the Abbe sine condition [3]. In other words, the sine condition is a necessary condition for the absence of coma in an optical system which has no spherical aberration. This condition can be described by the following:

$$\frac{n \sin\theta}{n' \sin\theta'} = \beta = \frac{y'}{y} \tag{1.46'}$$

where

β = Lateral magnification
y = Height of object
y' = Height of image
θ = Angle of the incident ray to the optical axis
θ' = Angle of the imaging ray to the optical axis
n = Refractive index of medium surrounding object
n' = Refractive index of medium surrounding image.

The lateral magnification β can be expressed as $\beta = \frac{s'/n'}{s/n} = \frac{ns'}{n's}$. Then, Equation (1.46′) can then be rewritten as:

$$s \sin\theta = s' \sin\theta' \tag{1.51}$$

We can now explain the significance of Equation (1.51). In the previous section, we examined the significance of the primary and the secondary principal plane in paraxial theory. However, as θ and θ'

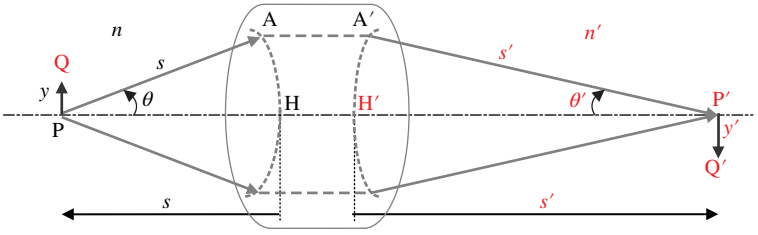


Figure 1.22 Abbe sine condition

get larger, we have to go beyond the scope of paraxial theory. Under these more general conditions, instead of using the term "principal planes," we should speak of the primary principal surface (which is a sphere centered at the object P, on whose surface the primary principal point lies), and the secondary principal surface (which is a sphere centered at the image P′, on whose surface the secondary principal point lies). If the sine condition is satisfied, then the line connecting the point A (where the incident ray proceeding from the on-axis object P intersects with the primary principal surface) to the point A′ (where the imaging ray traveling to the on-axis image P′ intersects with the secondary principal surface) will always run parallel to the optical axis (Figure 1.22).

1.7.2 The Sine Condition for an Off-Axis Object and Its Off-Axis Image

The sine condition described above holds for an object and its image, on the optical axis. However, a similar condition can also be satisfied by an off-axis object and its image [3]. In Figure 1.23, if there is no aberration for the object P on the axis and also no aberration for the off-axis object Q which lies on a plane which is perpendicular to the optical axis and passes through the on-axis object P, then the following equation will be satisfied.

$$\frac{n(\sin\theta - \sin\theta_0)}{n'(\sin\theta' - \sin\theta_0')} = \beta_M \equiv \frac{\delta Y'}{\delta Y} \tag{1.52}$$

where

θ_0 = Angle of the principal ray of the beam proceeding from the off-axis object Q
θ_0' = Angle of the principal ray of the imaging beam to the off-axis Image Q′
θ = Angle of the meridional ray proceeding from the off-axis object Q
θ' = Angle of the meridional ray of the imaging beam to the off-axis image Q′
β_M = Lateral magnification for a small object δY on Q, with respect to the meridional ray ($=\delta Y'/\delta Y$).

If the image height Y' is small and there is no distortion, then β_M can be assumed to equal β:

$$\beta_M \cong \beta \tag{1.53}$$

We can apply the concept of the principal surface to off-axis objects, just as we did for objects lying on the optical axis. In Figure 1.24, the primary principal surface will touch the point where the principal ray

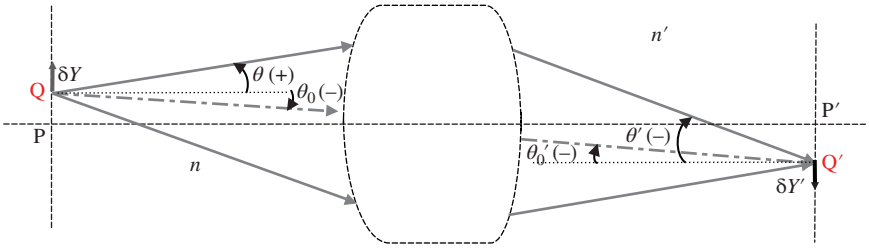


Figure 1.23 Sine condition for off-axis object and its image

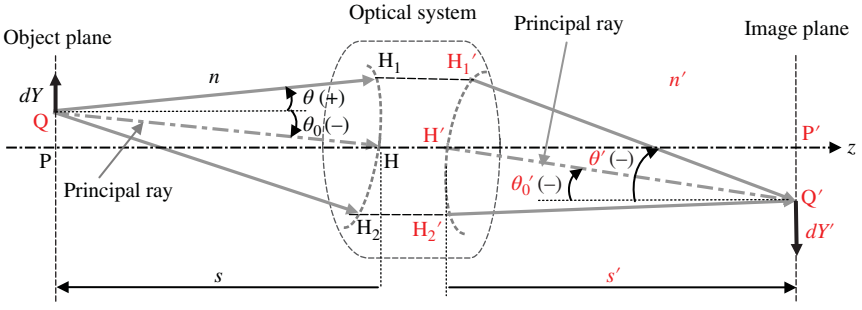


Figure 1.24 Principal surfaces for an off-axis object and its image

from the object Q intersects with the optical axis, while the secondary principal surface will touch the point where the principal ray traveling to the image Q′ intersects with the optical axis. If the optical system is surrounded by air and the image height is small, it can be assumed that the primary and secondary principal surfaces lie on the primary and secondary principal points, and whose curvature centers are situated at the object Q and its image Q′, respectively. (In this case, the principal ray will pass through the principal points.)

We can understand why Equation (1.52) and Equation (1.53) will yield similar results to the Sine condition. As in Figure 1.24, if the optical system has no aberration, an arbitrary ray proceeding from the object at Q will intercept the primary principal surface at H_1, and then intercept the secondary principal surface at H_1'. The ray H_1H_1' will always be parallel to the optical axis.

If the incident beam is collimated as shown in Figure 1.25, the primary principal surface will be a plane which is normal to the incident beam and is located at the intersection of the principal ray and the optical axis. In this case, the lines H_1H_1' and H_2H_2' in Figure 1.25 will always be parallel to the optical axis, as in Figure 1.24.

1.8 Aplanatic Lenses

A lens which is free from spherical and coma aberration is referred to as an aplanatic lens [3]. The theory of geometrical optics predicts that meniscus lenses can be used for configuring optical systems that have both a low aberration and a large NA (numerical aperture).

An aplanatic lens has one aplanatic surface (a sphere). In Figure 1.26a, if the object P and its image P′ are related to each other as in Equation (1.54a) and Equation (1.54b), then spherical aberration and

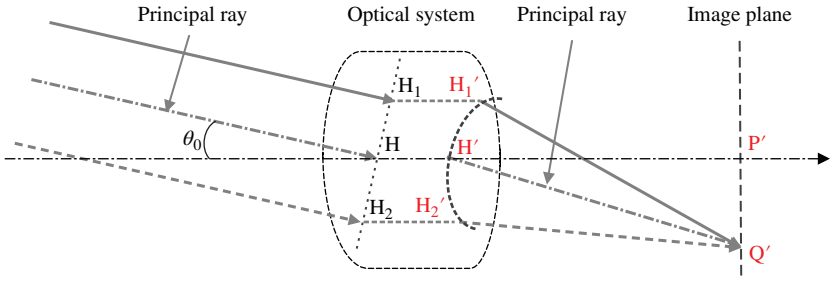


Figure 1.25 Principal surfaces for an off-axis object and its image (collimated beam incidence)

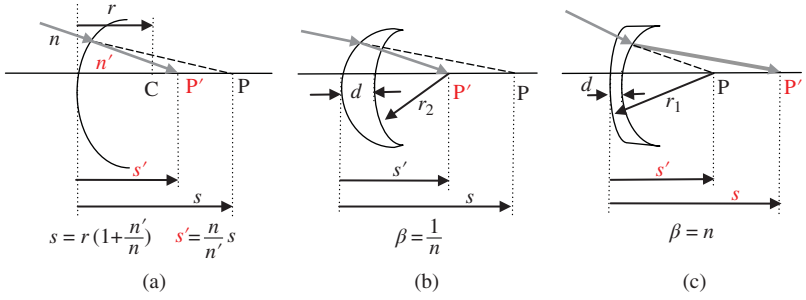


Figure 1.26 Aplanatic sphere and aplanatic lenses. (a) Aplanatic sphere, (b) convex lens, and (c) concave lens

coma aberration will both vanish.

$$s = r\left(1 + \frac{n'}{n}\right) \tag{1.54a}$$

$$s' = r\left(1 + \frac{n}{n'}\right) = \frac{n}{n'}s \tag{1.54b}$$

The lateral magnification β will then be:

$$\beta = \frac{s'/n'}{s/n} = \frac{s'}{s}\frac{n}{n'} = \left(\frac{n}{n'}\right)^2 \tag{1.54c}$$

Both convex and concave lenses can be regarded as aplanatic lenses which satisfy Equation (1.54a–c). An aplanatic lens surrounded by air can be described as follows:

- *Convex lens*: Let $n = 1$ and let $n' = n$. Then we can describe the first surface of the lens as an aplanatic sphere, while the second surface is a sphere centered at point P′, and having a radius of r_2. Let d denote the thickness of the lens (Figure 1.26b). The radii of the first and the second surfaces will then be:

$$r_1 = \frac{s}{n+1} \tag{1.55a}$$

$$r_2 = \frac{s}{n} - d \tag{1.55b}$$

$$\beta = \frac{1}{n} \quad \beta : \text{Lateral magnification} \tag{1.55c}$$

- *Concave lens*: Let $n = n$ and let $n' = 1$. Then we can describe the second surface as an aplanatic sphere, while the first surface is a sphere centered at P, and having a radius of r_1. Let d denote the thickness of the lens (Figure 1.26c). The radii of the first and the second surfaces will then be:

$$r_1 = s \tag{1.56a}$$

$$r_2 = (s - d)\frac{n}{(n+1)} \tag{1.56b}$$

$$\beta = n \quad \beta : \text{Lateral magnification} \tag{1.56c}$$

Aplanatic surfaces can reduce spherical aberration and coma to zero, no matter what the value of the NA. For this reason, they are very effective when used in optical systems with large NA, such as a microscope having both a large magnification and a large NA. However, aplanatic lenses always have a positive lateral magnification – that is, they form virtual images. For real-life applications, we need to use aplanatic lenses in combination with lenses that form actual images. We can use this combination to alter the focal length of an optical system, without increasing the aberration.

1.9 Reflection and Transmission

Reflectance and transmittance refer to the ratios of reflected and transmitted light to the total amount of incident light respectively, at the boundary between different media [2, 4]. Reflectance and transmittance values depend on the polarization angle. In this section, we will briefly examine how reflectance and transmittance vary in relation to the direction of polarization. A polarized ray can be thought of as being composed of two orthogonal linearly polarized elements: one parallel to the plane which contains the incident, reflected and refracted rays (p), and the other, perpendicular to that (s).

In Figure 1.27, A_i, A_r, and A_t represent the amplitudes of the incident wave, the reflected wave, and the transmitted wave, respectively. Each amplitude can be broken down into "p" and "s" components, where "p" denotes the polarization component which is parallel to the plane containing the incident, reflected, and refracted rays (i.e., parallel to the page), while "s" denotes the polarization component which is vertical to the plane described above (i.e., perpendicular to the page).

1.9.1 Angles of Reflection and Refraction

In Figure 1.27, the relationship between the angle of incidence θ_i, the angle of reflection θ_r, and the angle of transmission θ_t will be:

$$\theta_r = \pi - \theta_i \tag{1.1'}$$

$$n_1 \sin\theta_i = n_2 \sin\theta_t \tag{1.2}$$

where

n_1 = Refractive index of the medium through which the incident light propagates
n_2 = Refractive index of the medium through which the transmitted light propagates.

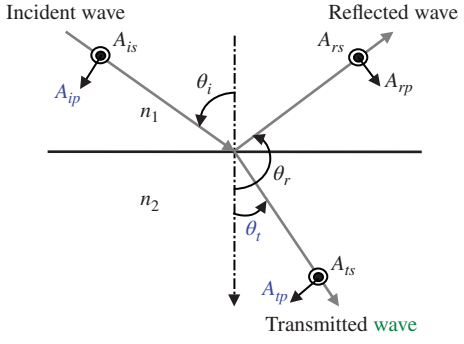


Figure 1.27 Reflection and refraction of light

1.9.2 Amplitude Reflection and Transmission Coefficients

In Figure 1.27, the amplitude reflection coefficients r_p, r_s and the amplitude transmission coefficients t_p, t_s will be as follows:

$$r_p \equiv \frac{A_{rp}}{A_{ip}} = \frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)} = \frac{n_2 \cos\theta_i - n_1 \cos\theta_t}{n_2 \cos\theta_i + n_1 \cos\theta_t} \tag{1.57}$$

$$r_s \equiv \frac{A_{rs}}{A_{is}} = -\frac{\sin(\theta_i - \theta_t)}{\sin(\theta_i + \theta_t)} = \frac{n_1 \cos\theta_i - n_2 \cos\theta_t}{n_1 \cos\theta_i + n_2 \cos\theta_t} \tag{1.58}$$

$$t_p \equiv \frac{A_{tp}}{A_{ip}} = \frac{2\sin\theta_t \cos\theta_i}{\sin(\theta_i + \theta_t)\cos(\theta_i - \theta_t)} = \frac{2n_1 \cos\theta_i}{n_2 \cos\theta_i + n_1 \cos\theta_t} \tag{1.59}$$

$$t_s \equiv \frac{A_{ts}}{A_{is}} = \frac{2\sin\theta_t \cos\theta_i}{\sin(\theta_i + \theta_t)} = \frac{2n_1 \cos\theta_i}{n_1 \cos\theta_i + n_2 \cos\theta_t} \tag{1.60}$$

1.9.3 Reflectance and Transmittance

The reflectance R and transmittance T are as follows:

$$R = \frac{|A_r|^2}{|A_i|^2} = |r|^2 \tag{1.61}$$

$$T = \frac{n_2 \cos\theta_t |A_t|^2}{n_1 \cos\theta_i |A_i|^2} = \frac{n_2 \cos\theta_t}{n_1 \cos\theta_i} |t|^2 \tag{1.62}$$

R can be simply expressed as the square of r; however, as we can see, the expression of T is not quite that simple. Another expression for T is:

$$T = 1 - R \tag{1.63}$$

The above equations are derived from the boundary condition. The theoretical explanation has been omitted here.

Calculations for r, t, R, and T in the case where a ray of light traveling through the air is incident upon a glass surface (assume that $n_1 = 1$, $n_2 = 1.5$) are shown in Table 1.1.

Table 1.1 Calculations for r, t, R, and T

$n_1 = 1$		$n_2 = 1.5$							
θ_i degrees	θ_t degrees	r_p	r_s	t_p	t_s	R_p	R_s	T_p	T_s
0	0	0.2	−0.2	0.8	0.8	0.04	0.04	0.96	0.96
5	3.3309802	0.1989814	−0.201018	0.7993209	0.7989818	0.0395936	0.0404083	0.9604064	0.9595917
10	6.6477771	0.1958864	−0.204107	0.7972576	0.7958934	0.0383715	0.0416595	0.9616285	0.9583405
15	9.9358837	0.190595	−0.209368	0.79373	0.7906317	0.0363265	0.0438351	0.9636735	0.9561649
20	13.180142	0.1828976	−0.216981	0.7885984	0.7830186	0.0334515	0.0470809	0.9665485	0.9529191
25	16.36441	0.1724799	−0.227208	0.7816533	0.7727919	0.0297493	0.0516235	0.9702507	0.9483765
30	19.471221	0.1588998	−0.240408	0.7725999	0.7595918	0.0252491	0.0577961	0.9747509	0.9422039
35	22.48145	0.1415526	−0.257058	0.7610351	0.7429421	0.0200371	0.0660788	0.9799629	0.9339212
40	25.373994	0.1196225	−0.277773	0.746415	0.7222272	0.0143095	0.0771577	0.9856905	0.9228423
45	28.125506	0.0920134	−0.303337	0.7280089	0.696663	0.0084665	0.0920134	0.9915335	0.9079866
50	30.710221	0.0572497	−0.334736	0.7048332	0.6652637	0.0032775	0.1120484	0.9967225	0.8879516
56.31	33.690098	−7.09E − 07	−0.384616	0.6666662	0.615384	5.031E − 13	0.1479295	1	0.8520705
60	35.26439	−0.042449	−0.420204	0.6383672	0.5797959	0.0018019	0.1765715	0.9981981	0.8234285
65	37.17167	−0.113857	−0.477559	0.5907621	0.5224406	0.0129634	0.228063	0.9870366	0.771937
70	38.789556	−0.206132	−0.547352	0.5292453	0.4526476	0.0424904	0.2995947	0.9575096	0.7004053
75	40.087037	−0.326749	−0.631946	0.4488338	0.3680538	0.1067651	0.399356	0.8932349	0.600644
80	41.036411	−0.486635	−0.73389	0.3422432	0.2661097	0.2368138	0.5385949	0.7631862	0.4614051
85	41.615601	−0.70232	−0.855772	0.1984531	0.1442281	0.4932538	0.7323455	0.5067462	0.2676545
90	41.810315	−1	−1	0	0	1	1	0	0

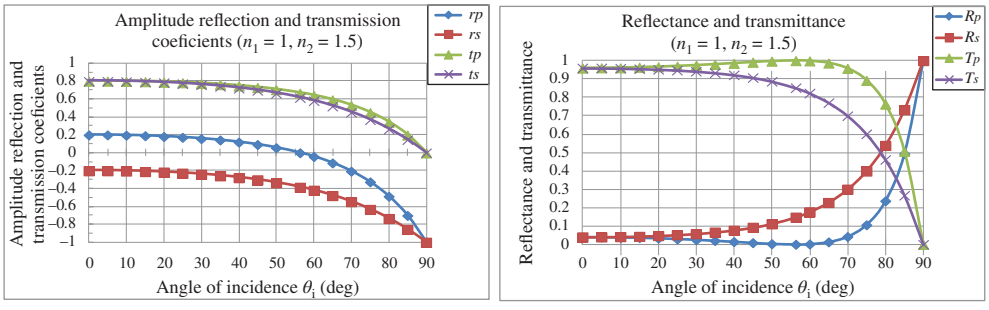


Figure 1.28 (a) Calculation results for r and t and (b) calculation results for R and T

The graphs in Figure 1.28 show that the amplitude reflection coefficients r_p, r_s, and the reflectances R_p, R_s of glass ($n = 1.5$) when surrounded by air will be as follows:

when $\theta_i = 0$, $R_p = 0.04$, $R_s = 0.04$, $r_p = 0.2$, $r_s = -0.2$ and
when $\theta_i = 56.31°$, $R_p = 0$, $R_s = 0.148$, $r_p = 0$, $r_s = 0.385$.

The results calculated above mean that when light is incident vertically on the glass, the reflectance will be 4%, and the amplitude reflection coefficient will be 20%.

When the angle of incidence is 56.31°, there will be no reflection for light rays polarized parallel to the incident plane, that is, there will be perfect (100%) transmission. This angle is referred to as Brewster's angle.

References

[1] Miyake, K. (1979) *Kika-kogaku*, Kyoritsu Publishing, Japan (in Japanese).
[2] Sagai, J. (1997) *Kogaku no Kiso*, Corona Publishing, Co. Ltd, Japan (in Japanese).
[3] Hayamizu, Y. (1989) *Hikari-kiki no kogaku I*, Optomechatronics Association, Japan (in Japanese).
[4] Suhara, T. (1998) *Koha-kogaku*, Corona Publishing Co. Ltd, Japan (in Japanese).

2

Examples of Simple Optical Design Using Paraxial Theory

In Chapter 1, we discussed the Gaussian lens formula $1/s' = 1/f + 1/s$ [Equation (1.24)], which is widely used in paraxial theory. This equation readily yields the position of the image in an optical system. In this chapter, we will investigate the practical applications of some simple optical design methods. First, we will look at the various types of lenses that are used in real-life imaging optical systems.

2.1 Types of Lenses

Lenses can be classified into simple lenses and composite lenses. They can also be classified according to shape (e.g., spherical lenses, aspherical lenses, Fresnel lenses, and rod lenses) and composition (glass, plastic, etc.). Lenses are usually designed to optimize certain desirable characteristics, such as a low level of aberration (whether chromatic or achromatic), high transmittance, an antireflection (AR) coating, and so on. In the field of optical design, it is very important to have a detailed understanding of lenses, in order to choose the most suitable kind of lens for a given practical application.

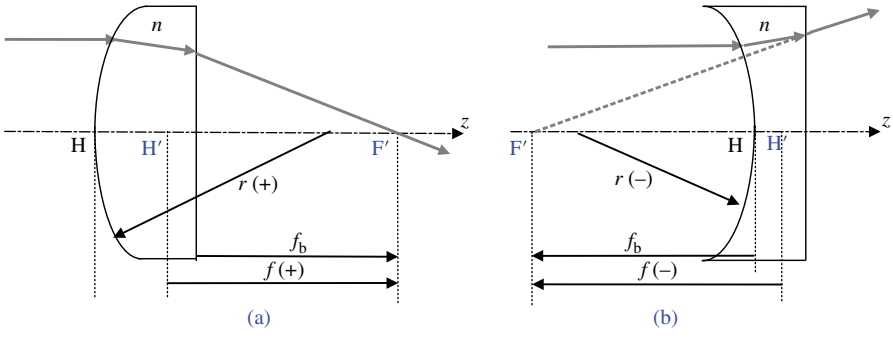
2.1.1 Plano-Convex Lens and Plano-Concave Lens

As shown in Figure 2.1a,b, plano-convex and plano-concave lenses have a spherical surface on one side, and a planar surface on the other side. In order to reduce the aberration for incident rays traveling parallel to the axis, it is advisable to set the orientation of each lens as depicted in Figure 2.1a,b, with the spherical surface at the front, and the planar surface at the back. The primary and secondary principal points are labeled H and H′, respectively, in Figure 2.1a,b. For a plano-convex or a plano-concave lens, the principal point on the spherical side lies on the vertex of the sphere.

2.1.2 Biconvex Lens

As shown in Figure 2.2, a biconvex lens is a lens which has spherical surfaces on both sides. Generally speaking, a biconvex lens has high refractive power; however, it also creates greater aberrations than a plano-convex lens. A lens with concave surfaces on both sides is called a biconcave lens.

Optical Design Using Excel®: Practical Calculations for Laser Optical Systems, First Edition. Hiroshi Nakajima.
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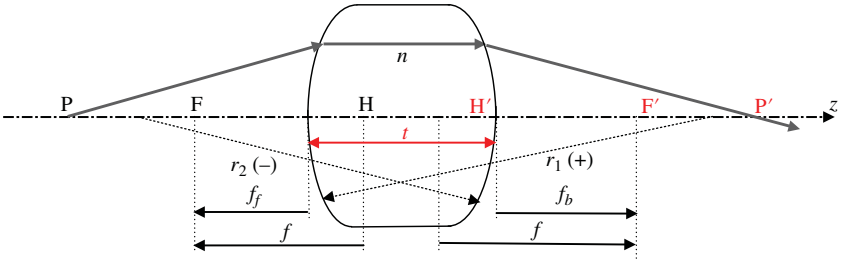


$$f = r/(n-1) \qquad (1.26')$$

where

f = Focal length f_b = Back focal length n = Refractive index
H = Primary principal point H′ = Secondary principal point
r = Curvature radius of the lens surface (positive for (a), negative for (b))

Figure 2.1 (a) Plano-convex lens and (b) plano-concave lens



$$\frac{1}{f} = (n-1)\left\{\frac{1}{r_1} - \frac{1}{r_2} + \frac{t\,(n-1)}{r_1 r_2 n}\right\} \qquad (1.34')$$

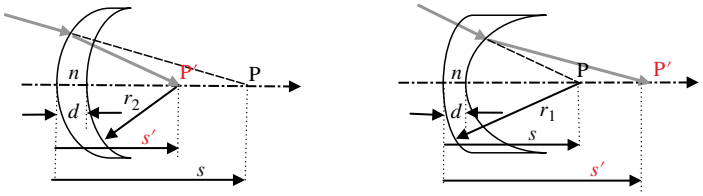
where

f = Focal length H = Primary principal point f_f = Front focal length
$r_1 > 0$, $r_2 < 0$: Radii of the spherical surfaces of the lens
H′ = Secondary principal point f_b = Back focal length
n = Refractive index t = Thickness of the lens (distance between the vertices)

Figure 2.2 Bi-convex lens

2.1.3 Meniscus Lens

As shown in Figure 2.3a,b, a meniscus lens is a lens which has a convex surface on one side, and a concave surface on the other side. A positive meniscus lens is thicker in the middle than at the edges and serves as a convex lens, while a negative meniscus lens thickens toward the edges and serves as a diverging or concave lens. Generally, this kind of lens is used as an aplanatic lens, which can alter the focal length of a system without causing an increase in either spherical aberration or coma aberration.



$$r_1 = \frac{s}{n+1} \quad (1.55a) \qquad r_2 = \frac{s}{n} - d \quad (1.55b) \qquad \beta = \frac{1}{n} \quad (1.55c) \qquad \text{for a positive lens}$$

$$r_1 = s \quad (1.56a) \qquad r_2 = (s-d)\,\frac{n}{(n+1)} \quad (1.56b) \qquad \beta = n \quad (1.56c) \qquad \text{for a negative lens}$$

where

P = Object point P′ = Image point s = Distance to object s' = Distance to image

d = Thickness of lens $r_1, r_2 > 0$ are the radii of the spherical surfaces of the lenses

n = Refractive index

Figure 2.3 (a) Positive meniscus lens and (b) negative meniscus lens

This lens has to be used in combination with other lenses to make a real image. In Figure 2.3a,b, we can alter the focal length of the system from s to s' without causing an increase in either spherical aberration or coma aberration.

2.1.4 Cylindrical Lens

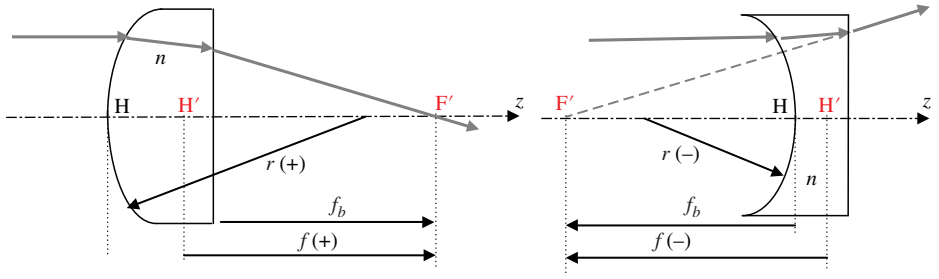
Because of its shape, a cylindrical lens can refract light only in one direction – along either the x–z plane or the y–z plane, which is perpendicular to the cylinder axis. Generally, this kind of lens has a cylindrical surface on one side and a planar surface on the other side (Figure 2.4a,b).

2.1.5 Achromatic Lens

As shown in Figure 2.5, an achromatic lens is a lens that corrects for chromatic aberration by using a combination of two lenses made of different kinds of glass, whose dispersions cancel each other out. This lens has a lower degree of aberration than other kinds of lenses. This is true not only for chromatic aberration, but also for spherical aberration.

2.1.6 Aspheric Lens

As shown in Figure 2.6, an aspheric lens is a lens whose surface profile is symmetrical around the optical axis, but not a portion of a sphere. An aspheric lens's more complex surface can eliminate spherical aberration and also reduce other forms of aberration, when compared with a simple lens. A single aspheric lens can often replace a much more complex multi-lens system. The resulting device is smaller and lighter, and possibly cheaper than the multi-lens design. Recently, aspheric lenses have been used in a wider variety of applications, such as eyeglasses, laser diode (LD) beam collimation or focusing, and condenser lenses for illumination. Figure 2.6 depicts two applications: these lenses can be used to generate a collimated beam from a point light source, or conversely, to focus a collimated beam onto a very tiny area.



$$f = r / (n-1) \qquad (1.26')$$

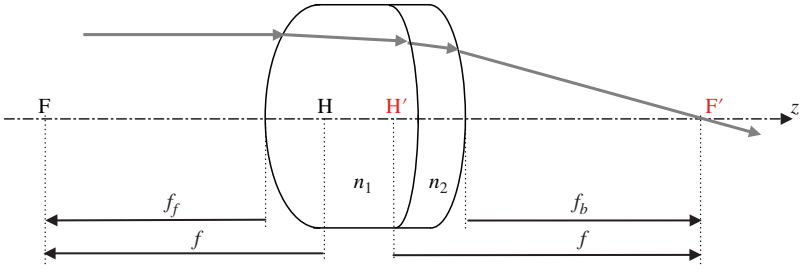
where

f = Focal length n = Refractive index

f_b = Back focal length H = Primary principal point H′ = Secondary principal point

r = Curvature radius of the lens surface (positive for Fig. 2.6, negative for Fig. 2.7)

Figure 2.4 (a) Cylindrical convex lens and (b) cylindrical concave lens



where

f = Focal length, f_f = Front focal length, f_b = Back focal length,

F = First focal point, F′ = Second focal point,

H = Primary principal point, H′ = Secondary principal point

n_1 = Refractive index of the first medium n_2 = Refractive index of the second medium

Figure 2.5 Achromatic lens

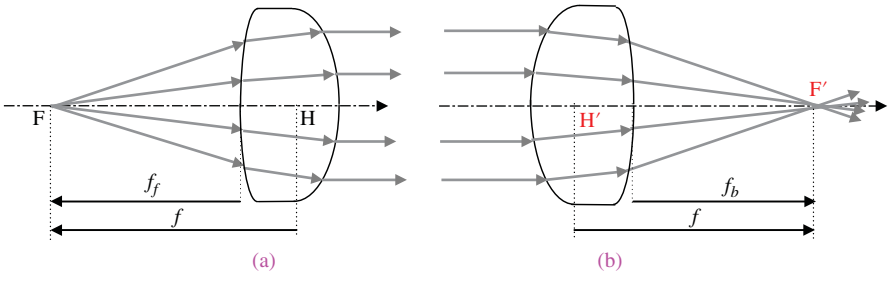


Figure 2.6 Aspheric lens. (a) Collimating and (b) focusing

2.1.7 Microscope Objective Lens

Figure 2.7 shows an optical system which uses a microscope objective lens and an imaging lens. The conjugate point of the objective lens lies at infinity. The lateral magnification β in the figure will then be:

$$\beta = -f_i/f_o$$

When using a microscope, we observe the image by using a lens to magnify the image generated by the objective lens. Usually, the conjugate point of an objective lens lies at infinity. Thus rays that originate at the front focal point of the objective lens, are collimated after passing through the objective lens, and are then focused onto the image sensor by the imaging lens. In general, microscope objective lenses have very small aberrations, which are minimized to the diffraction limit. Microscope objective lenses are used in a wide range of high-precision optical systems.

Numerical aperture

The numerical aperture (NA) denotes a dimensionless number that characterizes the range of angles over which the system can accept or emit light (Figure 2.8). NA is defined in Equation (2.1), and is typically used in relation to microscope objective lenses. In microscopy, NA is important because it indicates the resolving power of a lens. The resolution δ (the smallest resolvable distance between two objects) is defined as in Equation (2.2). (The coefficient 0.61 is a constant multiplication factor.)

$$NA = n \sin\theta \tag{2.1}$$

$$\delta = \frac{0.61\lambda}{NA} \tag{2.2}$$

where

n = Refractive index of the medium surrounding the object P
θ = Half-angle of the maximum cone of light that can enter (or exit) the lens
λ = Wavelength of the light being imaged.

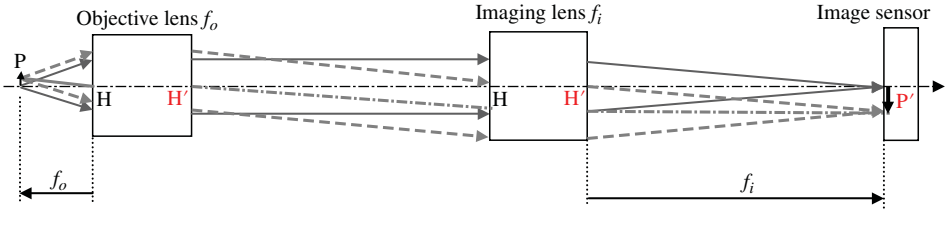


Figure 2.7 An imaging system using a microscope objective lens

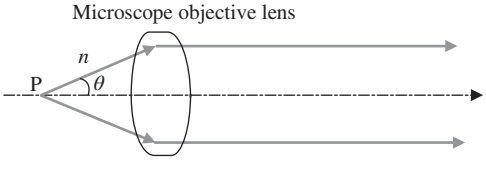


Figure 2.8 Numerical aperture

2.1.8 Camera Lens

Usually, camera lenses are manufactured with aberration compensation suitable for an object located at an infinite distance or an object which forms an image with a lateral magnification between 1/40 and 1/50. For these reasons, great care needs to be taken when positioning a camera lens in practical applications. Figure 2.9 illustrates a beam expander, consisting of a combination of a microscope objective lens and a camera lens designed for experimental use. The camera lens's second focal point should be set to coincide with the objective point of the microscope objective lens.

F-number

When an observer looks inside a camera lens from the front, a diaphragm aperture can be seen. However, the apparent size of the aperture when viewed from the front is different from the actual size of the diaphragm aperture. The apparent size of the aperture when viewed from the front is referred to as the entrance pupil, while the apparent size when viewed from the back (the film side) is referred to as the exit pupil.

We can now define D in Figure 2.10 as the diameter of the entrance pupil (or effective diameter), while the F-number of the camera is defined by Equation (2.3). The F-number gives us a rough idea of the amount of exposure required when taking a photo, as well as the depth of focus.

$$\text{F-number} = f/D \tag{2.3}$$

2.1.9 f-θ Lens

An f-θ lens is typically used for laser beam scanning, where a rotating polygonal mirror scans the laser beam. In Figure 2.11 the beam reflected by the mirror travels at a constant angular velocity, which depends on the rotation speed of the mirror. An f-θ lens is a lens designed so as to position the scanning spot at a height h which is proportional to the incident beam angle, θ (Figure 2.11).

$$h \approx f\theta \tag{2.4}$$

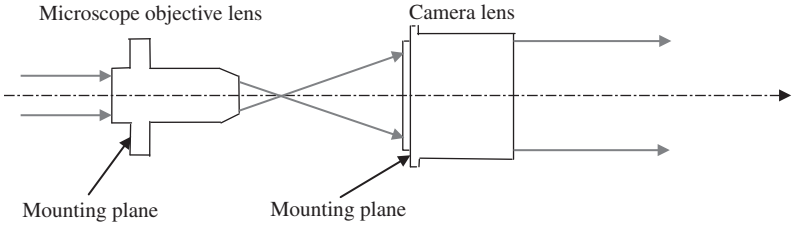


Figure 2.9 A beam expander consisting of a microscope objective lens and a camera lens

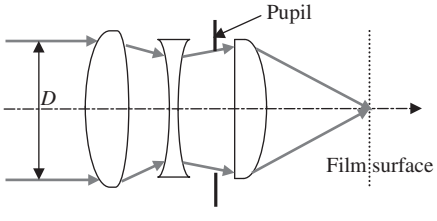


Figure 2.10 Entrance pupil of a camera

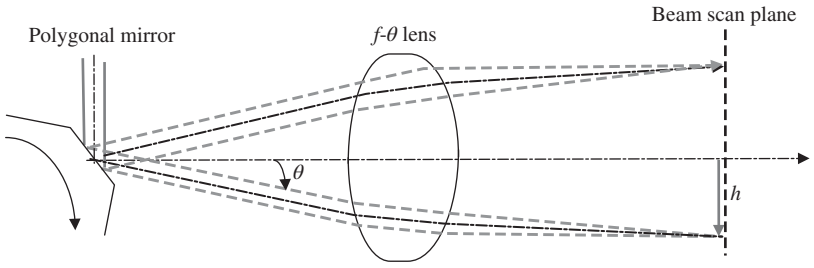


Figure 2.11 Beam scanning using an *f-θ* lens

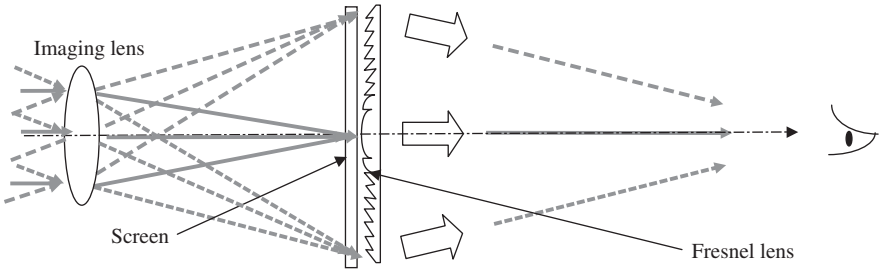


Figure 2.12 An example of a Fresnel lens application

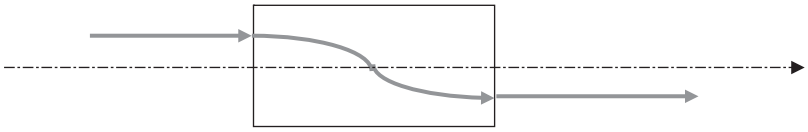


Figure 2.13 Rod lens

2.1.10 Fresnel Lens

A Fresnel lens requires a smaller amount of material than a conventional spherical lens, making it lighter in weight, smaller, and more economical. It achieves this by breaking up the surface of a conventional lens into a set of concentric annular sections. For each of these zones, the overall thickness of the lens is decreased, effectively chopping the continuous surface of a standard lens into a set of surfaces with the same curvature, but separated by discontinuities lying between them. Figure 2.12 shows an example of how this kind of lens might be used in a projection TV display screen, to increase peripheral brightness when the screen is viewed from directly in front.

2.1.11 Rod Lens

As shown in Figure 2.13, a rod lens is shaped like a cylinder, whose refractive index n gradually decreases from the central axis to the periphery, and whose front and back surfaces are planar. Hence, an incident ray parallel to the axis will not travel in a straight line, but will travel forwards along a curved path. Photocopying machines are a familiar example of a practical application of rod lenses. In photocopying machines, arrays of rod lenses are used to copy images onto a photosensitive drum.

2.2 Applied Calculations for Simple Optical Systems

Problem 2.1 An imaging system using a camera lens

Using a camera lens, configure an optical imaging system having a lateral magnification of $\beta = -1/40$ in which an object of height $w_0 = 400$ mm, standing approximately 2 m in front of the lens, will form an inverted image of height $w = -10$ mm.

Answer. Focal length $f = 50$ mm, Object distance $s = -2050$ mm, Image distance $s' = 51.25$ mm.

Solution

After making the calculations as shown below, we can configure the imaging system using a camera lens as in Figure 2.14.

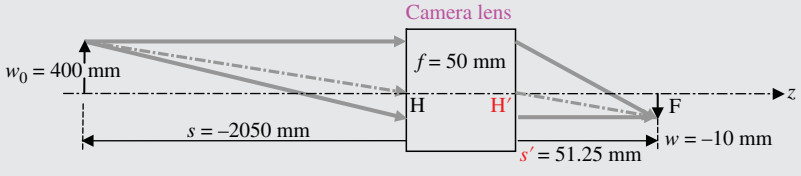


Figure 2.14 An imaging system using a camera lens

Using the Gaussian lens formula and lateral magnification equation from Chapter 1,

$$1/f = 1/s' - 1/s \tag{1.24'}$$

$$\beta = s'/s \tag{1.27'}$$

we can substitute the following values of s and β into the formula:

$s = -2000$ mm (where the minus sign denotes an object in front of the lens)
$\beta = -1/40$ (where the minus sign denotes an inverted image).

We can then derive s' and f:

$$s' = s\beta = 50 \text{ mm} \quad f = 1/(1/s' - 1/s) = 48.78 \text{ mm}$$

In real life, people commonly purchase camera lenses with a focal length $f = 50$ mm, which is very similar to the calculated value $f = 48.78$ mm. Using Equation (1.24′) and Equation (1.27′), and substituting the values $f = 50$ mm and $\beta = s'/s = -1/40$, yields the result:

$$s' = 51.25 \text{ mm} \quad s = -2050 \text{ mm}$$

Note

Camera lenses are suitable for this kind of application, because their object distance and lateral magnification are typically similar to the values specified in this example.

Problem 2.2 An imaging system using a microscope objective lens

Using a microscope objective lens with a focal length of $f = 4$ mm, configure an optical imaging system having a lateral magnification of $\beta = -50$, in which an object of height $w_0 = 0.1$ mm will form an inverted image of height $w = -5$ mm.

Answer. Focal length $f = 4$ mm, Object distance $s = -4.08$ mm, Image distance $s' = 204$ mm.

Solution

After making the calculations as shown below, we can configure the imaging system using a microscope objective lens as in Figure 2.15.

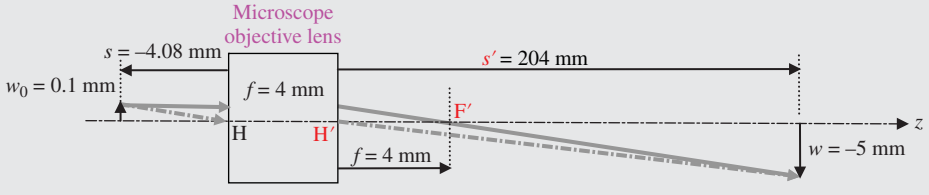


Figure 2.15 An imaging system using a microscope objective lens

Using the Gaussian lens formula and the lateral magnification equation from Chapter 1,

$$1/s' = 1/f + 1/s \tag{1.24}$$

$$\beta = s'/s \tag{1.27′}$$

we can substitute the following values of β and f into the formula:

$\beta = -50$ (where the minus sign denotes an inverted image), and $f = 4$ mm.

From Equation (1.27′) and Equation (1.24), we obtain:

$$s' = s\,\beta = -50\,s \quad 1/s' = 1/(-50\,s) = 1/4 + 1/s$$

This yields the result:

$$s = -4.08 \text{ mm} \quad s' = 204 \text{ mm}$$

Note

The above calculation yields an object distance of $s = -4.08$ mm, which is very similar to the normal value for microscope objective lenses of $f = 4$ mm, so we can expect this setting to have ideal aberration characteristics.

Problem 2.3 An imaging system using a microscope objective lens and an imaging lens

Using a microscope objective lens L_1 ($f_1 = 4$ mm) and an imaging lens L_2 ($f_2 = 200$ mm), configure an optical imaging system with a lateral magnification of $\beta = -50$.

Answer. Object distance $s_1 = -4$ mm, Image distance $s_2' = 200$ mm, Lateral magnification $\beta = -50$.

Solution

After making the calculations as shown below, we can configure the imaging system using a microscope objective lens and an imaging lens as in Figure 2.16.

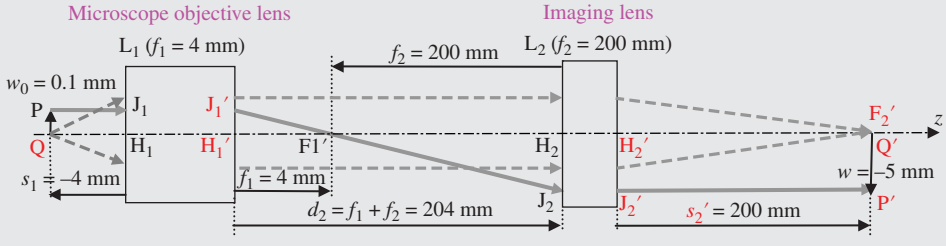


Figure 2.16 An imaging system using a microscope objective lens and an imaging lens

In Figure 2.16, we can define s_1 and d_2 as follows:

$$s_1 = -f_1 = -4 \text{ mm}, \qquad d_2 = f_1 + f_2 = 204 \text{ mm}$$

Rays emitted from the object at point Q, which lies on the axis ($s_1 = -4$ mm), will travel parallel to the axis after passing through L_1 and will finally converge at point Q′ (the focal point of L_2), after passing through L_2. A ray emitted from the object at point P and traveling parallel to the axis will follow path $J_1, J_1', F_1', J_2, J_2'$ and will then travel parallel to the axis after passing through L_2 before ending up at the image point P′. In other words, the object PQ on the object plane will generate an image P′Q′ on the image plane.

$$w_0 = \text{QP} = \text{H}_1'\text{J}_1', \quad w = \text{Q}'\text{P}' = \text{H}_2\text{J}_2, \quad \Delta\text{F}_1'\text{H}_1'\text{J}_1' \propto \Delta\,\text{F}_1'\text{H}_2\text{J}_2$$

From the similarity between $\Delta F_1' H_1' J_1'$ and $\Delta F_1' H_2 J_2$, we can derive the following result:

$$\beta = w/w_0 = -f_2/f_1 = -50$$

The optical imaging system in Figure 2.16 has a lateral magnification of $\beta = -50$. Even if we assign d_2 a different value from that used above, the optical imaging system will still have the same value of $\beta = -50$.

Using the Gaussian lens formula $1/s' = 1/f + 1/s$ and lateral magnification equation $\beta = s'/s$ from Chapter 1, we can calculate the lateral magnifications β_1 and β_2 for the lenses L_1 and L_2 individually.

$$s_1' = s_1 f_1/(s_1 + f_1) \quad \therefore \ \beta_1 = s_1'/s_1 = f_1/(s_1 + f_1)$$
$$s_2' = s_2 f_2/(s_2 + f_2) \quad \therefore \ \beta_2 = s_2'/s_2 = f_2/(s_2 + f_2)$$

The overall lateral magnification β can be calculated as follows,

$$\beta = \beta_1 \beta_2 = f_1 f_2/[(s_1 + f_1)(s_2 + f_2)] \tag{2.5}$$

Here, we can show that

$$s_2 = -(d_2 - s_1') \tag{2.6}$$

where d_2 is the distance between the lenses.

Note

We can expect similar performances from the imaging systems described in Problems 2.2 and 2.3. Problem 2.2 has a simpler configuration than Problem 2.3; however, Problem 2.3 has greater versatility and more degrees of freedom in its design.

Problem 2.4 Correcting for the insertion of a glass plate into an imaging system

When a glass plate of thickness t and refractive index n is inserted into an optical imaging system, it will be necessary to adjust the lens positions behind the glass plate to compensate for the defocus. Calculate the corrected value, using paraxial theory.

Answer. After the glass plate has been inserted, it will be necessary to move the lenses behind the glass plate back by a distance $\Delta d = t(1 - 1/n)$.

Solution

In Figure 2.17, the glass plate causes us to mistakenly perceive rays emitted from the object at point P as if they had come from point P′ instead. It will therefore be necessary to move the lenses backwards after the glass plate has been inserted, by a distance $\Delta d = \mathrm{PP}'$. Using Figure 2.17, distance Δd can be calculated as follows:

$$\Delta d = t - ((w_2 - w_1)/\tan\theta_i) = t - (t \tan\theta_t/\tan\theta_i) = t(1 - \tan\theta_t/\tan\theta_i) \tag{2.7}$$

In paraxial theory, we assume that θ_i has a low value, close to zero. Applying the refraction equation, we obtain:

$$\tan\theta_t / \tan\theta_i \approx \sin\theta_t / \sin\theta_i \approx 1/n \tag{2.8}$$

If we substitute Equation (2.8) into Equation (2.7), then we obtain the following result for Δd:

$$\Delta d = t(1 - 1/n) \tag{2.9}$$

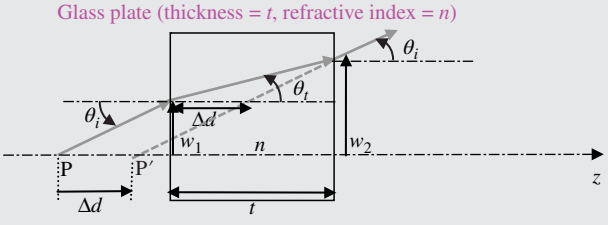


Figure 2.17 Insertion of a glass plate into an imaging system

Problem 2.5 Laser diode beam collimation

Configure an optical system to obtain a collimated beam with a diameter $D_\perp = \sim 4.2$ mm and $D_\parallel = \sim 1.4$ mm (at half maximum intensity) using a laser diode (LD) point source with a radiation angle of $\theta_\perp = 12°$ and $\theta_{||} = 4°$ (half angle at half maximum intensity).

The $_\perp$ sign in $\theta_\perp$ and $D_\perp$ denotes the direction perpendicular to the junction plane of the diode element.

The $_\parallel$ sign in $\theta_\parallel$ and $D_\parallel$ denotes the direction parallel to the junction plane of the diode element.

Solution

In Figure 2.18, if we set d_0 to equal the value of f, then rays emitted from the laser diode will travel parallel to the axis after passing through the lens, and a ray with a radiation angle θ emitted from the LD will travel parallel to the axis at a height w after passing through the lens.

$$\text{If } d_0 = f \quad \text{then} \quad w = f\sin\theta$$

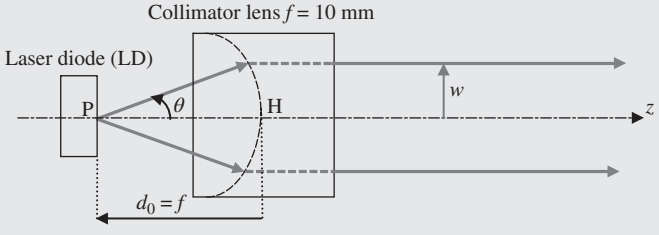


Figure 2.18 Laser diode beam collimation

The focal length f of the collimator lens should be:

$$f = w/\sin\theta$$

If we substitute the values $w_{\perp} = 2.1$ mm, $w_{||} = 0.7$ mm (radius of the beam), $\theta_{\perp} = 12°, \theta_{||} = 4°$ (half angle at half maximum intensity) into the above equation, then we obtain the following:

$$f_{\perp} = 10.10044 \text{ mm}, \quad f_{||} = 10.03491 \text{ mm}$$

This is only a rough calculation, which ignores the change in light intensity that occurs when the diverging beam with a ray angle of θ becomes a parallel beam.

In real-life applications, we would normally use a lens with a focal length $f = 10$ mm, which is close to the value specified above. In such a case, the collimated beam diameter $D_{\perp}$ and $D_{\|}$ will be as follows:

$$D_{\perp} = 2w_{\perp} = 2f\sin\theta_{\perp} = 4.15823 \text{ mm}, \; D_{||} = 2w_{||} = 2f\sin\theta_{||} = 1.39513 \text{ mm}$$

(diameter at half maximum intensity)

Answer. The focal length of the lens: $f = 10$ mm,

The distance between the LD and the lens: $d_0 = 10$ mm

Notes

1. To achieve greater precision, it is better to use an aspheric lens, a combination lens or a microscope objective lens.
2. The lens should have a large value for its NA, that is, about 0.5 or more.

Problem 2.6 Beam expander using cylindrical lenses

Configure an optical system which generates a collimated circular beam from the collimated oval beam described in Problem 2.5 ($D_{\perp} = 4.2$ mm, $D_{||} = 1.4$ mm), expanding the diameter of the beam by a factor of three, but only in a parallel direction.

Solution 1

Using a combination of a cylindrical-concave lens with a focal length f_1 and a cylindrical-convex lens with a focal length f_2 as shown in Figure 2.19, we can define f_2 and d_2 as follows:

$$f_2 = -3f_1, \qquad d_2 = f_2 + f_1$$

We can generate an expanded collimated beam from an incident collimated beam. The diameter of the beam in a parallel direction after traveling a distance d_2 and passing through the second lens will be three times greater than that of the incident beam.

Answer. $f_1 = -50$ mm, $f_2 = 150$ mm, $d_2 = 100$ mm or $f_1 = -100$ mm, $f_2 = 300$ mm, $d_2 = 200$ mm

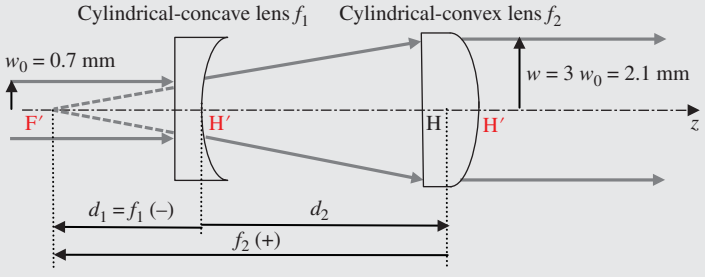


Figure 2.19 Beam expander using cylindrical lenses

Notes

1. In this configuration, the concave lens's negative aberration and the convex lens's positive aberration compensate for each other, so the total aberration is comparatively low.
2. In this configuration, the lenses need to be precisely aligned, so that they both rotate around the axis at the same angle; otherwise significant aberrations will be generated.

Solution 2

A combination of two prisms can convert a beam from an oval beam to a circular beam. Beam converters such as the one shown in Figure 2.20 are commercially available.

Note

In theory, a beam converter such as described above should not create any optical aberration. In practice, however, the shifting of the optical axis needs to be taken into account.

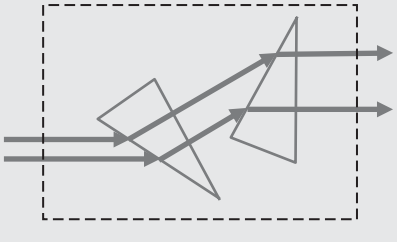


Figure 2.20 Beam converter (oval beam to circular beam)

Problem 2.7 Beam expander

Configure a beam expander which expands an incident collimated beam by a factor of three from 1.4 mm to 4.2 mm in diameter.

Solution 1

In the configuration in Figure 2.21, which consists of a microscope objective lens f_1 and a camera lens f_2, if we select a camera lens whose focal length f_2 is equal to $3f_1$, we can generate an expanded beam whose diameter is three times greater than that of the incident beam. Taking into account the range of lenses that are commercially available at the time of writing, either of the following selections would be satisfactory for our purposes.

$$f_1 = 20 \text{ mm}, \ f_2 = 60 \text{ mm} \ (M = 3) \text{ or } f_1 = 18 \text{ mm}, \ f_2 = 55 \text{ mm} \ (M = 3.06)$$

Answer. $f_1 = 20$ mm, $f_2 = 60$ mm ($M = 3$) or $f_1 = 18$ mm, $f_2 = 55$ mm ($M = 3.06$)

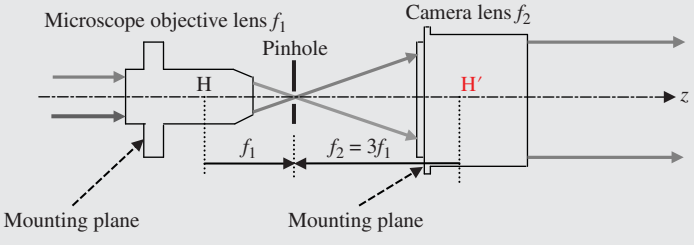


Figure 2.21 Beam expander using a microscope objective lens and a camera lens

Notes

1. For the configuration described above, the intensity of the beam will be somewhat diminished after passing through multiple lenses.
2. Another alternative would be to purchase a beam expander meeting the above requirements, which is designed for a specific purpose.

Solution 2

As in Figure 2.22, a combination of a plano-concave lens f_1 and a plano-convex lens f_2 can also be used to configure a beam expander. If we set the focal length f_2 to equal $-3f_1$, we will obtain an expanded beam whose diameter is three times greater than that of the incident beam. In real-life applications, we also need to consider aberrations and other factors affecting the performance of the optical system, when selecting the focal length. In our example, either of the following combinations of focal lengths would be suitable.

Answer. $f_1 = -50$ mm, $f_2 = 150$ mm, or $f_1 = -100$ mm, $f_2 = 300$ mm

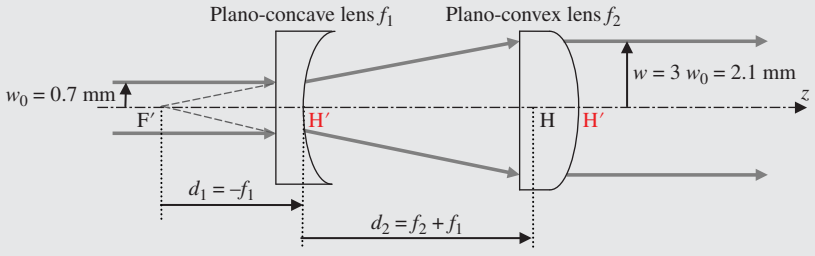


Figure 2.22 Beam expander using a plano-concave lens and a plano-convex lens

We will also need to make sure that the aberration falls within an acceptable range. The aberration can be calculated, using a ray trace calculation (see Chapters 4 and 5).

2.3 Considerations Relating to the Design of Laser Optical Systems

In this section we will investigate some of the factors which need to be taken into consideration when designing laser optical systems.

2.3.1 Design Safety

Laser beams have a very high intensity. Directly exposing the human body to a laser beam may have harmful consequences. When designing devices or systems equipped with lasers, various safety considerations need to be taken into account. Most importantly, these devices should never leak any radiation, even when they malfunction or are not being operated in accordance with specifications. If any leakage does occur, it must be kept below the legally specified limit. There are international safety standards relating to lasers. Devices that are equipped with lasers must comply with the following safety standards in developed countries:

USA: FDA (Food and Drug Administration) 21 CFR 1040
Europe: CE Mark EN60825 [equivalent to IEC68025 (International Standard)]
Japan: JIS (Japan Industrial Standard) JIS C 6802.

Designers of laser optical systems need to be thoroughly familiar with these laser safety standards.

2.3.2 Polarization

Usually laser beams are linearly polarized at a specific angle. When a laser beam is incident upon a boundary surface between different media, the irradiance (or power) of the transmitted beam and the reflected beam will vary depending on the polarization angle (see Section 1.9). The polarization angle of the laser therefore needs to be taken into careful consideration when designing laser systems.

2.3.3 *Antireflection Coatings*

As discussed in Section 1.9, the reflectance of glass is about 4% for vertically incident rays. Sometimes rays reflected from the surface may have adverse effects on optical systems, such as interference. (Even if the reflectance is only 4%, this still corresponds to an amplitude reflection coefficient of 20%. Reflected rays can thus generate interference relatively easily, which may have unanticipated negative consequences for optical systems.) Additionally, reflection from multiple surfaces will reduce overall light intensity. In order to minimize these negative consequences, all surfaces of lenses, prisms, and glass plates will need to be coated with optimized antireflection (AR) coatings.

2.3.4 *Problems Caused by Interference of Light*

A laser can easily generate interference, which can have adverse effects on optical systems. Interference therefore needs to be taken into account, when designing laser optical systems. The following examples illustrate how interference from lasers can negatively affect optical systems.

2.3.4.1 Destructive Interference Caused by Beams Reflected between Two Concave Surfaces, Two Plane Surfaces or between a Concave Surface and a Plane Surface

Consider the case of an optical imaging system with a concave surface L_1 and a surface L_2 which may be either a plane or a concave surface, as shown in Figure 2.23. If the incident ray A is reflected from the surface L_2, it will travel back to L_1 and be reflected so that it ends up at point B′ on the image plane, which is exactly where the incident ray B arrives. This will generate destructive interference at B′ between the incident ray B and the reflected ray A, which may obstruct image sensing. In order to avoid this outcome, reduce the reflectance of each surface, both surfaces will need to be coated with AR coatings. If, on the other hand, the surfaces are convex–convex, interference is not likely to occur, because the reflected rays will diverge from the imaging rays, and will therefore probably not interfere with the original image.

2.3.4.2 Interference Caused by Beams Reflected from the Front and Rear Surfaces of a Plate Beam Splitter

In Figure 2.24a, there are two beams that travel to the image sensor plane: one which is reflected by the front surface of a plate beam splitter, and another which is reflected by the rear surface. This may result in interference on the image sensor plane between the two beams. One simple way to avoid this problem is to change the shape of the plate to a wedge shape, so that the beam reflected from the back surface bypasses the image sensor and does not cause interference (Figure 2.24b).

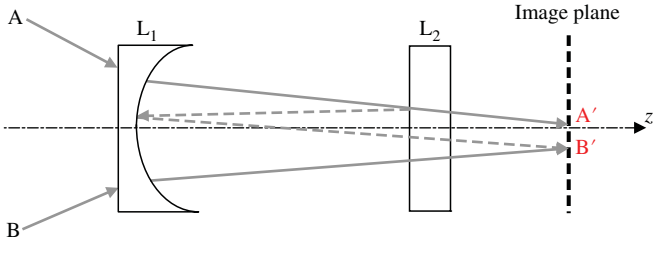


Figure 2.23 Interference between a concave surface and a plane surface

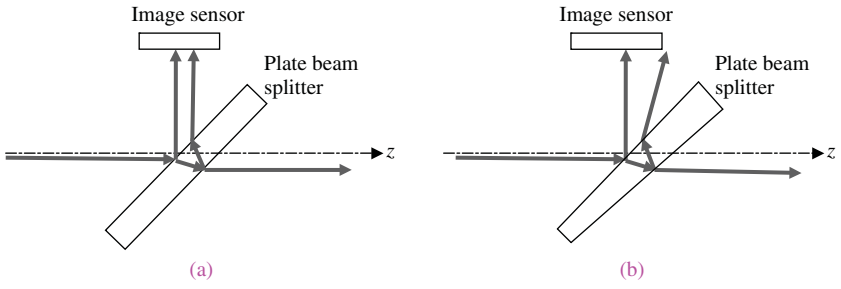


Figure 2.24 Interference caused by a plate beam splitter. (a) Parallel plate beam splitter and (b) wedge shape beam splitter

2.3.4.3 Surface Measurement of a Transparent Object

When making a surface measurement of a transparent object as in Figure 2.25, we need to take into account the beam reflected by the rear surface. Unfortunately, the beam reflected by the rear surface can never be entirely eliminated, even if the beam is focused directly upon the front surface. Hence the beams reflected by the front and rear surfaces may destructively interfere with each other. The beam reflected by the rear surface always needs to be taken into account in optical design applications.

2.3.4.4 Laser Diode Collimator Lenses

In the case where a LD collimator lens has a concave surface curving toward the LD, whose center of curvature coincides with the radiation point as shown in Figure 2.26, the reflected beam on the concave surface will return to the radiation point and interfere with the normal operation of the system. With this kind of application, the AR coating on the lens needs to be optimized, in order to avoid this problem.

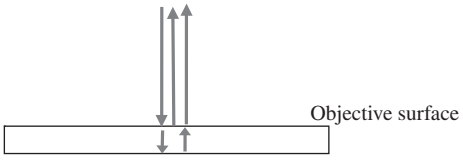


Figure 2.25 Interference between beams reflected by the front and rear surfaces.

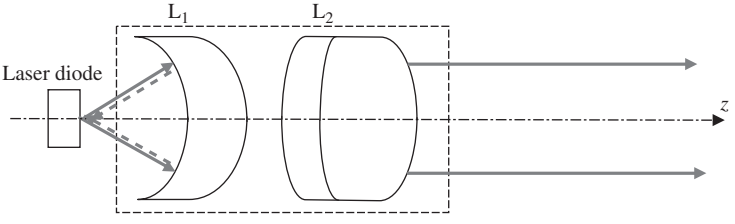


Figure 2.26 Interference caused by rays reflected by the first surface

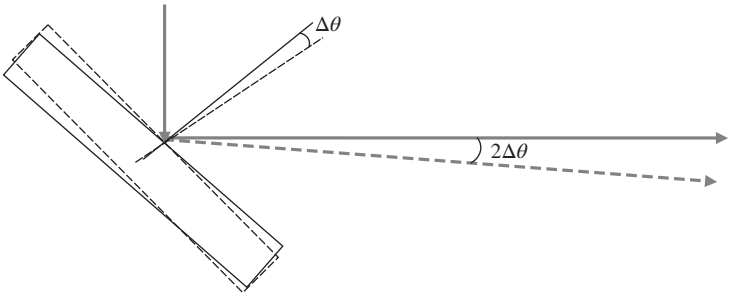


Figure 2.27 Change in the path of a beam reflected by a mirror

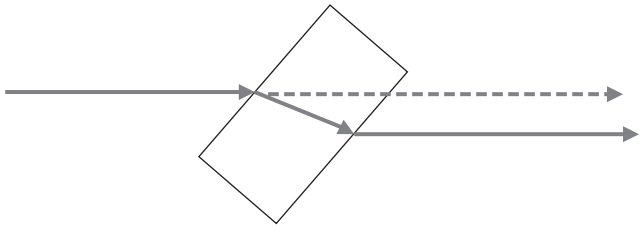


Figure 2.28 Beam shift using a prism

2.3.5 Mirrors

When the angle of a mirror changes by $\Delta\theta$, the beam reflected by the mirror will undergo a change of direction of $2\Delta\theta$ (Figure 2.27). If the beam travels a distance r after being reflected, the beam will diverge by $2r\Delta\theta$ from its original position. The further the reflected beam has to travel, the greater the divergence will be. The following factors therefore need to be taken into consideration, when using mirrors in optical systems:

1. Mirrors should be thick enough that they will not warp. Otherwise, a flat mirror may end up turning into a concave or a convex mirror, due to warping.
2. Mirrors should not be installed on a moving base. The vibration of a mirror will cause vibrations in the reflected beam. The further the reflected beam has to travel, the larger the amplitude of the vibration will be. If it is necessary to alter the path of a beam, a prism can be used instead of a mirror reflection as shown in Figure 2.28.

2.3.6 Considerations Relating to a Combination of Cylindrical Lenses

In a laser system consisting of two or more cylindrical lenses, the lenses need to be precisely aligned, so that they both rotate around the axis at the same angle (or at a right angle); otherwise significant aberrations will appear in the optical imaging system (see Chapter 5).

2.3.7 Reducing or Eliminating Stray Rays

In an application designed for highly sensitive signal detection, it is necessary to collect as much signal light as possible, and reduce any stray light (noise light) to a minimum level, so as to keep the

signal-to-noise ratio high. In other words, optical systems should be placed in a dark environment, and stray rays should be reduced or deflected. The following examples illustrate some methods for reducing or deflecting stray rays entering the sensor.

2.3.7.1 Beam Trap

Sometimes rays originating from an optical system are reflected by the walls and return to the sensor as stray rays. This increases the noise level in signal detection. To avoid this problem, stray beams need to be eliminated. Figure 2.29 shows one kind of beam trap which eliminates any rays that enter, by causing them to be reflected so many times by the horn-shaped interior wall that their intensity is effectively reduced to zero. The outside of the glass should either be coated black or covered in black tape.

2.3.7.2 Side Walls of Optical System Components

Rays reflected by the side wall of a lens may generate stray light, which has the undesirable consequence of increasing the level of noise in signal detection. In order to reduce reflection of rays, the surface of the side wall needs to be rough, and should be covered with a black coating. The coating material should have a high refractive index; otherwise, the rays hitting the side wall will be totally reflected and will not be reduced at all (Figure 2.30).

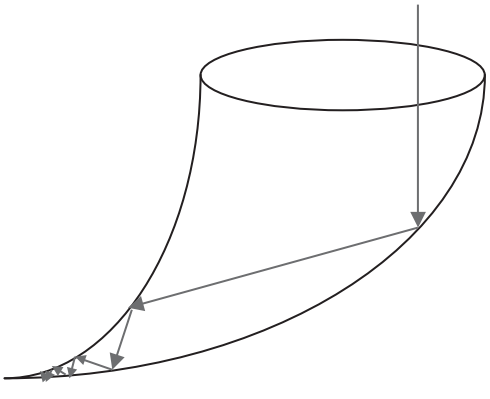


Figure 2.29 A beam trap using a horn-shaped hollow piece of glass

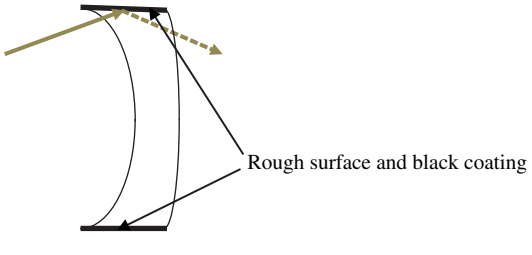


Figure 2.30 Side wall of a lens designed to reduce stray light

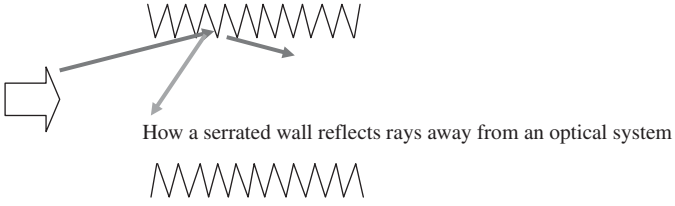


Figure 2.31 A serrated wall

2.3.7.3 Preventing the Entry of Stray Rays Reflected by a Wall

Rays reflected by a wall can have the undesirable consequence of increasing the level of stray light. To minimize this problem, the entire inner wall of an imaging optical system should be coated black and its surface also needs to be rough. If the optical system has a hood, then the inside wall of the hood should be serrated as shown in Figure 2.31, in order to reflect rays away from the system.

2.3.8 Mechanical Requirements of Optical Systems

The following list gives the mechanical requirements:

1. It is important to use mechanical parts with a high degree of stiffness or rigidity, whose surface is as perfectly flat as possible. Temperature variations may also need to be taken into consideration.
2. Mechanical parts which have been manufactured with a high degree of accuracy should always be used.
3. Vibration needs to be eliminated.

An imaging system is extremely sensitive to vibration. To avoid this problem, we need to consider ways of eliminating vibration, for example, by means of mounting it on an air-suspension. For an optical system, the greater the level of accuracy required, the more accurate will be the alignments that need to be made to the system. When assembling and adjusting an optical system, the components need to be assembled in precise order. Assembly procedures should be executed very carefully, one step at a time. Special measuring equipment may also need to be used.

3

Ray Tracing Applications of Paraxial Theory

In this chapter we will study a method of ray tracing which applies the Gaussian lens formula, $1/s' = 1/s + 1/f$ [Equation (1.24)], using a simplifying approximation that holds for rays when the angles are small. Applying this method (known as *paraxial theory*), we can easily calculate the path of the ray and the height of the image.

3.1 Deriving the Equations for Ray Tracing Using Paraxial Theory

When we perform ray tracing using paraxial theory (Figure 3.1), we have to first determine the object height w_0 and the angle θ_0 of the ray which proceeds from the object. Then we can calculate w_1, θ_1 for Lens L_1 and w_2, θ_2 for Lens L_2, and so on. Finally, we can derive the vertical height of the image, on the image plane.

Assuming that the ray heights w_o, w_1, w_2 and the ray angles θ_0, θ_1, θ_2 are small, we can calculate the ray height w_i for the ith lens (L_i) and the ray angle θ_i after passing through L_i as in Equation (3.1) and Equation (3.2).

$$w_i = w_{i-1} + d_i\theta_{i-1} \tag{3.1}$$

$$\theta_i = \theta_{i-1} - \frac{w_i}{f_i} \tag{3.2}$$

where

w_i = Ray height on L_i, whose sign is positive when w_i is in the upper axis and negative when it is in the lower axis

θ_i = Angle of ray after passing through L_i, whose sign is positive when the ray is traveling upwards and negative when traveling downwards

d_i = Distance between L_{i-1} and $L_i (= H_{i-1}{}' H_i)$

f_i = Focal length of L_i, whose sign is positive when the lens is convex and negative when it is concave.

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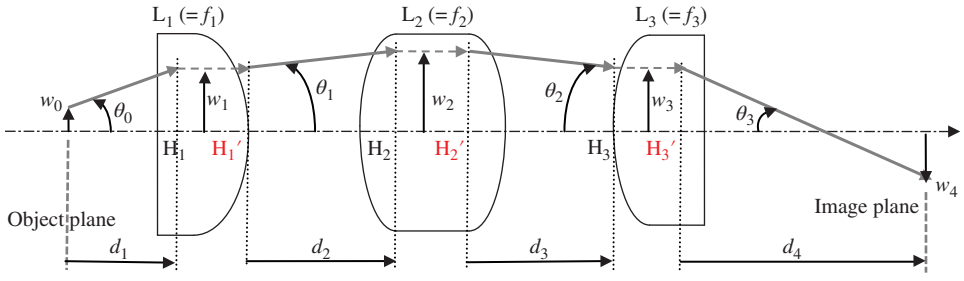


Figure 3.1 Ray tracing using paraxial theory

Equation (3.1) and Equation (3.2) can be obtained by assuming that the ray angle θ is small. When θ is small, we can use the following approximation:

$$\tan\theta\ (\approx \sin\theta) \approx \theta$$

Thus we can calculate w_i in Figure 3.1 as:

$$w_i = w_{i-1} + d_i\theta_{i-1} \tag{3.1}$$

In Chapter 1, we discussed the Gaussian lens formula:

$$\frac{1}{s'} = \frac{1}{s} + \frac{1}{f} \tag{1.24}$$

If we multiply both sides of the equation by the ray height *w,* then Equation (1.24) reads as follows:

$$\frac{w}{s'} = \frac{w}{s} + \frac{w}{f} \tag{1.24′}$$

Given our assumption that the ray angle θ is small, we can assume that w/s and w/s' represent the ray angle before and after passing through the lens, respectively. We can now rewrite Equation (1.24′) as follows, after taking into account their signs:

$$-\theta' = -\theta + \frac{w}{f} \tag{3.2′}$$

If we multiply both sides by −1, then we obtain:

$$\theta_i = \theta_{i-1} - \frac{w_i}{f_i} \tag{3.2}$$

Performing ray tracing calculations with the aid of paraxial theory is a very useful method for calculating the image height. The following sample problems illustrate some real-life applications of ray tracing.

3.2 Problems of Ray Tracing Calculations Using Paraxial Theory

Please download the Excel® files from the companion web site, which show calculation details for all problems in this book as listed in Appendix G

Problem 3.1 Imaging using a microscope objective lens

Let us consider Problems 2.2 and 2.3 from Section 2.2. Applying paraxial theory, we can readily solve these problems by performing a ray tracing for the image, using a microscope objective lens.

Solution

Optical System A (see Section 2.2, Problem 2.2)
Imaging using a microscope objective lens

(1) Configuration: see Figure 3.2a

Optical System B (see Section 2.2, Problem 2.3)
Imaging using a microscope objective lens and an imaging lens

(1) Configuration: see Figure 3.2b

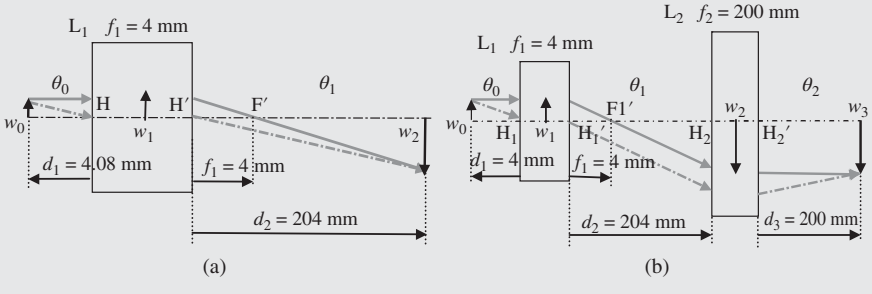


Figure 3.2 (a) Optical system A and (b) optical system B

(2) Equations

$$w_1 = w_0 + d_1\theta_0 \quad (3.1)$$
$$\theta_1 = \theta_0 - w_1/f_1 \quad (3.2)$$
$$w_2 = w_1 + d_2\theta_1 \quad (3.1')$$

(2) Equations

$$w_1 = w_0 + d_1\theta_0 \quad (3.1)$$
$$\theta_1 = \theta_0 - w_1/f_1 \quad (3.2)$$
$$w_2 = w_1 + d_2\theta_1 \quad (3.1')$$
$$\theta_2 = \theta_1 - w_2/f_2 \quad (3.2')$$
$$w_3 = w_2 + d_3\theta_2 \quad (3.1'')$$

(3) Configuration and ray settings proceeding from the object

	Optical system A	Optical system B
Focal length of L_1, f_1 (mm)	4	4
Focal length of L_2, f_2 (mm)	–	200
Distance of L_1 from object, d_1 (mm)	4.08	4
Distance of L_2 from L_1, d_2 (mm)	–	204
Numerical aperture of L_1, NA_1	0.55	
Incident ray angle, θ_0 (rad)	$0 - \pm f_1 NA_1/d_1$	
Object height, w_0 (mm)	0, ±0.1	

Let us set the incident ray angle θ_0 to the value of the subtended angle of the aperture size of L_1 ($a = f_1 NA_1 = 2.2$ mm). If $w_0 = 0$ mm, the value of the subtended angle θ_0 will be:

$$\theta_0 \approx \pm a/d_1 = \pm 2.2 \text{ mm}/4.08 \text{ mm} \approx \pm 0.5392157 \text{ rad (Optical system A)}$$

$$\theta_0 \approx \pm a/d_1 = \pm 2.2 \text{ mm}/4 \text{ mm} \approx \pm 0.55 \text{ rad (Optical system B)}$$

We can apply this value of θ_0 to the object height $w = 0$ mm and also to $w = \pm 0.1$ mm as an approximation.

For example, ray tracing for optical system A, where the object height $w_0 = 0.1$ mm and the incident ray angle $\theta_0 = 0.5392157$ rad can be calculated as follows. From Equation (3.1),

$$w_1 = w_0 + d_1\theta_0 = 0.1 + 4.08 \times 0.5392157 = 2.3 \text{ mm}$$

From Equation (3.2),

$$\theta_1 = \theta_0 - w_1/f_1 = 0.5392157 - 2.3/4 = -0.0357843 \text{ rad}$$

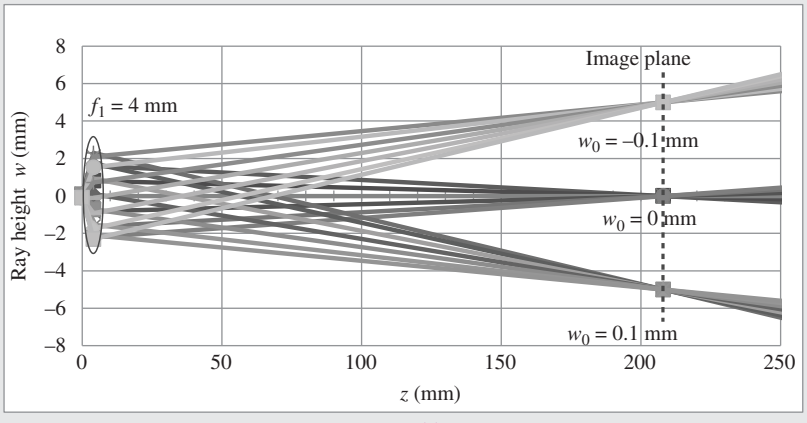
From Equation (3.1′),

$$w_2 = w_1 + d_2\theta_1 = 2.3 + 204 \times (-0.0357843) = -5 \text{ mm}$$

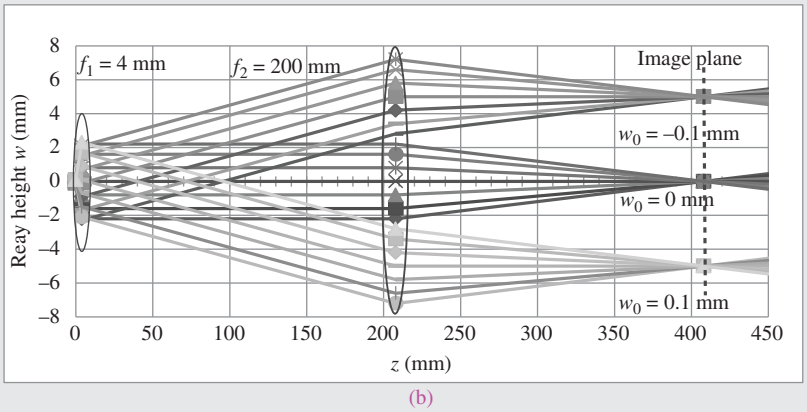
Thus we can derive the image height $w_2 = -5$ mm.

(4) Ray tracing calculation results: see Figure 3.3a, b

The graphs in Figure 3.3a,b show rays focusing on the image plane, at a horizontal distance of z from the object, where the ray height equals w, measured along the vertical axis.



(a)



(b)

Figure 3.3 (a) Ray tracing for the imaging system A and (b) ray tracing for the imaging system B

(5) **Notes**

1. Both optical systems have the same lateral magnification $\beta = -50$ and generate an image of height ∓ 5 mm from an object of height ± 0.1 mm. However, the paths of these rays are extremely divergent.
2. The latter optical system allows any value of d_1 (the distance between L_1 and L_2), but it is advisable to follow the manufacturer's recommended range. In this example, we set the value d_1 as $d_1 = f_1 + f_2$. In this case, the principal ray-making image will travel parallel to the axis. (We refer to this case as a telecentric system – one in which the image height will not vary, even when the distance of the image screen falls outside the focus range.)

Problem 3.2 Laser beam scanning system

Perform ray tracing calculations for an optical system which scans a laser spot onto a screen using a collimated laser beam, a rotating polygon mirror (or a Galvanometer mirror), and a scanning lens (f = 100 mm).

Solution

A simple configuration which satisfies this problem would be an optical system that positions the rotating polygon mirror and the screen at the first and the second focal points of the scanning lens, respectively (Figure 3.4).

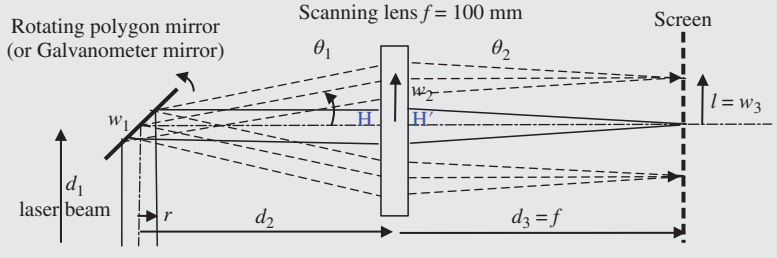


Figure 3.4 Laser beam scanning system

Equations

$$w_2 = w_1 + d_2\theta_1 \tag{3.1}$$

$$\theta_2 = \theta_1 - w_2/f \tag{3.2}$$

$$l = w_3 = w_2 + d_3\theta_2 \tag{3.1'}$$

Settings: To obtain a scanning range of $l = w_3 = 0$ to ± 10 mm, set the values as follows: f = 100 mm, d_1 = 50 (100) mm, d_2 = 100 (50) mm, d_3 = 100 mm, r = 5 mm (laser beam radius), $\theta_1 = 0$ to ± 0.1 rad.

Calculation results

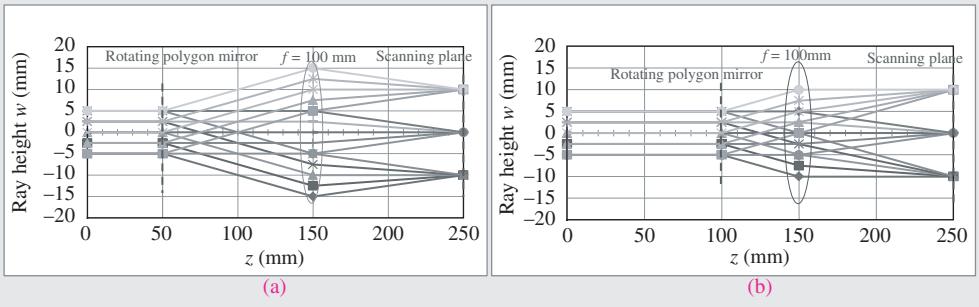


Figure 3.5 Ray tracing for a laser scanning system (a) when d_2 = 100 mm and (b) when d_2 = 50

As shown on the graphs of the calculation results in Figure 3.5a,b, the scanning range w_2 is ± 10 mm when $\theta_1 = \pm 0.1$ rad, which can be calculated by using Equation (3.1), Equation (3.2), and Equation (3.1′) as follows:

$$w_3 = f\theta_1 = 100 * (\pm 0.1) = \pm 10 \text{ (mm)}$$

$$\because w_3 = w_2 + f\theta_2 = d_2\theta_1 + f(\theta_1 - d_2\theta_1/f) = f\theta_1$$

The above equation is true regardless of the value of d_2.

Notes

1. This optical system scans the laser spot 0 to ± 10 mm onto a screen at a distance 100 mm from the lens, which corresponds to the incident beam angle of $\theta_1 = 0$ to ± 0.1 rad.
2. The relation between the incident beam angle θ_1 and the scanning position w_3 can be expressed as $w_3 = f\theta_1$, regardless of the distance of the rotating polygon mirror from the scanning lens, where f is the focal length of the scanning lens.

4

Two-Dimensional Ray Tracing

In this chapter, we will examine a method for calculating the degree of aberration in an optical system, using the ray tracing technique. Ray tracing is a straightforward task, if we follow a few simple steps. First, we begin by drawing the rays emitted from the object, and then we calculate the ray paths. We do this by calculating the ray heights for rays entering the lens and calculating the slopes of the rays exiting the boundary surface, applying the law of refraction (commonly known as Snell's law9). After repeating this process for each boundary surface, we can finally derive the ray heights on the image plane. If the optical system has zero aberration, then all of the rays emitted from the (point) object should reach the image plane, converging precisely at a single point. The object and image are referred to as conjugate points. In a real-life optical system, however, there is always a small degree of aberration. Hence the rays on the image plane do not converge exactly at one point, but are instead dispersed within a small radius of the point where they would converge under ideal conditions.

By calculating the optical path lengths (OPLs) between the boundary surfaces of the lenses, we can also obtain the cumulative OPL from the object to the reference wave front, which is a sphere whose center of curvature is situated at the ideal image point. (The reference wave front is the wave front when an optical system has no aberration.) We can then use the cumulative OPLs to calculate the wave front aberration of the optical system.

4.1 Ray Tracing for a Spherical Surface

In an optical system consisting of multiple lenses, we can designate the axis along which we measure the ray height as the y-axis and the direction of ray propagation as the z-axis, and we can then proceed as follows. First, we begin with an object point at an arbitrary height w_0 and a ray emitted from the object, at an arbitrary angle θ_0 to the optical axis (i.e., the z-axis). We then calculate the ray height w_1, where the ray meets the surface of the first lens and the slope angle of the refracted ray θ_1 exiting the surface, by applying Snell's law. Next, we calculate the ray height w_2 for the second surface and the ray slope angle θ_2 for the ray exiting the second surface. If we repeat this process, we finally obtain the ray height w_n on the image plane.

Figure 4.1 is a ray tracing diagram for the kth boundary surface and the two surfaces, one located in front of it and the other behind it. In Figure 4.1, d_k is the distance between the vertices of the $(k-1)$th and kth surfaces, of_k is the vertex offset of the kth surface from the $(k-1)$th surface, r_k is the radius of curvature of the kth surface (which is positive when the vertex is on the left), n_{k-1} and n_k are the refractive indices of the media in front of and behind the kth surface, respectively, w_{k-1} and w_k are the ray heights

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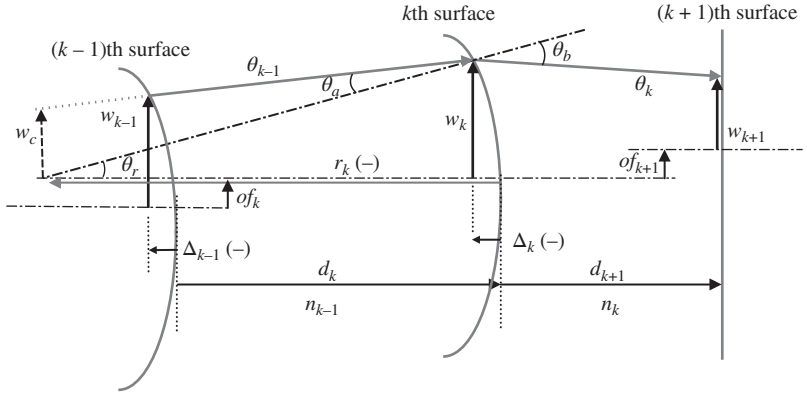


Figure 4.1 Ray tracing for a spherical surface

on the $(k-1)$th and kth surfaces, respectively, Δ_{k-1} and Δ_k are the z-positions of the ray on the surface, measured from the vertex (which is positive to the right of the vertex), and θ_{k-1} and θ_k are the slope angles of the rays in front of and behind the kth surface, respectively.

If we designate the center of curvature as the origin, then the equation of the kth surface is as follows:

$$y^2 + z^2 = {r_k}^2 \tag{4.1}$$

The equations for the incident ray will then be:

$$y = z\tan\theta_{k-1} + w_c \tag{4.2a}$$

$$w_c = w_{k-1} - of_k + (d_k - \Delta_{k-1} + r_k)\tan\theta_{k-1} \tag{4.2b}$$

where w_c is the ray height at the z-position of the surface's center of curvature.

We can calculate the intersection coordinates (Z,Y) for the incident ray and kth surface, by substituting Equation (4.2a) into Equation (4.1).

$$Z = \frac{-\tan\theta_{k-1}w_c \pm \sqrt{\tan^2\theta_{k-1}{w_c}^2 - (1+\tan^2\theta_{k-1})({w_c}^2 - {r_k}^2)}}{1+\tan^2\theta_{k-1}} \tag{4.3}$$

The sign in front of the $\sqrt{}$ is + when the vertex lies to the right of the surface. Thus Δ_k will be:

$$\Delta_k = Z + r_k = \frac{-\tan\theta_{k-1}w_c \pm \sqrt{\tan^2\theta_{k-1}{w_c}^2 - (1+\tan^2\theta_{k-1})({w_c}^2 - {r_k}^2)}}{1+\tan^2\theta_{k-1}} + r_k \tag{4.4}$$

The ray height $Y\,(\equiv w_k)$ on the kth surface is given by the following equation:

$$Y \equiv w_k = w_{k-1} - of_k + (d_{k-1} - \Delta_{k-1} + \Delta_k)\tan\theta_{k-1} \tag{4.5}$$

The slope angle θ_r of the normal to the surface at the incident point (*Z*,*Y*) is:

$$\theta_r = \sin^{-1}(-w_k/r_k) \tag{4.6}$$

The incident angle θ_a, and the refractive angle θ_b for the surface are as follows:

$$\theta_a = \theta_{k-1} - \theta_r \tag{4.7a}$$

$$\theta_b = \sin^{-1}\left(\frac{n_{k-1}}{n_k}\sin\theta_a\right) \tag{4.7b}$$

The slope angle θ_k of the ray exiting the surface will thus be:

$$\theta_k = \theta_r + \theta_b = \theta_r + \sin^{-1}\left(\frac{n_{k-1}}{n_k}\sin\theta_a\right) \tag{4.8}$$

The optical path length L_k between the $(k-1)$th and kth surfaces will be:

$$L_k = \frac{n_{k-1}(d_k - \Delta_{k-1} + \Delta_k)}{\cos\theta_{k-1}} \tag{4.9}$$

By repeating our calculations of w_k, Δ_k, $\tan\theta_k$, and L_k, surface by surface, we can finally obtain the ray height on the image plane, and by calculating the cumulative OPL from the object to the reference wave front, we can obtain the wave front aberration.

4.2 Ray Tracing for a Plane Surface

Plane surfaces are used in plano-convex lenses, plano-concave lenses, and cylindrical lenses, and also used for the image plane. Figure 4.2 depicts ray tracing for a plane surface. In Figure 4.2, α is the surface angle measured from the vertical (where the sign is positive when the surface is inclined as in Figure 4.2), while the other parameters are held the same as for Figure 4.1.

4.2.1 Vertical Plane ($\alpha = 0$)

w_k, Δ_k, and θ_k in Figure 4.2 can be described as follows:

$$w_k = w_{k-1} - of_k + (d_k - \Delta_{k-1} + \Delta_k)\tan\theta_{k-1} \tag{4.5}$$

$$\Delta_k = 0 \tag{4.10}$$

$$\theta_k = \sin^{-1}\left(\frac{n_{k-1}}{n_k}\sin\theta_{k-1}\right) \tag{4.7b′}$$

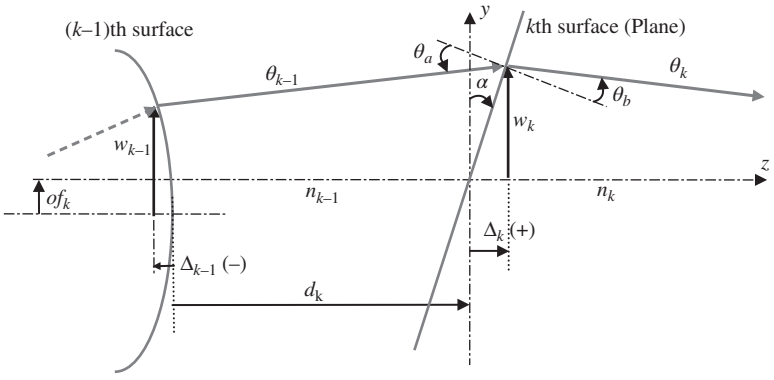


Figure 4.2 Ray tracing for a plane surface

4.2.2 Inclined Plane ($\alpha \neq 0$)

The equations of the incident ray and the plane surface are as shown below in Equation (4.11a) and (4.11b), where the origin is defined as the intersection of the plane and the z-axis.

$$y - (w_{k-1} - of_k) = \{z - (-d_k + \Delta_{k-1})\} \tan\theta_{k-1} \tag{4.11a}$$

$$y = \frac{1}{\tan\alpha} z \tag{4.11b}$$

The intersection coordinates of the ray and the plane (Z,Y), that is, Δ_k, w_k, can be calculated from Equation (4.12a,b).

$$Z \equiv \Delta_k = \frac{\tan\theta_{k-1} d_k - \tan\theta_{k-1}\Delta_{k-1} + w_{k-1} - of_k}{1/\tan\alpha - \tan\theta_{k-1}} \tag{4.12a}$$

$$Y \equiv w_k = \frac{1}{\tan\alpha} z \tag{4.12b}$$

From Figure 4.2, the incident angle θ_a will be:

$$\theta_a = \theta_{k-1} + \alpha$$

The refractive angle θ_b will then be:

$$\theta_b = \sin^{-1}\left(\frac{n_{k-1}}{n_k}\sin\theta_a\right) \tag{4.13}$$

Hence the angle θ_k made by the ray after it exits the surface will be:

$$\theta_k = -\alpha + \sin^{-1}\left(\frac{n_{k-1}}{n_k}\sin\theta_a\right) \tag{4.14}$$

The OPL can be calculated using Equation (4.9).

4.3 Ray Tracing for an Aspheric Surface (Using VBA Programming)

An aspheric lens is a lens whose surface profile is symmetrical around the optical axis, but not a portion of a sphere. The asphere's more complex surface profile can reduce or eliminate spherical aberration and also reduce other aberrations when compared with a simple lens. A single aspheric lens can often replace a much more complex multi-lens system. Here, we shall examine an aspheric lens which is used in LD (laser diode) beam collimation.

4.3.1 An Aspheric Lens Used for Collimating Light Rays Emitted from a Laser Diode

Figure 4.3 shows an aspheric lens used for LD beam collimation. In Figure 4.3, the incident collimated light rays converge at the focal point F. Conversely, if the point from which the LD rays are emitted is set at point F, the rays will propagate to the left and will exit the lens in parallel (as a collimated beam). This lens has aspheric surfaces on both sides, and their curvatures are designed to obtain a precisely collimated beam from the rays emitted from the LD. The curvatures are also designed on the assumption that the LD window glass lies directly in the optical path. For example, the aspheric surface equation can be expressed as a function of $z(y)$ [1].

$$z(y) = \frac{y^2}{R\left[1 + \sqrt{1 - \dfrac{(1+k)\,y^2}{R^2}}\right]} + A_4 y^4 + A_6 y^6 + A_8 y^8 + A_{10} y^{10} + A_{12} y^{12} \tag{4.15}$$

Table 4.1 shows some example values for the coefficients R, k, A_4, A_6, A_8, A_{10}, A_{12} of an aspheric lens, $f = 8.00$ mm, NA $= 0.5$.

The calculation results for the above aspheric surfaces are shown in Figure 4.4a,b. (The horizontal and the vertical axes on the graph represent the y-axis and z-axis in Figure 4.3, respectively.)

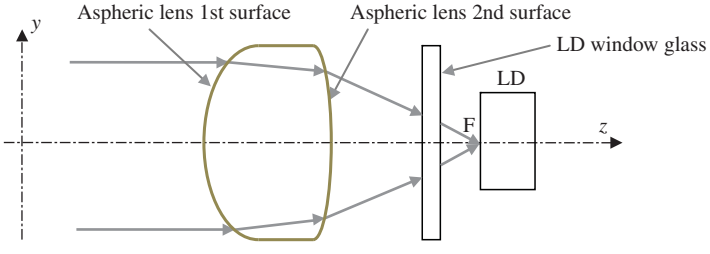


Figure 4.3 Aspheric lens used for collimating an LD beam

Table 4.1 Example values for the aspheric surface coefficients

	R	k	A_4	A_6	A_8	A_{10}	A_{12}
First surface	5.09239	−0.47317	0	-2.257×10^{-6}	2.412×10^{-9}	-8.263×10^{-9}	0
Second surface	−56.20310	0	5.124×10^{-4}	-2.266×10^{-5}	3.33×10^{-7}	0	0

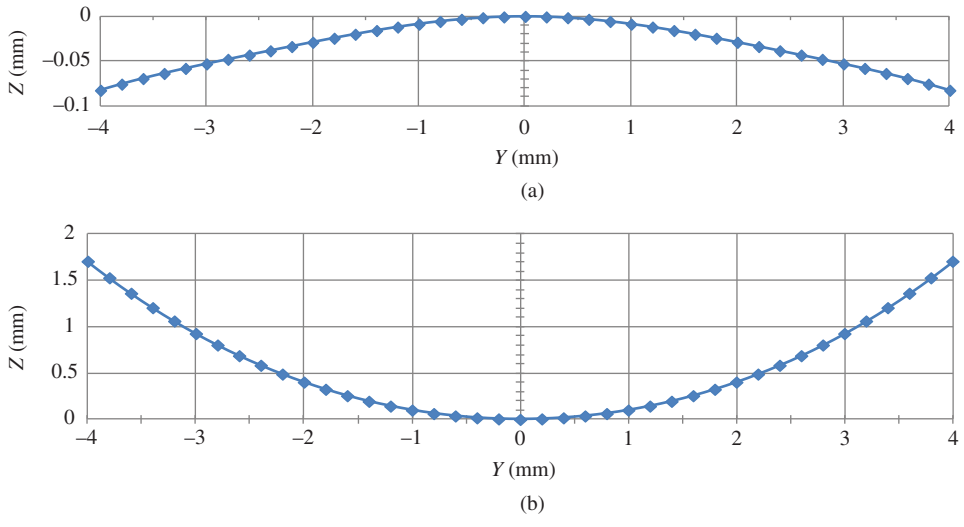


Figure 4.4 Examples of aspheric surfaces. (a) Second surface of an aspherical lens and (b) first surface of an aspherical lens

4.3.2 Ray Tracing for an Aspheric Lens Using a VBA Program

4.3.2.1 Obtain the Intersection of the Incident Ray and the Aspheric Surface Using VBA Program

In Figure 4.5, the equation of the incident ray $z_A(y)$ is as follows (where the origin is defined as the vertex of the aspheric surface):

$$y - w + of = m(z + d - \Delta) = mz + md - m\Delta \tag{4.16a}$$

$$z_A(y) = \frac{y - w + of - md + m\Delta}{m} \tag{4.16b}$$

The equation of the aspheric surface $z_B(y)$ is as follows:

$$z_B(y) = \frac{y^2}{R\left[1 + \sqrt{1 - \dfrac{(1+k)\,y^2}{R^2}}\right]} + A_4 y^4 + A_6 y^6 + A_8 y^8 + A_{10} y^{10} + A_{12} y^{12} \tag{4.17}$$

Using VBA (visual basic for application) programming, we can calculate the intersection coordinates of the incident ray $z_A(y)$ in Equation (4.16b) and the aspheric surface $z_B(y)$ in Equation (4.17). By applying Newton's method, we can obtain the value of y which makes $s(y) = z_A(y) - z_B(s) \rightarrow 0$, and this value of y represents the intersection.

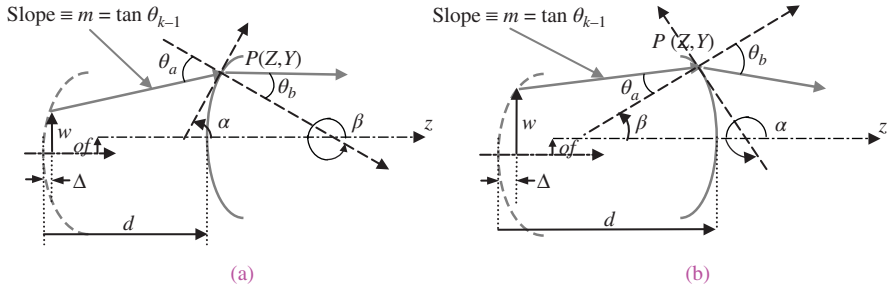


Figure 4.5 Ray tracing for an aspheric lens. (a) First surface (convex on the left) and (b) second surface (convex on the right)

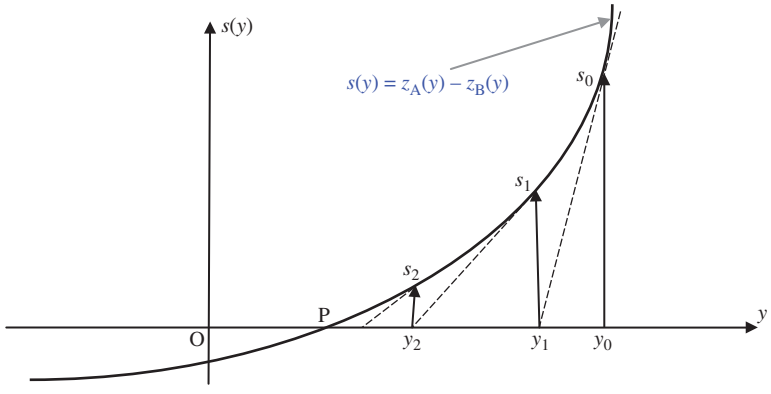


Figure 4.6 Newton's method

Newton's Method

If we know $s'(y)$, the derivative of the function $s(y)$ (i.e., the slope of the function $s(y)$), we can use Newton's method for obtaining the y-value which makes $s(y) \to 0$. In Figure 4.6,

$$s'(y_n) = \frac{s(y_n)}{-(y_{n+1} - y_n)}$$

Hence,

$$y_{n+1} = -\frac{s(y_n)}{s'(y_n)} + y_n$$

First, we begin with y_0, an arbitrary value of y. Next, we calculate $s\,(y_0)$ and $s'(y_0)$, and then we calculate y_1 (the next value of y), using the above equation. After repeating this process many times, we reach y_n which is arbitrarily close to P (the point where $s(y) = 0$). Thus, we can obtain the value of y which is the correct solution to $s(y) = 0$, with a sufficient level of accuracy.

The Derivative of the Function $s(y) = z_B(y) - z_A(y)$

The derivative of the function $s(y) = z_B(y) - z_A(y)$ = Equation (4.17) – Equation (4.16b) can be calculated by applying the following standard differentiation formulae:

$$\frac{d}{dx}x^n = nx^{n-1} \text{ and } \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

The results of our calculations are as follows:

$$s' = \frac{2y}{R\left(1+\sqrt{1-\frac{(1+k)y^2}{R^2}}\right)} + \frac{(1+k)y^3}{R^3\left(1+\sqrt{1-\frac{(1+k)y^2}{R^2}}\right)^2\sqrt{1-\frac{(1+k)y^2}{R^2}}}$$
$$+4A_4y^3 + 6A_6y^5 + 8A_8y^7 + 10A_{10}y^9 + 12A_{12}y^{11} - \frac{1}{m}$$

4.3.2.2 Obtain the Slope of the Refracted Ray

In Figure 4.5, the derivative of the curve $\Delta y/\Delta z$ tangential to the surface is tan α. Hence the slope angle β of the normal to the surface will be:

$$\beta = \alpha \pm \pi/2$$

(where the sign is positive when $\alpha < 0$, and negative when $\alpha \geq 0$).

The incident angle θ_a, the refractive angle θ_b, and the angle of the refracted ray θ_k will thus be:

$$\theta_a = \theta_{k-1} - \beta \tag{4.18a}$$
$$\theta_b = \sin^{-1}(1/n\ \sin\theta_a) \tag{4.18b}$$
$$\theta_k = \beta + \theta_b \tag{4.18c}$$

4.3.2.3 Ray Tracing

We can input the results for the position of the ray (Δ,w) and the slope of the ray tan θ_k (as calculated above in Sections 4.3.2.1 and 4.3.2.2) into the two-dimensional ray tracing table, and then perform ray trace calculations for the next surface. The OPL can be calculated using:

$$L_k = \frac{n_{k-1}(d_k - \Delta_{k-1} + \Delta_k)}{\cos\theta_{k-1}} \tag{4.9}$$

4.4 Ray Tracing for an Aberration-Free Lens

In the case of an optical system containing a lens where the focal length is given but the other parameters remain unspecified, we can no longer perform ray tracing calculations, as we lack vital data relating to the surface of the lens. In this section, we will discuss a method of performing a ray trace calculation for an optical system containing a lens whose parameters are unspecified, by assuming it to be an aberration-free lens with a focal length of *f*.

We can also analyze the quality of the collimation beam by examining the spot diagram after focusing the beam with an aberration-free lens.

Definition of an Aberration-free Lens

Assume that the lens satisfies the assumptions of paraxial theory (as described in Chapter 3) for both near-axis and off-axis rays.

$$\frac{1}{s'} = \frac{1}{s} + \frac{1}{f} \tag{1.24}$$

1. If the object lies on a plane vertical to the axis, the image must also lie on a vertical plane.
2. We can regard the lateral magnification equation [Equation (4.19)] as being applicable to both near-axis and off-axis rays.

$$\beta = \frac{y'}{y} \cong \frac{s'/n'}{s/n} = \frac{s'}{s} \quad (\because n = n' = 1 \text{ in air}) \tag{4.19}$$

In Chapter 3, we examined the ray tracing method for paraxial rays. This method can be successfully applied if we are given the focal length f and the distance d between the lenses. And the calculation results obtained are free from aberration. (In real life, of course, lenses are free from aberration only for paraxial rays. However, we can generalize to include off-axis rays, for a hypothetical ideal lens.) In this section, we will apply paraxial theory to the case of an aberration-free lens, for the purposes of performing a two-dimensional ray tracing.

4.4.1 Ray Tracing Procedure for an Aberration-Free Lens

4.4.1.1 Calculation of the Ray Height for the Incident Rays

Figure 4.7 depicts an aberration-free lens which is inserted into a two-dimensional ray tracing set-up. The left and right sides of the figure show the two-dimensional ray tracing, and the aberration-free lens is situated in the center.

In Figure 4.7, the ray height w_k at the point where the ray meets the aberration-free lens can be calculated by performing a two-dimensional ray tracing.

$$w_k = w_{k-1} + (d_k - \Delta_{k-1}) \tan\theta_{k-1} - of_k \tag{4.5}$$

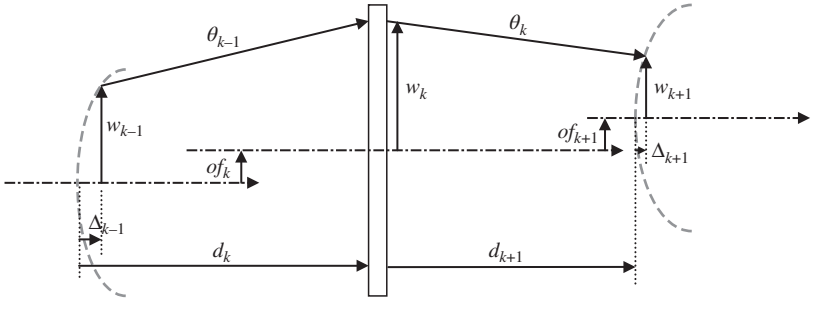


Figure 4.7 Ray tracing for an aberration-free lens

The ray height w_k can also be calculated for an aberration-free lens, using paraxial theory.

$$w_k = w_{k-1} + (d_k - \Delta_{k-1})\theta_{k-1} - of_k \tag{3.1′}$$

Comparing Equation (4.5) and Equation (3.1′), we find that $\tan\theta_{k-1}$ in Equation (4.5) has been replaced with θ_{k-1} in Equation (3.1). When we do the calculations for an aberration-free lens using Equation (3.1′), we substitute the value of $\tan\theta_{k-1}$ in Equation (4.5) (which is calculated by performing a two-dimensional ray tracing) for θ_{k-1} in Equation (3.1′). [The ray exiting the $(k-1)$th surface must satisfy Equation (4.5) and the incident ray onto the kth surface must satisfy Equation (3.1′), and both of them must be identical.]

4.4.1.2 Calculation of the Slope of the Ray Exiting the Aberration-Free Lens

The slope of the ray exiting the aberration-free lens can be calculated by applying Equation (3.2), which is identical with that used in paraxial theory.

$$\theta_k = \theta_{k-1} - \frac{w_k}{f_k} \tag{3.2}$$

Here the value of $\tan\theta_{k-1}$, which was calculated by performing a two-dimensional ray tracing on the surface in front of the lens, can be substituted for θ_{k-1}.

4.4.1.3 Calculation of Ray Height for Rays after Exiting the Aberration-Free Lens

In Figure 4.7, the ray height for the next surface (after the rays exit the aberration-free lens) w_{k+1} can be calculated for the aberration-free lens, using the principles of paraxial theory.

$$w_{k+1} = w_k + (d_{k+1} + \Delta_{k+1})\theta_k - of_{k+1} \tag{4.20}$$

The ray height w_{k+1} for the next surface can also be calculated by performing a two-dimensional ray tracing.

$$w_{k+1} = w_k + (d_{k+1} + \Delta_{k+1})\tan\theta_k - of_{k+1} \tag{4.21}$$

When we calculate the next surface using Equation (4.21), we can substitute the value of θ_k in Equaion (3.2) (which is calculated for an aberration-free lens) for $\tan\theta_k$. [The ray exiting the kth surface must satisfy Equation (4.20) and the incident ray onto the $(k+1)$th surface must satisfy Equation (4.21), and both of them must be identical.]

4.5 Optical Path Length Calculation for an Aberration-Free Lens

4.5.1 Calculate the Coordinates of the Virtual Object and Its Virtual Image for an Aberration-Free Lens

In Figure 4.8, we first obtain the intersection of two incident rays entering the aberration-free lens and we treat this as the virtual object P for the aberration-free lens. Then, using this virtual object P, we can calculate the position of the virtual image P′ where the rays exiting the aberration-free lens should converge.

We can calculate the intersection P(z,y) of one incident ray with another incident ray[1] (e.g., the principal ray). In Figure 4.8, if we define the origin as the primary principal point H of the aberration-free lens,

[1] In Excel calculation sheets which include an aberration-free lens, we set the above "another incident ray" at the top of the incident ray settings, as a ray which is very close to the principal ray.

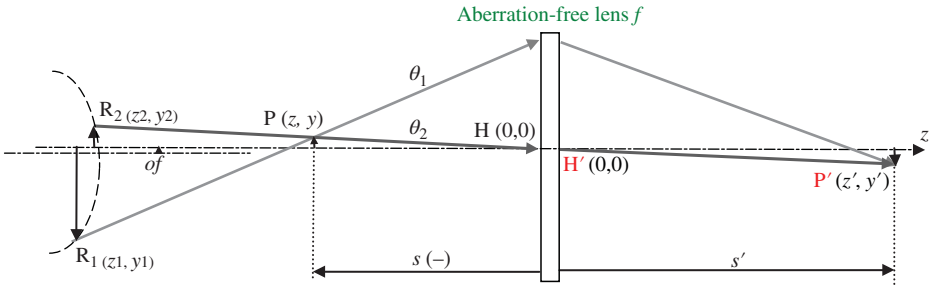


Figure 4.8 Virtual object point P(z,y) and virtual image point P′(z',y') for an aberration-free lens

then the equations for the two incident rays can be written as:

$$(y - y_1) = \tan\theta_1 (z - z_1) \quad \text{for incident ray 1, and} \tag{4.22a}$$

$$(y - y_2) = \tan\theta_2 (z - z_2) \quad \text{for incident ray 2 (another incident ray}^{1}\text{)} \tag{4.22b}$$

4.5.2 Obtain the Imaginary Object P(z,y)

The intersection of these two rays P(z,y) can be calculated as the solution of the two equations listed in Equation (4.22a,b):

$$z = s = \frac{\tan\theta_1 z_1 - \tan\theta_2 z_2 - y_1 + y_2}{\tan\theta_1 - \tan\theta_2} \tag{4.23a}$$

$$y = \tan\theta_1 (z - z_1) + y_1 - of \tag{4.23b}$$

4.5.3 Obtain the Imaginary Image P′(z′,y′)

The imaginary image P′(z',y') can be calculated from P(z,y) using the lens formula $1/s' = 1/s + 1/f$ [Equation (1.24)]. (Here, we locate the origin at the secondary principal point H′ of the aberration-free lens.)

$$z' = s' = \frac{sf}{s+f} \tag{4.24a}$$

$$y' = \frac{s'y}{s} \tag{4.24b}$$

[1] In Excel calculation sheets which include an aberration-free lens, we set the above "another incident ray" at the top of the incident ray settings, as a ray which is very close to the principal ray.

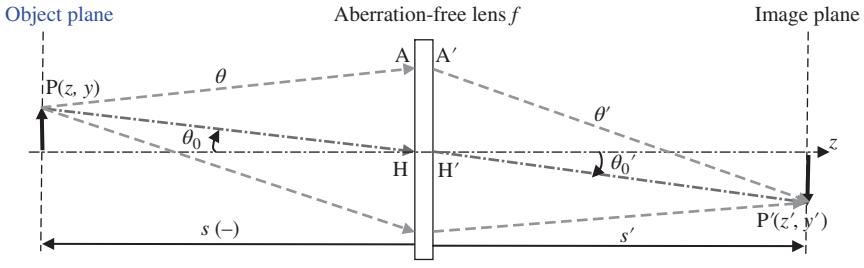


Figure 4.9 Optical path length correction for an aberration-free lens

4.5.4 *Optical Path Length Adjustment for an Aberration-Free Lens*

The optical path length for an aberration-free lens could be calculated as the total length of the ray paths between the surfaces, using the same method as for other ray tracings, that is, using:

1. the length of the ray path from the surface lying in front of the aberration-free lens to the lens itself;
2. the length of the ray path from the aberration-free lens to the next surface.

However, this method would not yield accurate results for aberration-free lenses, so it needs to be corrected by subtracting an adjustment value from each of the two ray path lengths.

In Figure 4.9, if the lens has no aberration, the OPL from the object P to the image P′ through any route must be equal to the length of path PH + H′P′. Thus we will need to subtract adjustments ΔL_i and ΔL_0 from the ray path length calculation. These are calculated as follows.

The required adjustment to the optical path length before the ray enters the lens ΔL_i is:

$$\Delta L_i = \mathrm{PA} - \mathrm{PH} = \frac{-s}{\cos\theta} - \frac{-s}{\cos\theta_0} = -s\left(\frac{1}{\cos\theta} - \frac{1}{\cos\theta_0}\right) \tag{4.25}$$

The required adjustment after the ray exits the lens ΔL_0 will be:

$$\Delta L_0 = \mathrm{A'P'} - \mathrm{H'P'} = \frac{s'}{\cos\theta'} - \frac{s'}{\cos\theta_0'} = s'\left(\frac{1}{\cos\theta'} - \frac{1}{\cos\theta_0'}\right) \tag{4.26}$$

Here,

$$\theta_0 \approx \theta_0' \approx \tan^{-1}\left(\frac{y}{s}\right)$$

Optical Path Length Correction when the Incident Rays are Parallel

When the incident rays are in parallel, the object distance s can no longer be meaningfully defined, so Equation (4.27) will have to be used instead of Equation (4.25) (Figure 4.10a).

$$\Delta L_i = w_L \sin\theta_i \tag{4.27}$$

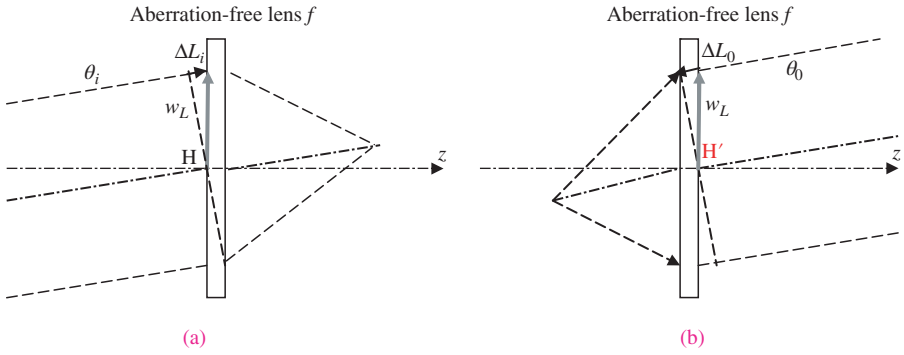


Figure 4.10 Optical path length correction for parallel incident rays and parallel exiting rays. (a) Incident rays are parallel and (b) exiting rays are parallel

Optical Path Length Correction when Output Rays are Parallel
If the object lies on the first focal plane, then the rays exiting the lens will be in parallel. In this case, we can use Equation (4.28) instead of Equation (4.26) (Figure 4.10b):

$$\Delta L_o = -w_L \sin\theta_o \tag{4.28}$$

Paraxial theory has been used here to illustrate the principles of ray tracing for an aberration-free lens. This lens will not generate any aberrations when a ray tracing is performed. However, it needs to be kept in mind that the calculated ray heights for this lens will be significantly different from the actual values observed, when it has a large numerical aperture.

4.6 Ray Tracing for an Optical System Which Is Set at a Tilt

Lenses should be set vertically to the optical axis. If lenses are not set vertically to the optical axis, there will be some aberrations in the focused image. Here, we will study the ray tracing for lenses which are not set vertically to the axis. Consider a lens which is tilted at an angle α as shown in Figure 4.11a. To make the calculation easier, we will change the (calculation base) optical axis from that of the incident rays to that of the lens (which is rotated from the incident ray optical axis by an angle α as shown in Figure 4.11b). After passing through the lens, we set the optical axis back to the incident ray optical axis (which is rotated from the lens axis by an angle $-\alpha$) and calculate the path of the exiting rays that travel on to the next surface.

We use the incident ray coordinate system for the rays traveling to the incident plane IQ as in Figure 4.11a. We use the lens coordinate system for the rays passing through the lens (from the incident plane IQ to the exiting plane JR) whose axis is rotated from that of the incident ray by an angle α as shown in Figure 4.11b. After passing through the exiting plane JR, we set the axis back to that of the incident ray coordinate system. In the figure, PQ(①), QR(②), and RS(③) represent the axes for calculating the incident rays, the rays passing through the lens and the rays exiting the lens, respectively, and HQ and KR represents the plane normal to the optical axis at the incident plane and that at the exiting plane, respectively.

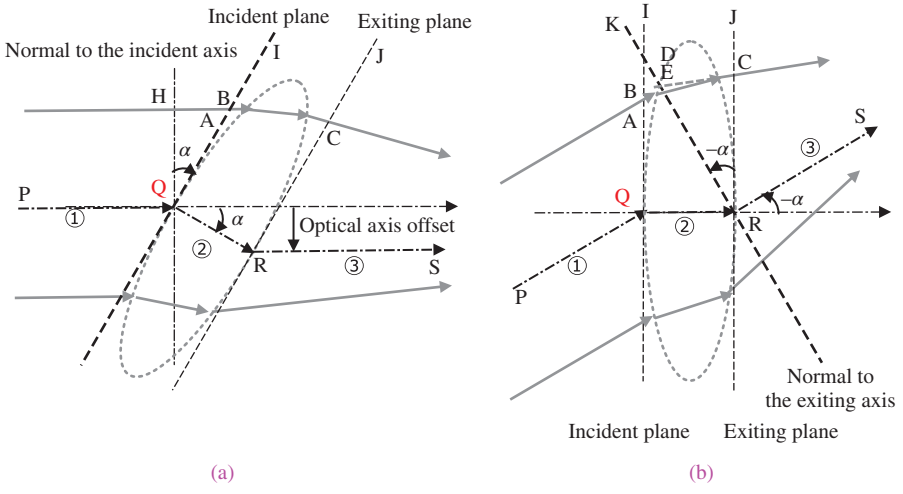


Figure 4.11 Ray tracing for a tilted lens. (a) A tilted lens in the incident ray coordinates (expressed in terms of the incident ray coordinate system). (b) A tilted lens in the lens axis coordinates (expressed in terms of the lens axis coordinate system)

4.6.1 Ray Tracing for the Rays Traveling to the Incident Plane IQ (Tilted at an Angle α)

Calculate the ray position $A(\Delta_A, w_A)$ and ray slope $\tan\theta_A$ on the incident plane IQ which is tilted at an angle α, using the same method as in Section 4.2.

1. When the tilt angle $\alpha = 0$

$$\Delta_A = 0 \tag{4.29}$$

$$w_A = w_{k-1} - of_A + (d_A - \Delta_{k-1} + \Delta_A)\tan\theta_{k-1} \tag{4.30}$$

$$\tan\theta_A \equiv \tan\theta_k = \tan\theta_{k-1} \tag{4.31}$$

2. When the tilt angle $\alpha \neq 0$

$$Z \equiv \Delta_A = \frac{\tan\theta_{k-1} d_A - \tan_{k-1}\Delta_{k-1} + w_{k-1} - of_A}{1/\tan\alpha - \tan\theta_{k-1}} \tag{4.32}$$

$$Y \equiv w_A = w_{k-1} - of_A + (d_k - \Delta_{k-1} + \Delta_A)\tan\theta_{k-1} \tag{4.30}$$

$$\tan\theta_A \equiv \tan\theta_k = \tan\theta_{k-1} \tag{4.31}$$

3. Optical path length

$$L_A = n_{k-1}(d_A - \Delta_{k-1} + \Delta_A)/\cos\theta_{k-1} \tag{4.33}$$

Here, of_A, d_A, w_A, Δ_A, and $\tan\theta_A$ are identical to of_k, d_k, w_k, Δ_k, and $\tan\theta_k$ shown in Figure 4.2. Also, $of_A = 0$, and $\tan\theta_k = \tan\theta_{k-1}$ (we shall not perform any refraction calculations here).

4.6.2 Transformation of the Coordinate to the Lens Coordinates (Rotation of the Axis through Angle α)

Transform the coordinates of the above ray position (Δ_A, w_A) and the ray slope $\tan\theta_A$ into the lens coordinates (Δ_B, w_B), and slope $\tan\theta_B$ by rotating the axis through an angle α.

$$\Delta_B = 0 \tag{4.34}$$

$$w_B = w_A / \cos\alpha \tag{4.35}$$

$$\tan\theta_B = \tan[\arctan(\tan\theta_A) + \alpha] \tag{4.36}$$

4.6.3 Ray Tracing for Rays Passing through the Lens

Calculate the ray tracing for the rays passing through the lens using the lens coordinate system. Starting with the values Δ_B, w_B, $\tan\theta_B$ which lie on the incident plane IQ, compute the values Δ_C, w_C, $\tan\theta_C$ which lie on the exiting plane JR.

4.6.4 Calculate the Exiting Ray Δ_D, w_D, $\tan\theta_D$ on the Plane KR Which Is Tilted at an Angle $-\alpha$ to the Lens Coordinate System (Which Is Normal to the Exiting Optical Axis)

Calculate the ray tracing from C(Δ_C,w_C) on the exiting plane JR to D (Δ_D,w_D) on the plane KR which is tilted at angle $-\alpha$ to the lens axis coordinates.

1. When the tilt angle $\alpha = 0$,

$$\Delta_D = 0 \tag{4.37a}$$

$$w_D = w_C - of_D + (d_D - \Delta_C + \Delta_D)\tan\theta_C \tag{4.37b}$$

$$\tan\theta_D = \tan\theta_C \tag{4.37c}$$

2. When the tilt angle $\alpha \neq 0$,

$$Z \equiv \Delta_D = \frac{\tan\theta_C d_D - \tan\theta_C \Delta_C + w_C - of_D}{1/\tan(-\alpha) - \tan\theta_C} \tag{4.38a}$$

$$Y \equiv w_D = \frac{1}{\tan(-\alpha)} Z \tag{4.38b}$$

$$\tan\theta_D = \tan\theta_C \tag{4.38c}$$

3. Optical path length

$$L_D = n_{k-1}(d_D - \Delta_{k-1} + \Delta_D)/\cos\theta_{k-1} \tag{4.39}$$

In the above equation, $of_D = 0$, and $\tan\theta_D = \tan\theta_C$ (we will not perform any refraction calculations here).

4.6.5 Transform the Coordinates Back into the Incident Ray Coordinate System (Rotate Axis through an Angle $-\alpha$)

Transform the coordinates of the above ray position (Δ_D,w_D) and ray slope $\tan\theta_D$ into the exiting ray coordinates (Δ_E,w_E), $\tan\theta_E$ by rotating the axis through an angle $-\alpha$.

$$\Delta_E = 0 \tag{4.40}$$

$$w_E = w_D/\cos(-\alpha) + \mathrm{QR}\sin(-\alpha) \tag{4.41}$$

$$\tan\theta_E = \tan[\arctan(\tan\theta_D) + (-\alpha)] \tag{4.42}$$

The term $+\mathrm{QR}\sin(-\alpha)$ in Equation (4.41) represents a term compensating for the optical axis offset between the incident rays and the exiting rays. Here we have obtained the exiting ray from a tilted lens, as in Equation (4.40), Equation (4.41) and Equation(4.42), which are expressed using the incident ray coordinate system.

Notes

1. In this book, the actual ray trace calculations for a tilted lens are performed using the surface types "TiltStart" and "TiltEnd" units. "TiltStart" is for the calculations in Sections 4.6.1 and 4.6.2, and "TiltEnd" is for the calculations in Sections 4.6.4 and 4.6.5. Place "TiltStart" just before the lens and place "TiltEnd" just after the lens at a distance $d = 0$, and an offset $of = 0$.
2. TiltStart and TiltEnd do not include calculations for refracting rays. Set the refractive index n to the same value as that of the medium in front of TiltStart (or TiltEnd).
3. If we use "TiltStart" and "TiltEnd" in the ray trace calculation, the value of the distance along the axis will be slightly different from the actual distance, because of the slant axis between "TiltStart" and "TiltEnd."

4.7 How to Use the Ray Trace Calculation Table

4.7.1 Calculation Table

Figure 4.12 is an illustration of a ray trace calculation table (Units: mm). If we specify the optical parameters (n, d, r, and *offset*) for each surface (first surface, second surface, etc.) in the top rows, and specify the incident rays (height w_0, slope $\tan\theta_0$) emitted from the object in the left columns, then the ray height w, the ray slope $\tan\theta$ and the optical path length (OPL) for each ray at each surface will be automatically calculated in each cell in the calculation units. (If an aberration-free lens is included, we need to add a ray which is very close to the principal ray at the top of the incident ray settings.) We can calculate the ray trace for any given incident ray passing through an arbitrarily specified optical surface, step by step, until we finally obtain the image height on the image plane. When we formulate a new calculation table, we can copy and paste the incident ray settings and the calculation units with their optical parameters (i.e., "Sphere" for the front surface of the lens, "Sphere" for the back surface of the lens, "Plane" for the next lens surface, etc.) in the order in which we need them, from the original table (which we now possess) to the new sheet. The row numbers of the cells must not be changed when copied and pasted. The reason is that while the column numbers use relative addresses, absolute addresses are used for the row numbers in some parts. It should also be noted that the aberration-free lens unit includes the parameters for the optical path length compensation ΔL_i and ΔL_o and that ΔL_o will be referenced by the "OPL" cell, in the next calculation unit. Thus the OPL cells next to an aberration-free lens unit should be different from the OPL cells of other units.

Optical Parameters →

Calculations →

									f=	100																			zb =
									p=	1.994813																			
	Units: mm	λ (mm)= 0.0005461		za=	0				p'=	-1.994881	za'=	10				za'=	16											za =	
	Surface no.			0_a surface	Object plane				1_{ff} surface (f1 in)		Sphere					2_{ff} surface (f1 out)		Sphere										3_{ff} surface (f1vertex)	
	n=			n=	1					n=	1.519						n=	1										n=	
	d=									d=	10						d=	6										d=	
	r=									r=	102.76406						r=	-102.7647										n'[*]=	
	of=									of=	0						of=	0										of=	
		$\theta 0$ (degrees)		$\Delta 0$	w^0	tan00		wc	0b	$\Delta 1$	w^1	tan01	*OPL1*	wc	0b	$\Delta 2$	w^2	tan02	OPL2	0b	$\Delta 3$								
	Ray1	0		0	-10	0		-10	0.0641056	0.4877059	-10	0.0333707	10.487706	-13.24538	-0.196946	-0.471898	-9.831782	0.1014732	7.6613832	0.101127	0								
	Ray2	0		0	-8	0		-8	0.0512718	0.3118643	-8	0.026611	10.311864	-10.58817	-0.157097	-0.30075	-7.856366	0.0807472	8.1863468	0.0805724	0								
	Ray3	0		0	-6	0		-6	0.0384465	0.175307	-6	0.0199752	10.175307	-7.936397	-0.117558	-0.168762	-5.887022	0.060313	8.5930733	0.06024	0								
	Ray4	0		0	-4	0		-4	0.0256275	0.0778773	-4	0.013307	10.077877	-5.288986	-0.078247	-0.074876	-3.922191	0.0400923	8.8827543	0.0400708	0								
	Ray5	0		0	-2	0		-2	0.0128127	0.0194638	-2	0.0066506	10.019464	-2.643671	-0.039086	-0.0187	-1.96035	0.0200115	9.05623	0.0200088	0								
	Ray6	0		0	0	0		0	0	0	0	0	10	0	0	0	0	0	9.114	0	0								
	Ray7	0		0	2	0		2	-0.012813	0.0194638	2	-0.006651	10.019464	2.643706	0.0390861	-0.0187	1.9603503	-0.020011	9.05623	-0.020009	0								
	Ray8	0		0	4	0		4	-0.025627	0.0778773	4	-0.013307	10.077877	5.288686	0.0782468	-0.074876	3.9221905	-0.040092	8.8827543	-0.040071	0								
	Ray9	0		0	6	0		6	-0.038446	0.175307	6	-0.019975	10.175307	7.9363972	0.1175578	-0.168762	5.8870215	-0.060313	8.5930733	-0.06024	0								
	Ray10	0		0	8	0		8	-0.051272	0.3118643	8	-0.026661	10.311864	10.588168	0.1570972	-0.30075	7.8563663	-0.080747	8.1863468	-0.080572	0								
	Ray11	0		0	10	0		10	-0.064106	0.4877059	10	-0.033371	10.487706	13.245383	0.1969464	-0.471898	9.8317816	-0.101473	7.6613832	-0.101127	0								

Settings for incident rays ↑ 1st surface ↑ 2nd surface ↑ 3rd↑

calculation unit calculation unit

Figure 4.12 Calculation table for the two-dimensional ray tracing

Distance z_a →

Calculation results for ray height w →

Surface no.	0	1	2	3	4	5	6	7	8	9	10	11	12
za=	0	10	60	63	260.3705	264.3705	314.3705	318.6705	321.0705	321.0705	496.0705	497.62828	499.62828
Ray1	0	-0.699268	-0.699268	-0.699268	-2.079608	-2.09796	-2.097961	-2.068399	-2.05808	-2.057995	-0.018164	-5.99E-06	0.0233064
Ray2	0	-0.524078	-0.524078	-0.524078	-1.558533	-1.572308	-1.572307	-1.550042	-1.542336	-1.5423	-0.013604	3.928E-06	0.0174747
Ray3	0	-0.349208	-0.349208	-0.349208	-1.038464								0116474
Ray4	0	-0.174551	-0.174551	-0.174551	-0.519065								005823
Ray5	0	0	0	0	0								0
Ray6	0	0.1745506	0.1745506	0.1745506	0.5190651								005823
Ray7	0	0.3492077	0.3492077	0.3492077	1.0384645								011647
Ray8	0	0.5240778	0.5240778	0.5240778	1.5585332								017475
Ray9	0	0.6992681	0.6992681	0.6992681	2.0796082								023306

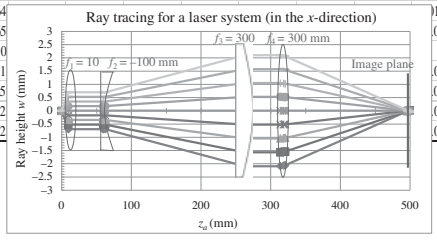


Figure 4.13 Calculation results for the two-dimensional ray tracing

4.7.2 Calculation Results

4.7.2.1 Ray Tracing from the Object to the Image

As shown in Figure 4.13, we can formulate a table of distances z and ray heights w, taking our data from our calculation results (whose units are in mm). Next, we graph the table, with the horizontal axis representing the distance z and the vertical axis representing the ray height w, as shown in Figure 4.13. The total ray tracing from the object to the image on the graph will then appear on the screen, and also, we can readily verify whether the rays will pass through the lens aperture or not.

Note

In the graph, the horizontal position includes a small error of the value Δ_k at the spherical surface. Nevertheless, the graph still provides a very useful way to view the ray tracing as a whole.

4.7.2.2 Ray Tracing Details Near the Image Plane

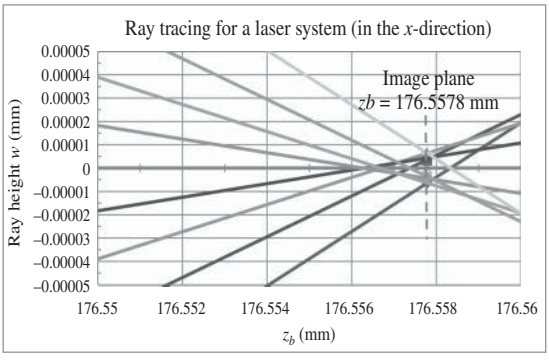
First, we set up three plane surface units in advance: one on the image plane itself, one in front of the image plane, and one behind it. Next, we formulate a table of distances z and ray heights w, taking our

Distance $z_b \rightarrow$

Calculation results for ray height $w \rightarrow$

Surface no.	10	11	12
zb=	175	176.55778	178.55778
Ray 1	-0.018164	-5.99115E-06	0.0233064
Ray 2	-0.013604	3.92807E-06	0.0174747
Ray 3	-0.00906	6.57505E-06	0.0116474
Ray 4	-0.004527	4.46373E-06	0.005823
Ray 5	0	0	0
Ray 6	0.0045275	-4.46373E-06	-0.005823
Ray 7	0.0090603	-6.57505E-06	-0.011647
Ray 8	0.0136039	-3.92807E-06	-0.017475
Ray 9	0.0181638	5.99115E-06	-0.023306

(a)



(b)

Figure 4.14 Calculation results for rays near the image plane. (a) Calculation results and (b) graph

data from the calculated results for these three planes as shown in Figure 4.14a. Finally, we graph these results, with z on the horizontal axis and w on the vertical axis as shown in Figure 4.14b. A detailed ray tracing will then be visible on the graph, near the image. This example shows that the rays on the image plane converge at a single point, to an accuracy of ±6.42 nm.

4.7.2.3 Phase (Difference) on the Image Plane and Wave-Front Aberration

The phase (difference) on the image plane shows the differences in the cumulative optical path lengths of the rays from the object to the image plane (we use a positive sign when the phase leads that of the principal ray).

On the other hand, the wave-front aberration is the phase difference between the wave front after passing through the last lens and the reference wave front (the wave front when the optical system has no aberration). When the optical system has no aberration, rays emitted from the object point diverge (forming a spherical wave front), and then pass through the optical system. After passing through the last lens, the rays (which have a spherical wave front) will converge precisely at a single point: the ideal image point. Thus the center of curvature for the reference wave front should be located at the ideal image point, and the radius can have any value which is significantly greater than the wavelength λ.

Figure 4.15 illustrates the principles of wave-front aberration. $\mathbf{S_0}$ is the reference wave-front surface exiting the last lens, which includes the center of the exiting pupil C, and whose center of curvature is located at the ideal image point Q_0 $(x_0,y_0,0)$. $\mathbf{S'}$ is the surface of the wave front exiting the last lens, which includes C, and Q(x,y,z) is its image point. In Figure 4.15 the ζ-axis and the z-axis are situated on the optical axis, and d is the distance between the $\xi-\eta$ plane (the exiting plane) and the $x-y$ plane (the ideal image plane). A_0 (ξ_0,η_0,ζ_0) is the point of intersection between the exiting ray and the reference wave front $\mathbf{S_0}$, and A′ (ξ',η',ζ') is the point of intersection between the exiting ray and the wave front $\mathbf{S'}$.

In Figure 4.15, the wave-front aberration W can be calculated as follows:

$$W = \frac{n\mathrm{A}_0\mathrm{A}'}{\lambda} \quad [\lambda]$$

where n is the refractive index ($n = 1$ in air) and λ is the wavelength.

In this text, we will adopt the sign convention that W is positive when A′ is to the right of A_0 (i.e., when its phase leads that of the reference). In an optical system using multiple lenses, the total wave-front aberration can then be calculated as the sum of the aberrations generated by each lens.

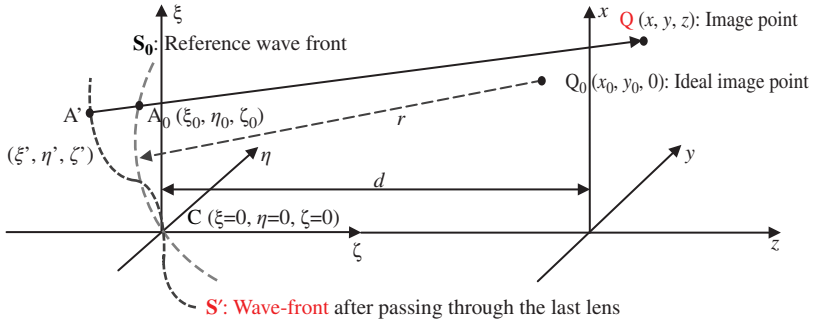


Figure 4.15 Wave-front aberration

Surface no.	6	6		
	tanθ6	θ6 (degrees)	OPL ΣL'	Phase (λ)
Ray1	0.0556504	3.1852496	197.07047	0.0586305
Ray2	0.0444936	2.547616	197.0705	0.0024007
Ray3	0.0333542	1.9103486	197.07051	–0.009206
Ray4	0.0222284	1.2733847	197.0705	–0.00633
Ray5	0.0111119	0.6366365	197.0705	–0.001809
Ray6	0	0	197.0705	0
Ray7	–0.011112	–0.636637	197.0705	–0.001809
Ray8	–0.022228	–1.273385	197.0705	–0.00633
Ray9	–0.033354	–1.910349	197.07051	–0.009206
Ray10	–0.044494	–2.547616	197.0705	0.0024007
Ray11	–0.05565	–3.18525	197.07047	0.0586305

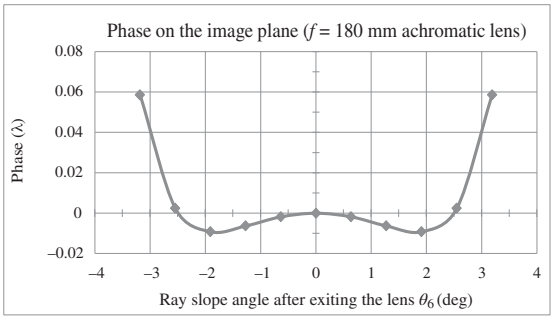


Figure 4.16 A calculation example for the phase (difference) on the image plane

Calculation of the Phase (Difference) on the Image Plane

Phase (difference) on the image plane can be obtained by calculating the cumulative OPL from the object to the image plane for each ray. The OPL difference for each ray from that of the principal ray will yield the phase difference on the image plane as shown in the equation below. Here, we set the sign to positive when the phase leads that of the principal ray.

$$\text{Phase (difference) on image plane} = -\left(\sum \text{OPL} - \sum \text{OPL}_{\text{Principal ray}}\right)/\lambda \quad [\lambda]$$

where λ is the wavelength.

Figure 4.16 is a calculation example of the phase (difference) on the image plane.

Here, the horizontal axis on the graph represents the ray slope angle θ (in degrees) at the image plane and the vertical axis on the graph represents the phase (difference) (λ) at the image plane.

Calculation of the Wave-Front Aberration

In this text, we defined the reference wave front as a sphere centered at the ideal image point, on whose surface the point of intersection between the principal ray and the last vertex plane (a vertical plane which includes the vertex of the last surface of the last lens) lies. In the problems given at the end of this chapter, the image plane is set at a distance so as to minimize the spot size. We then treat the curvature center of the reference wave front (the ideal image point) as the point of intersection between the principal ray and the image plane.

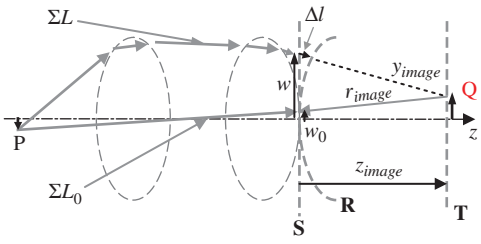


Figure 4.17 Wave-front aberration calculation

In Figure 4.17, **S**, **R**, and **T** are the last vertex plane, the reference wave front and the image plane, respectively, and P and Q (z_{image}, y_{image}) is the object point and the ideal image point which is situated at the point of intersection between the principal ray and the image plane. Here,

z_{image} = Distance from the last vertex plane **S** to the image plane **T**
y_{image} = Ray height of the principal ray at the image plane **T**
r_{image} = Radius of curvature of the reference wave front
ΣL = Cumulative optical path length from the object to the last vertex plane **S**
Δl = Optical path length from the last vertex plane **S** to the reference wave front **R**
ΣL_0 = Cumulative optical path length of the principal ray traveling from the object P to the reference wave front **R**
w = Ray height at the plane **S**
w_0 = Principal ray height at the plane **S**.

r_{image} can be calculated as:

$$r_{image} = \sqrt{{z_{image}}^2 + (y_{image} - w_0)^2}$$

The cumulative optical path length $\Sigma L'$ from the object to the reference wave front **R** is:

$$\sum L' = \sum L + \Delta l$$

$\Delta\, l$ is the optical path length from the plane **S** to the sphere **R** for each ray. We regarded this ray as heading toward the ideal image point Q, since

$$r_{image} >> |w_{image} - y_{image}|$$

where w_{image} = ray height at the image plane.

Here, Δl can be approximated as follows:

$$\Delta l \approx \sqrt{{z_{image}}^2 + (y_{image} - w)^2} - r_{image}$$

The wave-front aberration W will then be:

$$W \approx -\frac{\Sigma L' - \Sigma L_0}{\lambda} = -\frac{\Sigma L + \Delta l - \Sigma L_0}{\lambda} \quad [\lambda]$$

Figure 4.18 shows a calculation example for the wave-front aberration.

4.7.3 Aspheric Lens Ray Tracing

Ray trace calculations for aspheric lenses can be performed using VBA programming. The VBA program has a "Start Calculation" button on its worksheet, and by clicking on this button, the calculation can be

Wave front aberration $= \sqrt{(r^2+w^2)} - r = -(\Sigma L_i + \Delta l - \Sigma L_0)/\lambda$

Surface no.	4	4	4	4		WF Aberration
	tan06	06 (degrees)	OPL ΣL	w4	Δl	W (λ)
Ray1	0.0556504	3.1852496	20.223257	−9.826787	0.2732313	0.0213262
Ray2	0.0444936	2.547616	20.321805	−7.856179	0.1746832	0.0216281
Ray3	0.0333542	1.9103486	20.398313	−5.889099	0.0981794	0.0147121
Ray4	0.0222284	1.2733847	20.452886	−3.92463	0.0436101	0.0071268
Ray5	0.0111119	0.6366365	20.4856	−1.961886	0.0108988	0.0018482
Ray6	0	0	20.4965	0	0	0
Ray7	−0.011112	−0.636637	20.4856	1.9618859	0.0108988	0.0018482
Ray8	−0.022228	−1.273385	20.452886	3.9246297	0.0436101	0.0071268
Ray9	−0.033354	−1.910349	20.398313	5.8890986	0.0981794	0.0147121
Ray10	−0.044494	−2.547616	20.321805	7.856179	0.1746832	0.0216281
Ray11	−0.05565	−3.18525	20.223257	9.8267866	0.2732313	0.0213262

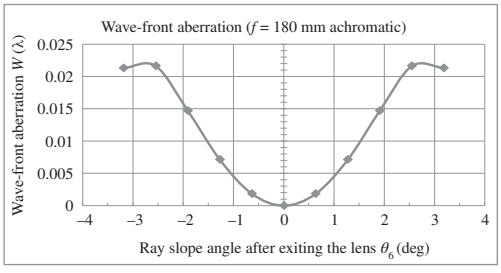


Figure 4.18 Calculation results for the wave-front aberration

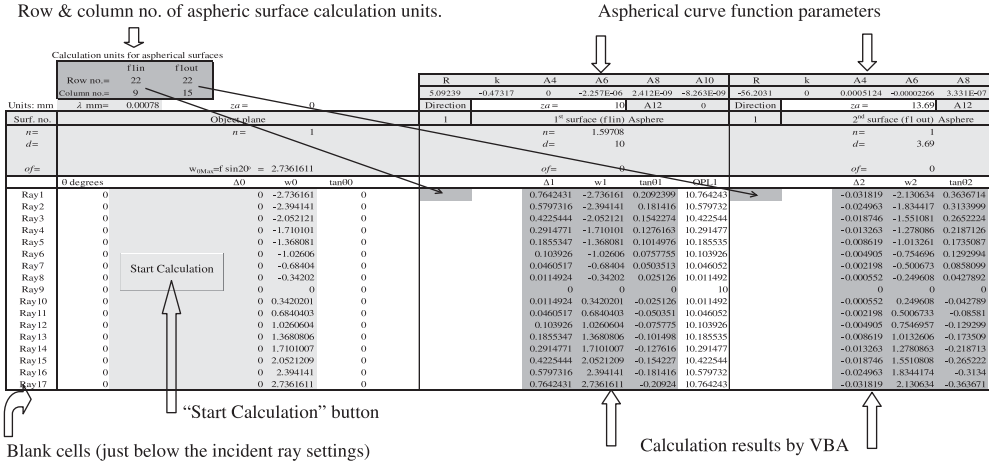


Figure 4.19 Calculation table for the aspheric lens ray tracing

initiated. When creating a new calculation table, the user can generate a copy of the sheet he/she already has, by using the "Move or copy" function, which appears when the user right-clicks on the sheet's name bar, at the bottom of the screen. After copying the sheet, the calculation table can then be altered in the manner outlined above (see Section 4.7.1).

In Figure 4.19, first input the address (row and column) of the header cell for the aspherical surface calculation unit. Additional header addresses may be input to the right of the first address, if multiple calculations are desired. The cells to the right of the last row and column number input should be left blank, so that the program can recognize the end of the row and column number settings.

Aspheric lens ray tracing requires certain parameters for the aspherical curve (R, k, A_4, A_6, A_8, A_{10}, A_{12}, and the setting direction) which other ray tracings do not possess. These parameters can also be altered as required. The cells in the row just below the incident ray settings should be left blank, so that the program will realize that it has reached the last ray.

4.7.4 Explanation of Symbols Used

In the calculation table, we will be using the following symbols:

z_a = z-distance from the object (in mm)
z_b = z-distance from the vertex of the last surface of the last lens (in mm)
n_k = Refractive index of the medium immediately behind the kth surface
d_k = Distance from $(k-1)$th to kth surface (in mm)
r_k = Radius of curvature of the kth surface (in mm) (For a spherical surface, this will be positive when the vertex is on the left)
f_k = Focal length of the aberration-free lens at the kth surface (in mm)
α = Surface angle (measured from the vertical) (in deg) (only for a plane surface) or tilt angle of a lens (measured from the vertical) (in deg) (α for TiltStart, $-\alpha$ for TiltEnd)
of_k = Offset of the kth surface axis from the $(k-1)$th surface axis (in mm)
Δ_k = Ray's z-coordinate at the kth surface measured from the vertex (in mm)
w_k = Ray height at the kth surface (in mm)
$\tan\theta_k$ = Slope of the ray after exiting the kth surface
OPL_k = Optical path length from $(k-1)$th to the kth surface (in mm)

4.8 A Method for Generating a Ray Trace Calculation Table Using a VBA Program

Previously, we looked at ray trace calculations that use a calculation table upon which the Excel cells can perform their calculations directly. This text also provides readers with an alternative way of performing calculations, using VBA programming. The reader can choose to perform the calculation either directly on cells in an Excel spreadsheet or using a VBA program. In order to perform calculations with the aid of a VBA program, the following steps need to be carried out.

4.8.1 Two-Dimensional Ray Trace Calculation Table Using VBA Programming

Specify the following items on the calculation table as shown in Figure 4.20:

① **Row Number and column Number**
The first step is to specify the row number and the column number of the header cell in the optical configuration table. (This is the entry labeled ① in Figure 4.20.)

② **Wavelength and refractive index of the medium surrounding the object**
Next, we specify the wavelength λ (in mm) and refractive index n_0 of the medium surrounding the object.

③ **Optical configuration settings**
We can then determine the optical configuration by specifying the boundary surface parameters, in the following order.

1. **Surface no.:** Surfaces should be numbered from first to last. (This field may also be left blank.) If a wave-front aberration calculation is required, then we can insert a vertical plane at the vertex of the last surface of the last lens and type "Aberration" in the "Surf. no." cell. (The reference wave front will include the point of intersection between the principal ray and this plane, and its center of curvature will be located at the ideal image point, which can be calculated using paraxial theory.)

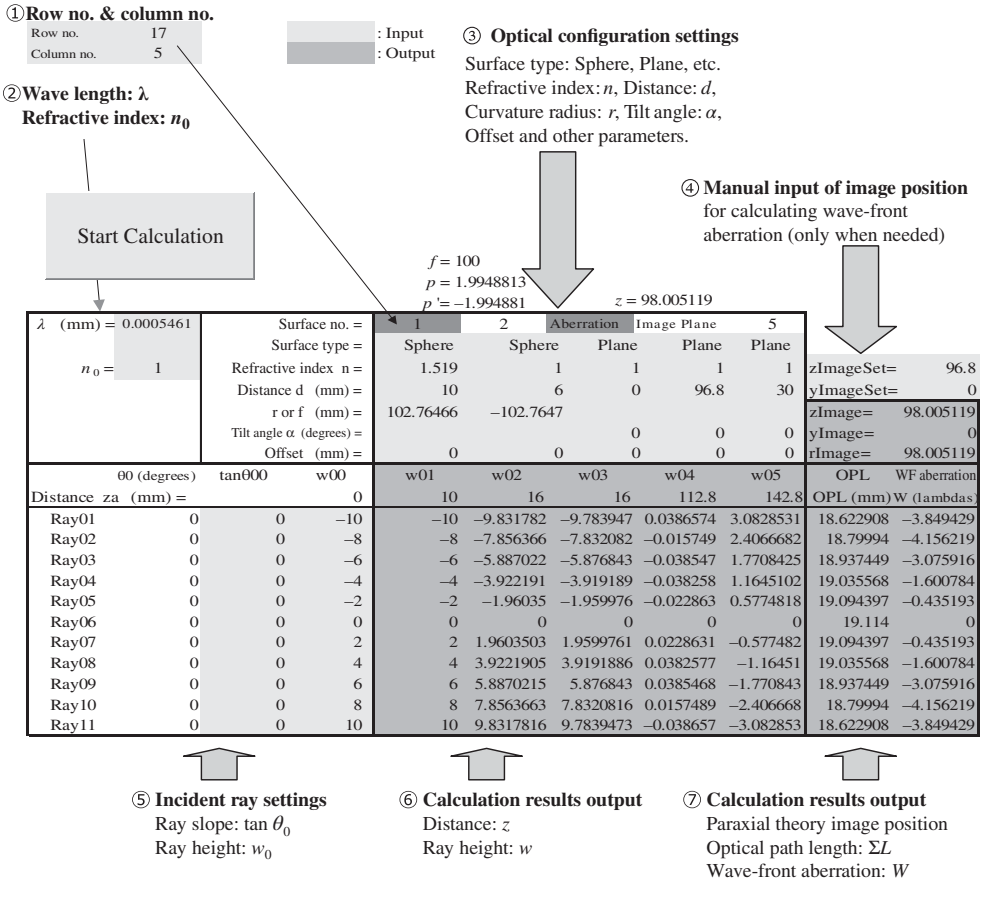


Figure 4.20 Calculation table for the VBA two-dimensional ray tracing

2. **Surface type:** "Sphere," "Plane," "AbeFreeLens," "TiltStart," or "TiltEnd." (This program does not support aspherical lenses.)
3. ***n:*** The refractive index of the medium lying behind the surface.
4. ***d:*** The distance between the vertices of the surface and the preceding surface (along the axis) (in mm).
5. ***r* or *f*:** The curvature radius of the surface (input for spherical surfaces only), or the focal length (input for aberration-free lenses only) (in mm).
6. ***α:*** The surface angle, as measured from the vertical (in deg). This is used for "Plane," "TiltStart," and "TiltEnd" also.
7. **Offset:** Offset of the vertex from the axis of the preceding surface (in mm). (Input for spherical surfaces and aberration-free lenses.)

④ **Manually specify the image position in order to calculate the wave-front aberration**

The y–z image coordinates (in mm) can also be manually input, when performing a wave-front aberration calculation. Usually, there is no need to specify these coordinates, as the wave-front aberration

will (by default) be calculated using the ideal image point as a given, which is calculated using paraxial theory. However, if we input the image coordinates into these cells, then the wave-front aberration will be calculated using this fixed image position as a datum.

⑤ **Incident ray settings**

Specify the following values for any rays emitted from an arbitrary object point.

1. Ray slope tan θ_0
2. Ray height (object height) w_0 (in mm).

Note

It is important to specify the details of the rays, listing each ray in the correct order, so that the principal ray is the middle one in the list of rays. If the principal ray is not the middle item, then the user will have to type "Principal ray" in the ray number cell, in order to identify it as such. (The image point can then be calculated from the principal ray and a ray traveling very close to it. The wave-front aberration can subsequently be derived, based on this principal ray with a value of zero.)

After specifying all the items above, click "Start Calculation." The program will then start performing the ray trace calculation. The calculation results will be displayed on the calculation table after the calculation has been completed.

4.8.2 Output of Calculation Results

The following calculation results are output on the calculation table as shown in Figure 4.20:

⑥ **Distance *z* and ray height *w***

The cumulative distance z from the object to the surface and the ray heights w will be output on the calculation table (in mm). We can also draw a ray trace graph from these z–w data with the ray heights w on the vertical axis, and the distance z on the horizontal axis.

Note

The z-position on the graph will contain a small error for a spherical surface (or a surface set at a tilt), due to the difference between the z-position of the vertex (or on the axis) and the point where the ray enters the surface.

⑦ **Ideal image point calculated by paraxial theory, optical path length, and wave-front aberration**

See Figure 4.17 which depicts the following items.

1. **Ideal image point calculated by paraxial theory ($\mathbf{z_{Image}}$, $\mathbf{y_{Image}}$, $\mathbf{r_{Image}}$)**

   The ideal image position Q (z_{image}, y_{image}, and r_{image}) (in mm) which is shown in Figure 4.17, is calculated using the principles of paraxial theory, from two rays (the principal ray and a ray traveling very close to it).

   z_{image} = Distance from the last vertex plane **S** to the image plane **T**
   y_{image} = Ray height of the principal ray at the image plane **T**
   r_{image} = Radius of curvature of the reference wave front = $\sqrt{[{z_{image}}^2 + (y_{image} - w_0)^2]}$
   w_0 = Principal ray height at the plane **S**.

   (Plane **S** = a vertical plane situated at the vertex of the last surface of the last lens.)
2. **Optical path length ΣL** (=OPL)

   This is the OPL (in mm) from the object P to the plane **S**.

3. **Wave-front aberration W**
   The wave-front aberration W (in rad) is the phase difference between the wave front of each ray and the reference wave front **R**.

$$W = -\left(\sum L + \Delta l - \sum L_0\right)/\lambda \quad (\lambda)$$

where,

$$\Delta l \approx \sqrt{{z_{image}}^2 + (y_{image} - w)^2} - r_{image}$$

w = Ray height at the plane **S**
$\sum L_0$ = Cumulative OPL of the principal ray traveling from the object P to the reference wave front **R**

(A positive sign for W indicates that the phase leads the reference wave front **R**.)

Notes

1. The cells just below the incident ray settings should be left blank, so that the program will realize that it has reached the last ray.
2. The cell immediately to the right of the rightmost "Surface type" should be left blank, so that the program will realize that it has reached the last surface. [2]

4.9 Sample Ray Tracing Problems

In this section, we shall begin by designing some simple lenses by applying paraxial theory, and then calculate the aberration for these lenses, using ray tracing techniques.

Please download the Excel files from the companion web site, which show calculation details for all problems in this book as listed in Appendix G

Problem 4.1 Designing a Bi-convex Lens

Design a bi-convex lens with a focal length of $f = 100$ mm.
Use BK7 optical glass ($n = 1.519$ when $\lambda = 546.1$ nm), where the lens diameter $D = 40$ mm and the center thickness $t = 6$ mm. Calculate the curvature radii and the principal points of the lens.

Solution

Calculate the curvature radius of the surfaces r and the position of the principal points p, p' in Figure 4.21, as follows:
(1) Calculate the radius of curvature for the surface
The radius of curvature r of a bi-convex lens with the same degree of curvature for the front and rear surfaces is given by the following equation:

$$r = \frac{n(n-1)f' + \sqrt{n^2(n-1)^2 f'^2 - n(n-1)^2 f' t}}{n} \quad (1.38)$$

[2] This program has been successfully tested using Microsoft Office Excel 2003, Microsoft Office Excel 2007, and Microsoft Office Excel 2010.

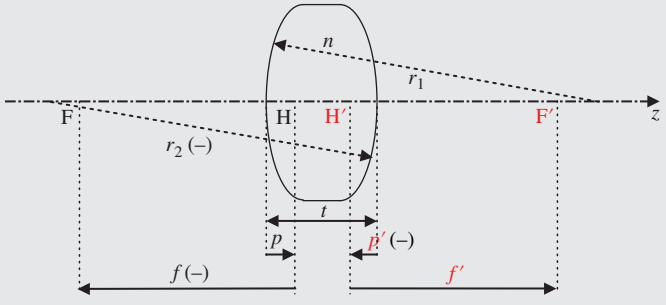


Figure 4.21 A bi-convex lens

If we substitute the values $n = 1.519$, $f' = 100$ mm, and $t = 6$ mm into Equation (1.38), then the radius of curvature r will be:

$$r = 102.7647 \text{ mm}$$

(2) Calculate the principal points

The distance p from the front surface vertex to the primary principal point, and the distance p' from the rear surface vertex to the secondary principal point will be as follows:

$$p = -\frac{r_1 t}{n(r_2 - r_1) + (n-1)t} \tag{1.35}$$

$$p' = -\frac{r_2 t}{n(r_2 - r_1) + (n-1)t} \tag{1.36}$$

If we substitute the given values, $n = 1.519$, $t = 6$ mm, and the values obtained above for r ($r_1 = r = 102.7647$ mm, $r_2 = -r = -102.7647$ mm) into Equation (1.35) and Equation (1.36), we obtain the following values for p and p':

$$p = 1.994881 \text{ mm}, \quad p' = -1.994881 \text{ mm}$$

Problem 4.2 Focusing a collimated beam using a bi-convex lens

(See Appendix B where refractive indices of BK7 are shown in a table.)

Use the bi-convex lens ($f = 100$ mm) designed in Problem 4.1 to focus a collimated beam 20 mm in diameter onto a single spot. Perform ray tracing calculations for the optical system and obtain the focused spot size and its distance from the lens.

Solution

(1) Input the optical configuration (Figure 4.22)

$\lambda = 546.1$ nm	Surface 1: **Sphere**	Surface 2: **Sphere**	Image plane: **Plane**
Refractive index n	1.519	1	1
Distance d (mm)	10 (arbitrary)	6	96.8
Radius of curvature r (mm)	102.76	−102.76	–

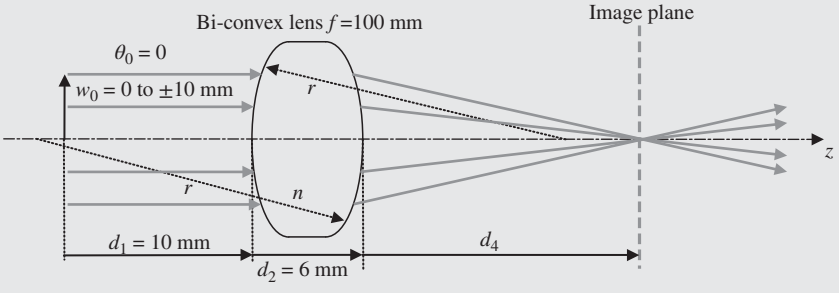


Figure 4.22 Beam focusing using a bi-convex lens

(2) Specify details for the incident rays (Figure 4.22)

w_0	0 to ±10 mm
θ_0	0°

(3) Perform ray tracing calculations:
The results of the ray tracing calculations are shown in the graph in Figure 4.23. The horizontal and vertical axes represent the distance (z_a: starting from the object, or z_b: starting from the rear surface vertex of the last lens) and ray height of the ray paths, respectively.

(4) Calculation results

Image distance d_4	Focused spot radius w	Wave-front aberration W
96.8 mm from the rear surface vertex	±0.039 mm	+0/−4.2 λ

Note

The radius of the focused beam is relatively large (±0.039 mm), and so is the wave-front aberration.

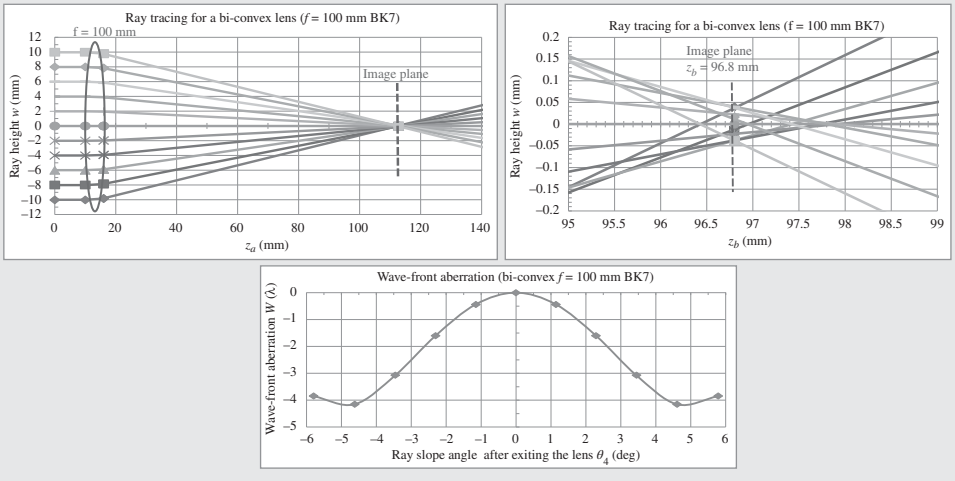


Figure 4.23 Ray tracing results for a bi-convex lens

Problem 4.3 Designing a plano-convex lens – Part 1

Design a plano-convex lens with a focal length of $f = 100$ mm. Use BK7 optical glass ($n = 1.519$ for $\lambda = 546.1$ nm), or SF6 ($n = 1.813$ for $\lambda = 546.1$ nm) where the lens diameter $D = 40$ mm, and the center thickness $t = 6$ mm. Calculate the curvature radius and the principal points of the lens.

Solution

Calculate the curvature radius of the surfaces r and the position of the principal points p, p' in Figure 4.24, as follows:

(1) Calculate the radius of curvature of the surface

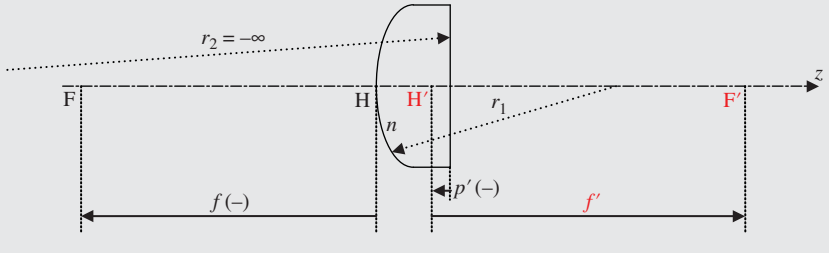


Figure 4.24 A plano-convex lens

The focal length of a simple lens can be represented by the following equation:

$$\frac{1}{f'} = -\frac{1}{f} = (n-1)\left\{\left(\frac{1}{r_1} - \frac{1}{r_2}\right) + \frac{(n-1)t}{nr_1r_2}\right\} \tag{1.34}$$

The surface of the plano-convex lens is flat on one side, so if we set $r_2 = -\infty$, then f' becomes:

$$f' = -f = \frac{1}{(n-1)}r \tag{1.26'}$$

If we substitute the values $f' = 100$ mm, $n = 1.519$ (or 1.813) into Equation (1.26′), we can derive r:

$$r = 51.9 \text{ mm (BK7)} \quad s = 81.3 \text{ mm (SF6)}$$

(2) Calculate the principal points

The positions of the primary and secondary principal points p and p' can be calculated as follows:

$$p = -\frac{r_1 t}{n(r_2 - r_1) + (n-1)t} \tag{1.35}$$

$$p' = -\frac{r_2 t}{n(r_2 - r_1) + (n-1)t} \tag{1.36}$$

If we substitute the values $r_1 = 51.8$ mm/81.3 mm (BK7/SF6), $r_2 = -\infty$, $n = 1.519/1.813$ (BK7/SF6), and $t = 6$ mm, into Equation (1.35) and Equation (1.36), we can derive the principal points p and p':

$$p = 0 \text{ mm}$$

$$p' = -3.95 \text{ mm}/-3.31 \text{ mm (BK7/SF6)}$$

Note

The primary principal point of a plano-convex lens will be at the front surface vertex, since $p = 0$.

Problem 4.4 Focusing a collimated beam using a plano-convex lens – Part 1

Use the plano-convex lens ($f = 100$ mm) designed in Problem 4.3 to focus a collimated beam 20 mm in diameter onto a single spot. Perform ray tracing calculations for the optical system and obtain the focused spot size and its distance from the lens.

Solution

(1) Input the optical configuration BK7(SF6) (Figure 4.25)

$\lambda = 546.1$ nm	Surface 1: **Sphere**	Surface 2: **Plane**	Image plane: **Plane**
Refractive index n	1.519 (1.813)	1	1
Distance d (mm)	10 (arbitrary figure)	6	95.22 (96.26)
Curvature radius r (mm)	51.9 (81.3)	–	–

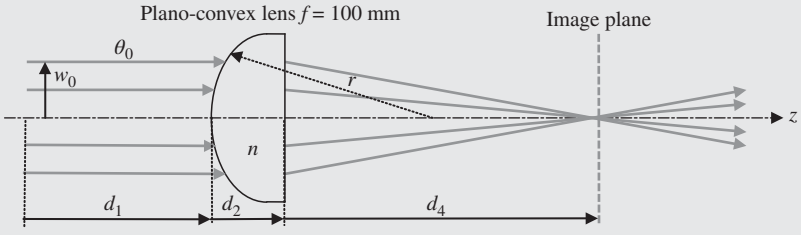


Figure 4.25 Beam focusing using a plano-convex lens

(2) Specify details for the incident rays (Figure 4.25)

w_0	0 to ±10 mm	θ_0	0°

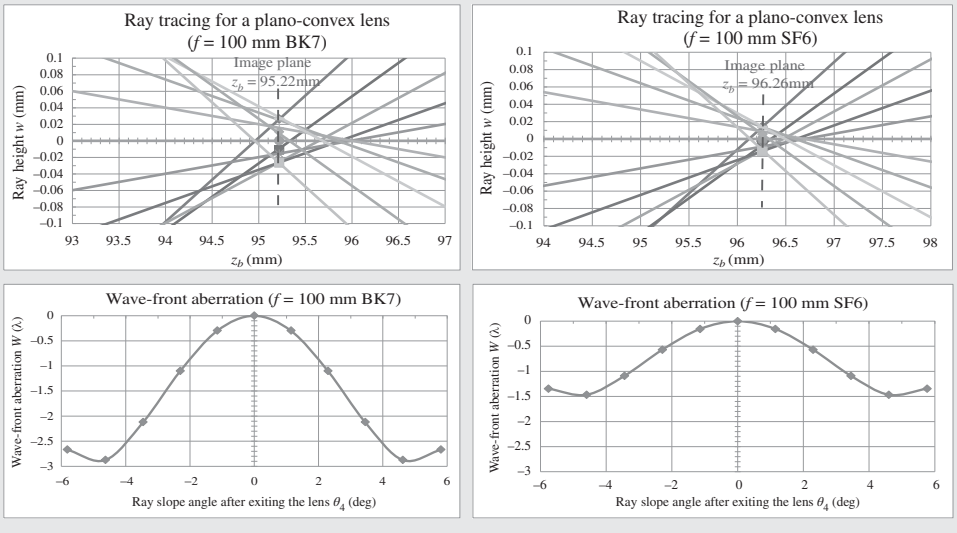


Figure 4.26 Ray tracing results for a plano-convex lens

(3) Perform ray tracing calculations (Figure 4.26)
(4) Calculation results

	BK7 (n = 1.518)	SF6 (n = 1.813)
Image distance d_4 (mm)	95.22	96.26
Focused spot radius w (mm)	±0.027	±0.014
Wave front aberration W (λ)	+0/−2.9	+0/−1.5

Notes

1. A plano-convex lens has less aberration than a bi-convex lens of the same focal length.
2. For a given focal length, lenses with a higher refractive index have less aberration.

Problem 4.5 Designing a plano-convex lens – Part 2

Design a plano-convex lens with a focal length $f = 180$ mm. Use BK7 optical glass ($n = 1.519$ for $\lambda = 546.1$ nm), or SF6 ($n = 1.813$ for $\lambda = 546.1$ nm) where the lens diameter $D = 40$ mm, and the center thickness $t = 4$ mm. Calculate the curvature radius and the principal points of the lens.

Solution

Calculate the curvature radius of the surfaces r and the position of the principal points p, p' in Figure 4.27, as follows:

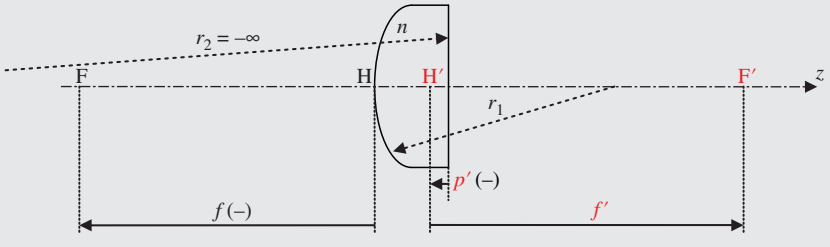


Figure 4.27 A plano-convex lens

(1) Calculate the radius of curvature of the surface

The equation for the radius of curvature for a plano-convex lens is the same as for Problem 4.3.

$$r = (n - 1) f' \tag{1.26''}$$

If we substitute the values $f' = 180$ mm, $n = 1.519$ (BK7) or 1.813 (SF6), into Equation (1.26″), we can derive r as follows:

$r = 93.42$ mm (BK7) $\quad$ $r = 146.34$ mm (SF6)

(2) Calculate the principal points

The position of the primary and secondary principal points p and p' can be calculated as follows:

$$p = -\frac{r_1 t}{n(r_2 - r_1) + (n - 1)t} \tag{1.35}$$

$$p' = -\frac{r_2 t}{n(r_2 - r_1) + (n - 1)t} \tag{1.36}$$

If we substitute the values $r_1 = 93.24$ mm/146.34 mm (BK7/SF6), $r_2 = -\infty$, $n = 1.518/1.813$ (BK7/SF6) and $t = 4$ mm, into Equation (1.35) and Equation (1.36), we can derive the principal points p and p' as follows:

$p = 0$ mm(BK7), $p' = -2.63331$ mm(BK7) $\quad p = 0$ mm(SF6), $p' = -2.20629$ mm (SF6)

Note

The primary principal point of a plano-convex lens will be at the front surface vertex, since $p = 0$.

Problem 4.6 Focusing a collimated beam using a plano-convex lens – Part 2

Use the plano-convex (BK7 or SF6) lens ($f = 180$ mm) designed in Problem 4.5 to focus a collimated beam 20 mm in diameter onto a single spot. Perform ray tracing calculations for the optical system and obtain the focused spot size and its distance from the lens.

Solution

(1) Input the optical configuration (Figure 4.28)

$\lambda = 546.1$ nm	First surface BK7 (SF6)	Second surface	Image plane
Type of surface	Sphere	Plane	Plane
Refractive index n	1.519 (1.813)	1	1
Distance d (mm)	10 (Arbitrarily set)	4 (=t)	176.9 (177.55)
Curvature radius r (mm)	−93.42 (−146.34)	–	–

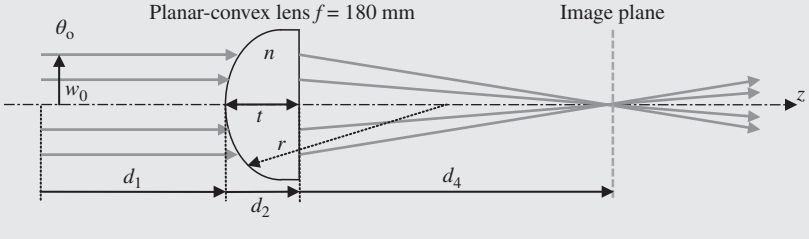


Figure 4.28 Beam focusing using a plano-convex lens

(2) Specify details of the incident rays (Figure 4.28)

w_0	0 to ±10 mm	θ_0	0°

(3) Perform ray tracing calculations (Figure 4.29)

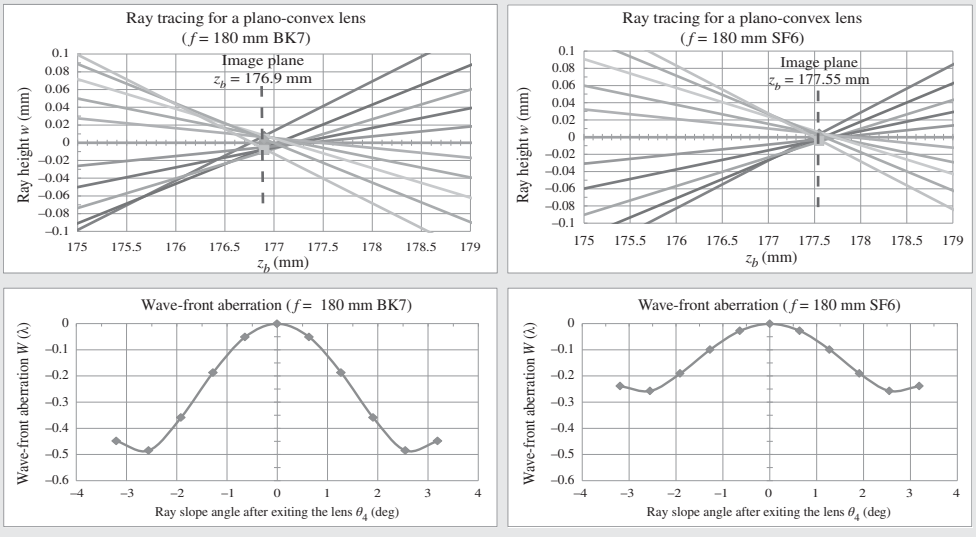


Figure 4.29 Ray tracing results for a plano-convex lens

(4) Calculation results

	BK7 (n = 1.519)	SF6 (n = 1.813)
Image distance d_4 (mm)	176.9	177.55
Focused spot radius w (mm)	±0.008	±0.004
Wave front aberration W (λ)	+0/−0.5	+0/−0.26

Note

The aberration is less than it was for Problem 4.4. The aberration for a focused spot is smaller in an optical system with a longer focal length.

Problem 4.7 Ray tracing for two planar plates

Perform ray tracing calculations for an optical system consisting of an incident collimated beam and two planar plates (refractive index n = 1.519, thickness t = 0.707 mm) 10 mm apart, set at angles of 45° and −45° to the vertical axis, respectively.

Solution

(1) Input the optical configuration (Figure 4.30)

	First surface	Second surface	Third surface	Fourth surface	Image plane
Type of surface	Plane	Plane	Plane	Plane	Plane
Refractive index n	1.519	1	1.519	1	1
Distance d (mm)	10	1	10	1	10
Surface angle α (°)	45	45	−45	−45	0

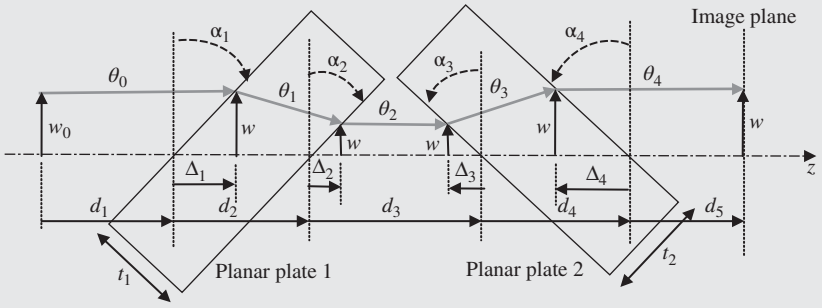


Figure 4.30 Two planar plates

(2) Specify details of the incident rays (Figure 4.30)

w_0	θ_0
0 to ±10 mm	0°

(3) Perform ray tracing calculations (Figure 4.31)

Notes

1. The incident rays entering the first plate were bent downwards, resulting in the output rays exiting the plate 0.237 mm below the incident rays, whereas the output rays exiting the second plate were bent upwards by 0.237 mm. Thus the net effect for rays passing through the entire optical system was zero.
2. A plane unit ($n = 1$, $\alpha = 0°$) can be used anywhere, without having any effect on the path of the exiting ray, and it tells us the ray height at that location.
3. Alternative solutions for this problem using "TiltStart" and "TiltEnd" are written in the Excel calculation files. See the files "Chapter 4: Two-dimensional ray tracing" and "Two-dimensional VBA ray tracing."

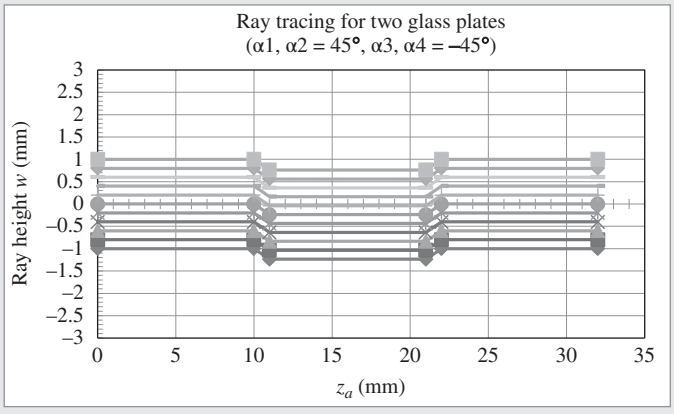


Figure 4.31 Ray tracing results for two planar plates

Problem 4.8 Focal lengths and principal points for a combination of two lenses

Calculate the focal length f and the principal point p, p' for a combination of two lenses which are in contact along a common surface, as described in the following table and Figure 4.32.

First lens	$n_1 = 1.519$ (BK7), $t_1 = 4.3$ mm, $r_1 = 103.1$ mm, $r_2 = -78.22$ mm
Second lens	$n_2 = 1.652$ (SF2), $t_2 = 2.4$ mm, $r_2 = -78.22$ mm, $r_3 = -291$ mm

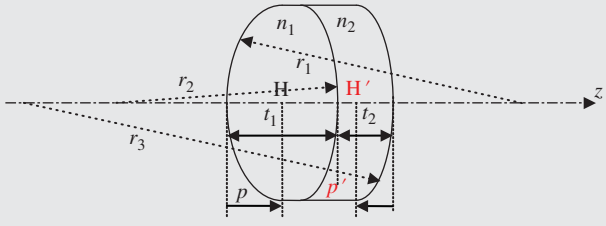


Figure 4.32 An achromatic lens

Solution

Calculate the focal length f' and the principal points p, p' of the lens after they are combined as follows (Figure 4.32).

(1) The focal lengths f_1, f_2, and the principal points p_1, p_1', p_2, p_2' for the simple lenses before they come into contact with each other will be:

$$\frac{1}{f'} = -\frac{1}{f} = (n-1)\left\{\left(\frac{1}{r_1} - \frac{1}{r_2}\right) + \frac{(n-1)t}{nr_1r_2}\right\} \tag{1.34}$$

$$p = -\frac{r_1 t}{n(f_2 - r_1) + (n-1)t} \tag{1.35}$$

$$p' = -\frac{r_2 t}{n(r_2 - r_1) + (n-1)t} \tag{1.36}$$

Substituting the values of n, r_1, r_2, t into Equation (1.34), Equation (1.35) and Equation (1.36), we get:

First lens : $f_1' = -f_1 = 86.3966$ mm, $p_1 = 1.62277$ mm, $p_1' = -1.2312$ mm
Second lens : $f_2' = -f_2 = -164.805$ mm, $p_2 = -0.5364$ mm, $p_2' = -1.9957$ mm

(2) Combination of two lenses (Figure 4.33)

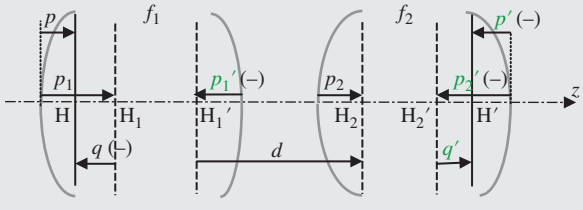


Figure 4.33 Combination of two lenses

The focal length f and the principal points H, H′ for the combination of the two lenses f_1, f_2 will be:

$$f = \frac{f_1 f_2}{d - f_1' + f_2} \tag{1.39}$$

$$f' = -\frac{f_1' f_2'}{d - f_1' + f_2} \tag{1.40}$$

$$q = \mathrm{H_1H} = \frac{f_1 d}{d - f_1' + f_2} \tag{1.41}$$

$$q' = \mathrm{H_2'H'} = \frac{f_2' d}{d - f_1' + f_2} \tag{1.42}$$

The value of d is equal to $d = p_2 - p_1' = 0.60472$ mm, when the two lenses come into contact with each other. If we substitute the values of f_1, f_2, d into Equation (1.39), Equation (1.40), Equation (1.41) and Equation (1.42), we obtain:

Combined focal length: $f' = 180.001$ mm,
Combined principal points: $q = -0.7588$ mm, $q' = -1.4474$ mm

If we rewrite the principal points as the distances from the vertex, p, p' ($p = q + p_1$, $p' = q' + p_2'$), we get:

Combined focal length: $f' = 180.001$ mm,
Combined principal points: $p = 0.8640$ mm, $p' = -3.4431$ mm

Note

Combining a lens where $f_1' = 86.4$ mm with a lens where $f_2' = -164.8$ mm results in a lens where $f' = 180$ mm. This is an example of an achromatic lens.

Problem 4.9 Focusing a collimated beam using an achromatic lens

Use the achromatic lens ($f = 180$ mm) designed in Problem 4.8 to focus a collimated beam 20 mm in diameter onto a single spot. Perform ray tracing calculations and obtain the focused spot size and its distance from the lens.

Solution

(1) Input the optical configuration: achromatic $f = 180$ mm (Figure 4.34)

$\lambda = 546.1$ nm	First surface BK7	Second surface SF2	Third surface	Image plane
Surface type	Sphere	Sphere	Sphere	Plane
Refractive index n	1.519	1.652	1	1
Distance d (mm)	10	4.3 ($= t_1$)	2.4 ($= t_2$)	176.574
Curvature radius r (mm)	103.1	−78.22	−291	–

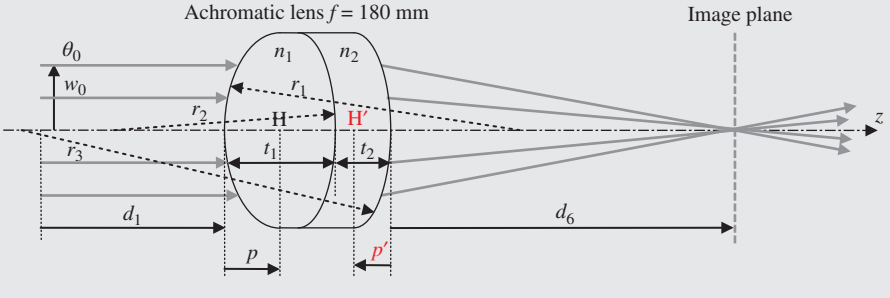


Figure 4.34 Beam focusing using an achromatic lens

(2) Specify details of the incident rays (Figure 4.34)

w_0	0 to ±10 mm	θ_0	0°

(3) Perform ray tracing calculations (Figure 4.35)

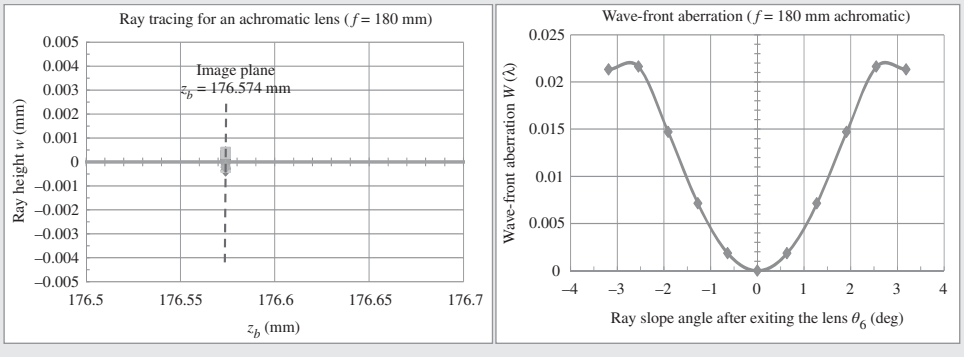


Figure 4.35 Ray tracing results for an achromatic lens

(4) Calculation results

Image distance d_5 (mm)	176.574
Focused spot radius w (mm)	±0.0004
Wave-front aberration W (λ)	+0.022/−0

Notes

1. An achromatic lens has very little aberration compared to a plano-convex lens. In this example, the achromatic lens has a small focused spot radius of ±0.4 μm, which is about 10 times smaller than that of the plano-convex lens.
2. The achromatic lens also displays very little wave-front aberration: only 0.022λ.

Problem 4.10 Designing an aplanatic lens – Part 1

Design an aplanatic lens (SF6: $n = 1.813$ for $\lambda = 546.1$ nm) which when placed to the right of the achromatic lens ($f_1 = 180\ mm$) described in Problem 4.9, will give the optical system a combined focal length of $f = 100$ mm.

Solution

(1) Specify the details of the achromatic lens to be used ($f = 180$ mm)(Figure 4.36)

Focal length $f_1' = -f_1 =$ (mm)	180.0006
Primary principal point H position p_1 (mm)	0.8640
Secondary principal point H′ position p_1' (mm)	−3.4431
Effective aperture radius w_L (mm)	10

(2) Design the aplanatic lens (Figure 4.36)

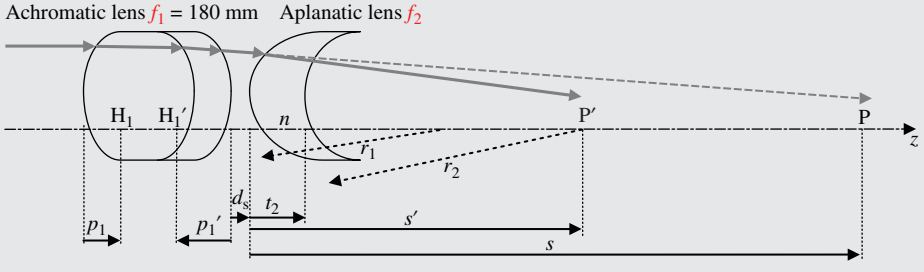


Figure 4.36 An aplanatic lens system

An aplanatic lens (+power) when used in combination with an existing lens, can reduce its focal length, without increasing spherical aberration and coma aberration. The relevant equations for an aplanatic lens are Equation (1.55a), Equation (1.55b), and Equation (1.55c).

$$r_1 = s/(n+1) \tag{1.55a}$$

$$r_2 = s/n - t \tag{1.55b}$$

$$\beta = s'/s = 1/n \tag{1.55c}$$

where r_1 = radius of curvature of the first surface, r_2 = radius of curvature of the second surface, t = center thickness, n = refractive index, and β = lateral magnification.

The focal length will be a fraction (1/1.8) of the original, after we insert an aplanatic lens with a refractive index $n \approx 1.8$ [Equation (1.55c)] into the system. Thus, adding an aplanatic lens ($n_1 \approx 1.8$) should enable us to reduce the focal length of the optical system from $f = 180$ to 100 mm.

For instance, suppose we let $n = 1.813$, and arbitrarily let the lens thickness $t_2 = 5$ mm, and the distance between lenses $d_s = 0.5$ mm. Substituting these values into Equation (1.55a), Equation (1.55b), and Equation (1.55c), we obtain the following results:

Refractive index n	Center thickness t	Object distance s	Radius of first surface r_1	Radius of second surface r_2	Magnification β
1.813	5 mm	176.0575 mm	65.5871 mm	92.1084 mm	0.55157

(3) Calculate the focal length and principal point position of the aplanatic lens (Figure 4.36)
The focal length f'_2 and the principal point position p_2, p_2' of the aplanatic lens itself can be obtained by substituting the values of r_1, r_2, and the other variables into Equation 1.34, Equation (1.35) and Equation (1.36).

$$\frac{1}{f'} = -\frac{1}{f} = (n-1)\left\{\left(\frac{1}{r_1} - \frac{1}{r_2}\right) + \frac{(n-1)t}{nr_1r_2}\right\} \tag{1.34}$$

$$p = -\frac{r_1 t}{n(r_2 - r_1) + (n-1)t} \tag{1.35}$$

$$p' = -\frac{r_2 t}{n(r_2 - r_1) + (n-1)t} \tag{1.36}$$

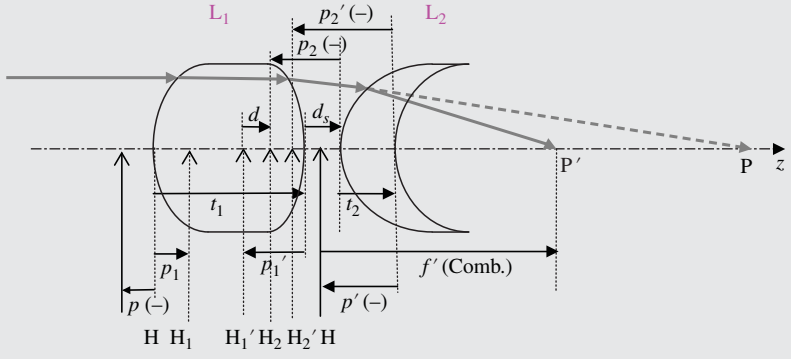
Focal length f_2'	Primary principal point p_2	Secondary principal point p_2'
223.237 mm	−5.4341 mm	−7.9973 mm

Problem 4.11 Designing an aplanatic lens – Part 2

Calculate the focal length and the principal points for the aplanatic lens system described in Problem 4.10, after the two lenses are combined.

Solution

(1) Calculate the focal length for a combination of two lenses (Figure 4.37)
The focal lengths f, f' and the principal points H,H′ of the two lenses (f_1, f_2) in combination can be calculated using Equation (1.39), Equation (1.40), Equation (1.41) and Equation (1.42).



Explanation of symbols

H_1, H_1', H_2, H_2' = Principal points of the two lenses L_1 and L_2
H, H′ = Principal points of the optical system after they are combined
p_1, p_1', p_2, p_2' = Positions of the principal points of L_1 and L_2 (distance from the vertex)
p, p' = Positions of the principal points of the optical system after they are combined (distance from the vertex)
d_s = Distance between the vertices of L_1 and L_2
d = $H_1'H_2$

Figure 4.37 An aplanatic lens system (a combination of two lenses)

If we substitute Equation (1.39) and Equation (1.40) with the values for the two lenses ($f_1' = 180.0006, f_2' = 223.2369$ mm), we obtain the focal length f' after we combine them.

$$f = \frac{f_1 f_2}{d - f_1' + f_2} \tag{1.39}$$

$$f' = -\frac{f_1' f_2'}{d - f_1' + f_2} \tag{1.40}$$

$$H_1H = \frac{f_1 d}{d - f_1' + f_2} \tag{1.41}$$

$$H_2'H' = \frac{f_2' d}{d - f_1' + f_2} \tag{1.42}$$

$$d = H_1'H_2 = d_s + p_2 - p_1' \quad \text{(distance from } H_1' \text{ to } H_2\text{)}$$

$$d = d_s + p_2 - p_1' = -1.491 \text{ mm} \quad f' = -f = 99.2833 \text{ mm}$$

(2) Calculate the principal point after combining the two lenses (Figure 4.37)
If we substitute the focal lengths for the two lenses ($f_1' = 180.0006$, $f_2' = 223.2369$ mm) into Equation (1.39b) and Equation (1.40b), we obtain the principal points H_1H, $H_2'H'$, and p, p' after we combine them.

$$H_1H = -0.6631 \text{ mm}, \quad H_2'H' = 0.8224 \text{ mm},$$
$$p = H_1H + p_1 = 0.20088 \text{ mm}, \quad p' = H_2'H' + p_2' = -7.1749 \text{ mm},$$

p, p': distance from vertex to principal point.

Note
We obtained a focal length $f \approx 100$ mm, which is the same as the value we previously stipulated.

Problem 4.12 Ray tracing for an aplanatic lens system

Use the aplanatic lens system described in Problems 4.10 and 4.11 ($f = 100$ mm), to focus a collimated beam 20 mm in diameter onto a single spot. Perform ray tracing calculations for the optical system and obtain the focused spot size and its distance from the lens.

Solution
(1) Input the optical configuration: aplanatic lens system (Figure 4.38) ($\lambda = 546.1$ nm)

Surface no.	First surface	Second surface	Third surface	Fourth surface	Fifth surface	Image plane
Surface type	Sphere	Sphere	Sphere	Sphere	Sphere	Plane
Focal length f (mm)	Achromatic $f = 180$			Aplanatic $f = 223.84$		–
Refractive index n	1.519(BK7)	1.652(SF2)	1	1.813(SF6)	1	1
Distance d (mm)	10	4.3	2.4	0.5	5	92.113
Radius of curvature r (mm)	103.1	−78.22	−291	62.58709	92.108375	–

(2) Specify details of the incident rays (Figure 4.38)

w_0	0 to ±10 mm	θ_0	0°

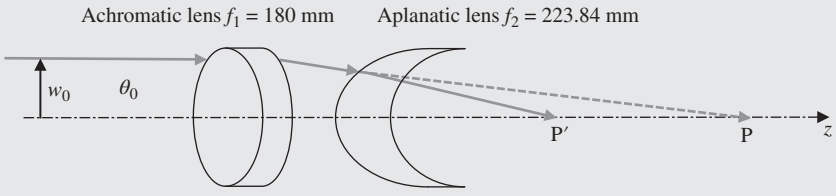


Figure 4.38 Beam focusing using an aplanatic lens system

(3) Perform ray tracing calculations (Figure 4.39)

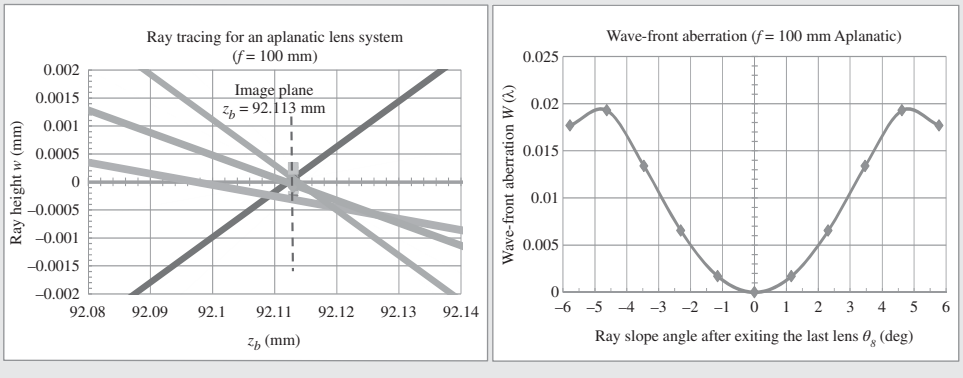


Figure 4.39 Ray tracing results for an aplanatic lens system

(4) Calculation results

Image distance d_1 (= f_b) (mm)	92.113
Focused spot radius w (mm)	±0.00024
Wave-front aberration W (λ)	+0.019/−0

Note

This aplanatic lens system has a focal length of about 100 mm, which is about 1.8 times shorter than the focal length of the basic achromatic lens (180 mm). It also has a wave-front aberration which is almost the same as that of the basic achromatic lens – that is, about +0.022/−0 λ.

Problem 4.13 Ray tracing for a beam expander

Perform ray tracing calculations for an optical system consisting of a collimated incident beam with a diameter of 6.667 mm and a 3× beam expander, which uses a plano-concave lens ($f = -100$ mm) and a plano-convex lens ($f = 300$ mm) to widen the beam, and an aberration-free lens ($f = 100$ mm) to focus the output beam onto a single spot. Obtain the focused spot size and its distance from the lens.

Solution

(1) Input the optical configuration: beam expander (Figure 4.40) ($\lambda = 546.1$ nm)

Surface no.	First surface	Second surface	Third surface	Fourth surface	Fifth surface	Image plane
Focal length f (mm)	$f_1 = -100$ mm		$f_2 = 300$ mm		$f_3 = 100$ mm	
Surface type	Plane	Sphere	Plane	Sphere	AbeFreeLens	Plane
Refractive index n	1.519	1	1.519	1	–	1
Distance d (mm)	100	2	198.2225	2.7	100	100.0095
Curvature radius r (mm)	–	51.9	–	−155.70	–	–

If we set up the plano-concave lens and the plano-convex lens as in Figure 4.40, for a beam expander application, the distance between f_1 (the second principal point) and f_2 (the first principal point) will then be the value of $f_1 + f_2$ which generates a collimated beam when the beam exits the plano-convex lens.

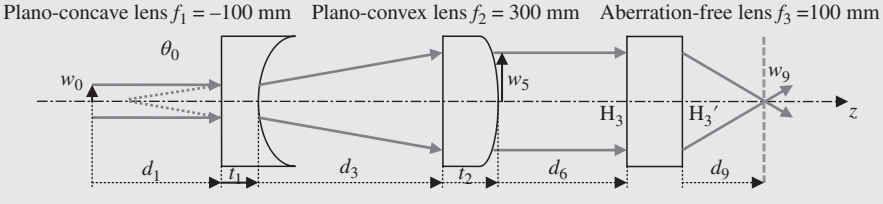


Figure 4.40 A beam expander

(2) Specify details of the incident rays (Figure 4.40)

w_0	0 to ±3.333 mm	θ_0	0°

(3) Perform ray tracing calculations (Figure 4.41)

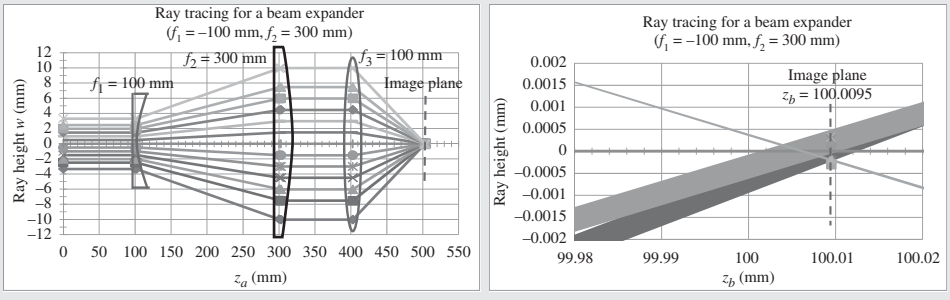


Figure 4.41 Ray tracing results for a beam expander

(4) Calculation results

The rays exiting the planar convex lens are almost perfectly collimated (see graph in Figure 4.41).

Expanded beam radius w_5 (mm)	±10.01303
Focused spot radius w_9 (mm)	±0.000314 at z_b = 100.0095
Wave front aberration W (λ)	+0.031/−0

Note

We obtained a very narrowly focused ray (radius = 0.0003 mm), because the "−" and "+" aberrations caused by the plano-concave lens and plano-convex lens canceled each other out.

Problem 4.14 A laser optical system

Perform ray tracing calculations for a laser optical system which employs a LD, the collimator lens and the beam expander described in Section 2.2 (Problems 2.5 and 2.6) to generate a collimated laser beam, and which focuses the beam onto a small spot, using the achromatic lens (f = 180 mm) shown in Problem 4.8. Perform the ray tracing in both the x- and y-directions.

Solution

(1) Input the optical configuration of the laser optical system (Figure 4.42) (λ = 546.1 nm). Units: mm

Surface no. [Units: mm]	First surface	Second surface	Third surface	Fourth surface	Fifth surface	Sixth surface	Seventh surface	Eighth surface	Image plane
Focal length f	$f_1 = 10$	Cylindrical $f_2 = -100$		Cylindrical $f_3 = 300$		Achromatic $f_4 = 180$			–
Surface type	AbeFreelens.	Plane	Sphere	Plane	Sphere	Sphere	Sphere	Sphere	Plane
Refractive index n	1	1.519	1	1.519	1	1.519	1.652	1	1
Distance d	10	50	3	197.3705	4	50	4.3	2.4	176.55778
Curvature radius r	–	–	51.9	–	−155.70	103.1	−78.22	−291	–

The above configuration shows an aberration-free lens f_1, a beam expander using cylindrical lenses f_2 and f_3, and a focusing lens f_4. In order to obtain parallel rays leaving f_3, d_4 should be set to the value where the distance between principal points of f_2 and f_3 is equal to $f_2 + f_3$. Here we have made a small adjustment to the value d_4 in order to minimize the focused spot of the beam leaving f_4.

(2) Specify the details of the incident rays (rays being emitted from the LD) (Figure 4.42)

w_0	0 mm	θ_0	θ_{0x}: 0 to ±4°, θ_{0y}: 0 to ±11.8477°

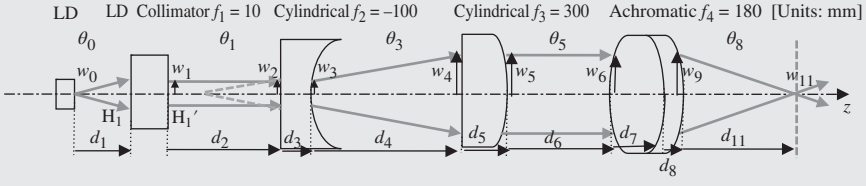


Figure 4.42 A laser optical system

We have set the incident ray slopes in the y-direction ($\tan\theta_{0y}$) to be three times greater than those in the x-direction ($\tan\theta_{0x}$).

(3) Perform ray tracing calculations (Figure 4.43)

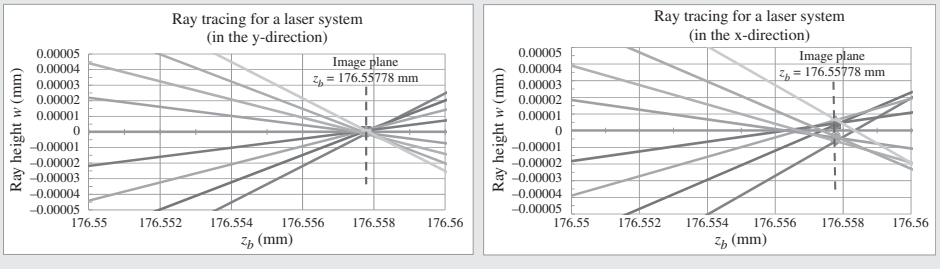


Figure 4.43 Ray tracing results for a laser optical system

(4) Calculation results

Beam radius after leaving f_3 w_5 (mm)	$w_{5x} = 2.09796$, $w_{5y} = 2.09780$
Focused radius of rays leaving f_4 w_{11} (mm)	$w_{11x} = 0.0000066$, $w_{11y} = 0.0000012$ at $d_{11} = 176.55778$

Notes

1. Rays exiting the beam expander are converted into a circular collimated beam.
2. The focused radius of the rays after passing through achromatic lens f_4 is very small ($w_x = 6.6$ nm, $w_y = 1.2$ nm). This shows that the optical system has almost zero aberration.
3. In real-life cases, the focused beam is never this small, because the light from the LD is not emitted from a point source but from an area with a finite size. Additionally, light cannot be focused onto an infinitesimal point, due to diffraction (see Chapters 8 and 9).

Problem 4.15 Ray tracing for an imaging system using a plano-convex lens

Design an imaging system using a plano-convex lens ($f = 100$ mm) as specified in the table below, which magnifies an object by a factor of two ($\beta = -2$), where the height of the object equals 0 to ± 10 mm. Perform ray tracing calculations for the optical system. (Assume that the light source is monochromatic, with a wavelength of $\lambda = 546.1$nm.)

Solution

(1) Specifications for the plano-convex lens ($f = 100$ mm, BK7)

Focal length f	100 mm	Center thickness t	6 mm
Refractive index n	1.519 at 546.1 nm	Principal point p	3.95 mm
Radius of curvature r	51.9 mm	Aperture radius w_L	10 mm

(2) Calculate the object distance and the image distance (Figure 4.44)

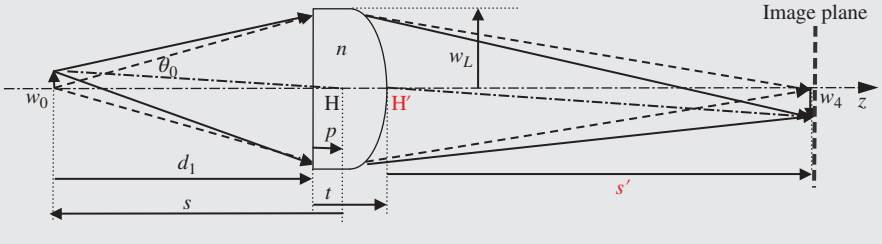


Figure 4.44 An imaging system using a plano-convex lens

If we substitute the values $f = 100$ mm and $\beta = -2$ into lens equations Equation (1.24) and Equation (1.29), we can derive the object distance s and the image distance s'.

$$1/s' = 1/f + 1/s \tag{1.24}$$

$$\beta = s'/s \tag{1.29}$$

$$s = -150 \text{ mm},\ s' = 300 \text{ mm},\ d_1 = -s - p = 146.05 \text{ mm}$$

(3) Configuration

$\lambda = 546.1$ nm	First surface	Second surface	Image plane
Surface type	Plane	Sphere	Plane
Refractive index n	1.519	1	1
Distance d (mm)	$146.05 = -s - p$	6	300 ($= s'$)
Radius of curvature r (mm)	–	−51.9	–

(4) Specify details of the incident rays

We set five object points with a height w_0 of −10, −5, 0, 5, and 10 mm. Each objective point emits a ray that travels to the lens (aperture size of 20 mm in diameter). Here we define the principal ray as the ray starting from each objective point and traveling through the principal point H of the lens.

Object height w_0 (mm)	−10	−5	0	5	10
Slope of principal ray θ_0 (deg)	3.814	1.909	0	−1.909	−3.814
Range of values for ray slopes θ_0 (deg)	0 to 7.595	−1.909 to 5.711	−3.814 to 3.814	−5.711 to 1.909	−7.595 to 0

Slope of principal ray $\theta_0 = \tan^{-1}[(-w_0)/(-s)]$
Range of ray slopes $\theta_0 = 0$ to $\tan^{-1}[(\pm w_L - w_0)/(-s)]$
w_L = aperture radius of the lens

(5) Perform ray tracing calculations (Figure 4.45)

(6) Calculation results

Object height w_0 (mm)	−10	−5	0	5	10
Image height of the principal ray w_4 (mm)	20.0007	10.0001	0	−10.0001	−20.0007
Focused size of rays on image plane (mm)	+1.15 −0.40	+0.77 −0.39	+0.51 −0.51	+0.39 −0.77	+0.40 −1.15

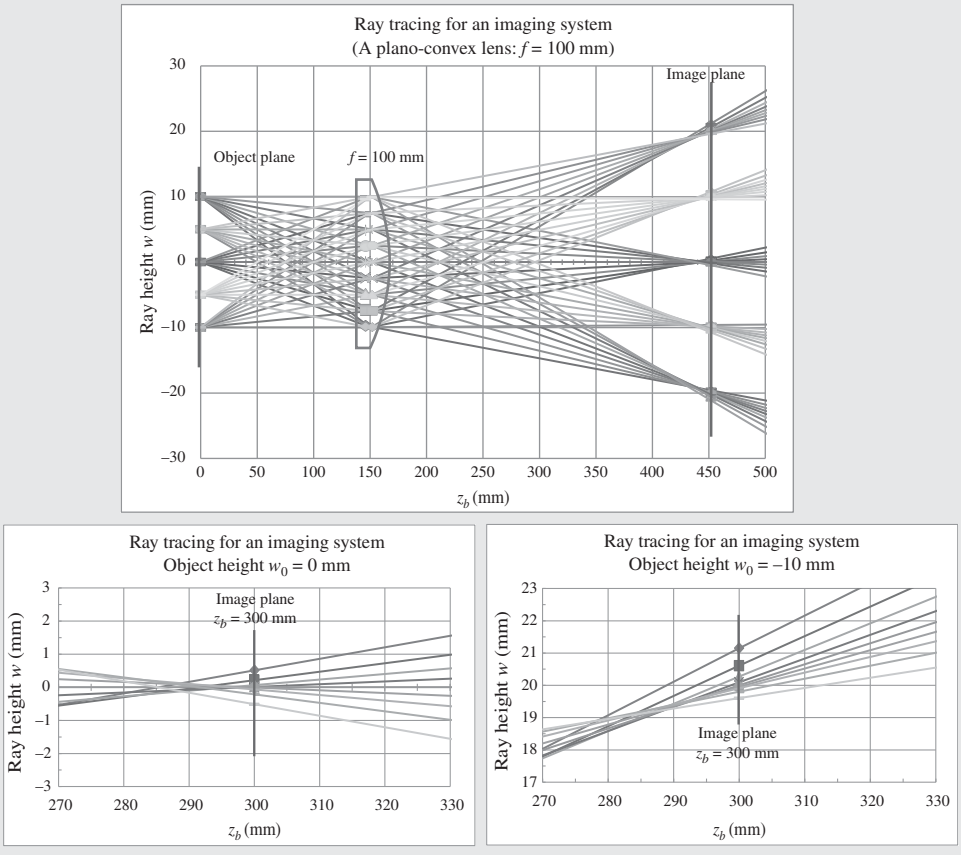


Figure 4.45 Ray tracing results for an imaging system (a plano-convex lens)

Notes

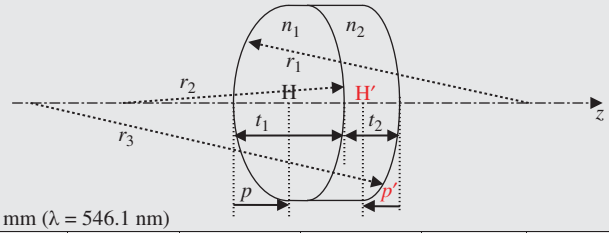
In this problem, we set the image plane at the image point obtained using paraxial theory ($s' = 300$ mm). Our calculation results highlighted the following points:

1. Each principal ray height at the image plane is almost exactly double the object height (but with the sign reversed).
2. Each image point on the image plane is not a precisely defined point, but an area with a radius of ±0.51 mm, when the image height is zero. For an image height of +20 mm, the image point is much larger and more asymmetrical: +0.40/−1.15 mm. This result is due to the simultaneous occurrence of five Seidel aberrations within the optical system, generating large aberrations.

This problem illustrates the fact that a single plano-convex lens generates a large degree of aberration. A typical optical imaging system, such as a camera lens or a microscope lens, uses a complex configuration of multiple lenses, in order to minimize these aberrations.

Problem 4.16 Ray tracing for an imaging system using two achromatic lenses

If we use two achromatic lenses instead of the single plano-convex lens used in Problem 4.15, the results improve dramatically. To illustrate this, we will design an imaging system equivalent to the optical system in Problem 4.15, using two achromatic lenses ($f_A = 150$ mm, $f_B = 300$ mm) as shown in Figure 4.46. We will then perform ray tracing calculations [2].



Units: mm (λ = 546.1 nm)

	r_1	r_2	r_3	t_1	t_2	n_1 (BK7)	n_2 (SF2)
f_A = 150 mm	88.888	–65	–222.3	5.1	3	1.519	1.652
f_B = 300 mm	161	–135	–588.8	3.6	3	1.519	1.652

Figure 4.46 Specifications of the two achromatic lenses

Solution

(1) Focal length and the principal points for the combination of the two lenses (Figure 4.47)

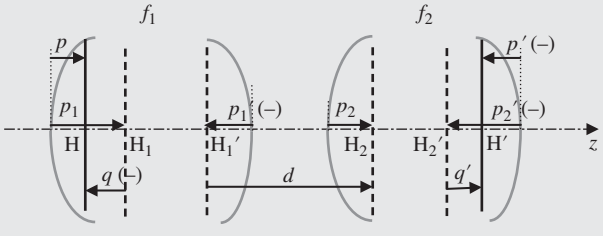


Figure 4.47 Combination of two lenses

The focal length and the principal points of each simple lens will be as follows:

$$\frac{1}{f'} = -\frac{1}{f} = (n-1)\left\{\left(\frac{1}{r_1} - \frac{1}{r_2}\right) + \frac{(n-1)t}{nr_1r_2}\right\} \tag{1.34}$$

$$p = -\frac{r_1 t}{n(r_2 - r_1) + (n-1)t} \tag{1.35}$$

$$p' = -\frac{r_2 t}{n(r_2 - r_1) + (n-1)t} \tag{1.36}$$

The focal length f and principal points H, H′ for the combination of the two lenses f_1, f_2 will be:

$$f = \frac{f_1 f_2}{d - f_1' + f_2} \tag{1.39}$$

$$f' = -\frac{f_1' f_2'}{d - f_1' + f_2} \tag{1.40}$$

$$q = \mathrm{H_1H} = \frac{f_1 d}{d - f_1' + f_2} \tag{1.41}$$

$$q' = \mathrm{H_2'H'}\frac{f_1 d}{d - f_1' + f_2} \tag{1.42}$$

$p = q + p_1$ (Primary principal point) $\quad p' = q' + p_2'$ (Secondary principal point)
The calculation results are shown below. Units: mm.

Single lens	f'	p	p'	Combination	f'	p	p'
f_A first	73.1695	1.961537	−1.43439	f_A combination	149.5254	1.247084	−3.97197
f_A second	−141.957	−0.7561	−2.58585				
f_B first	142.0722	1.294456	−1.08541	f_B combination	299.3739	0.690085	−3.5082
f_B second	−269.354	−0.54165	−2.36283				

(2) Configuration
We use f_A as the objective lens and f_B as the imaging lens, as in Figure 4.48

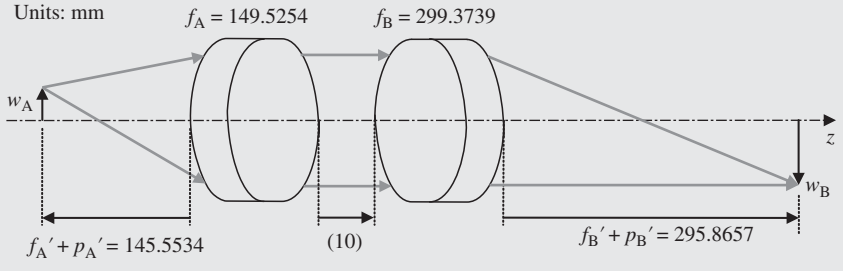


Figure 4.48 Imaging using two achromatic lenses

(3) Perform ray tracing calculations (Figure 4.49)
(4) Calculation results

Object height w_A (mm)	−10	−5	0	5	10
Principal ray image height w_B (mm)	20.01612	10.01012	0	−10.01012	−20.01612
Focused size of rays on image plane (mm)	+0.248 −0.245	+0.0626 −0.0637	+0.00209 −0.00209	+0.0637 −0.0626	+0.245 −0.248

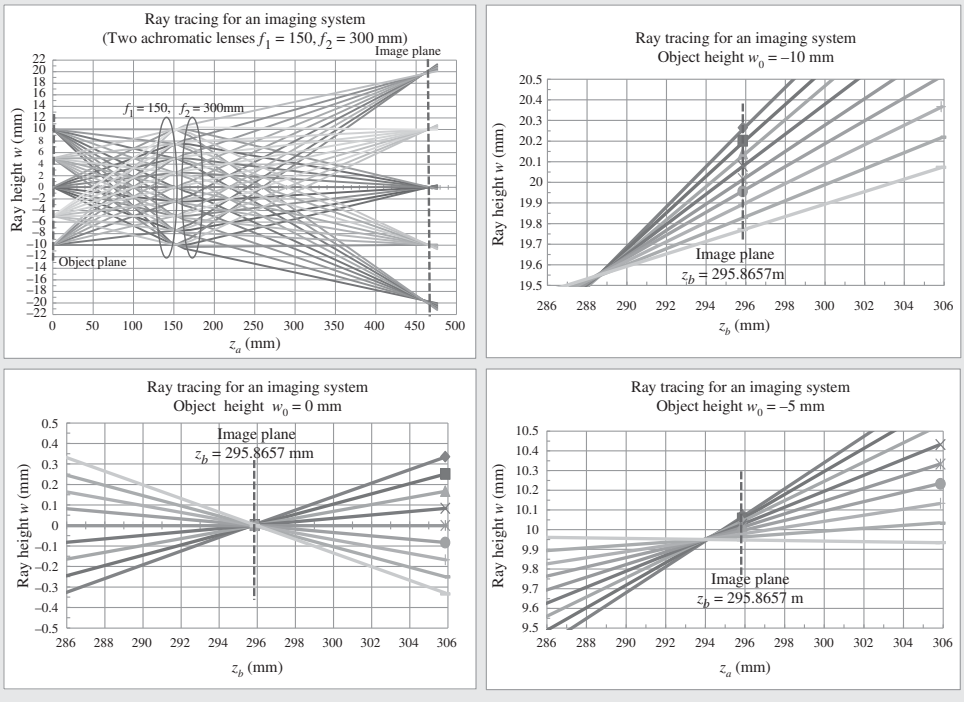


Figure 4.49 Ray tracing results for an imaging system (two achromatic lenses)

Notes

1. The image height for the principal rays is a close approximation to the value obtained from paraxial theory ($\beta = -f_B/f_A = -2.002$).
2. The focused radius of the rays on the image plane is very small for an image height of zero, however it becomes increasingly large for an image height of greater value.
3. We may conclude that achromatic lenses have very small aberrations, which are much smaller than those of plano-convex lenses.

Problem 4.17 Ray tracing for a focusing system using an aspheric lens

Perform ray tracing calculations for a focusing system using an aspheric lens, as shown below. Obtained the focused spot size and its distance from the lens.

Note: Place a cover glass ($t = 0.25$ mm, BK7 $n = 1.51072$, at $\lambda = 780$ nm) in front of the image plane [1].

Solution

(1) Aspheric lens: NA = 0.5, f = 8.00 mm, n = 1.59708 at λ = 780 nm (Figure 4.50) [1]

	R	k	A_4	A_6	A_8	A_{10}	A_{12}
First surface	5.09239	−0.47317	0	-2.257×10^{-6}	2.412×10^{-9}	-8.263×10^{-9}	0
Second surface	−56.20310	0	5.124×10^{-4}	-2.266×10^{-5}	3.33×10^{-7}	0	0

$$z(y) = \frac{y^2}{R\left[1+\sqrt{1-\frac{(1+k)y^2}{R^2}}\right]} + A_4y^4 + A_6y^6 + A_8y^8 + A_{10}y^{10} + A_{12}y^{12} \tag{4.26}$$

Numerical aperture	Effective focal length	Design wavelength	Center thickness	Cover glass t, n
NA = 0.5	f = 8.00 mm	λ = 780 nm	t = 3.69 mm	0.250 mm, 1.51072

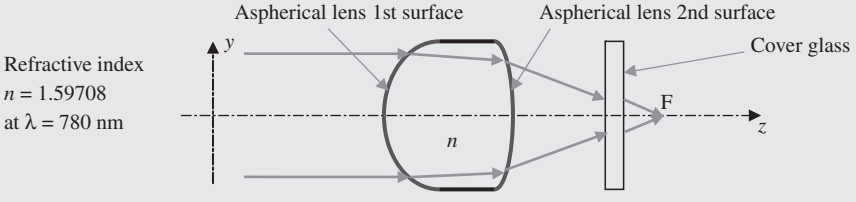


Figure 4.50 Beam focusing using an aspheric lens

(2) Specify details of the incident rays (Figure 4.50)

w_0	0 to ±2.736 mm	θ_0	0°

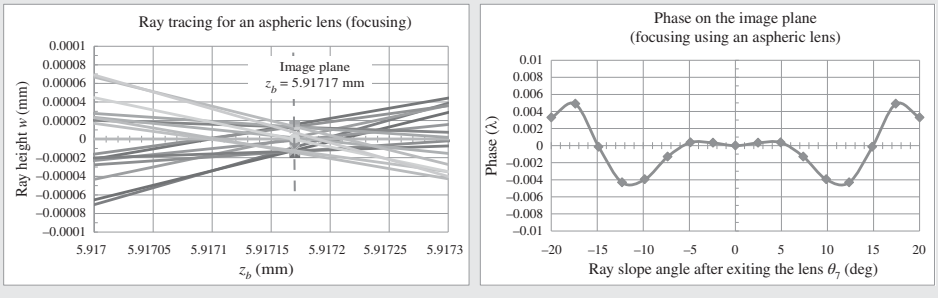


Figure 4.51 Ray tracing results for an aspheric lens

The above ray height $w = \pm 2.736$ mm corresponds to a converging angle of $\pm 20°$ ($w = f \sin 20° = 2.736$ mm).

(3) Perform ray tracing calculations (Figure 4.51) (the image distance is 5.91717 mm behind the lens)

(4) Calculation results

Image distance d (mm)	5.91717
Focused spot radius w (mm)	±0.000015
Phase difference at the image plane (λ)	+0.0049/−0.0043

Notes

1. The calculated value for the focused spot radius is very small: ±15 nm.
   The theoretical aberration of the aspheric lens is very small (almost zero), even though a small, light-weight single lens is used.

Problem 4.18 A laser collimator using an aspheric lens

Perform ray tracing calculations for a collimator optical system consisting of a LD ($\lambda = 780$ nm, emission angle ±20°) and an aspheric lens, as shown in Problem 4.17. After collimation, focus the beam using either an aberration-free lens ($f = 8$ mm) or the aspheric lens ($f = 8$ mm).

Solution

(1) Configuration (Figure 4.52). $d_1 = 5.91717$ mm, $d_5 = 5$ mm, $d_8 = 8$ mm ($d_9 = 5.91717$ mm for the figure on the right)

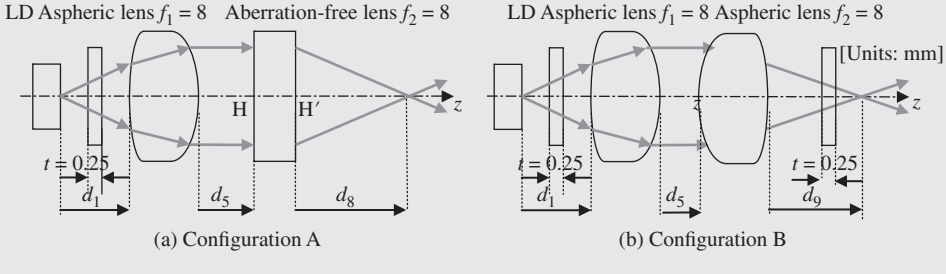
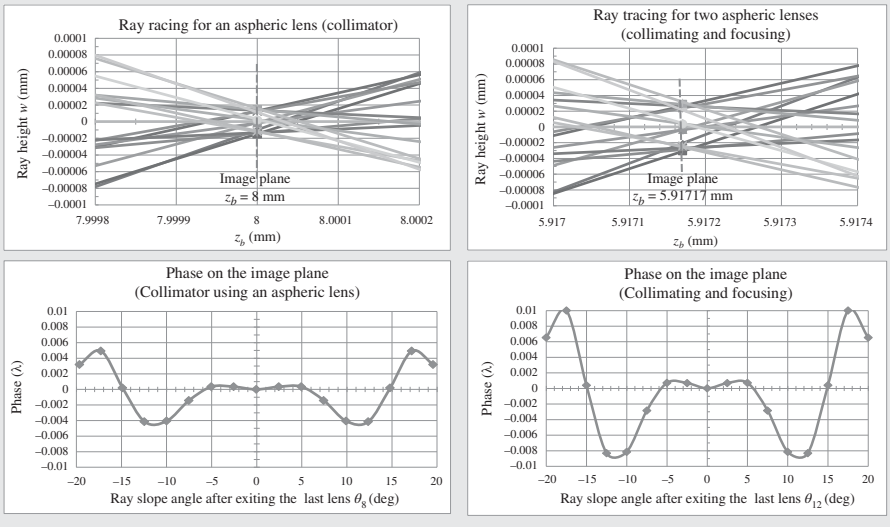


Figure 4.52 A laser collimator using an aspheric lens

(2) Specify details of the incident rays (Figure 4.52)

w_0	0 mm	θ_0	0 to ±20°

(3) Perform ray tracing calculations (Figure 4.53)



(a) Configuration A (b) Configuration B

Figure 4.53 Ray tracing results for a laser collimator using an aspheric lens

Perform ray tracing calculations for the "collimation-focusing" optical system with the two configurations illustrated in Figure 4.52.

(4) Calculation results

	Configuration A	Configuration B
Image distance d (mm)	8	5.91717
Focused spot radius w (mm)	±0.000015	±0.000031
Phase difference at the image plane (λ)	+0.0049/−0.0041	+0.01/−0.0083

Notes

1. **Configuration A:** *Aspheric lens–aberration-free lens*. After the collimated rays have been focused, the focused radius will be similar to that obtained in Problem 4.17.
2. **Configuration B:** *Aspheric lens–aspheric lens*. After focusing the collimated rays, the focused radius will be double that obtained in Problem 4.17. The doubling in size is due to the aberration generated by the extra aspheric lens (during focusing).

Problem 4.19 A laser beam scanning system

Configure a laser beam scanning system and perform ray tracing calculations for the system, which consists of a LD, a collimator lens, a rotating polygon mirror (or a galvanometer mirror), and a focusing lens. Use the following specifications for each component.

1. **Laser diode:** $\lambda = 780$ nm, beam emission angle $\theta_{\perp} = \pm 20.68°$ (at $1/e^2$).
2. **Collimator lens:** aspheric lens, $f_1 = 8$ mm, as in Problems 4.17 and 4.18.
3. **Focusing lens:** achromatic lens (Figure 4.54).

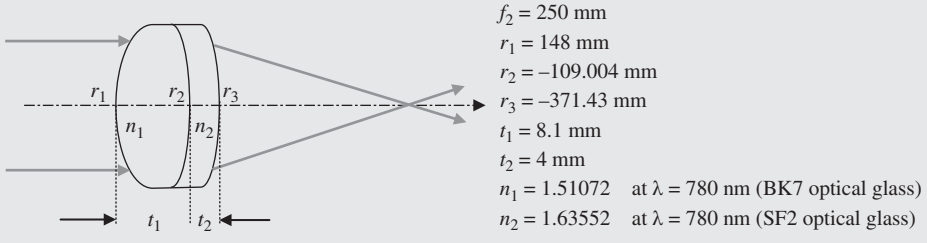


Figure 4.54 An achromatic lens for focusing a scanned beam

Solution

(1) Configuration (Figure 4.55)

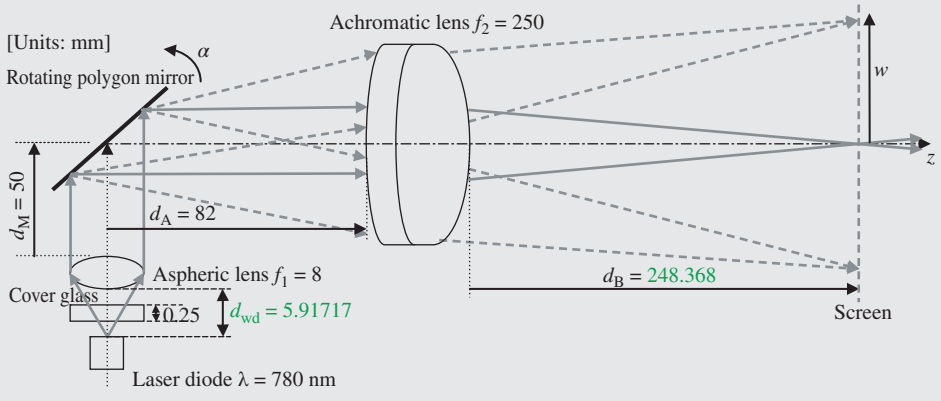


Figure 4.55 A laser beam scanning system

(2) Specify details of the incident rays starting from the LD

$w_0 = 0$ mm, $\theta_0 = -20.68°, -15°, -10°, -5°, 0°, 5°, 10°, 15°, 20.68°$

(3) Generate a calculation table on the Excel spreadsheet

Using units of surface type "Plane," "Asphere" (refer to Problems 4.17 and 4.18), "Plane mirror" [as in point (4)] and "Sphere," we can generate a calculation table on the Excel spreadsheet.

Set the working distance of the collimator lens f_1 to $d_{wd} = 5.91717$ mm as in Problem 4.17.

Set d_M, d_A, and d_B in Figure 4.55 to $d_M = 50$ mm, $d_A = 82$ mm, and $d_B = 248.368$ mm, respectively. d_B is the value which minimizes the focused spot size for the beam, when $\alpha = 0°$.

(4) Plane mirror unit

We can use a plane mirror unit for our polygon mirror. A detailed explanation can be found on the Excel file of this problem on the companion web site. In Figure 4.55, the incident angle to the mirror is 45° and the optical axis changes its direction by 90° after being reflected by the mirror. In the calculations, we will define the optical axis as a straight line, ignoring the ray's 90° direction change after it is reflected by the mirror.

(5) Relation between the rotation angle α and the scanning position w on the screen

When the mirror rotates α (in rad), the reflection beam changes direction by 2α (in rad), and the spot on the screen changes position by w. w can be approximated by the equation $w \approx 2\alpha f_2$, which we studied in Problem 3.2 in Chapter 3. However in this problem, we will not use the above equation in order to calculate the expected scanning position from the mirror rotation angle α. Here, we treat the relation between α and w as linear, and use the ratio of the two, when the scan height w display a value of 41 mm.

(6) Perform ray tracing calculations (on the Excel spreadsheet) (Figure 4.56)

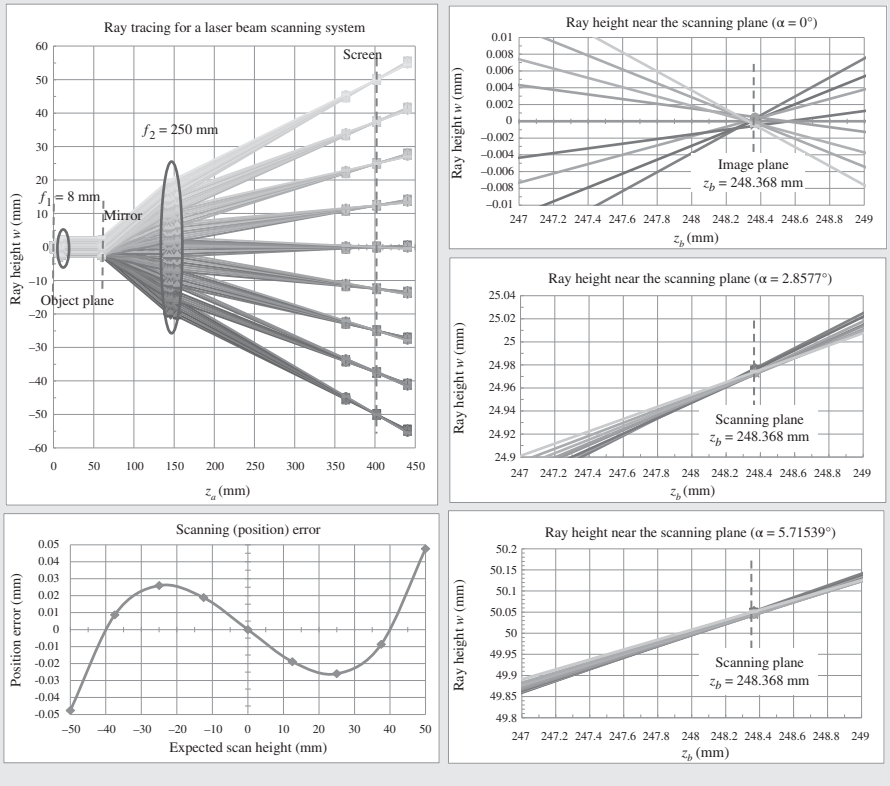


Figure 4.56 Ray tracing results for a laser beam scanning system calculated directly on Excel cells

(7) Ray trace calculations (using two-dimensional VBA ray tracing program) (Figure 4.57)

We can calculate ray tracing not only directly, on the Excel spreadsheet, but also by running the VBA program. However, the two-dimensional VBA ray tracing program does not support aspheric lenses. Here, we calculate the VBA ray tracing excluding the aspheric lens f_1, assuming that the aspheric lens has no aberration. After calculating the ray tracing, we obtain the VBA calculation results

which are almost the same as the calculations performed directly on the Excel spreadsheet (which includes the aspheric lens), as the graphs in Figure 4.57 illustrate. The reason for this almost-perfect agreement is that the theoretical aberration of the aspheric lens $f_1 = 8$ mm is very small (almost zero), as we showed in our solution to Problems 4.17 and 4.18.

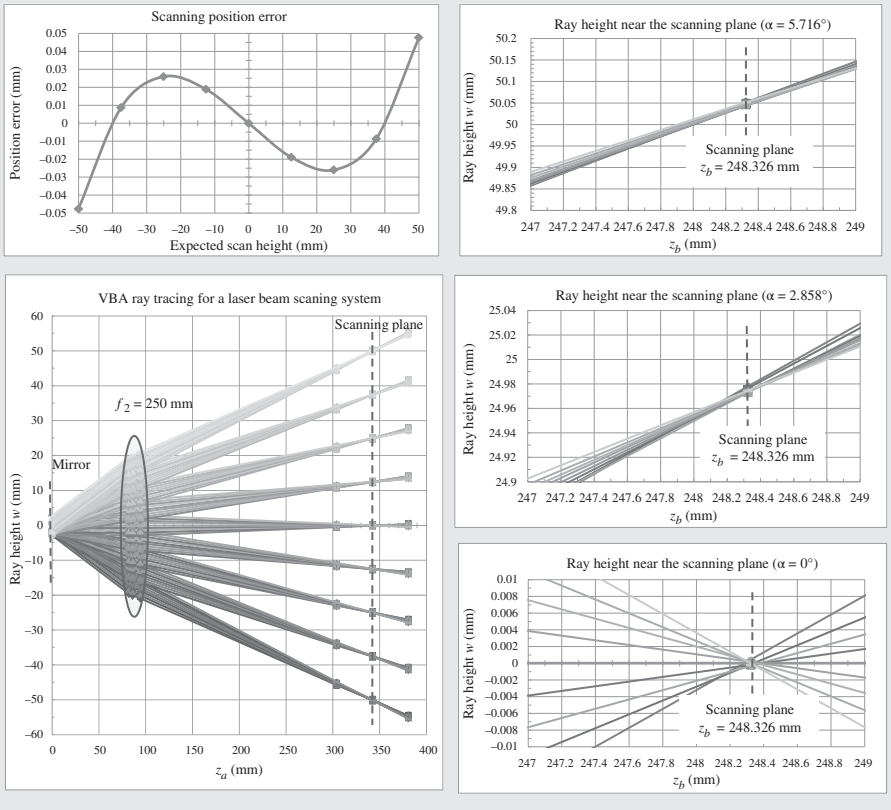


Figure 4.57 Ray tracing results for a laser beam scanning system calculated by the VBA program

Notes

1. We treated the relation between the mirror angle α (rad) and the scan position w (mm) as linear, and approximated it as $\boldsymbol{w = 501.241\ \alpha}$ (or $w = \mathbf{501.1869}\ \alpha$ in the calculation using two-dimensional VBA ray tracing) which was obtained by using the values of α and w at the scan position $w = 41$ mm. (The proportional constant computed above, 501.241 (or 501.1869), is similar to the value calculated using paraxial theory, as $w \approx 2\alpha f_2 = 500.4134\alpha$, where $f_2 = 250.2067$ mm.)
2. The calculated scanning position difference from the above linear equation (which shows expected scanning positions) is less than ±0.05 mm, for a scanning range of ±50 mm. In other words, the scanning position error is less than ±0.1% of the scanning range.
3. The spot size (i.e., aberration) in the scanning (x) direction on the screen is no more than 9 μm, for a scanning range of ±50 mm. The aberration in the y-direction (normal to the scanning) can be calculated by performing a three-dimensional ray tracing on which we can draw an X-Y spot diagram. This technique will be discussed at length in Chapter 5.

4. The actual spot size will be greater than the calculated aberration size, for two reasons: first, the LD light emission point is not really a point, as it has a finite area; and secondly, light diffraction causes the spot to spread out over a small area.
5. When using a polygon mirror, it is important to keep the following points in mind:
   (a) The precise location of the reflection point will vary slightly, depending on the mirror rotation angle. Here, we have ignored this variation.
   (b) If the surfaces of the polygon mirror do not fit each other perfectly, each scanning line will have a slightly different y-coordinate (vertical to the scanning direction) on the screen. We can prevent this effect by adopting the following countermeasures:
      (i) Using a mirror manufactured according to the most precise industrial standards.
      (ii) Inserting two cylindrical convex lenses into the optical system – one which projects the image of the LD light emission point onto the mirror's surface, and the other which subsequently projects this image onto the screen, but only in the y-direction (Figure 4.58).
6. In this problem, we set the distance of d_M to 82 mm, which is shorter than the value of f_2. This was done in order to reduce system aberration.

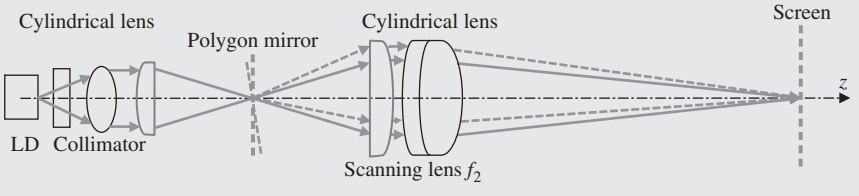


Figure 4.58 How to prevent scanning line variation in the y-direction. (The figure lies on the $y-z$ plane)

Problem 4.20 Focusing a beam using an achromatic lens which is set at a tilt

Calculate the ray tracing required to focus a beam using the achromatic lens in Problem 4.9 (f = 180 mm), where the lens is tilted at an angle of 2° to the vertical.

Solution

(1) Input the optical configuration: achromatic lens (f = 180 mm) (Figure 4.59). Units: mm

(λ = 546.1 nm) Type of surface	TiltStart	First surface Sphere	Second surface Sphere	Third surface Sphere	TiltEnd	Image Plane
Refractive index n	1	1.519	1.652	1	1	1
Distance d	10	0	4.3 ($=t_1$)	2.4 ($=t_2$)	0	176.3
Curvature radius r	–	103.1	– 78.22	−291	–	–
Tilt angle α (deg)	2	–	–	–	−2	–

(2) Specify the details of the incident rays (Figure 4.59)

w_0	0 to ±10 mm
θ_0	0°

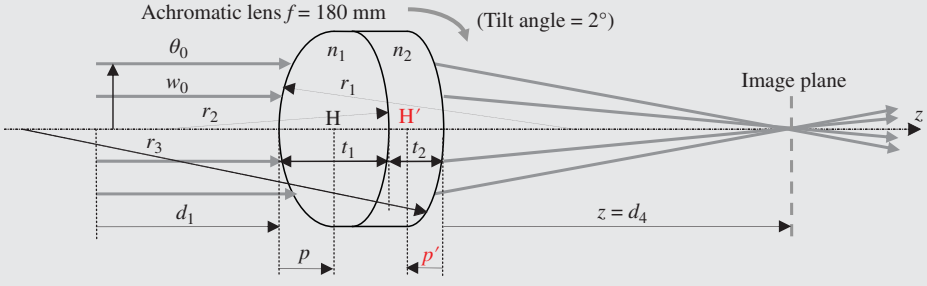


Figure 4.59 Beam focusing using an achromatic lens set at a tilt

(3) Perform ray tracing calculations (Figure 4.60)
(4) Calculation results

Items	Tilt angle $\alpha = 0°$	Tilt angle $\alpha = 2°$
Image distance d_4 (mm)	176.574	176.3
Image height w (mm)	0	−0.114
Focused spot diameter D (mm)	0.0008	0.0062

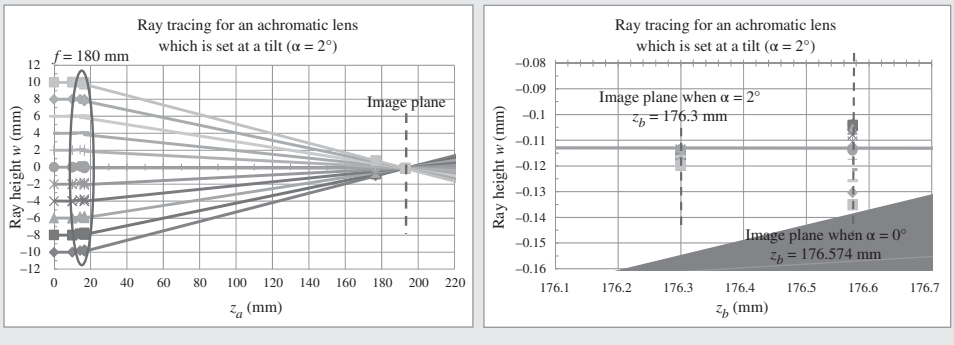


Figure 4.60 Ray tracing results for an achromatic lens set at a tilt

Notes

1. A ray tracing for a lens (or an optical system) which is set at a tilt can be performed by inserting "TiltStart" just before the lens and "TiltEnd" just after the lens.
2. Tilting a lens increases the aberration. (In the above case, the aberration increases the focused spot diameter from 0.0008 to 0.0062 mm.)

References

[1] Edmund Optics Inc. (2003) *Optical Parts Catalogue*, Edmund Optics Inc., Japan.
[2] Sigma-Koki Co., Ltd (2003) *Optical Parts Catalogue*, Sigma-Koki Co., Ltd, Japan.

5

Three-Dimensional Ray Tracing

In Chapter 4, we studied two-dimensional ray tracing. Using this technique, we were able to estimate the amount of aberration for a given optical system. However, two-dimensional ray tracing will not enable us to obtain a precise estimate of the degree of aberration for rays which are asymmetrical around the optic axis, such as rays from an off-axis object or rays exiting a cylindrical lens. In such cases, we need to use a technique called three-dimensional ray tracing in order to calculate the exact amount of aberration. In three-dimensional ray tracing, the z-axis, the y-axis, and the x-axis are used to represent the optical axis, the longitudinal ray height and the latitudinal ray height, respectively. We will explore this technique in this chapter.

Three-dimensional ray tracing can be performed using the same procedures as for two-dimensional ray tracing, except that the dimensionality of the space increases from two to three dimensions.

5.1 Three-Dimensional Ray Tracing for a Spherical Surface

We can use three-dimensional coordinates to describe any given ray propagating through an optical system. We designate the optical axis, along which the ray propagates, as the z-axis, and then we measure the ray height along both the x- and y-axes. First, we begin with an object point at an arbitrary height w_{0x}, w_{0y} on the x–y object plane, and an arbitrary ray emitted from the object, whose slope is $\tan\theta_{0x}$ when projected onto the x–z plane and $\tan\theta_{0y}$ when projected onto the y–z plane. We can then calculate the ray height w_{1x}, w_{1y} where the ray intersects with the first surface of the optical system and the slope angle $\tan\theta_{1x}$, $\tan\theta_{1y}$ of the refracted ray exiting the surface, by applying Snell's law. If we repeat this process for each surface, we will finally obtain the ray height w_{nx}, w_{ny} on the image plane.

Figure 5.1 shows a ray entering and exiting the kth surface. In the figure, d_k is the distance between the vertexes of the $(k-1)$th and kth surfaces; of_{kx} and of_{ky} are the offsets of the kth surface axis from the $(k-1)$th surface axis; r_k is the radius of curvature for the kth surface (which is defined as positive when the vertex lies on the left); n_{k-1} and n_k are the refractive indices of the media entering and exiting the kth surface; $(w_{(k-1)x}, w_{(k-1)y})$ and (w_{kx}, w_{ky}) are the ray heights for the $(k-1)$th and kth surfaces; Δ_{k-1} and Δ_k are the distances along the z-axis from the vertex to the ray intersection at the $(k-1)$th and kth surfaces, respectively (defined as positive when to the right of the vertex) and $(\tan\theta_{(k-1)x}, \tan\theta_{(k-1)y})$ and $(\tan\theta_{kx}, \tan\theta_{ky})$ are the x- and y-components of the slope of the rays entering and exiting the kth surface, respectively.

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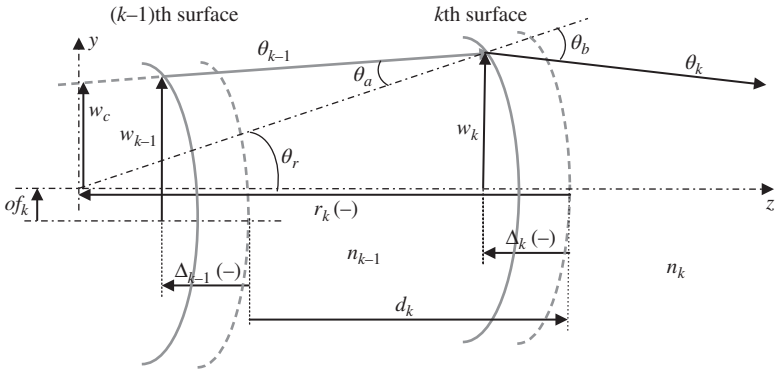


Figure 5.1 Ray refraction on a boundary surface

5.1.1 Calculate the Intersection Coordinates (w_{kx}, w_{ky}, Δ_k) of the Ray with the kth Surface

The equation of the spherical surface can be described by the following equation, if we define the center of curvature for the *k*th surface as the origin.

$$x^2 + y^2 + z^2 = {r_k}^2 \tag{5.1}$$

The equations of the incident ray can then be expressed as follows:

$$x = z \tan \theta_{(k-1)x} + w_{cx} \quad (x = az + b) \tag{5.2}$$

$$y = z \tan \theta_{(k-1)y} + w_{cy} \quad (y = cz + d) \tag{5.3}$$

where

$$w_{cx} = w_{(k-1)x} - of_{kx} + (d_{k-1} - \Delta_{k-1} + r_k) \tan \theta_{(k-1)x} \tag{5.4}$$

$$w_{cy} = w_{(k-1)y} - of_{ky} + (d_{k-1} - \Delta_{k-1} + r_k) \tan \theta_{(k-1)y} \tag{5.5}$$

Here, w_{cx}, w_{cy} are the incident ray heights along the *x*- and *y*-axes, at the point where the ray's *z*-coordinate is equivalent to the *z*-coordinate of the *k*th surface's center of curvature. If we substitute Equation (5.2) and Equation (5.3) into Equation (5.1), and then calculate the *z*-coordinate of the point where the incident ray intersects with the surface, we obtain:

$$Z = \frac{-(ab + cd) \pm \sqrt{(ab + cd)^2 - (1 + a^2 + c^2)(b^2 + d^2 - r^2)}}{(1 + a^2 + c^2)} \tag{5.6}$$

The sign in front of the $\sqrt{}$ will be positive when the vertex of the surface is to the right of the surface, and negative when the vertex is to the left of the surface. Δ_k will thus be:

$$\Delta_k = Z + r_k \tag{5.7}$$

The *x*- and *y*-coordinates of the intersection $X(\equiv w_{kx})$ and $Y(\equiv w_{ky})$ are given by the following equations:

$$X \equiv w_{kx} = w_{(k-1)x} - of_{kx} + (d_x - \Delta_{k-1} + \Delta_k)\tan\theta_{(k-1)x} \tag{5.8}$$

$$Y \equiv w_{ky} = w_{(k-1)y} - of_{ky} + (d_x - \Delta_{k-1} + \Delta_k)\tan\theta_{(k-1)y} \tag{5.9}$$

5.1.2 *Calculate the Incident Angle θ_a and the Refraction Angle θ_b for the Surface*

Here, we will express the direction of propagation of the ray using the x- and y-components of the slope of the ray ($\tan\theta_x$ and $\tan\theta_y$, respectively). In Figure 5.2, we designated the point where the ray intersects with the surface as the origin. We defined the incident ray, the refraction ray, and the normal to the surface (at the ray's point of intersection) as vectors $\boldsymbol{L_i}$, $\boldsymbol{L_o}$, and $\boldsymbol{L_r}$, respectively, where each vector has a length of 1 unit along the z-axis. Thus $m_x(=\tan\theta_x)$ and $m_y(=\tan\theta_y)$ will be the lengths of the ray's x- and y-components, when projected onto the x–y plane at $z = 1$. In Figure 5.2, $m_{ix}(=\tan\theta_{ix})$ and $m_{iy}(=\tan\theta_{iy})$ are the incident ray vector's x- and y-components, and $m_{rx}(=\tan\theta_{rx})$, and $m_{ry}(=\tan\theta_{ry})$ are the normal vector's x- and y-components ($m_{rx} = w_{kx}/Z$, $m_{ry} = w_{ky}/Z$). θ_a is the incident angle, while θ_b is the refraction angle. Both θ_a and θ_b lie on the plane including the incident ray vector $\boldsymbol{L_i}$ and the normal vector $\boldsymbol{L_r}$. $\boldsymbol{L_o}$ (=OR) is the refraction ray vector.

The lengths of the incident ray vector $\boldsymbol{L_i}$ and the normal vector $\boldsymbol{L_r}$ are as follows:

$$L_i = \sqrt{(1 + {m_{ix}}^2 + {m_{iy}}^2)} = \sqrt{(1 + \tan^2\theta_{(k-1)x} + \tan^2\theta_{(k-1)y})} \tag{5.10}$$

$$L_r = \sqrt{(1 + {m_{rx}}^2 + {m_{ry}}^2)} = \sqrt{(1 + \tan^2\theta_{rx} + \tan^2\theta_{ry})} \tag{5.11}$$

The length of PQ ($\equiv a$) in Figure 5.2 will therefore be:

$$a = \sqrt{(m_{ix} - m_{rx})^2 + (m_{iy} - m_{ry})^2} = \sqrt{(\tan\theta_{(k-1)x} - \tan\theta_{rx})^2 + (\tan\theta_{(k-1)y} - \tan\theta_{ry})^2} \tag{5.12}$$

Applying the cosine rule to ΔOPQ, we obtain:

$$\cos\theta_a = ({L_i}^2 + {L_r}^2 - a^2)/(2L_iL_r) \tag{5.13}$$

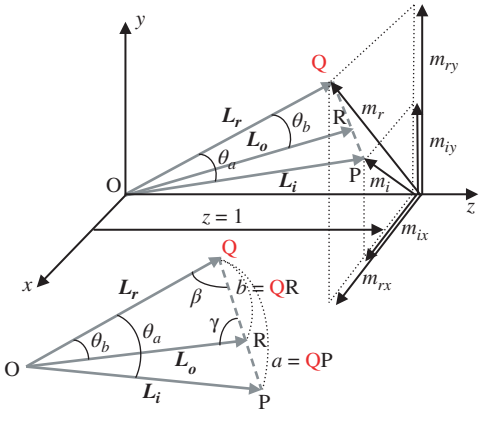


Figure 5.2 Three-dimensional expression of incident angle θ_a and refraction angle θ_b

Finally, applying Snell's law, Equation (1.2), we get the following value for $\sin\theta_b$:

$$\sin\theta_b = n_{k-1}/n_k \sin\theta_a \tag{5.14}$$

5.1.3 Calculate the Slopes $\tan\theta_{kx}$, $\tan\theta_{ky}$ of the Refracted Ray

Applying the cosine rule to ΔOPQ, and the sine rule to ΔORQ, we obtain the values of β and b in Figure 5.2.

$$\beta = \cos^{-1}[({L_r}^2 + a^2 - {L_i}^2)/(2L_r a)] \tag{5.15a}$$

$$\gamma = \pi - \theta_b - \beta \tag{5.15b}$$

$$b = (L_r \sin\theta_b)/\sin\gamma \tag{5.16}$$

Given the vector components in the $z = 1$ plane in Figure 5.2, we can derive $\tan\theta_{kx}$ and $\tan\theta_{ky}$ (the x- and y-components of the slope of the ray after exiting the surface):

$$\tan\theta_{kx} = \tan\theta_{rx} + b/a(\tan\theta_{(k-1)x} - \tan\theta_{rx}) \tag{5.17}$$

$$\tan\theta_{ky} = \tan\theta_{ry} + b/a(\tan\theta_{(k-1)y} - \tan\theta_{ry}) \tag{5.18}$$

Here we have obtained the output ray angles θ_{kx} and θ_{ky}.

5.1.4 Calculate the Optical Path Length

In Figure 5.1, the optical path length (OPL) of the ray traveling between the $(k-1)$th surface and the kth surface, L_k will be:

$$L_k = n_{k-1}(d_{k-1} - \Delta_{k-1} + \Delta_k)/\cos\theta_{k-1} \tag{5.19}$$

Here, θ_{k-1} is defined as follows:

$$\tan\theta_{k-1} = \sqrt{\tan^2\theta_{(k-1)x} + \tan^2\theta_{(k-1)y}} \tag{5.20}$$

5.2 Three-Dimensional Ray Tracing for a Cylindrical Surface

Two-dimensional ray tracing for a cylindrical lens fails to provide us with a complete analysis of the ray path, even if the object lies on the optical axis. In this case, we need to use three-dimensional ray tracing. Three-dimensional ray tracing for a cylindrical lens can be performed by following the same procedures as for a spherical lens, except that we have to change the boundary surface from a sphere to a cylinder. Here, we shall examine the ray tracing for a cylindrical surface whose curvature lies along the x-axis.

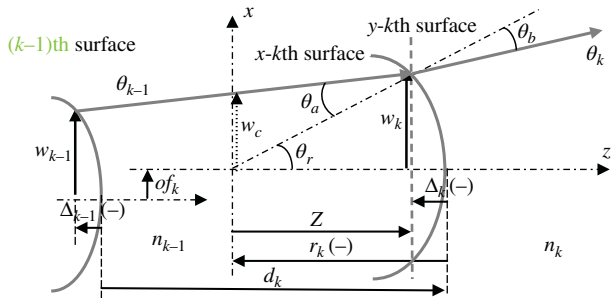


Figure 5.3 Cylindrical surface

5.2.1 *Calculate the Intersection Coordinates (w_{kx}, w_{ky}, Δ_k) of the Ray with the Surface*

When the surface is a cylinder with a radius of r which is curved in the x-direction as shown in Figure 5.3, it can be described by the following equation, if we define the center of curvature for the kth surface as the origin:

$$x^2 + z^2 = {r_k}^2 \text{ (cylindrical surface)} \tag{5.21}$$

The equations for the x-components of the incident ray are:

$$x = z \tan\theta_{(k-1)x} + w_{cx} \quad (x = az + b) \tag{5.2}$$

$$\text{where } w_{cx} = w_{(k-1)x} - of_{kx} + (d_{kx} - \Delta_{k-1} + r_k)\tan\theta_{(k-1)x} \tag{5.4}$$

Here, w_{cx} is the incident ray height along the x-axis, at the point where the ray's z-coordinate is equivalent to the z-coordinate of the kth surface's center of curvature. If we substitute Equation (5.2) into Equation (5.21), and then calculate the z-coordinate of the point where the incident ray intersects with the surface, we obtain:

$$Z = \frac{-ab \pm \sqrt{(ab)^2 - (1 + a^2)(b^2 - r^2)}}{1 + a^2} \tag{5.22}$$

The sign in front of the $\sqrt{}$ will be positive when the vertex of the surface is to the right of the surface, and negative when the vertex is to the left of the surface. Δ_k and the ray height at the $X(\equiv w_{kx})$ and $Y(\equiv w_{ky})$ intersections will then be:

$$\Delta_k = Z + r_k \tag{5.7}$$

$$X \equiv w_{kx} = w_{(k-1)x} - of_{(k-1)x} + (d_k - \Delta_{k-1} + \Delta_k)\tan\theta_{(k-1)x} \tag{5.23}$$

$$Y \equiv w_{ky} = w_{(k-1)y} - of_{(k-1)y} + (d_k - \Delta_{k-1} + \Delta_k)\tan\theta_{(k-1)y} \tag{5.24}$$

5.2.2 *Calculate the Incident Angle θ_a and the Refraction Angle θ_b for the Surface*

We can calculate θ_a and θ_b by following the same procedure as for a spherical lens. The only difference here is that the normal to the surface has no y-component.

$$\tan\theta_{rx} \equiv m_{rx} = w_{rx}/Z \tag{5.25a}$$

$$\tan\theta_{ry} \equiv m_{ry} = 0 \tag{5.25b}$$

The following equations, which are used to derive θ_a and θ_b, are the same as for a spherical lens:

$$L_i = \sqrt{(1 + {m_{ix}}^2 + {m_{iy}}^2)} = \sqrt{(1 + \tan^2\theta_{(k-1)x} + \tan^2\theta_{(k-1)y})} \tag{5.10}$$

$$L_r = \sqrt{(1 + {m_{rx}}^2 + {m_{ry}}^2)} = \sqrt{(1 + \tan^2\theta_{rx} + \tan^2\theta_{ry})} \tag{5.11}$$

$$a = \sqrt{(m_{ix} - m_{rx})^2 + (m_{iy} - m_{ry})^2} = \sqrt{(\tan\theta_{(k-1)x} - \tan\theta_{rx})^2 + (\tan\theta_{(k-1)y} - \tan\theta_{ry})^2} \tag{5.12}$$

$$\cos\theta_a = ({L_i}^2 + {L_r}^2 - a^2)/(2L_iL_r) \tag{5.13}$$

$$\sin\theta_b = n_{k-1}/n_k \sin\theta_a \tag{5.14}$$

5.2.3 *Calculate the x- and y-Components of the Slope of the Output Ray, tan θ_{kx}, and tan θ_{ky}*

(These will be the same as for a spherical lens.)

$$\beta = \cos^{-1}[({L_r}^2 + a^2 - {L_i}^2)/(2L_ra)] \tag{5.15a}$$

$$\gamma = \pi - \theta_b - \beta \tag{5.15b}$$

$$b = (L_r \sin\theta_b)/\sin\gamma \tag{5.16}$$

$$\tan\theta_{kx} = \tan\theta_{rx} + b/a(\tan\theta_{(k-1)x} - \tan\theta_{rx}) \tag{5.17}$$

$$\tan\theta_{ky} = \tan\theta_{ry} + b/a(\tan\theta_{(k-1)y} - \tan\theta_{ry}) \tag{5.18}$$

5.2.4 *Calculate the Optical Path Length*

(This will be the same as for a spherical lens.)

$$L_k = \frac{n_{k-1}(d_{k-1} - \Delta_{k-1} + \Delta_k)}{\cos\theta_{k-1}} \tag{5.19}$$

$$\tan\theta_{k-1} = \sqrt{\tan^2\theta_{(k-1)x} + \tan^2\theta_{(k-1)y}} \tag{5.20}$$

5.3 Simulation for Two Cylindrical Lenses Which Are Fixed Longitudinally (or Laterally) but Allowed to Rotate Slightly around the Optical Axis

In an optical system having two or more cylindrical lenses which are fixed longitudinally (or laterally), if one of them is allowed to rotate by a small angle ϕ around the optical axis, it will generate a significant

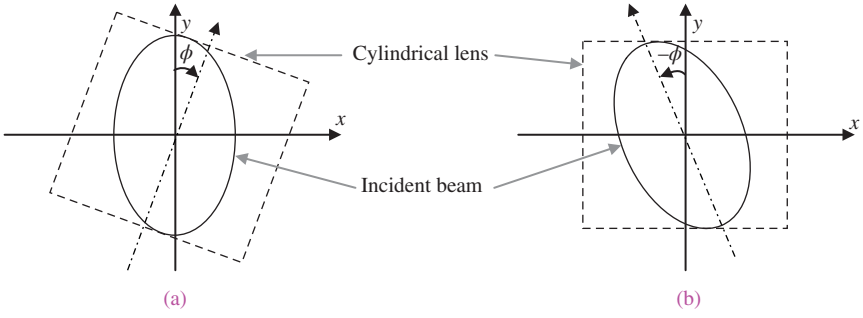


Figure 5.4 (a) Cylindrical lens (rotated by ϕ) and (b) incident beam (rotated by $-\phi$)

degree of aberration. To calculate a ray tracing in this case, we use a method that involves rotating the incident rays, instead of rotating the lens. Figure 5.4(a) shows a cylindrical lens rotated by an angle ϕ and the incident beam (not rotated), while Figure 5.4(b) shows the cylindrical lens (not rotated) and the incident beam, which is rotated by an angle $-\phi$. Inspecting the two figures, we can readily perceive that the relationship between the cylindrical lens and the incident beam is the same for both figures.

Thus instead of rotating the lens, we can rotate the incident rays by an angle $-\phi$ before the rays enter the lens and then calculate the ray tracing for rays passing through the lens. After exiting the lens, we rotate the rays by an angle ϕ, [thereby resetting the net rotation angle to its original value (zero)], and we can subsequently perform a similar ray tracing at the next surface, and so on. Using this method, we can easily calculate the ray tracing for a cylindrical lens which is fixed longitudinally (or laterally), but is allowed to rotate around the optical axis. Here, we can obtain the optical path length by performing the same calculations as for other surfaces, regardless of the ray rotation angle.

5.3.1 Rotate Rays by an Angle ϕ around the Optical Axis

Figure 5.5 shows the slope of the incident ray, which is projected onto the $x-y$ plane. The optical axis (z-axis) is vertical to the page. In the figure, OP is the incident ray slope vector $\boldsymbol{R_i}$ (where the ray propagates 1 unit length in z-axis and changes its height by OP). OQ is the ray slope vector $\boldsymbol{R_o}$ obtained after

R = length of vectors OP and OQ
P_x, P_y = x- and y-components of vector OP
Q_x, Q_y = x- and y-components of vector OQ
ψ = Angle of orientation of the incident ray slope
ϕ = Rotation angle

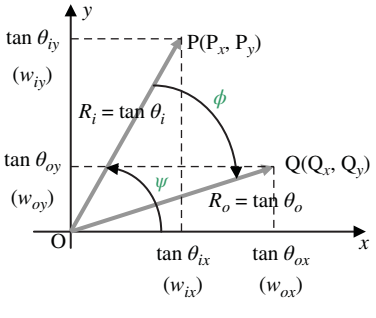


Figure 5.5 Rotation of a ray

rotating the incident ray clockwise by an angle ϕ. The coordinates of point Q(Q_x,Q_y) can be expressed as follows:

$$Q_x = R\cos(\psi - \phi) = R\cos\psi\cos\phi + R\sin\psi\sin\phi = P_x\cos\phi + P_y\sin\phi \tag{5.26}$$

$$Q_y = R\sin(\psi - \phi) = R\sin\psi\cos\phi - R\cos\psi\sin\phi = P_y\cos\phi - P_x\sin\phi \tag{5.27}$$

where

R = Length of vectors OP and OQ
ψ = Angle of orientation of the incident ray slope
P_x, P_y = x- and y-components of vector OP
ϕ = Rotation angle
Q_x, Q_y = x- and y-components of vector OQ

The x- and y-components of the ray slope after the incident ray is rotated by an angle of ϕ (i.e., $\tan\theta_{ox}$, $\tan\theta_{oy}$) can be derived from the x- and y-components of the original incident ray slope (i.e., $\tan\theta_{ix}$, $\tan\theta_{iy}$) as follows:

$$\tan\theta_{ox} = \tan\theta_{ix}\cos\phi + \tan\theta_{iy}\sin\phi \tag{5.28}$$

$$\tan\theta_{oy} = -\tan\theta_{ix}\sin\phi + \tan\theta_{iy}\cos\phi \tag{5.29}$$

Likewise, Equation (5.30) and Equation (5.31) show how we can calculate the ray heights w_{ox}, w_{oy} after the ray is rotated by an angle of ϕ, from the values of w_{ix} and w_{iy}, which are the x- and y-components of the incident ray height:

$$w_{ox} = w_{ix}\cos\phi + w_{iy}\sin\phi \tag{5.30}$$

$$w_{oy} = -w_{ix}\sin\phi + w_{iy}\cos\phi \tag{5.31}$$

5.4 Three-Dimensional Ray Tracing for a Plane Surface Which Is Perpendicular to the Optical Axis

When we are performing a three-dimensional ray tracing, we can still use plane surfaces for a plano-convex lens, a plano-concave lens and a cylindrical lens. We can also use them when defining an image plane. The ray tracing procedure for a plane surface which is perpendicular to the optical axis is outlined below.

5.4.1 Calculate the x- and y-Components of the Ray Height w_{kx}, w_{ky} at the Boundary Surface

In Figure 5.6, w_{kx} and w_{ky}, the x- and y-components of the incident ray height, can be expressed as follows:

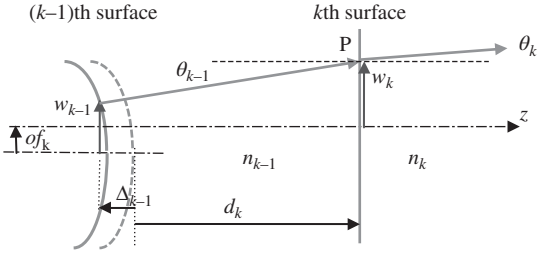


Figure 5.6 Ray tracing for a plane surface medium

$$w_{kx} = w_{(k-1)x} - of_{kx} + (d_k - \Delta_{k-1}) \tan\theta_{(k-1)x} \tag{5.32}$$

$$w_{ky} = w_{(k-1)y} - of_{ky} + (d_k - \Delta_{k-1}) \tan\theta_{(k-1)y} \tag{5.33}$$

5.4.2 *Calculate the Incident Angle θ_a and the Refraction Angle θ_b at the Boundary Surface*

Figure 5.7 shows the incident ray vector $\boldsymbol{L_i}$ and the refraction ray vector $\boldsymbol{L_o}$ at a boundary (plane) surface which is vertical to the axis. If the boundary surface is a vertical plane, then the normal to the plane will be parallel to the optical axis; hence the refracted ray will lie on the plane which includes the incident ray and the optical axis. In Figure 5.7, OR is a vector which represents the refracted ray. The refraction angle θ_b (which is equivalent to the output angle θ_k of the ray exiting the kth surface) will be:

$$\theta_b \equiv \theta_k = \sin^{-1}\left(\frac{n_{k-1}}{n_k}\sin\theta_a\right) \tag{4.7′}$$

where

$$\theta_a \equiv \theta_{k-1} = \tan^{-1}\left(\sqrt{\tan^2\theta_{(k-1)x} + \tan^2\theta_{(k-1)y}}\right) \tag{5.20′}$$

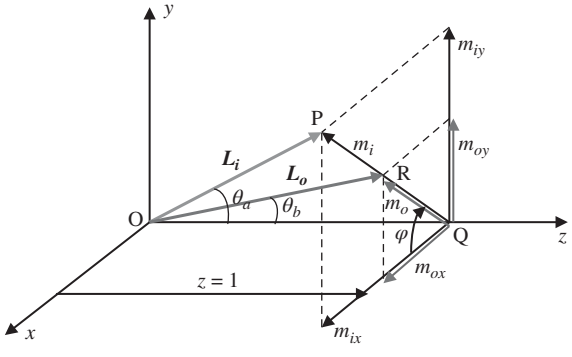


Figure 5.7 Incident and refracted ray vector on a plane boundary surface

The x- and y-components of the refracted ray slope ($\tan\theta_{kx}$, $\tan\theta_{ky}$) will be as follows:

$$\tan\theta_{kx} = \tan\theta_k \cos\varphi \tag{5.34}$$

$$\tan\theta_{ky} = \tan\theta_k \sin\varphi \tag{5.35}$$

where φ is:

$$\varphi = \tan^{-1}\left(\frac{\tan\theta_{(k-1)y}}{\tan\theta_{(k-1)x}}\right) \tag{5.36}$$

We can thus derive the x- and y-components of the height (w_{kx}, w_{ky}) and slope ($\tan\theta_{kx}$, $\tan\theta_{ky}$) of the output ray.

5.5 Three-Dimensional Ray Tracing for an Aberration-Free Lens

5.5.1 Three-Dimensional Expression of Ray Refraction by an Aberration-Free Lens

For an aberration-free lens, we can use the ray tracing procedure based on the principles of paraxial theory (outlined in Chapter 3), in the same way as we did for the two-dimensional ray tracing in Chapter 4. In Chapter 3, we showed that the angle of the ray exiting the lens θ_o can be expressed as follows:

$$\theta_o = \theta_i - w/f \tag{3.2}$$

where θ_i is the incident ray slope angle (rad), w is the ray height at the lens and f is the focal length of the lens. In Equation (3.2), the ray exiting the lens will always be refracted toward the center of the lens, and thus the change $-w/f$ in the angle of the exiting ray will always be in this direction. (If we use a concave lens instead of a convex lens, the change will be in the reverse direction, and the ray will be refracted away from the center of the lens.) We can generalize the expression derived in Equation (3.2) from two to three dimensions, as follows:

$$\theta_{ox} = \theta_{ix} - w_x/f \tag{5.37}$$

$$\theta_{oy} = \theta_{iy} - w_y/f \tag{5.38}$$

Figure 5.8a is a diagram of the incident ray entering the lens and the refraction ray exiting it, where both of them are projected onto the lens pupil. (In this diagram, the rays run from front to back through the paper.) In Figure 5.8a, the incident ray traveling parallel to the x–z plane from the left which makes an angle θ_{ix} with the z-axis passes through point P on the lens pupil, with a height of w_y above the lens center ($w_x = 0$). For the ray exiting the lens, the angle will only change in the y-direction, by $-w_y/f$. As a result, the ray exiting the lens moves downwards and to the right, as shown in Figure 5.8a (for the case of a convex lens).

Figure 5.8b illustrates the addition of ray vectors. The ray vector θ_o after the ray exits the lens can be obtained by adding the incident ray vector θ_i and the refraction vector $\Delta\theta$ $(= -w_y/f)$. Figure 5.8c depicts the general case for a ray vector.

5.5.2 Three-Dimensional Ray Tracing for an Aberration-Free Lens

Three-dimensional ray tracing for an aberration-free lens which is perpendicular to the optical axis can be performed using the same method as for two-dimensional ray tracing as shown in Figure 5.9, except that the number of dimensions has increased from two to three.

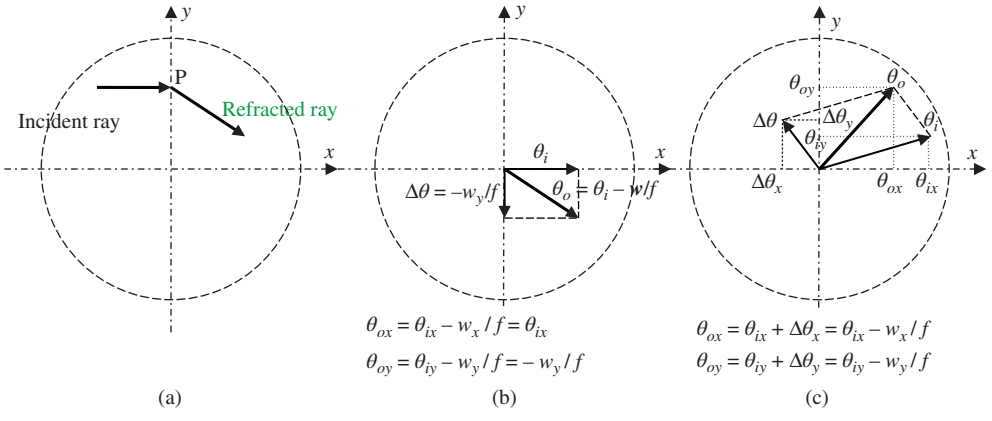


Figure 5.8 Vector representation of ray slopes. (a) Incident ray and refracted ray projected onto the lens pupil, (b) vector representation of the refracted ray slope, and (c) vector representation of the refracted ray slope (general case)

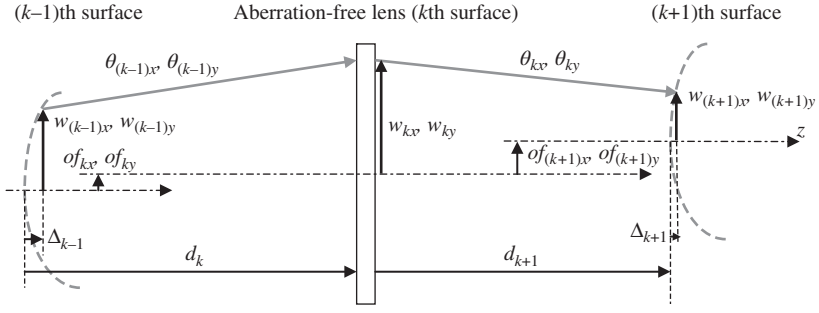


Figure 5.9 Ray tracing for an aberration-free lens

5.5.2.1 Path of the Incident Ray from the $(k-1)$th to the kth Surface

The x- and y-components w_{kx}, w_{ky} of the incident ray height on the aberration-free lens can be calculated from the ray parameters at the $(k-1)$th surface, which lies to the left of the aberration-free lens.

$$w_{kx} = w_{(k-1)x} + (d_k - \Delta_{k-1})\,\theta_{(k-1)x} - of_{kx} \tag{5.39}$$

$$w_{ky} = w_{(k-1)y} + (d_k - \Delta_{k-1})\,\theta_{(k-1)y} - of_{ky} \tag{5.40}$$

Here we will directly substitute the x- and y-components $\tan\theta_{(k-1)x}$, $\tan\theta_{(k-1)y}$ of the ray slope for the corresponding components of the ray angles $\theta_{(k-1)x}$, $\theta_{(k-1)y}$ in Equation (5.39) and Equation (5.40), respectively.

5.5.2.2 Calculation of the Ray Angle after Exiting the Aberration-Free Lens

The x- and y-components θ_{kx}, θ_{ky} of the ray angle after exiting the aberration-free lens can be calculated by applying Equation (5.37′) and Equation (5.38′):

$$\theta_{kx} = \theta_{(k-1)x} - w_{kx}/f \tag{5.37′}$$

$$\theta_{ky} = \theta_{(k-1)y} - w_{ky}/f \tag{5.38′}$$

Here, we will directly substitute $\tan\theta_{(k-1)x}$ and $\tan\theta_{(k-1)y}$ (which we calculated at the $(k-1)$th surface) for the corresponding components of $\theta_{(k-1)x}$ and $\theta_{(k-1)y}$.

5.5.2.3 Path of the Output Ray to the Next Surface

After the ray exits the aberration-free lens, the x- and y-components $w_{(k+1)x}, w_{(k+1)y}$ of the ray height at the next surface will be:

$$w_{(k+1)x} = w_{kx} + (d_{k+1} + \Delta_{k+1})\tan\theta_{kx} - of_{(k+1)x} \tag{5.41}$$

$$w_{(k+1)y} = w_{ky} + (d_{k+1} + \Delta_{k+1})\tan\theta_{ky} - of_{(k+1)y} \tag{5.42}$$

Here, we will directly substitute θ_{kx} and θ_{ky}, which were calculated in Equation (5.37′) and Equation (5.38′), for the corresponding components of $\tan\theta_{kx}$ and $\tan\theta_{ky}$, in order to obtain the ray height at the next surface.

5.5.3 *Optical Path Length Correction for an Aberration-Free Lens*

As shown in Figure 5.10, we can think of the incident ray as being part of an incident beam, and we can suppose that this beam has a convergence point (or virtual object point) from which rays are emitted (provided that the beam is not collimated). (In three-dimensional terminology, rays, after passing through an optical system containing aberrations, may not converge exactly on a single point. Nevertheless, to a good approximation, we can treat the rays as if they converged at a single point, as long as the optical system does not contain any significant sources of aberration.) Using paraxial theory, we can then calculate the virtual image point where the output rays all converge, after exiting the aberration-free lens. Below, we will describe a method for making an optical path length adjustment, where appropriate.

5.5.3.1 Obtain a Virtual Object Point *P*(*x*,*y*,*z*)

Let us consider two rays in an incident beam: Ray #1, for which we will perform a ray tracing; and a dummy ray, Ray #2 (e.g., the principal ray). We can calculate the point of intersection for these two rays,

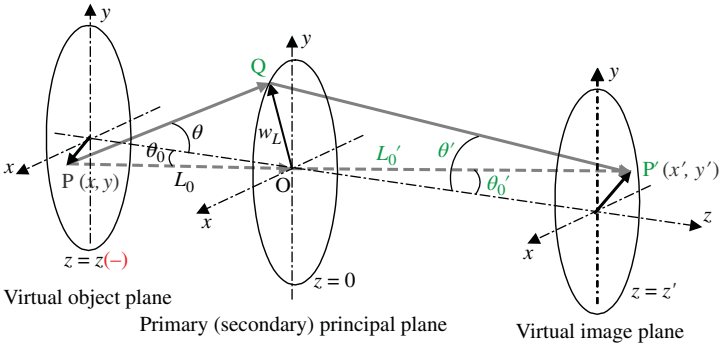


Figure 5.10 Virtual object point and virtual image point for an aberration-free lens

and treat this point as an imaginary object point.

$$\text{Ray \#1}: x - x_1 = m_{x1}(z - z_1) \tag{5.43a}$$

$$y - y_1 = m_{y1}(z - z_1) \tag{5.43b}$$

$$\text{Ray \#2}: x - x_2 = m_{x2}(z - z_2) \tag{5.44a}$$

$$y - y_2 = m_{y2}(z - z_2) \tag{5.44b}$$

Here, we set the origin on the primary principal point of the aberration-free lens, while (x_1, y_1, z_1), (x_2, y_2, z_2) are the intersection point coordinates of Ray #1, Ray #2 with the $(k-1)$th surface, and m_{x1}, m_{y1}, m_{x2}, m_{y2} are the x- and y-components of the slopes of Ray #1 and Ray #2, respectively, after exiting the $(k-1)$th surface.

For the above two rays, which are described by Equation (5.43a,b) and Equation (5.44a,b), we can define the virtual objective point P as the point where they have the same value of x (or y) for the same value of z. The virtual object point (x, y, z) will be:

$$s \equiv z = z_x(m_{x2}z_2 - m_{x1}z_1 - x_2 + x_1)/(m_{x2} - m_{x1}) \tag{5.45a}$$

or

$$s \equiv z = z_y(m_{y2}z_2 - m_{y1}z_1 - y_2 + y_1)/(m_{y2} - m_{y1}) \tag{5.45b}$$

$$x = m_{x1}z - m_{x1}z_1 + x_1 = \tan\theta_{(k-1)x}z - \tan\theta_{(k-1)x}(-d + \Delta_{k-1}) + w_{(k-1)x} - of_{kx} \tag{5.46}$$

$$y = m_{y1}z - m_{y1}z_1 + y_1 = \tan\theta_{(k-1)y}z - \tan\theta_{(k-1)y}(-d + \Delta_{k-1}) + w_{(k-1)y} - of_{ky} \tag{5.47}$$

5.5.3.2 Obtain the Virtual Image Point $P'(x', y', z')$

The image point can be calculated by applying the Gaussian lens formula, Equation (1.24), since the lens is an aberration-free lens in this example. Thus the image distance s' $(\equiv z')$ will be:

$$s' \equiv z' = zf/(z + f) \tag{5.48}$$

The x- and y-components of the image height, x' and y', can be calculated from the equation for lateral magnification, Equation (1.27′).

$$x' = xz'/z \tag{5.49}$$

$$y' = yz'/z \tag{5.50}$$

5.5.3.3 Calculation of the Optical Path Length Adjustment Value

In Figure 5.10, if the lens has no aberration, then all rays emitted from the object P will converge precisely at the image point P′. Moreover, the optical path lengths for all rays traveling from the object P to the image P′ will always be identical, regardless of the route by which they travel (Fermat's principle). Here, we define the optical path length from P to P′ as the straight line length PO + OP′ (which corresponds to PH + H′P′ in Figure 4.9, where H and H′ are the primary and secondary principal points of the aberration-free lens, respectively). Thus, if we calculate the optical path length as the straight line length of PQ + QP′, we will need to subtract the difference between PQ + QP′ and PO + OP′ from the calculated value. The optical path length adjustment value ΔL for the incident ray is:

$$\Delta L = \text{PQ} - \text{PO} \tag{5.51}$$

The optical path length correction $\Delta L'$ for the ray after it exits the lens is:

$$\Delta L' = \mathrm{QP}' - \mathrm{OP}' \tag{5.52}$$

where PQ, PO, QP′, OP′ are calculated as follows:

$$\mathrm{PQ} = -\frac{z}{\cos\theta_{k-1}} = -\frac{z}{\cos(\tan^{-1}\sqrt{\tan^2\theta_{(k-1)x} + \tan^2\theta_{(k-1)y}})} \tag{5.53}$$

$$\mathrm{PO} = -\frac{z}{\cos\theta_O} = -\frac{z}{\cos(\tan^{-1}\sqrt{(x/z)^2 + (y/z)^2})} \tag{5.54}$$

$$\mathrm{QP}' = -\frac{z'}{\cos\theta_k} = -\frac{z'}{\cos(\tan^{-1}\sqrt{\tan^2\theta_{kx} + \tan^2\theta_{ky}})} \tag{5.55}$$

$$\mathrm{OP}' = -\frac{z'}{\cos\theta_O} = -\frac{z'}{\cos(\tan^{-1}\sqrt{(x/z)^2 + (y/z)^2})} \tag{5.56}$$

Note

We will directly substitute the values of θ_{kx}, θ_{ky} which were calculated for the aberration-free lens ray tracing for $\tan\theta_{kx}$ and $\tan\theta_{ky}$ into the above equations.

Optical Path Length Adjustment Value for the Case of Parallel Incident Rays

If the incident rays are parallel, the value of z in Equation (5.53) and Equation (5.54) tends to infinity. In this case, we use a special procedure to calculate the optical path length adjustment value. First, we calculate the incident ray slope $\tan\theta_i$ and its direction (or orientation) φ_i. Then we calculate the height w_L and angle (orientation) ψ of the ray at the lens, from the ray's x- and y-components w_{Lx} and w_{Ly}. Finally we calculate the φ_i-component of w_L, $w_{L\varphi i}$.

In Figure 5.11a, the slope $\tan\theta_i$ and its angle of orientation φ_i of the incident ray can be represented as follows:

$$\tan\theta_i = \sqrt{\tan^2\theta_{ix} + \tan^2\theta_{iy}} \tag{5.57}$$

$$\phi_i = \tan^{-1}\left(\frac{\tan\theta_{iy}}{\tan\theta_{ix}}\right) \tag{5.58}$$

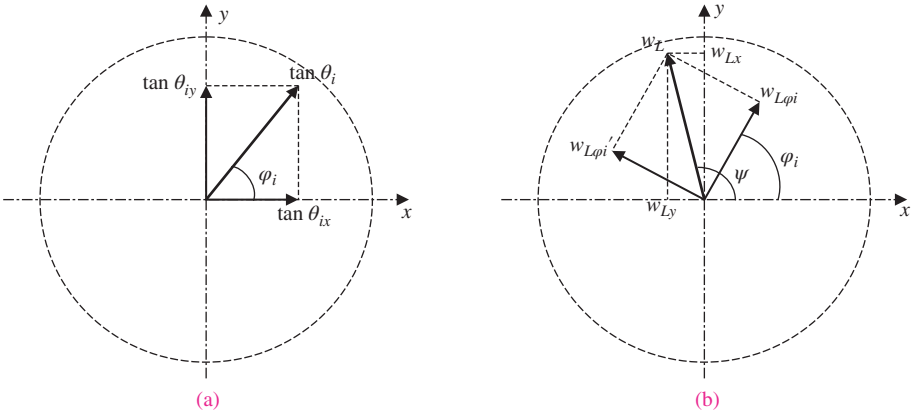


Figure 5.11 (a) Incident ray slope vector and (b) incident ray height at the lens

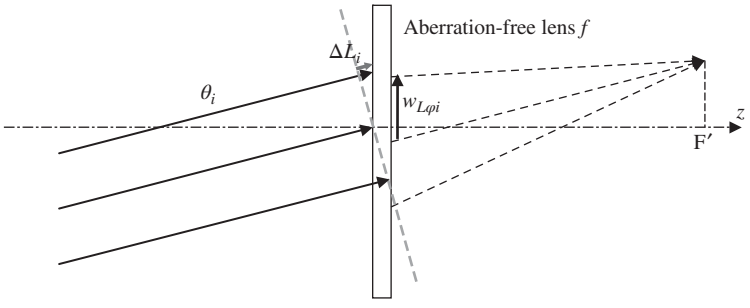


Figure 5.12 Parallel incident rays onto an aberration-free lens (cross-sectional view in the direction of φ_i)

In Figure 5.11b, the height w_L and direction (orientation) ψ of the incident ray at the lens, and the φ_i-component $w_{L\varphi i}$ of the ray height w_L can be calculated as follows:

$$w_L = \sqrt{{w_{Lx}}^2 + {w_{Ly}}^2} \tag{5.59}$$

$$\psi = \tan^{-1}\left(\frac{w_{Ly}}{w_{Lx}}\right) \tag{5.60}$$

$$w_{L\varphi i} = w_L \cos(\psi - \varphi_i) \tag{5.61}$$

The diagram in Figure 5.12 is a cross-sectional view of a parallel incident beam onto an aberration-free lens, in the direction of φ_i. In this figure, Equation (5.61) allows us to calculate the ray height $w_{L\varphi i}$ for a ray traveling in the direction of φ_i. The OPL correction value ΔL_i will be:

$$\Delta L_i = w_{L\varphi i} \sin\theta_i \tag{5.62}$$

Optical Path Length Correction Value for the Case of Parallel Output Rays

If the output rays are parallel, then the value of z' in Equation (5.55) and Equation (5.56) will tend to infinity. In such a case, we will have to calculate the optical path length correction value by following a procedure similar to the one we employed for parallel incident rays. First, we calculate the slope $\tan\theta_o$ and the angle of orientation φ_o of the output ray. Then we calculate the height w_L and the angle of orientation ψ of the ray at the lens, from the ray's x- and y-components w_{Lx} and w_{Ly}. Finally we calculate the φ_o-component of w_L, $w_{L\varphi o}$.

In Figure 5.13a, the slope $\tan\theta_o$ and the angle of orientation φ_o of the output ray can be represented as follows:

$$\tan\theta_O = \sqrt{\tan^2\theta_{Ox} + \tan^2\theta_{Oy}} \tag{5.63}$$

$$\varphi_o = \tan^{-1}\left(\frac{\tan\theta_{Oy}}{\tan\theta_{Ox}}\right) \tag{5.64}$$

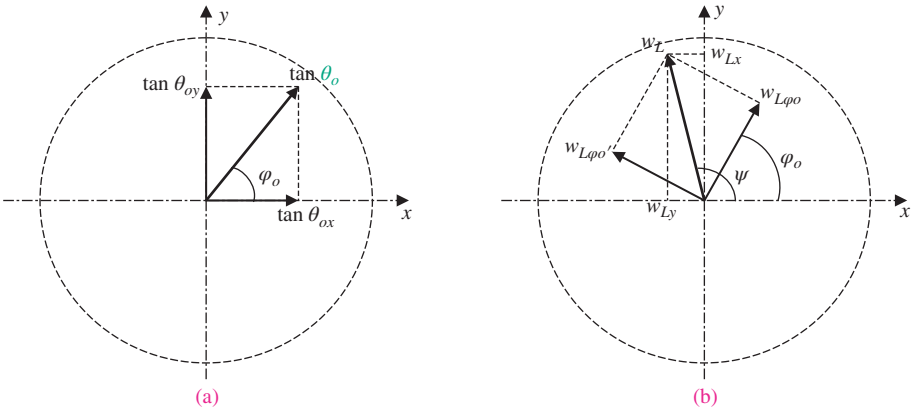


Figure 5.13 (a) Exiting ray slope vector and (b) exiting ray height at the lens

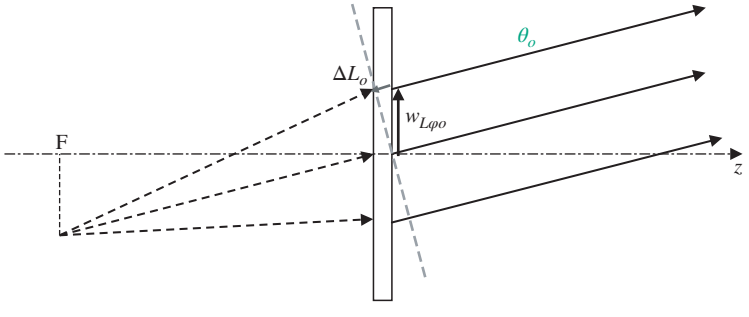


Figure 5.14 Parallel beam exiting from an aberration-free lens (cross-sectional view in the direction of φ_o)

In Figure 5.13b, the height w_L and the direction (or orientation) ψ of the ray at the lens, and the φ_o-component $w_{L\varphi o}$ of the ray height w_L can be calculated as follows:

$$w_L = \sqrt{{w_{Lx}}^2 + {w_{Ly}}^2} \tag{5.59}$$

$$\psi = \tan^{-1}\left(\frac{w_{Ly}}{w_{Lx}}\right) \tag{5.60}$$

$$w_{L\varphi o} = w_L \cos(\psi - \varphi_o) \tag{5.65}$$

Figure 5.14 is a diagram showing the rays of a parallel beam exiting from an aberration-free lens along a cross-section in the direction of φ_o. In this figure, the φ_o-component of the ray height $w_{L\varphi o}$ can be calculated from Equation (5.65). The optical path length correction value ΔL_o is:

$$\Delta L_o = w_{L\varphi o} \sin\theta_o \tag{5.66}$$

This aberration-free lens does not generate any aberration. However if we use this lens with a large numerical aperture (NA), the calculated ray height at the lens may differ significantly from the actual ray height.

5.6 Three-Dimensional Ray Tracing for a Lens Which Is Set at a Tilt

In order to perform a three-dimensional ray tracing for a lens which is set at a tilt, we will apply the same method as the one we discussed when doing two-dimensional ray tracing, but we will increase the number of dimensions from two to three. That is, for rays passing through the lens at a tilt angle α, we change the optical axis in our calculation base from that of the incident rays to that of the lens (which is rotated from the incident rays' optical axis by an angle α, as shown in Figure 5.16 (a), (b)). After passing through the lens, we restore the optical axis to that of the incident ray and then calculate the path of the exiting rays traveling to the next surface.

However, in three-dimensional space, we have to consider not only the tilt angle α but also the lens's tilt orientation angle ϕ. If we set the tilt orientation angle ϕ to 0 (i.e., the lens is tilted along the y-axis), then we can calculate the ray tracing for the lens set at a tilt by using the method discussed in Chapter 4, and by increasing the dimensions from two to three. If we rotate the incident rays around the optical axis through an angle $-\phi$ before they impinge on the lens, then we can assume that the tilt orientation of the lens will lie along the y-axis (Figure 5.15a,b). We can then calculate the ray tracing for the rays passing through the lens, assuming its tilt orientation angle $\phi = 0$. After exiting the lens, we rotate the rays back through an angle ϕ to their original orientation and calculate the path of the exiting rays traveling to the next surface.

5.6.1 Rotation of the Incident Rays by an Angle-ϕ

As shown in Figure 5.15a,b, if the tilt orientation angle of the lens is ϕ, we can rotate the incident rays by an angle $-\phi$ around the axis of propagation before they impinge on the lens. Then we can calculate the ray tracing, assuming that the lens is set at a tilt, with its orientation lying along the y-axis. Here, we can calculate the ray rotation by using Equation (5.28), Equation (5.29), Equation (5.30) and Equation (5.31).

5.6.2 Ray Tracing for a Lens Tilted at an Angle α

As we showed in Chapter 4, we can use the incident ray coordinates system for the rays traveling toward the incident plane IQ as in Figure 5.16a. We can also use the lens coordinates system for the rays passing through the lens (from the incident plane IQ to the exiting plane JR) whose axis is rotated from that of the incident ray by an angle α as shown in Figure 5.16b. After passing through the exiting plane JR, we

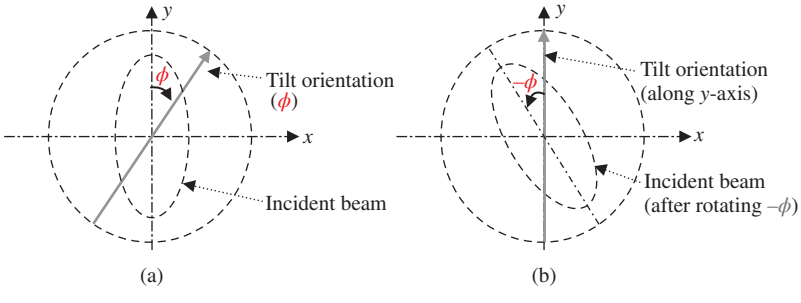


Figure 5.15 (a) Lens tilt orientation ϕ and (b) lens tilt orientation $\phi \rightarrow 0°$ (for rays passing through the lens)

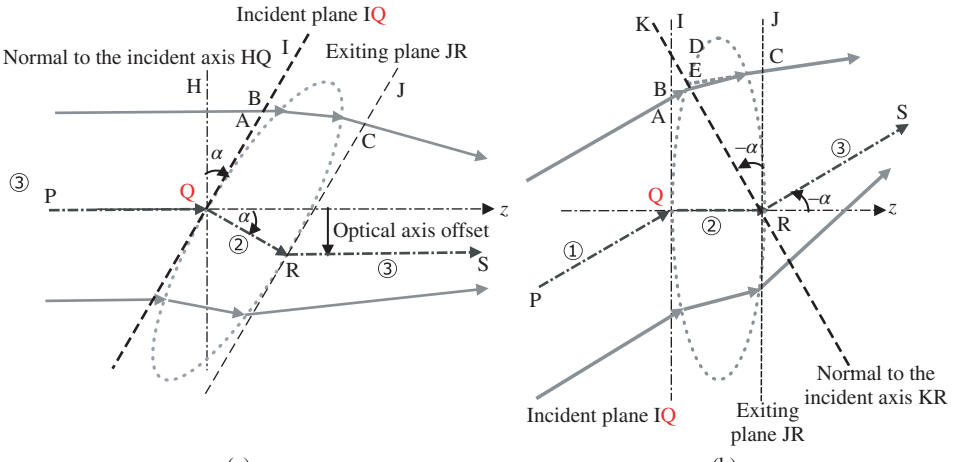


Figure 5.16 (a) A tilted lens (represented using incident ray coordinate system) and (b) a tilted lens (represented using lens axis coordinate system)

restore the axis to that of the incident ray coordinate system. In the figure, PQ(①), QR(②), and RS(③) show the axes for calculating the incident rays, the rays passing through the lens and the rays after exiting the lens, respectively, and HQ, KR depict the plane normal to the optical axis at the incident plane and that at the exiting plane, respectively.

5.6.2.1 Ray Tracing for the Rays Traveling to the Incident Plane IQ (Tilted at an Angle α)

Calculate the ray position A (Δ_A, w_{Ax}, w_{Ay}) and ray slope $\tan\theta_{Ax}$, $\tan\theta_{Ay}$ on the incident plane IQ, which is tilted at an angle α. Use the same methodology as in Section 4.6.

1. When the tilt angle $\alpha = 0$

$$\Delta_A \equiv \Delta_k = 0 \tag{5.67}$$

$$w_{Ax} \equiv w_{kx} = w_{(k-1)x} - of_{Ax} + (d_A - \Delta_{k-1} + \Delta_A)\tan\theta_{(k-1)x} \tag{5.68}$$

$$w_{Ay} \equiv w_{ky} = w_{(k-1)y} - of_{Ay} + (d_A - \Delta_{k-1} + \Delta_A)\tan\theta_{(k-1)y} \tag{5.69}$$

$$\tan\theta_{Ax} \equiv \tan\theta_{kx} = \tan\theta_{(k-1)x} \tag{5.70}$$

$$\tan\theta_{Ay} \equiv \tan\theta_{ky} = \tan\theta_{(k-1)y} \tag{5.71}$$

2. When the tilt angle $\alpha \neq 0$

$$Z \equiv \Delta_A \equiv \Delta_k = \frac{\tan\theta_{(k-1)y} d_A - \tan\theta_{(k-1)y}\Delta_{k-1} + w_{(k-1)y} - of_{Ay}}{1/\tan\alpha - \tan\theta_{(k-1)y}} \tag{5.72}$$

$$w_{Ax} \equiv w_{kx} = w_{(k-1)x} - of_{Ax} + (d_A - \Delta_{k-1} + \Delta_A)\tan\theta_{(k-1)x} \tag{5.68}$$

$$w_{Ay} \equiv w_{ky} = w_{(k-1)y} - of_{Ay} + (d_A - \Delta_{k-1} + \Delta_A)\tan\theta_{(k-1)y} \tag{5.69}$$

$$\tan\theta_{Ax} \equiv \tan\theta_{kx} = \tan\theta_{(k-1)x} \tag{5.70}$$

$$\tan\theta_{Ay} \equiv \tan\theta_{ky} = \tan\theta_{(k-1)y} \tag{5.71}$$

3. Optical path length L_A

$$L_A = n_{k-1}(d_A - \Delta_{k-1} + \Delta_A)/\cos\theta_{k-1} \tag{5.73}$$

where of_A, d_A, Δ_A, and $\tan\theta_A$ are equivalent to of_k, d_k, Δ_k, and $\tan\theta_k$, respectively, and where $\tan\theta_{kx} = \tan\theta_{(k-1)x}$, $\tan\theta_{ky} = \tan\theta_{(k-1)y}$ (we will not perform the refraction calculations here).

5.6.2.2 Coordinate Transformation to the Lens Coordinates (Rotation of the Axis by an Angle α)

Transform the coordinates of the above ray position (Δ_A, w_{Ax}, w_{Ay}) and ray slope $\tan\theta_{Ax}$, $\tan\theta_{Ay}$ to the lens coordinates (Δ_B, w_{Bx}, w_{By}), $\tan\theta_{Bx}$, $\tan\theta_{By}$ by rotating the coordinate system around the x-axis by an angle α.

$$\Delta_B = 0 \tag{5.74}$$

$$w_{Bx} = w_{Ax} \tag{5.75}$$

$$w_{By} = w_{Ay}/\cos\alpha \tag{5.76}$$

$$\tan\theta_{Bx} = \tan\theta_{Ax}/(\cos\alpha - \tan\theta_{Ay}\sin\alpha) \tag{5.77}$$

$$\tan\theta_{By} = \tan\left[\arctan\left(\tan\theta_{Ay}\right) + \alpha\right] \tag{5.78}$$

Here, $\tan\theta_{Bx}$ and $\tan\theta_{By}$ show the ray slope after rotating the coordinate system for the lens axis. (See "Ray slopes after rotating the coordinates" at the end of Section 5.6.3.)

5.6.2.3 Ray Tracing for Rays Passing through the Lens

Calculate the ray tracing for the rays passing through the lens using the lens coordinate system, starting with the values Δ_B, w_{Bx}, w_{By}, $\tan\theta_{Bx}$, $\tan\theta_{By}$ which relate to the incident plane IQ, and obtain the values Δ_C, w_{Cx}, w_{Cy} $\tan\theta_{Cx}$, $\tan\theta_{Cy}$ which relate to the exiting plane JR.

5.6.2.4 Calculate the Exiting Ray Δ_D, w_{Dx}, w_{Dy}, $\tan\theta_{Dx}$, $\tan\theta_{Dy}$ on the Plane KR, Which Is Tilted at an Angle $-\alpha$ in the Lens Coordinate System (Which Is Normal to the Exiting Optical Axis)

Calculate the ray tracing from C(Δ_C, w_{Cx}, w_{Cy}) on the exiting plane JR to D(Δ_D, w_{Dx}, w_{Dy}) on the plane KR which is tilted at an angle $-\alpha$ in the lens coordinate system.

1. When the tilt angle $\alpha = 0$,

$$\Delta_D = 0 \tag{5.79}$$

$$w_{Dx} = w_{Cx} - of_{Dx} + (d_D - \Delta_C + \Delta_D)\tan\theta_{Cx} \tag{5.80}$$

$$w_{Dy} = w_{Cy} - of_{Dy} + (d_D - \Delta_C + \Delta_D)\tan\theta_{Cy} \tag{5.81}$$

2. When the tilt angle $\alpha \neq 0$,

$$Z \equiv \Delta_D = \frac{\tan\theta_{Cy}d_D - \tan\theta_{Cy}\Delta_C + w_{Cy} - of_{Dy}}{1/\tan(-\alpha) - \tan\theta_{Cy}} \tag{5.82}$$

$$w_{Dx} = w_{Cx} - of_{Dx} + (d_D - \Delta_C + \Delta_D)\tan\theta_{Cx} \tag{5.80}$$

$$w_{Dy} = w_{Cy} - of_{Dy} + (d_D - \Delta_C + \Delta_D)\tan\theta_{Cy} \tag{5.81}$$

3. Optical path length L_D

$$L_D = n_C(d_D - \Delta_C + \Delta_D)/\cos\theta_C \tag{5.83}$$

where $d_D = of_{Dx} = of_{Dy} = 0$, $\tan\theta_{Dx} = \tan\theta_{Cx}$ and $\tan\theta_{Dy} = \tan\theta_{Cy}$ (we will not perform the refraction calculations here).

5.6.2.5 Coordinate Transformation Back to the Incident Ray Coordinate System (Rotation of the Axis by an Angle $-\alpha$)

Transform the coordinates of the above ray position (Δ_D, w_{Dx}, w_{Dy}) and ray slope $\tan\theta_{Dx}$(= $\tan\theta_{Cx}$), $\tan\theta_{Dy}$(= $\tan\theta_{Cy}$) back to the incident ray coordinates (Δ_E, w_{Ex}, w_{Ey}), $\tan\theta_{Ex}$, $\tan\theta_{Ey}$ by rotating the coordinate system around the x-axis by an angle $-\alpha$.

$$\Delta_E = 0 \tag{5.84}$$

$$w_{Ex} = w_{Dx} \tag{5.85}$$

$$w_{Ey} = w_{Dy}/\cos(-\alpha) + \mathrm{QR}\sin(-\alpha) \tag{5.86}$$

$$\tan\theta_{Ex} = \tan\theta_{Cx}/[\cos(-\alpha) - \tan\theta_{Cy}\sin(-\alpha)] \tag{5.87}$$

$$\tan\theta_{Ey} = \tan\,[\arctan\,(\tan\theta_{Cy}) + (-\alpha)] \tag{5.88}$$

$+\mathrm{QR}\sin(-\alpha)$ in Equation (5.86) is a term for compensating for the optical axis offset between the incident rays and the exiting rays. Here we obtained the exiting ray from a lens which is set at a tilt, as in Equation (5.84), Equation (5.85), Equation (5.86), Equation (5.87) and Equation (5.88), which are expressed in terms of the incident ray coordinate system.

5.6.3 Rotating the Exiting Rays (by an Angle ϕ) Back to Their Original Orientation

Lastly, as shown in Figure 5.17a,b, we rotate the exiting rays around the z-axis (by an angle ϕ) back to their original orientation. We have finally obtained the exiting rays which passed through a lens at a tilt, and we can calculate the exiting rays traveling to the next surface. We can calculate the ray rotation by applying the Equation (5.28), Equation (5.29), Equation (5.30) and Equation(5.31).

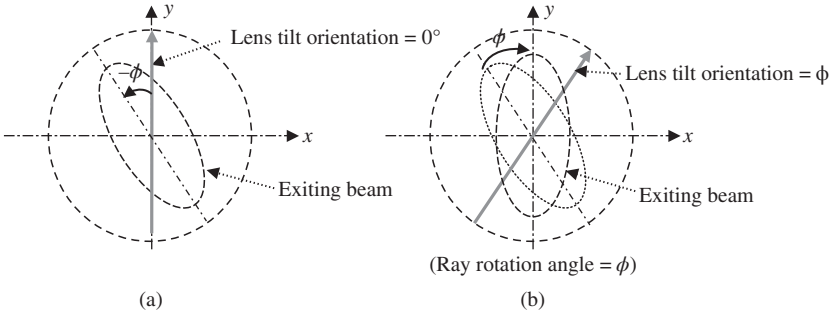


Figure 5.17 (a) Lens tilt orientation = 0° and (b) lens tilt orientation 0° → ϕ (restore the beam orientation to the original setting)

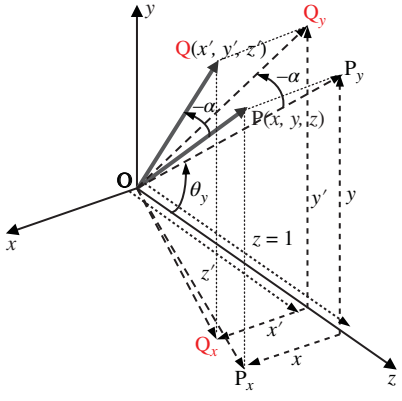


Figure 5.18 Rotation of the ray vector around the x-axis (counterclockwise through an angle $-\alpha$)

Ray Slopes After Rotating the Coordinates

Here, we will examine Equation (5.77) and Equation (5.78), which calculate the ray slopes, after rotating their coordinate system clockwise around the x-axis by an angle α.

In Figure 5.18, OP depicts a ray vector (the direction of a ray) which has a unit length in the z-direction ($z = 1$). OP_x and OP_y are the projections of OP onto the x–z plane and the y–z plane, respectively. (They represent the ray slopes $\tan\theta_x$ and $\tan\theta_y$, respectively.) After a vector's co-ordinate system is rotated clockwise around the x-axis (by an angle α), it will be equivalent to a vector which is rotated counterclockwise around the x-axis (by an angle $-\alpha$). In Figure 5.18, OQ depicts the vector after OP is rotated counterclockwise around the x-axis (by an angle $-\alpha$).

When the coordinates of P are (x, y, z) and $z = 1$, the ray slopes $\tan\theta_x$, $\tan\theta_y$ will be:

$$\tan\theta_x = x/z = x \tag{5.89a}$$

$$\tan\theta_y = y/z = y \tag{5.89b}$$

When the coordinates of Q are (x',y',z'), the ray slopes after rotation, $\tan\theta_x'$ and $\tan\theta_y'$, will be:

$$\tan\theta_x' = x'/z' \tag{5.90a}$$

$$\tan\theta_y' = y'/z' \tag{5.90b}$$

The coordinates of Q (x', y', z') after rotation can be calculated in the same manner as for Equation (5.26) and Equation (5.27).

$$x' = x \tag{5.91}$$

$$y' = z\sin\alpha + y\cos\alpha = \sin\alpha + y\cos\alpha \tag{5.92}$$

$$z' = z\cos\alpha - y\sin\alpha = \cos\alpha - y\sin\alpha \tag{5.93}$$

Substituting Equation (5.89 (a),(b)), Equation (5.91), Equation (5.92), Equation (5.93) and $z = 1$ into Equation (5.90 (a),(b)), we obtain the ray slopes after rotation, $\tan\theta_x'$ and $\tan\theta_y'$

$$\tan\theta_x' = x'/z' = \tan\theta_x/(\cos\alpha - \tan\theta_y\sin\alpha) \tag{5.94}$$

$$\tan\theta_y' = y'/z' = (\sin\alpha + \tan\theta_y\cos\alpha)/(\cos\alpha - \tan\theta_y\sin\alpha) \tag{5.95}$$

From inspecting Figure 5.18, we perceive that $\tan\theta_y'$ can also be calculated as follows:

$$\tan\theta_y' = \tan(\theta_y + \alpha) \tag{5.96}$$

Equation (5.94) and Equation (5.96) are equivalent to Equation (5.77) and Equation (5.78), respectively.

Notes

1. In this book, the actual ray trace calculation for a tilted lens is performed using the "TiltStart" and "TiltEnd" units. "TiltStart" is used for the calculations described in Sections 5.6.2.1 and 5.6.2.2 and "TiltEnd" is used for the calculations described in Sections 5.6.2.4 and 5.6.2.5. Input "TiltStart" just before the lens and input "TiltEnd" just after the lens with a distance $d = 0$ and an offset $of_x = of_y = 0$.
2. TiltStart and TiltEnd do not include calculations for refracting rays. Therefore we have to set the refractive index n to the same value as the medium in front of TiltStart (or TiltEnd).
3. We can calculate a spot diagram on any inclined plane, by using this tilt-calculation technique.
4. If we use "TiltStart" and "TiltEnd" in our ray trace calculation, we will find that the value of the calculated distance along the axis will be slightly different from the actual distance, due to the slant axis between "TiltStart" and "TiltEnd".

5.7 How to Use the Three-Dimensional Ray Trace Calculation Table

5.7.1 Calculation Table

Figure 5.19 shows a three-dimensional ray trace calculation table. First, we specify the optical parameters (n, d, r, and $offset_x$, $offset_y$) (in mm) for each surface (first surface, second surface, etc.) in the top rows. Then we specify the incident rays [height w (mm) and slope $\tan\theta$] emitted from the object in the left columns. In this three-dimensional ray tracing, when we calculate the height and slope of the ray, we calculate the x-components w_x, $\tan\theta_x$ and y-components w_y, $\tan\theta_y$ separately. We define the ray height and the ray slope in term of their x- and y-components w_x, w_y, $\tan\theta_x$, $\tan\theta_y$. The rays (ray height w_x, w_y, and ray slope $\tan\theta_x$, $\tan\theta_y$, etc.) exiting each surface (first surface, second surface, etc.) will be calculated in the calculation units which are placed in the right columns of the incident ray settings. We can calculate the ray trace from an arbitrary incident ray passing through an arbitrary optical surface, step by step, until we finally obtain the image height on the image plane. When we make a new calculation table, we can copy and paste the incident ray settings and the calculation units with their optical parameters (for the first surface, second surface, etc.) in the order in which we need them, from the original table, which we already possess, onto the new sheet. The row number of each cell must not be allowed to change when it is pasted, as absolute addresses are used for row number, while relative addresses are used for column number. It should also be noted that the aberration-free lens unit has ΔL_i and ΔL_o (in mm) as parameters for its optical path length correction and ΔL_o is referred by the optical path length "OPL" cell in the next calculation unit. Thus "OPL" cells next to the aberration-free lens unit should be different from the other cells.

5.7.2 Calculation Results

5.7.2.1 Spot Diagram on the Image Plane

As shown in Figure 5.20, we make a ray height table using the values of w_x, w_y (in mm) which we obtained from the ray tracing table. Then we draw a diagram with w_x along the x-axis and w_y along the y-axis, depicting the spot. In Figure 5.20, the spot diagram on the image plane (at the bottom) shows the spot size as being +/−0 mm in the y-direction (length) and +/−0.00023 mm in the x-direction (width).

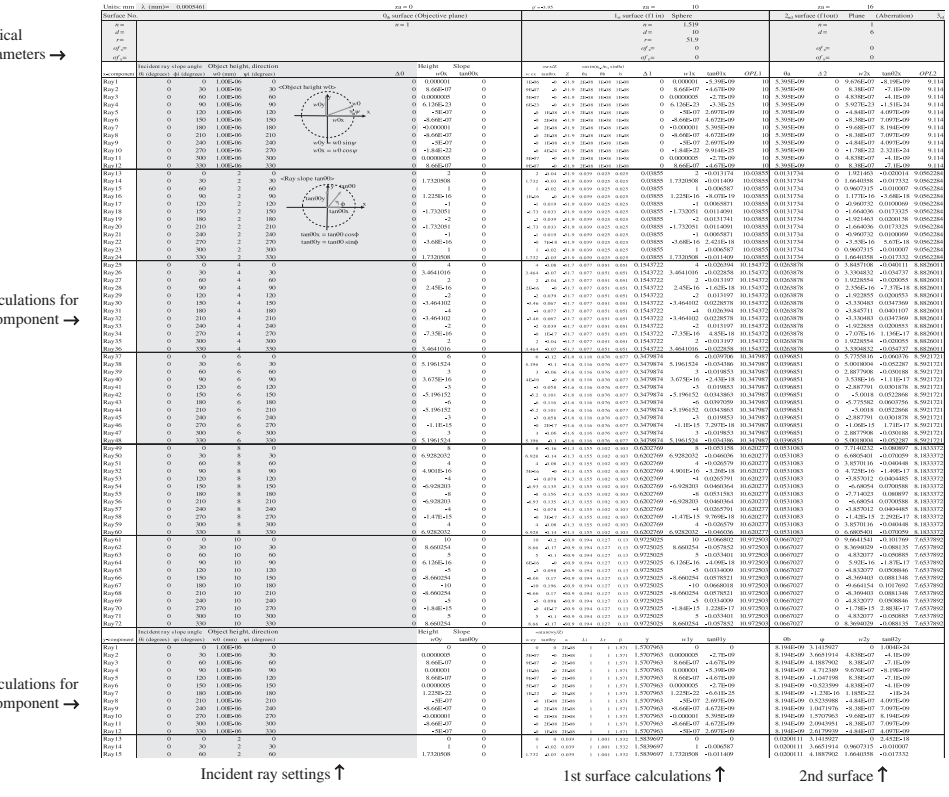


Figure 5.19 A calculation table for the three-dimensional ray tracing

5.7.2.2 Phase (Difference) on the Image Plane

The phase (difference) P_D on the image plane is calculated as follows.

$$P_D = -\left(\sum L - \sum L_0\right)/\lambda \quad [\lambda]$$

where

ΣL = Total optical path length from the object to the image plane for each ray
ΣL_0 = Total optical path length from the object to the image plane for the principal ray
λ = Wavelength.

Figure 5.21 shows a graph of the phase difference P_D against the ray angle θ.

$$\tan\theta = \sqrt{\tan^2\theta_x + \tan^2\theta_y}$$

5.7.2.3 Wave-Front Aberration

In this text, we have defined the reference wave front as a sphere centered at the ideal image point, on whose surface the point of intersection between the principal ray and the last vertex plane (a vertical

Results:	Incident beam		After expansion		After focusing	
	w0x	w0y	w7x	w7y	w10x	w10y
Ray0	0.001	0	0.0030007	0	-2.32E-07	3.677E-19
Ray1	0.5	0	1.5003937	0	-0.000111	1.838E-16
Ray2	0.4330127	0.75	1.2993688	0.75	-9.73E-05	0
Ray3	0.25	1.2990381	0.7501792	1.2990381	-5.72E-05	0
Ray4	3.063E-17	1.5	-8.74E-17	1.5	-6.79E-18	0
Ray5	-0.25	1.2990381	-0.750179	1.2990381	5.723E-05	0
Ray6	-0.433013	0.75	-1.299369	0.75	9.728E-05	0
Ray7	-0.5	1.838E-16	-1.500394	1.854E-16	0.0001113	6.126E-17
Ray8	-0.433013	-0.75	-1.299369	-0.75	9.728E-05	0
Ray9	-0.25	-1.299038	-0.750179	-1.299038	5.723E-05	0
Ray10	-9.19E-17	-1.5	2.623E-16	-1.5	3.879E-16	0
Ray11	0.25	-1.299038	0.7501792	-1.299038	-5.72E-05	0
Ray12	0.4330127	-0.75	1.2993688	-0.75	-9.73E-05	0
Ray13	1	0	3.00107	0	-0.000197	3.677E-16
Ray14	0.8660254	1.5	2.5989212	1.5	-0.000178	0
Ray15	0.5	2.5980762	1.5003937	2.5980762	-0.000111	0
Ray16	6.126E-17	3	-1.75E-16	3	-1.36E-17	0
Ray17	-0.5	2.5980762	-1.500394	2.5980762	0.0001113	0
Ray18	-0.866025	1.5	-2.598921	1.5	0.000178	0
Ray19	-1	3.675E-16	-3.00107	3.707E-16	0.000197	1.225E-16
Ray20	-0.866025	-1.5	-2.598921	-1.5	0.000178	0
Ray21	-0.5	-2.598076	-1.500394	-2.598076	0.0001113	0
Ray22	-1.84E-16	-3	5.247E-16	-3	7.758E-16	0
Ray23	0.5	-2.598076	1.5003937	-2.598076	-0.000111	0
Ray24	0.8660254	-1.5	2.5989212	-1.5	-0.000178	0
Ray25	2	0	6.0044057	0	-0.000189	7.356E-16
Ray26	1.7320508	3	5.1993132	3	-0.000223	0
Ray27	1	5.1961524	3.00107	5.1961524	-0.000197	0
Ray28	1.225E-16	6	-3.5E-16	6	-2.72E-17	0
Ray29	-1	5.1961524	-3.00107	5.1961524	0.000197	0
Ray30	-1.732051	3	-5.199313	3	0.0002231	0
Ray31	-2	7.351E-16	-6.004406	7.414E-16	0.0001894	2.453E-16
Ray32	-1.732051	-3	-5.199313	-3	0.0002231	0
Ray33	-1	-5.196152	-3.00107	-5.196152	0.000197	0
Ray34	-3.68E-16	-6	1.049E-15	-6	1.552E-15	0
Ray35	1	-5.196152	3.00107	-5.196152	-0.000197	0
Ray36	1.7320508	-3	5.1993132	-3	-0.000223	0
Ray37	3	0	9.0122946	0	0.0002271	1.104E-15
Ray38	2.5980762	4.5	7.8026572	4.5	-2.58E-06	0
Ray39	1.5	7.7942286	4.5023125	7.7942286	-0.000232	0
Ray40	1.838E-16	9	-5.25E-16	9	-4.07E-17	0
Ray41	-1.5	7.7942286	-4.502312	7.7942286	0.0002315	0
Ray42	-2.598076	4.5	-7.802657	4.5	2.578E-06	0
Ray43	-3	1.103E-15	-9.012295	1.112E-15	-0.000227	3.685E-16
Ray44	-2.598076	-4.5	-7.802657	-4.5	2.578E-06	0
Ray45	-1.5	-7.794229	-4.502312	-7.794229	0.0002315	0
Ray46	-5.51E-16	-9	1.574E-15	-9	2.327E-15	0
Ray47	1.5	-7.794229	4.5023125	-7.794229	-0.000232	0
Ray48	2.5980762	-4.5	7.8026572	-4.5	-2.58E-06	0

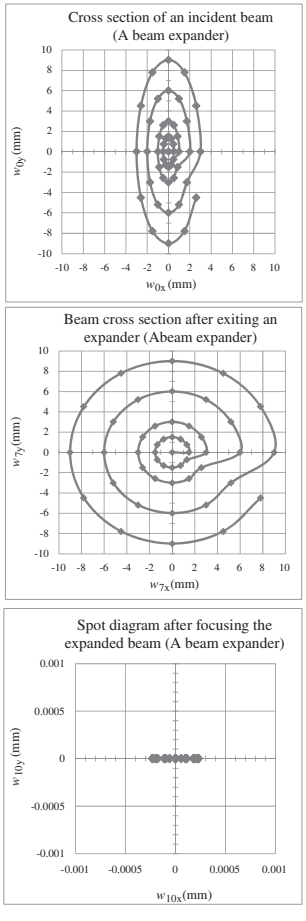


Figure 5.20 Calculation example for spot diagrams on the image plane

plane which includes the vertex of the last surface of the last lens) lies. In Figure 5.23, **S**, **R**, and **T** are the last vertex plane, the reference wave front and the ideal image plane, respectively, and P is the object point, and Q (z_{image}, x_{image}, y_{image}) is the ideal image point, which is situated at the point of intersection between the principal ray and the ideal image plane. The wave-front aberration W will be similar to the aberration calculated for two dimensions:

$$W = -\left(\sum L + \Delta l - \sum L_0\right)/\lambda \quad [\lambda]$$

where

ΣL = Total optical path length from the object P to the plane **S** for each ray
ΣL_0 = Total optical path length from the object P to the plane **S** for the principal ray
λ = Wavelength
Δl = Optical path length from the plane **S** to the reference wave front **R** for each ray.

Phase (difference) on image plane

	tan θ9	θ9 (degrees)	ΣOPL	Phase (λ)
Ray0	3.001E-05	0.0017194	348.069	0
Ray1	0.0150048	0.8596487	348.069	-0.001499
Ray2	0.0150035	0.8595749	348.069	-0.001141
Ray3	0.0150011	0.8594367	348.069	-0.000391
Ray4	0.015	0.8593722	348.069	5.194E-08
Ray5	0.0150011	0.8594367	348.069	-0.000391
Ray6	0.0150035	0.8595749	348.069	-0.001141
Ray7	0.0150048	0.8596487	348.069	-0.001499
Ray8	0.0150035	0.8595749	348.069	-0.001141
Ray9	0.0150011	0.8594367	348.069	-0.000391
Ray10	0.015	0.8593722	348.069	5.194E-08
Ray11	0.0150011	0.8594367	348.069	-0.000391
Ray12	0.0150035	0.8595749	348.069	-0.001141
Ray13	0.0300123	1.7190607	348.069	-0.004943
Ray14	0.0300086	1.7188475	348.069	-0.00397
Ray15	0.0300024	1.7184962	348.069	-0.001497
Ray16	0.03	1.718358	348.069	1.281E-06
Ray17	0.0300024	1.7184962	348.069	-0.001498
Ray18	0.0300086	1.7188475	348.069	-0.00397
Ray19	0.0300123	1.7190607	348.069	-0.004942
Ray20	0.0300086	1.7188475	348.069	-0.00397
Ray21	0.0300024	1.7184962	348.069	-0.001498
Ray22	0.03	1.718358	348.069	1.281E-06
Ray23	0.0300024	1.7184962	348.069	-0.001497
Ray24	0.0300086	1.7188475	348.069	-0.00397
Ray25	0.0600456	3.436232	348.069	-0.002909
Ray26	0.0600289	3.4352814	348.069	-0.00639
Ray27	0.0600061	3.4339808	348.069	-0.004924
Ray28	0.06	3.4336304	348.069	2.312E-05
Ray29	0.0600061	3.4339808	348.069	-0.004924
Ray30	0.0600289	3.4352814	348.069	-0.006391
Ray31	0.0600456	3.436232	348.069	-0.002909
Ray32	0.0600289	3.4352814	348.069	-0.006391
Ray33	0.0600061	3.4339808	348.069	-0.004924
Ray34	0.06	3.4336304	348.069	2.312E-05
Ray35	0.0600061	3.4339808	348.069	-0.004924
Ray36	0.0600289	3.4352814	348.069	-0.00639
Ray37	0.0901211	5.1496493	348.06897	0.0566597
Ray38	0.090073	5.1469147	348.06899	0.0210929
Ray39	0.0900125	5.1434744	348.069	-0.007096
Ray40	0.09	5.1427646	348.069	0.0001154
Ray41	0.0900125	5.1434744	348.069	-0.007096
Ray42	0.090073	5.1469147	348.06899	0.0211582
Ray43	0.0901211	5.1496493	348.06897	0.0566594
Ray44	0.090073	5.1469147	348.06899	0.0211582
Ray45	0.0900125	5.1434744	348.069	-0.007096
Ray46	0.09	5.1427646	348.069	0.0001154
Ray47	0.0900125	5.1434744	348.069	-0.007096
Ray48	0.090073	5.1469147	348.06899	0.0210929

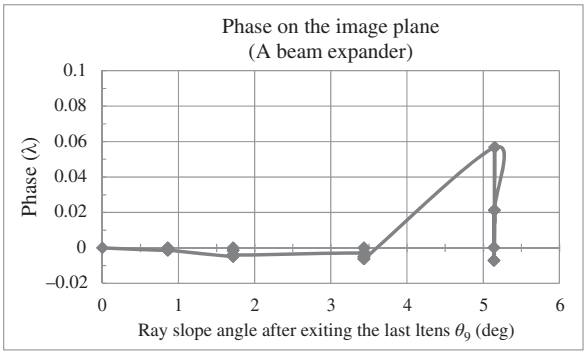


Figure 5.21 Calculation example for a phase (difference) on the image plane

$$\Delta l \cong \sqrt{{z_{image}}^2 + (x_{image} - w_x)^2 + (y_{image} - w_y)^2} - r_{image}$$

w_x, w_y = x- and y-components of the ray height on the plane **S**
z_{image} = Distance along the optical axis from the plane **S** to the ideal image plane **T**
x_{image}, y_{image} = x- and y-components of the ideal image height

$r_{image} = \sqrt{{z_{image}}^2 + (x_{image} - w_{Cx})^2 + (y_{image} - w_{Cy})^2}$ = Curvature radius of the reference wave front **R**

where

w_{Cx}, w_{Cy} = x- and y-components of the principal ray height at the plane S.

For three dimensions, the ray slope $\tan\theta$ and ray height w are calculated as follows:

$$\tan\theta = \sqrt{\tan^2\theta_x + \tan^2\theta_y} \qquad w = \sqrt{{w_x}^2 + {w_y}^2}$$

Figure 5.22 shows a calculation example of the wave-front aberration W against the ray angle θ.

Wave-front aberration $\Delta l = \sqrt{(z^2+w^2)} - z \quad W = -(\Sigma L + \Delta l - \Sigma L_0)/\lambda$

	tanθ9	θ9 (degrees)	ΣL	Δl	W (λ)
Ray0	3.001E-05	0.0017194	248.069	0	0
Ray1	0.0150048	0.8596487	248.05774	0.0112549	0.0015574
Ray2	0.0150035	0.8595749	248.05775	0.0112534	0.0011736
Ray3	0.0150011	0.8594367	248.05775	0.0112506	0.0003947
Ray4	0.015	0.8593722	248.05775	0.0112493	3.909E-08
Ray5	0.0150011	0.8594367	248.05775	0.0112506	0.000395
Ray6	0.0150035	0.8595749	248.05775	0.0112534	0.0011734
Ray7	0.0150048	0.8596487	248.05774	0.0112549	0.0015572
Ray8	0.0150035	0.8595749	248.05775	0.0112534	0.0011734
Ray9	0.0150011	0.8594367	248.05775	0.0112506	0.000395
Ray10	0.015	0.8593722	248.05775	0.0112493	3.909E-08
Ray11	0.0150011	0.8594367	248.05775	0.0112506	0.0003947
Ray12	0.0150035	0.8595749	248.05775	0.0112534	0.0011736
Ray13	0.0300123	1.7190607	248.02398	0.0450207	0.0058766
Ray14	0.0300086	1.7188475	248.02399	0.0450109	0.0044954
Ray15	0.0300024	1.7184962	248.024	0.0449954	0.001558
Ray16	0.03	1.718358	248.02401	0.0449898	1.268E-06
Ray17	0.0300024	1.7184962	248.024	0.0449954	0.0015575
Ray18	0.0300086	1.7188475	248.02399	0.0450109	0.0044954
Ray19	0.0300123	1.7190607	248.02398	0.0450207	0.0058768
Ray20	0.0300086	1.7188475	248.02399	0.0450109	0.0044954
Ray21	0.0300024	1.7184962	248.024	0.0449954	0.0015575
Ray22	0.03	1.718358	248.02401	0.0449898	1.268E-06
Ray23	0.0300024	1.7184962	248.024	0.0449954	0.001558
Ray24	0.0300086	1.7188475	248.02399	0.0450109	0.0044954
Ray25	0.0600456	3.436232	247.88889	0.1800999	0.0178831
Ray26	0.0600289	3.4352814	247.88899	0.1799999	0.0148121
Ray27	0.0600061	3.4339808	247.88913	0.1798691	0.0058804
Ray28	0.06	3.4336304	247.88916	0.1798382	2.31E-05
Ray29	0.0600061	3.4339808	247.88913	0.1798691	0.0058805
Ray30	0.0600289	3.4352814	247.88899	0.1799999	0.0148118
Ray31	0.0600456	3.436232	247.88889	0.1800999	0.0178834
Ray32	0.0600289	3.4352814	247.88899	0.1799999	0.0148118
Ray33	0.0600061	3.4339808	247.88913	0.1798691	0.0058805
Ray34	0.06	3.4336304	247.88916	0.1798382	2.31E-05
Ray35	0.0600061	3.4339808	247.88913	0.1798691	0.0058804
Ray36	0.0600289	3.4352814	247.88899	0.1799999	0.0148121
Ray37	0.0901211	5.1496493	247.6637	0.4052901	0.0193317
Ray38	0.090073	5.1469147	247.66415	0.4048377	0.0214597
Ray39	0.0900125	5.1434744	247.66471	0.4042847	0.0119169
Ray40	0.09	5.1427646	247.66482	0.4041831	0.0001154
Ray41	0.0900125	5.1434744	247.66471	0.4042847	0.011917
Ray42	0.090073	5.1469147	247.66415	0.4048377	0.021525
Ray43	0.0901211	5.1496493	247.6637	0.4052901	0.0193314
Ray44	0.090073	5.1469147	247.66415	0.4048377	0.021525
Ray45	0.0900125	5.1434744	247.66471	0.4042847	0.011917
Ray46	0.09	5.1427646	247.66482	0.4041831	0.0001154
Ray47	0.0900125	5.1434744	247.66471	0.4042847	0.0119169
Ray48	0.090073	5.1469147	247.66415	0.4048377	0.0214597

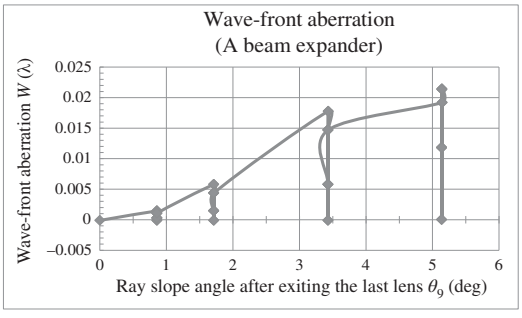


Figure 5.22 Wave-front aberration calculation

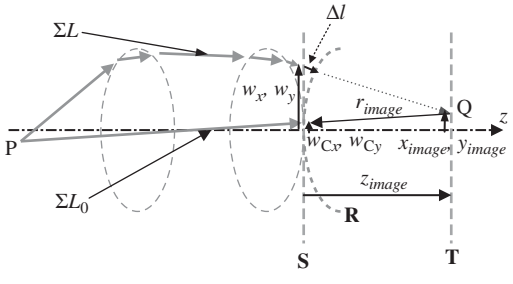


Figure 5.23 Calculation example for a wave-front aberration

5.7.3 Explanation of Symbols Used

In the calculation table, we will use the following symbols, which are similar to those used in Chapter 4:

z_a = z-distance from the object (in mm)
z_b = z-distance from the vertex of the last surface of the last lens (in mm)
n_k = Refractive index of the medium immediately behind the kth surface
d_k = Distance from the $(k-1)$th to the kth surface (in mm)
r_k = Radius of curvature of the kth surface (in mm) (only for a spherical or a cylindrical surface, positive when the vertex is on the left.)
f_k = Focal length of an aberration-free lens of the kth surface (in mm)
α = Tilt angle of the lens (measured from the vertical) (deg) (α for TiltStart, $-\alpha$ for TiltEnd)
ϕ = Rotation angle of the rays around the z-axis (deg)
of_{kx} = x-offset of kth surface axis from $(k-1)$th surface axis (in mm)
of_{ky} = y-offset of kth surface axis from $(k-1)$th surface axis (in mm)
Δ_k = Ray's z-coordinate on the kth surface from the vertex (in mm)
w_{kx} = x-component of the ray height on the kth surface (in mm)
w_{ky} = y-component of the ray height on the kth surface (in mm)
$\tan\theta_{ky}$ = x-slope of the ray after exiting the kth surface
$\tan\theta_{ky}$ = y-slope of the ray after exiting the kth surface
OPL_k = Optical path length from the $(k-1)$th to the kth surface (in mm)

5.8 Operating Instructions Using the Ray Trace Calculation Table, while Running the VBA Program

Previously we looked at three-dimensional ray trace calculations that use a calculation table, upon which Excel® cells can perform their calculations directly. This chapter also provides readers with an alternative way of performing calculations, using VBA programming. We can choose to perform this calculation either by calculating directly on cells on an Excel spreadsheet, or by using a VBA program. In order to perform calculations with the aid of a VBA program, the following steps need to be carried out.

5.8.1 Using the Three-Dimensional Ray Trace Calculation Table while Running the VBA Program

Specify the following items on the calculation table as shown in Figure 5.24.

① **Row Number, Column Number**
The first step is to specify the row number and the column number of the header cell in the optical configuration table.

② **Wavelength, refractive index of the medium surrounding the object, and other parameters**
Next, we specify the wavelength λ (mm), the refractive index n_0 of the medium surrounding the object, which intersection ("x- or y-intersection") of the imaging rays we will use to calculate the image coordinates (the default value is: "x-intersection"), and $K = b/a$ of the elliptical beam.

③ **Optical configuration settings**
We can then determine the optical configuration by setting the boundary surface parameters, in the following order.

1. **Surface Number:** Surfaces should be numbered from first to last. (This field may also be left blank.)

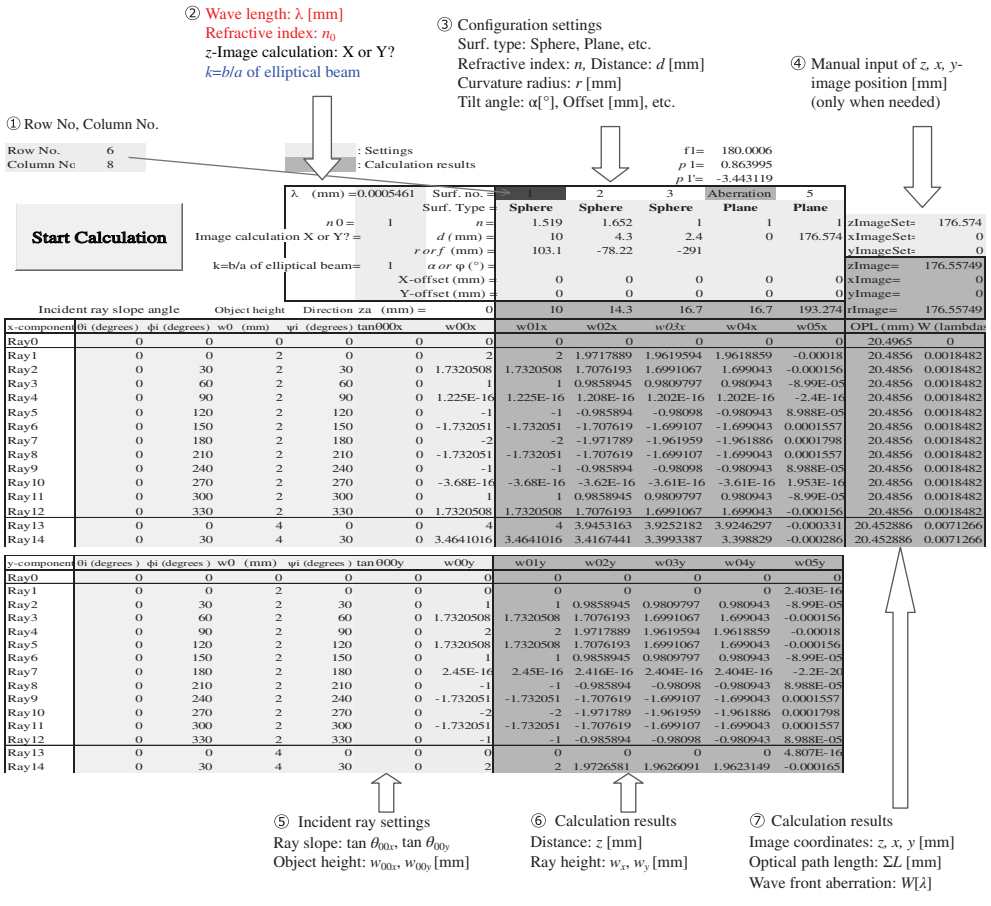


Figure 5.24 A calculation table for the three-dimensional ray tracing

If a wave-front aberration calculation is required, then we can insert an extra plane surface at the last surface vertex (of the last lens) and type "Aberration" in the Surface Number cell. (The reference wave front will include the intersection point between the principal ray and this plane, and its center of curvature will be located at the ideal image point, which can be calculated using paraxial theory.)

2. **Surface type:** "Sphere," "Plane," "XCylinder," or "YCylinder" (for cylindrical surfaces), "AbeFreeLens" (for an aberration-free lens), "TiltStart," "TiltEnd," or "Rotate."
3. ***n:*** The refractive index of the medium lying behind this surface.
4. ***d:*** The distance between the vertexes of the surface and the preceding surface (measured along the axis, in mm).
5. ***r* or *f:*** For spherical or cylindrical surfaces, input the radius of curvature r (in mm). If the surface is an aberration-free lens, input the focal length f (in mm) instead. Otherwise leave blank.
6. α or ϕ
   (a) **Tilt angle α:** For "TiltStart" or "TiltEnd," rotate the coordinate system clockwise by α (in deg) around the x-axis. Input "TiltStart" just before the lens (tilted at an angle α) with the value α and input "TiltEnd" just after the lens (tilted at an angle α) with the value $-\alpha$.

(b) **Ray rotation ϕ:** For "Rotate," rotate the incident rays clockwise around the z-axis. Do this in order to set the tilt orientation (ϕ [deg] clockwise from the y-axis) or in order to set the misalignment angle of a cylindrical lens, by inputting "Rotate" just before the cylindrical lens (or TiltStart) with the value $-\phi$ and input "Rotate" just after the cylindrical lens (or TiltEnd) with the value ϕ.

7. **X-offset, Y-offset:** The x- or y-offset (in mm) of the surface vertex from the (previous surface's) axis. Input this value when the surface type is "Plane," "Sphere," "AbeFreeLens," "XCylinder," or "YCylinder."

④ **Image position settings for calculating wave-front aberration**

Usually, the wave-front aberration is calculated based on the ideal image point, which is calculated using Paraxial theory. If you need to change the ideal image point which the wave front aberration is based on, you can force a change by manually inputting this image position (in mm).

⑤ **Incident ray settings**

Specify the incident rays for any rays emitted from an arbitrary object point.

1. Ray slopes $\tan\theta_{00x}$, $\tan\theta_{00y}$
2. Object heights w_{00x}, w_{00y}

Note

It is important to list each ray in the proper order, so that the principal ray is at the top of the list in the table. If the principal ray is not at the top of the list, the user will have to type "Principal ray" in the ray number cell, in order to identify it as such. The coordinates of the ideal image point can then be calculated using the principal ray and a nearby ray traveling very close to it.

After specifying the above items, click "Start Calculation." The program will then start performing the ray trace calculation. The calculation results will be displayed in the table after the calculation has been completed.

5.8.2 Calculation Results Table

⑥ **Range z, Ray height w_x, w_y**

The range (cumulative distance from the object) z (in mm) and the ray heights w_x, w_y (in mm) are then output. The range z is measured along the optical axis, from the object to the vertex of each of the surfaces; while the height w_x, w_y of the ray projected on the x–z and y–z planes are measured from the axis to the intersection of the ray with the surface. We can generate a beam cross-section diagram, depicting the x- and y-components w_x and w_y of the ray height.

Note

Range z is represented by the position of the vertex when the surface is a sphere or a cylinder, therefore, it's a little different from the ray's actual position.

⑦ **Image coordinates z_{image}, x_{image}, y_{image}, r_{image}, optical path length ΣL, and wave-front aberration W**

See Figure 5.23 which depicts the following items.

1. z_{image}, x_{image}, y_{image} are the ideal image coordinates calculated as the intersection point between the principal ray and a nearby ray traveling very close to it.
   (a) z_{image} = Distance along the optical axis from the plane **S** (the extra plane surface which we inserted at the last surface vertex, indicating "Aberration") to the ideal image plane **T**.
   (b) x_{image}, y_{image} = x- and y-components of the ideal image height.
   (c) $r_{image} = \sqrt{{z_{image}}^2 + (x_{image} - w_{Cx})^2 + (y_{image} - w_{Cy})^2} =$ Curvature radius of the reference wave front **R**, where, w_{Cx}, w_{Cy} = x- and y-components of the principal ray height on the plane **S**.
2. Optical path length ΣL = Total optical path length from the object P to the plane **S** for each ray.
3. Wave-front aberration W (in λ):

The wave- front aberration, that is, the cumulative optical path length difference from the object P to the reference wave front **R** between each ray and the principal ray, can be calculated using the following equations. (A positive sign indicates that the phase leads that of the principal ray.)

$$W = -\left(\sum L + \Delta l - \sum L_0\right)/\lambda \quad [\lambda], \text{ where}$$

$$\Delta l \cong \sqrt{z_{image}{}^2 + (x_{image} - w_x)^2 + (y_{image} - w_y)^2} - r_{image}$$

w_x, w_y = x- and y-components of the ray height on the plane **S** for each ray
ΣL_0 = Total optical path length from the object P to the plane **S** for the principal ray

Notes

1. The row just below the settings for the last ray should be left blank, so that the program will realize that it has reached the last ray.
2. The column immediately to the right of the settings for the last surface type should also be left blank, so that the program will realize that it has reached the last optical configuration setting.[1]

5.9 Three-Dimensional Ray Tracing Problems

In this section, we will work through some three-dimensional ray tracing problems which are similar to the two-dimensional ray tracing problems we examined in Chapter 4.

Please download the Excel files from the companion web site, which show calculation details for all problems in this book as listed in Appendix G

Problem 5.1 Beam focusing using a plano-convex lens

Perform three-dimensional ray tracing calculations for an optical system which uses a plano-convex lens (f = 100 mm) (the same lens as in Problem 4.4 of Chapter 4) to focus a parallel beam (20 mm in diameter) onto a single spot. Calculate the size of the focused spot and its distance from the lens.

Solution

(1) Input the optical configuration (Figure 5.25a)
We will use the same values as we used in Problem 4.4 of Chapter 4.

λ = 546.1 nm	First surface BK7	Second surface	Image plane
Refractive index n	1.519	1	1
Distance d (mm)	10 (arbitrary value)	6	95.22
Radius of curvature r (mm)	−51.9	–	–

(2) Specify details of the incident rays (Figure 5.25b)

Object height w_0	w_0 = 0 to ± 10 mm, ψ = 0 to 360°
Incident ray slope angle θ_0	0°

[1] This program has been checked using Microsoft Office Excel 2003, 2007, and 2010.

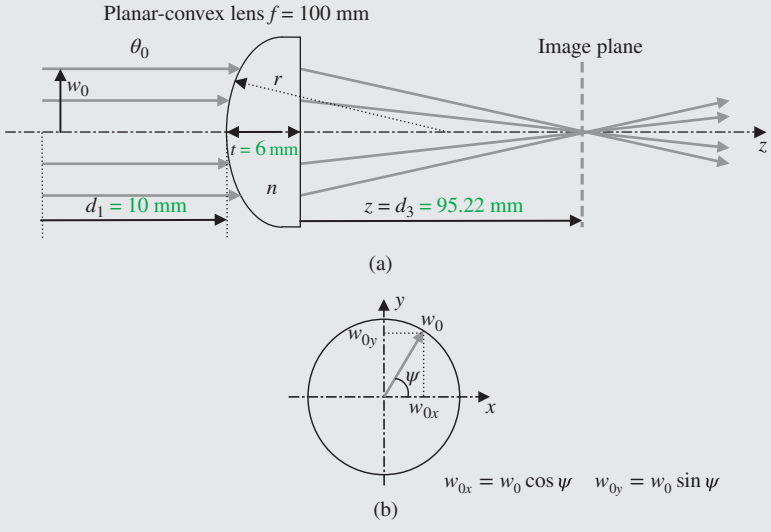


Figure 5.25 (a) Beam focusing using a plano-convex lens and (b) object height

(3) Perform ray tracing calculations (Figure 5.26)

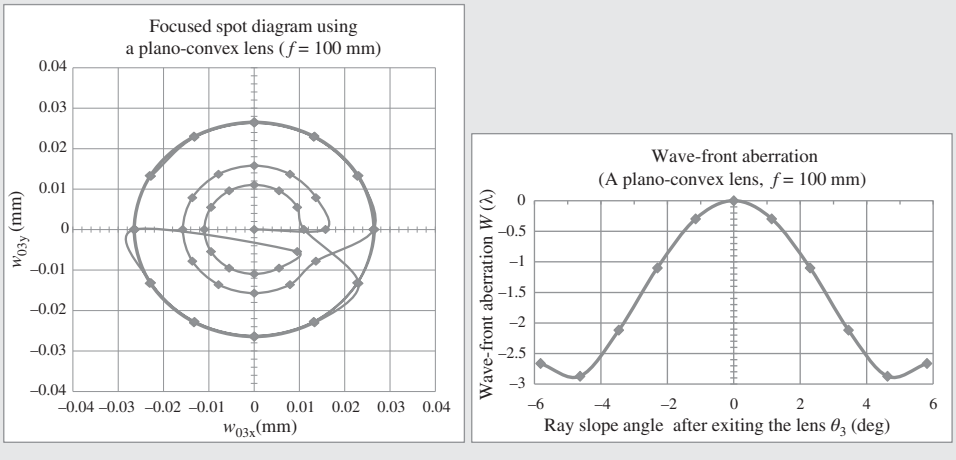


Figure 5.26 Ray tracing results for a plano-convex lens

(4) Calculation results

Focal distance z (mm)	Spot radius w (mm)	Wave-front aberration $W(\lambda)$
95.22	0.027	+0/−2.9

Notes

1. We have obtained the same results for the focused spot size in the three-dimensional ray tracing as for the two-dimensional ray tracing in Problem 4.4 of Chapter 4.
2. We generated the spot diagram above by performing a three-dimensional ray tracing.

Problem 5.2 Beam focusing using an achromatic lens

Perform three-dimensional ray tracing calculations for an optical system which uses an achromatic lens $f = 180$ mm (the same lens as in Problem 4.9 of Chapter 4) to focus a parallel beam ($D =$ 20 mm in diameter) onto a single spot. Calculate the size of the focused spot and its distance from the lens.

Solution

(1) Input the optical configuration ($f = 180$ mm) (Figure 5.27a)

$\lambda = 546.1$ nm	First: sphere	Second: sphere	Third: sphere	Fourth: plane
Refractive index n	1.519 (BK7)	1.652 (SF2)	1	1
Distance d (mm)	10 (arbitrary)	4.3 ($=t_1$)	2.4 ($=t_2$)	176.574
Radius of curvature r (mm)	103.1	−78.22	−291	–

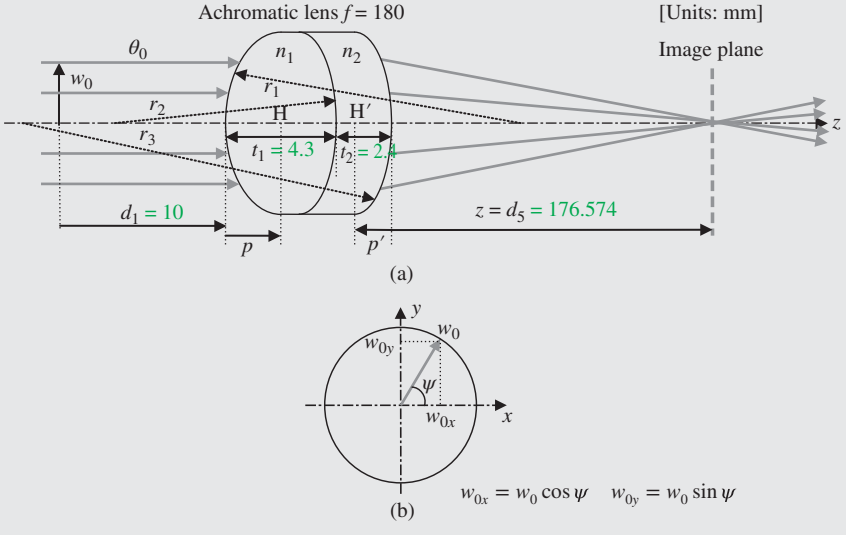


Figure 5.27 (a) Beam focusing using an achromatic lens and (b) object height

(2) Specify details of the incident rays (Figure 5.27b)

Object height w_0	0 to ± 10 mm, ψ = 0 to 360°
Incident ray slope angle θ_0	0°

(3) Perform ray tracing calculations (Figure 5.28)

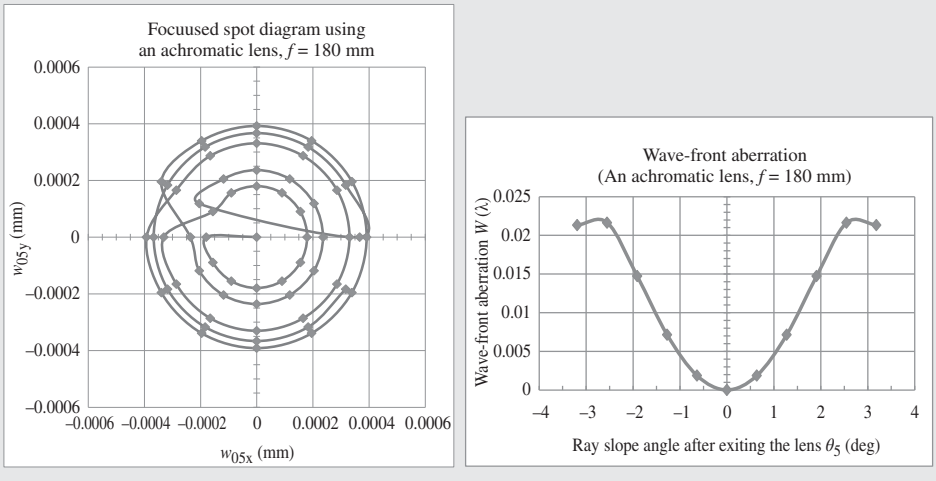


Figure 5.28 Ray tracing results for an achromatic lens

(4) Calculation results

Focal distance d_5 (mm)	Spot radius w (mm)	Wave-front aberration W (λ)
176.574	0.0004	+0.022/−0

Notes

1. Using the same focal distance z as for the two-dimensional ray tracing, we have managed to obtain the same results.
2. The results we obtained using an achromatic lens were much better than those we obtained with the plano-convex lens.

Problem 5.3 Triple beam expander using two cylindrical lenses

Perform three-dimensional ray tracing calculations for an optical system consisting of a collimated, elliptical incident beam (x-radius 3 mm × y-radius 9 mm); a triple beam expander, which uses two cylindrical lenses ($f_1 = -100$ mm, $f_2 = 300$ mm) to obtain a circular beam by widening the beam in the x-direction; and an aberration-free lens ($f_3 = 100$ mm) to focus the output beam onto a single circular spot.

Solution

(1) Input the optical configuration of the cylindrical beam expander (Figure 5.29a)

$\lambda = 532$ nm Surface type	First: Plane	Second: XCylinder	Third: Plane	Fourth: XCylinder	Fifth: AbeFreeLens	Image plane Plane
Refractive index n	1.519	1	1.519	1	1	1
Distance d (mm)	20	3	197.436	4	20	100
Radius of curvature r or focal length f (mm)	–	51.9	–	−155.70	100	–

The distance to the third surface d_5 has been optimized so as to minimize the size of the spot.

(2) Specify details of the incident rays for the elliptical beam (Figure 5.29b)

Object height w_{0x},w_{0y}	$w_0 = 0$ to 3 mm, $\psi = 0$ to 360°, $K = 3$
Incident ray angle θ_0	0°

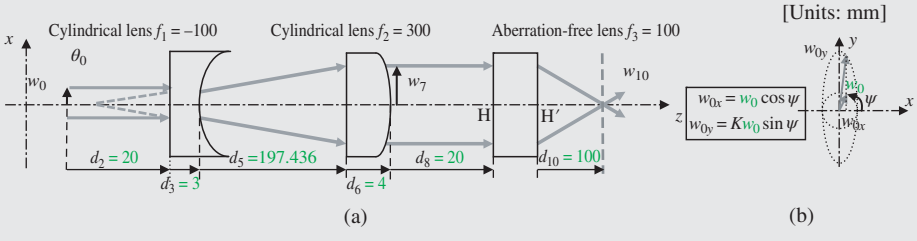


Figure 5.29 (a) Beam expander using two cylindrical lenses and (b) object height

(3) Perform ray tracing calculations (Figure 5.30)

(4) Calculation results

Beam diameter after exiting expander	Length of focused spot	Wave-front aberration
18 mm φ	0.00046 mm at $d_{10} = 100$ mm	+0.022/0 λ

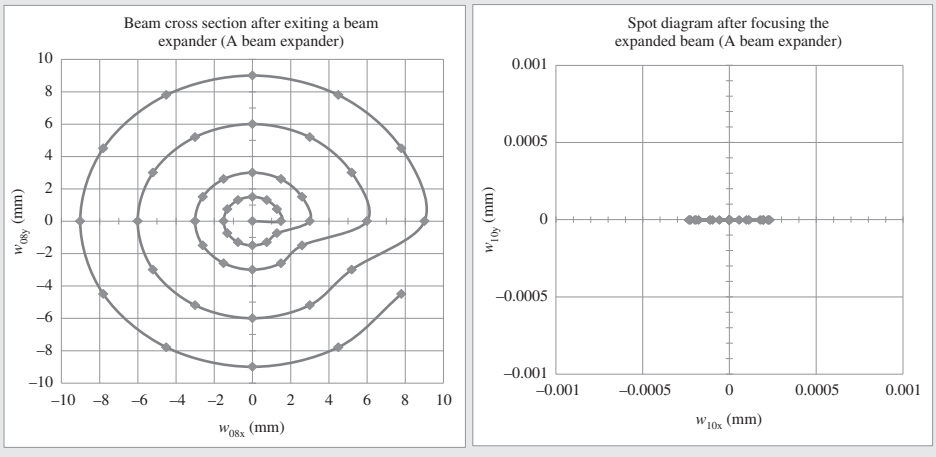


Figure 5.30 Ray tracing results for a beam expander

Notes

1. We can convert an elliptical incident beam into a circular output beam using two cylindrical lenses.
2. After focusing the beam, we obtained a line spot with a very short length of 0.00046 mm.

Problem 5.4 Beam expander using two misaligned cylindrical lenses

Perform a three-dimensional ray tracing calculation for the triple beam expander in Problem 5.3 ($f_1 = -100$ mm, $f_2 = 300$ mm), where the cylindrical lens f_1 is rotated $+1°$ around the optical axis. (This is equivalent to rotating the rays $-1°$ before they reach the x-cylindrical lens f_1 and $+1°$ after they exit the lens.)

Solution

(1) Input the optical configuration of the cylindrical beam expander (Figure 5.31a)

$\lambda = 532$ nm Surface type	Rotate	First: Plane	Second: XCylinder	Rotate	Third: Plane	Fourth: XCylinder	Fifth: AbeFreelens	Image plane Plane
Refractive index n		1.519	1		1.519	1	1	1
Distance d (mm)		20	3		197.436	4	20	100
Radius of curvature r or focal length f (mm)		–	51.9		–	−155.70	100	–
Ray rotation φ (°)	−1	–	–	+1	–	–	–	–

(2) Specify details of the incident rays for the elliptical beam (Figure 5.31b)

Object height w_{0x}, w_{0y}	$w_0 = 0$ to 3 mm, $\psi = 0$ to 360°, $K = 3$
Incident ray angle θ_0	0°

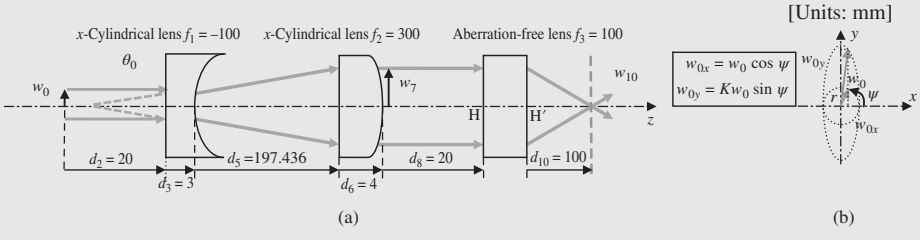


Figure 5.31 (a) Beam expander using two misaligned cylindrical lenses and (b) object height

(3) Perform ray tracing calculations (Figure 5.32)
(4) Calculation results

Beam diameter after exiting expander	Spot diameter after focusing	Wave-front aberration
18 mm φ	0.105 mm φ at $d_{10} = 100$ mm	+3.6/−4.0 λ

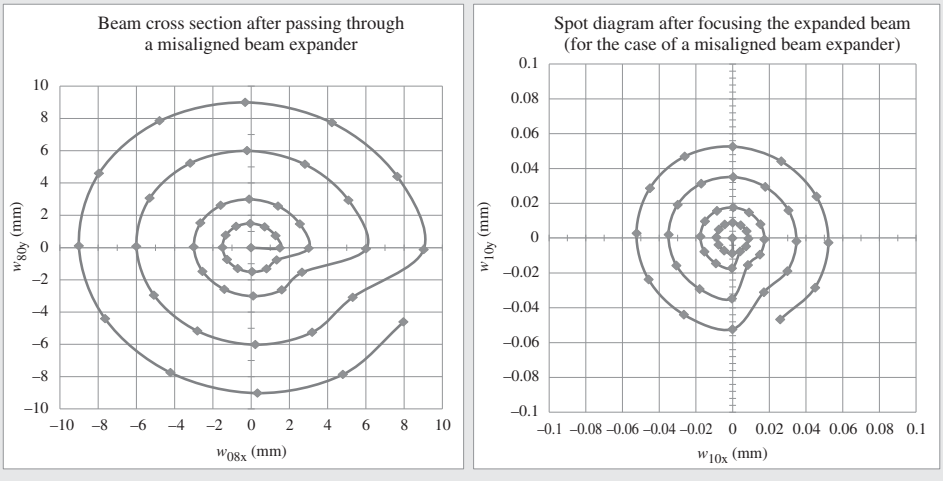


Figure 5.32 Ray tracing results for a misaligned beam expander

Notes

1. We can convert an elliptical incident beam into a circular output beam using two cylindrical lenses.

2. After focusing, we obtained a spot of diameter 0.105 mm, which was much larger than the spot in Problem 5.3, due to the angular misalignment between the two cylindrical lenses.

The example above demonstrates the importance of precisely aligning the two cylindrical lenses along the optical axis, in order to minimize the aberration.[2]

Problem 5.5 Laser beam collimator, expander, and focusing

Perform three-dimensional ray tracing calculations for the optical system described in Problem 4.14 of Chapter 4, which consists of: a laser diode; a laser beam collimator $f_1 = 10$ mm; two cylindrical lenses $f_2 = -100$ mm, $f_3 = 300$ mm; and an achromatic lens $f_4 = 180$ mm.

Solution

(1) Input the optical configuration (Figure 5.33a).
Here, we use the same settings as in Problem 4.14 of Chapter 4.

$\lambda = 546.1$ nm Surface type	First AbeFreeLens	Second Plane	Third XCylinder	Fourth Plane	Fifth XCylinder	Sixth Sphere	Seventh Sphere	Eighth Sphere	Image plane Plane
Refract. index n	1	1.519	1	1.519	1	1.519	1.652	1	1
Distance d (mm)	10	50	3	197.3705	4	50	4.3	2.4	176.55778
Curvature radius r (mm)	–	–	51.9	–	−155.70	103.1	−78.22	−291	–

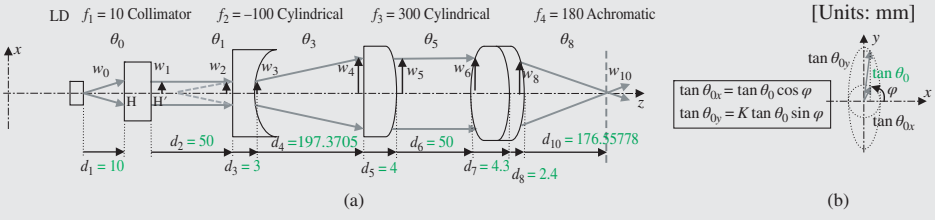


Figure 5.33 (a) Laser beam collimator, expander, and focusing and (b) incident ray slope

(2) Specify details of the incident rays (Figure 5.33b)

Object height w_0	0 mm
Incident ray angle θ_0	$\theta_0 = 0$ to 4°, $K = 3$, $\varphi = 0$ to 360°

[2] It is not possible to input tilt angles for cylindrical lenses. In this book, the calculation for cylindrical lenses does not allow tilted lenses.

(3) Perform ray tracing calculations (Figure 5.34)

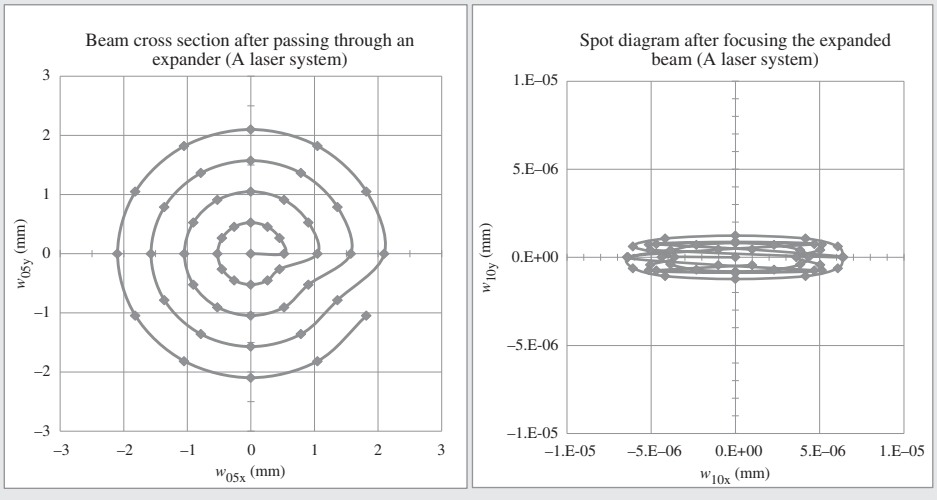


Figure 5.34 Ray tracing results for a laser optical system

We use the same image distance $d_{10} = 176.55778$ mm as we used in Problem 4.14 of Chapter 4.

(4) Calculation results

Beam radius after exiting expander w_5 (mm)	$w_x = 2.0980, w_y = 2.0978$
Spot radius after focusing w_{10} (mm)	$w_x = 0.0000066,\ \ w_y = 0.0000012$ at $d_4 = 176.55778$ mm

Notes

1. A circular beam was generated after passing through the expander.
2. The beam was then focused onto a very small spot ($w_x = 6.6$ nm, $w_y = 1.2$ nm), which means that this optical system has almost no aberration. (Our calculation results match those of Problem 4.14 in Chapter 4.)

Problem 5.6 Laser beam expander and focusing (independent expansion along the *x*- and *y*-axes)

Perform ray tracing calculations for the optical system shown in Figure 5.35a, which expands an incident beam (radius $w_0 = 0.34$ mm) by a factor of 5 along the *x*-axis and by a factor of 6.5 along the *y*-axis, and which focuses the beam onto a (line) spot with a length of 0.2 mm when measured along the *y*-axis using an achromatic lens $f_4 = 50$ mm. **Hint:** In order to obtain a spot with a length of 0.2 mm along the *y*-axis, you will need to defocus the spot slightly along the *y*-axis, by moving the third lens (thereby altering the value of d_5). Calculate the values of d_3, d_5, and d_{12} and generate a spot diagram on the image plane after performing a ray tracing.

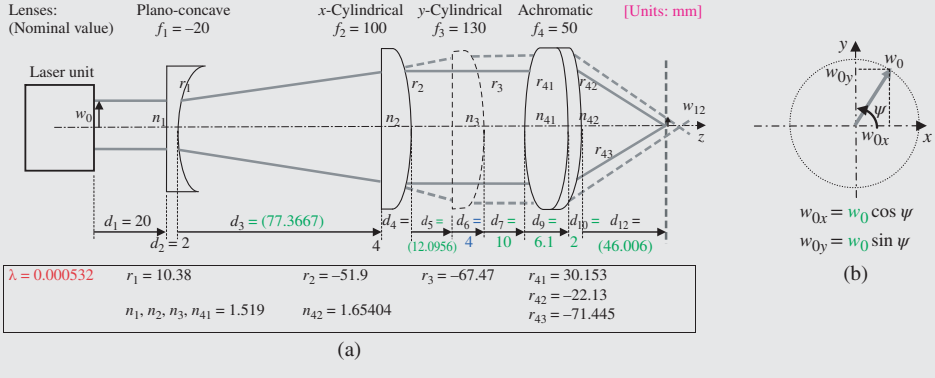


Figure 5.35 (a) Laser beam expander and focusing and (b) incident ray height

Solution

(1) Input the optical configuration (Figure 5.35a) [1]
(2) Calculate the values of d_3, d_5, and d_{12} (Figure 5.36)

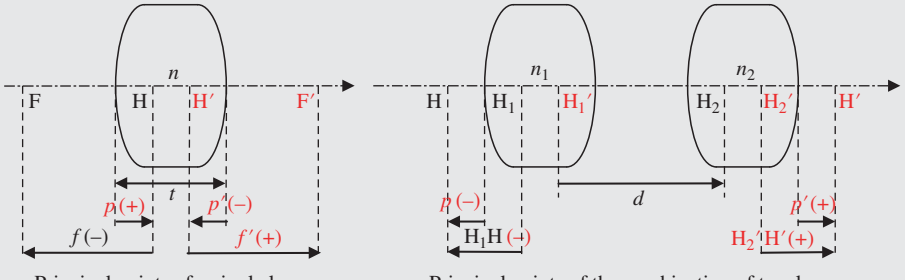


Figure 5.36 Principal points of a simple lens and the combination of two lenses

The focal length f', and the principal points p, p' for a single lens are calculated as follows:

$$\frac{1}{f'} = -\frac{1}{f} = (n-1)\left\{\left(\frac{1}{r_1} - \frac{1}{r_2}\right) + \frac{(n-1)t}{nr_1r_2}\right\} \tag{1.34}$$

$$p = -\frac{r_1 t}{n(r_2 - r_1) + (n-1)t} \tag{1.35}$$

$$p' = -\frac{r_2 t}{n(r_2 - r_1) + (n-1)t} \tag{1.36}$$

The focal length f' and the principal points H_1H, $H_2'H'$ for an achromatic lens can be obtained by treating the lens as a combination of two single lenses.

$$f = \frac{f_1 f_2}{d - f_1' + f_2} \tag{1.37}$$

$$f' = -\frac{f_1' f_2'}{d - f_1' + f_2} \tag{1.38}$$

$$H_1H = \frac{f_1 d}{d - f_1' + f_2} \tag{1.39}$$

$$H_2'H' = \frac{f_2' d}{d - f_1' + f_2} \tag{1.40}$$

The calculation results for f', p, p' of each lens are as follows (units: mm):

$f_1 = -20$	$f_2 = 100$	$f_3 = 130$	$f_4 = 50.18589$
$p_1 = 1.316656$	$p_2 = 2.633311$	$p_3 = 2.633311$	$p_4 = 1.184287$
$p_1' = 0$	$p_2' = 0$	$p_3' = 0$	$p_4' = -4.16868$

In order to obtain rays which are parallel in the x-direction after exiting the x-cylindrical lens f_2, d_3 needs to have the following value:

$$d_3 = f_1 + f_2 + p_1' - p_2 = 77.366689 \text{ mm}$$

The image distance d_{12} can be calculated as a focal point of f_4.

$$d_{12} = f_4 + p_4' = 46.017212 \text{ mm}$$

However, we can optimize the image position d_{12} so as to minimize the ray heights w_{5x} along x-axis, by performing a two-dimensional ray tracing. This would be the best image plane from the perspective of geometric optics.

Then we can perform a two-dimensional ray tracing in order to obtain a value for d_5 which gives us a ray height w_{5y} of 0.1 mm along the y-axis on the image plane.

The calculation results for d_{12} and d_5 are as follows:

$$d_{12} = 46.006 \text{ mm} \quad d_5 = 12.0956 \text{ mm}$$

(3) Specify the incident ray settings for a three-dimensional ray tracing (Figure 5.35b)

Object height w_{0x}, w_{0y}	$w_0 = 0$ to 0.34 mm, $\psi = 0$ to 360°
Incident ray angle θ_0	$\theta_0 = 0°$

(4) Perform a three-dimensional ray tracing (Figure 5.37)

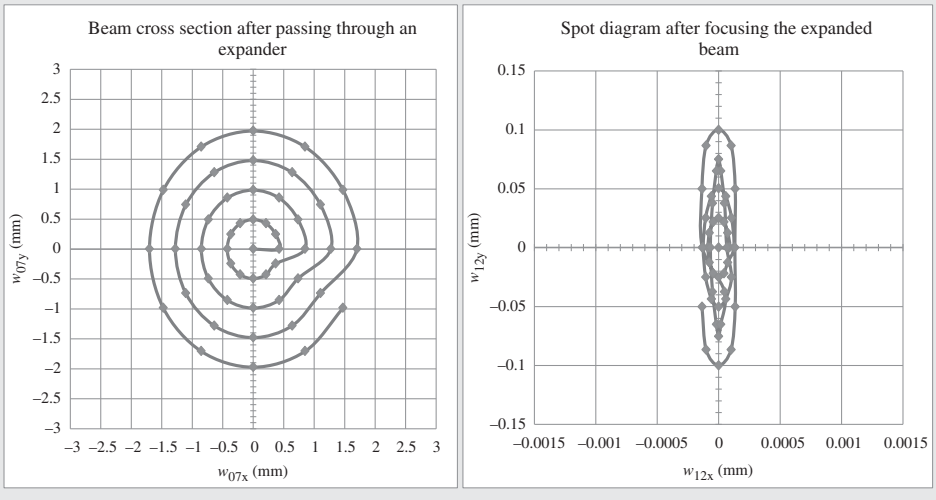


Figure 5.37 Ray tracing results for a beam expander and focusing

Notes

1. The calculated x-value for the focused spot radius was very small (±0.13 μm).
2. However, in real-life, the focused spot size in the x-direction will be larger than the theoretical value calculated above, due to diffraction. (Refer to Problem 7.4 in Chapter 7.)

Problem 5.7 Beam focusing using an achromatic lens which is set at a tilt

Perform three-dimensional ray tracing calculations for an achromatic lens $f = 180$ mm (the same lens as in Problem 5.2) to focus a parallel beam ($D = 20$ mm in diameter) onto a single spot, where the lens is tilted at an angle of two degrees with an orientation of 90° (along the x-axis).

Solution

(1) Input the optical configuration ($f = 180$ mm) (Figure 5.38a)

λ = 546.1 nm Surface type	Rotate	TiltStart	First: Sphere	Second: Sphere	Third: Sphere	TiltEnd	Rotate	Image plane Plane
Refractive index n	1	1	1.519	1.652	1	1	1	1
Distance d (mm)	–	10	0	4.3	2.4	0	–	176.3
Curvature radius r (mm)	–	–	103.1	−78.22	−291	–	–	–
Tilt/rotate angle (deg)	−90	2	–	–	–	−2	90	–

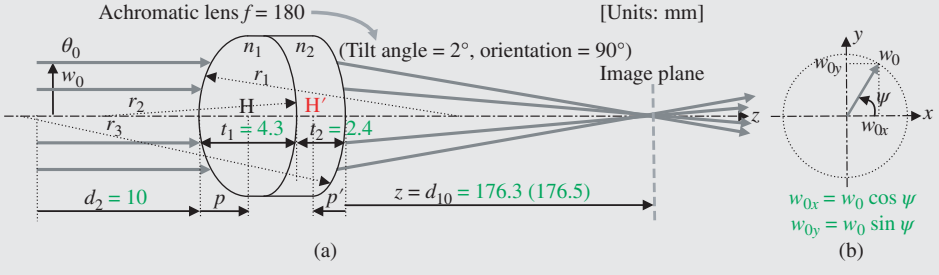


Figure 5.38 (a) Beam focusing using an achromatic lens which is set at a tilt and (b) object height

(2) Specify details of the incident rays (Figure 5.38b)

Object height w_0	0 to 10 mm, ψ = 0 to 360°	Incident ray angle θ_0	0°

(3) Perform ray tracing calculations (Figure 5.39)

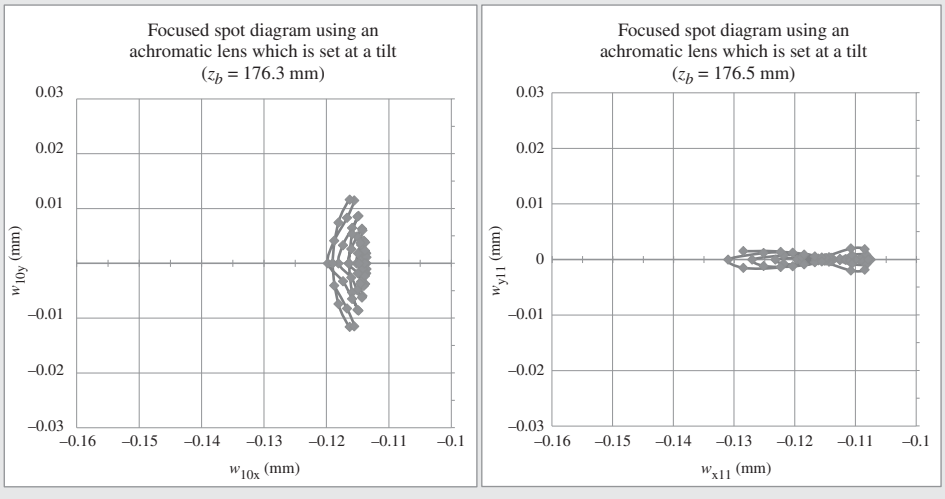


Figure 5.39 Ray tracing results for a beam expander and focusing

Notes

1. When performing a ray tracing for a lens which is tilted $\alpha°$ with an orientation of $\phi°$, input Rotate ($-\phi°$), and TiltStart ($\alpha°$) just before the lens and place TiltEnd ($-\alpha°$) and input Rotate ($\phi°$) just after the lens. Set the refractive index to the same value as that of the previous plane.
2. The aberration for a tilted lens has a greater value than that of a lens which is not tilted. The z-positions when the rays are focused in the x-direction (along the tilt-orientation) and in the y-direction (perpendicular to the tilt orientation) are different from each other. This means that a tilted lens will generate the aberration of astigmatism.

Reference

[1] Sigma-Koki Co., Ltd (2003) *Optical Parts Catalogue*, Sigma-Koki Co., Ltd, Japan.

6

Mathematical Formulae for Describing Wave Motion

Since light is a kind of electromagnetic wave, light exhibits the typical characteristics of waves. In this chapter, we shall examine the basic properties of waves. We shall also make use of a mathematical notation that has been created specifically for describing the behavior of waves, in order to arrive at a proper understanding of the various behaviors exhibited by light, such as interference and diffraction [1].

6.1 Mathematical Formulae for Describing Wave Motion

6.1.1 The Equations of One Dimensional Wave Motion

Figure 6.1a shows the up-and-down motion at the end of a rope that is shaken in a vertical direction at one end, while fixed at the other end. The disturbance created by the vertical motion travels along the rope, as shown in the figure. This is an example of wave behavior. We can treat this phenomenon from a mathematical perspective by representing the traveling wave U as a function of the horizontal position x and time t.

$$U = f(x, t) \tag{6.1}$$

We can represent the disturbance (or vertical displacement) at the instant $t = 0$ by the following equation (Figure 6.1b):

$$U(x, t)|_{t=0} = f(x, 0) = f(x) \tag{6.2}$$

After a time interval of t seconds has elapsed, the disturbance reaches point P′. The transverse wave at this point still retains the same shape as it had at time $t = 0$.

$$U = f(x') \tag{6.3}$$

More accurately, we can say that the wave function at point P is the same function as the wave function at point P′; the only thing that distinguishes the two is the amount of time that has elapsed. In other words:

$$x' = x - vt \tag{6.4}$$

where v is the propagation velocity of the wave.

Hence,

$$U(x, t) = f(x - vt) \tag{6.5}$$

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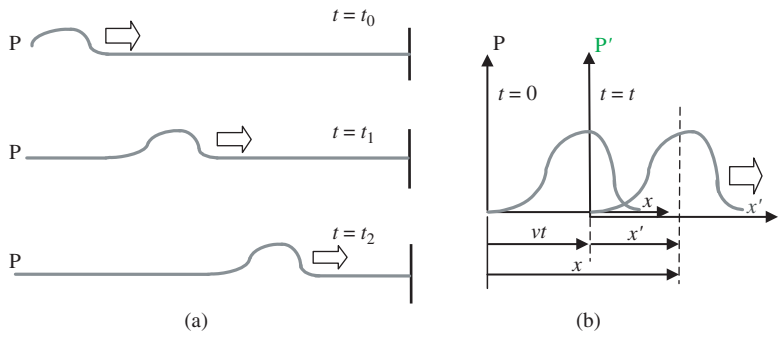


Figure 6.1 (a) Wave on a rope. (b) Moving reference coordinates

Equation (6.5) describes a one-dimensional wave function which specifies the value of the y-displacement in terms of the wave's horizontal position x and time t. The value after a time interval Δt has elapsed can be represented by letting $t = t + \Delta t$ and $x = x + v\Delta t$.

$$t \to t + \Delta t \quad x \to x + v\Delta t$$

If we substitute these values of t and x into Equation (6.5), we get:

$$f((x + v\Delta t) - v(t + \Delta t)) = f(x - vt)$$

The shape of the transverse wave described in Equation (6.5) does not vary over time. If the wave were traveling in a negative direction along the x-axis, Equation (6.5) would instead be:

$$U(x, t) = f(x + vt) \tag{6.5′}$$

Generalizing from Equation (6.5) and Equation (6.5′), we can express the argument of the function f as $(x \mp vt)$.

$$U(x, t) = f(x \mp vt) \tag{6.6}$$

6.1.2 Harmonic Waves

In its most basic form, a wave disturbance will take the form of a sine or cosine curve. This kind of wave is called a harmonic wave. Waves of any shape can be represented as a superposition of one or more harmonic waves. We can express the disturbance described in Section 6.1.1 as a simple harmonic wave, described by the equation:

$$U(x, t)|_{t=0} = U(x) = f(x) = A \sin kx \tag{6.7}$$

where k is the propagation number and A is the amplitude.

If the wave is traveling in a positive direction along the x-axis at velocity v and the time elapsed equals t, Equation (6.5) can be re-expressed in the following form:

$$U(x, t) = f(x - vt) = A \sin k(x - vt) \tag{6.8}$$

Figure 6.2a,b illustrate the wave motion described in Equation (6.8).

Regardless of whether we hold x or t constant, we will still obtain a sinusoidal wave. This indicates that the wave is periodic both in space and in time. The wave's period in space obviously corresponds to

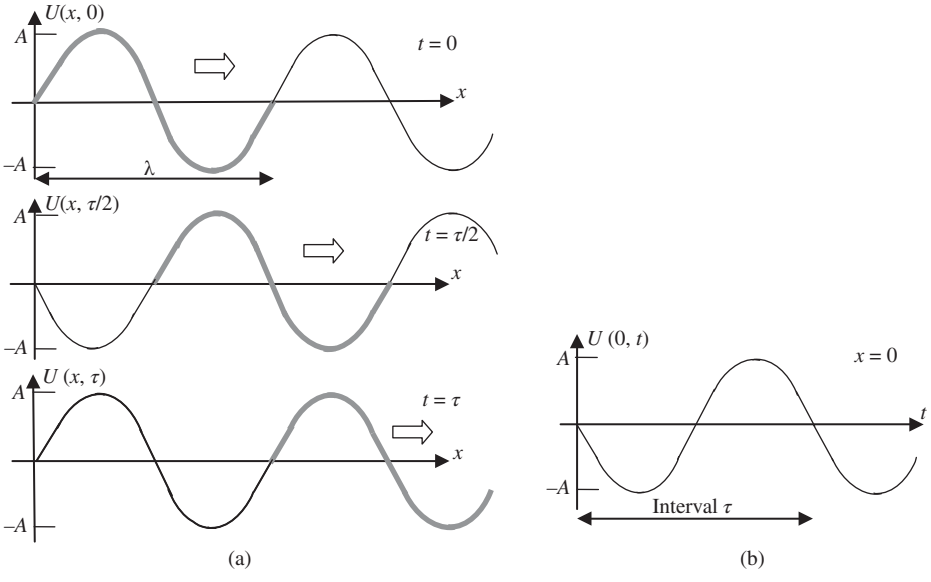


Figure 6.2 (a) Progression of wave (at $t = 0$, $t = \tau/2$, $t = \tau$). (b) Wave at $x = 0$

its wavelength λ. The value U would not change even if we altered the value of x, with the value λ either increasing or decreasing.

$$U(x, t) = U(x \pm \lambda, t) \tag{6.9}$$

That is,

$$\sin\ k(x - vt) = \sin\ k[(x \pm \lambda) - vt] = \sin\,[k(x - vt) \pm 2\pi] \tag{6.9'}$$

The propagation number k represents the rate of change of phase with distance. Equation (6.9′) tells us that the phase will change its value by 2π after the wave has traveled a distance λ. In other words, $|k\lambda| = 2\pi$. Hence,

$$k = \frac{2\pi}{\lambda} \text{ where } k = \text{propagation number} \tag{6.10}$$

Similarly, the period τ represents the length of time it takes for one complete wave to pass a stationary observer – in other words, the time between successive wave crests (or alternatively, successive wave troughs).

$$U(x, t) = U(x, t \pm \tau) \tag{6.11}$$

That is,

$$\sin k(x - vt) = \sin k(x - v(t \pm \tau)) = \sin\,(k(x - vt) \pm 2\pi) \tag{6.11'}$$

Thus

$$kv\tau = 2\pi \tag{6.12a}$$

or

$$(2\pi/\lambda)v\tau = 2\pi \tag{6.12b}$$

Hence,

$$\tau = \lambda / v \tag{6.13}$$

The frequency f is the inverse of τ. The velocity v can be expressed by Equation (6.14).

$$f \equiv 1/\tau \text{ (Hz)} \tag{6.14}$$

$$v = f\lambda \text{ (m/s)} \tag{6.15}$$

We can also define another periodic property of waves: the angular frequency ω.

$$\omega \equiv 2\pi/\tau = 2\pi f \text{ (rad/s)} \tag{6.16}$$

λ, τ, f, ω, and k all represent wave characteristics which are repeated periodically, both in space and in time. These can be applied not only to sinusoidal waves, but also to waves of any shape, as long as they have a periodically repeating pattern. We can express the sinusoidal wave in various forms, as shown by the following equations, which are all equivalent. For optical calculations, Equation (6.18) is generally used.

$$U = A \sin\ k(x \mp vt) \tag{6.17}$$

$$U = A \sin\ (kx \mp \omega t) \tag{6.18}$$

$$U = A \sin 2\pi f(x/v \mp t) \tag{6.19}$$

The above equations can all be used to express the basic equation of wave motion for a monochromatic wave, where the waves have only one frequency. We shall now discuss the phase, which was described in Equation (6.18): $U(x,t) = A\sin(kx \mp \omega t)$. The phase φ can be defined as:

$$\varphi = kx - \omega t \tag{6.20}$$

In the special case when $x = 0, t = 0, U(x,t)$ will be:

$$U(x,t)|_{x=0,t=0} = U(0,0) = 0$$

In the general case, the function can be described as having a phase constant δ which will be fixed, depending on how we define the origin for x and t. We can thus express the equation of wave motion as follows:

$$U(x,t) = A\sin(kx - \omega t + \varepsilon) \tag{6.21}$$

Here, ε is the initial phase (or epoch angle), which means that the phase φ is ε when $x = 0$ and $t = 0$.

6.1.3 Wave Equations

We shall now develop the one-dimensional wave function into a general format. In Equation (6.3) we defined the wave function $U(x,t) = f(x')$. In Equation (6.4), we defined $x' = x \mp vt$. The partial derivative of $U(x,t)$ with respect to x, with t held constant, will be:

$$\frac{\partial U}{\partial x} = \frac{\partial U}{\partial x'}\frac{\partial x'}{\partial x} = \frac{\partial f}{\partial x'} \tag{6.22}$$

The partial derivative of $U(x,t)$ with respect to t, with x held constant, will be:

$$\frac{\partial U}{\partial t} = \frac{\partial U}{\partial x'}\frac{\partial x'}{\partial t} = \mp v\frac{\partial f}{\partial x'} \tag{6.23}$$

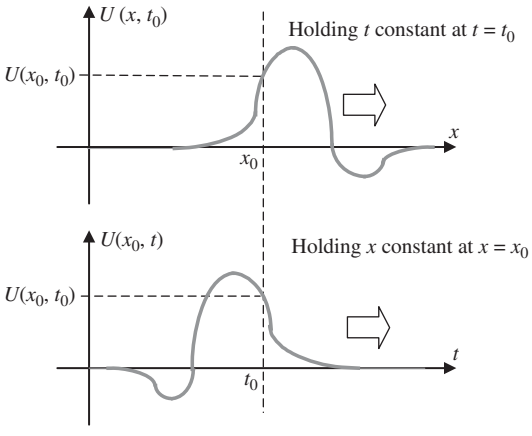


Figure 6.3 How U varies with x and t

Combining Equation (6.22) with Equation (6.23), we obtain:

$$\frac{\partial U}{\partial t} = \mp v \frac{\partial U}{\partial x} \tag{6.24}$$

Equation (6.24) shows that the rates at which U changes with respect to t and with respect to x are in direct proportion to one another (Figure 6.3). However, the sign of the ratio of the two rates may be negative or positive, depending on whether the wave is traveling in a positive or a negative direction, respectively.

It can be rather inconvenient to have to use two different equations for a wave, depending on which direction it traveling in, as Equation (6.24) shows. If we want to represent the wave by a single equation, regardless of whether it is traveling in a positive or negative direction, we will need to introduce second-order differential equations. The partial derivative of Equation (6.22) with respect to x, with t held constant, will then be:

$$\frac{\partial^2 U}{\partial x^2} = \frac{\partial^2 f}{\partial x'^2}$$

Similarly, the partial derivative of Equation (6.23) with respect to t, holding x constant, will be:

$$\frac{\partial^2 U}{\partial t^2} = \frac{\partial}{\partial t}\left(\mp v \frac{\partial f}{\partial x'}\right) = \mp v \frac{\partial}{\partial x'}\left(\frac{\partial f}{\partial t}\right) = \mp \frac{\partial}{\partial x'} v \left(\frac{\partial U}{\partial t}\right) = v^2 \frac{\partial^2 f}{\partial x'^2}$$

Combining the two equations above, we obtain:

$$\frac{\partial^2 U}{\partial t^2} = v^2 \frac{\partial^2 f}{\partial x'^2} \tag{6.25}$$

Rearranging, we obtain:

$$\frac{\partial^2 U}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 U}{\partial t^2} \tag{6.26}$$

Equation (6.26) is the one-dimensional wave equation we wanted. In fact, the electromagnetic wave equation is the three-dimensional form of Equation (6.26), which we can derive by applying Maxwell's equations. The general form of the solution to Equation (6.26) will be:

$$U = f(x - vt) + g(x + vt) \tag{6.27}$$

We can confirm Equation (6.27) as being a solution of Equation (6.26), by substituting Equation (6.27) into Equation (6.26).

Equation (6.27) shows a superposition of two waves, in which one is traveling in a positive direction and the other is traveling in a negative direction. If both $U_1(x, t)$ and $U_2(x, t)$ are solutions of Equation (6.26), then their superposition,

$$U = C_1 U_1(x,t) + C_2 U_2(x,t) \qquad (C_1, C_2 : \text{constant})$$

will also be a solution.

6.2 Describing Waves with Complex Exponential Functions

When discussing wave phenomena, it may sometimes be inconvenient to use sine or cosine functions. As an alternative, we can use complex numbers to describe these waves, thereby simplifying the mathematics involved. The complex exponential form of the wave equation is used in a variety of fields, including classical and quantum mechanics, electronics, and optics.

A complex number z can be expressed in the following form:

$$z = x + jy \tag{6.28}$$

$$j = \sqrt{-1} \tag{6.29}$$

Here, we will express z in polar coordinates, where x is a real number and y is an imaginary number as in Figure 6.4.

$$x = r\cos\theta \tag{6.30a}$$

$$y = r\sin\theta \tag{6.30b}$$

$$z = x + jy = r(\cos\theta + j\sin\theta) \tag{6.31}$$

Applying the Euler formula Equation (6.32), we can derive z, as shown in Equation (6.33).

$$e^{j\theta} = \cos\theta + j\sin\theta \tag{6.32}$$

$$z = re^{j\theta} = r\cos\theta + jr\sin\theta \tag{6.33}$$

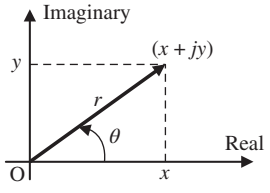


Figure 6.4 Argand diagram

In Equation (6.33), r represents the magnitude of z and θ represents the phase angle in radians. We refer to r as the absolute value of z (where $r = \sqrt{x^2 + y^2}$). Using this complex exponential form, we can calculate the motion of a wave much more easily than we could by using sine or cosine functions. The complex conjugate, which is indicated by an asterisk, is obtained by replacing the value j with $-j$, so that:

$$z^* = (x + jy)^* = x - jy \tag{6.34}$$

$$z^* = r(\cos\theta - j\sin\theta) \tag{6.35}$$

$$z^* = re^{-j\theta} \tag{6.36}$$

Addition and subtraction can be performed in a straightforward fashion:

$$z_1 \pm z_2 = (x_1 + jy_1) \pm (x_2 + jy_2) = (x_1 \pm x_2) + j(y_1 \pm y_2) \tag{6.37}$$

The above procedure is similar to the component addition of vectors. Multiplication and division can also be performed more easily, using polar coordinates.

$$z_1 z_2 = r_1 e^{j\theta 1} r_2 e^{j\theta 2} = r_1 r_2 e^{j(\theta 1 + \theta 2)} \tag{6.38}$$

$$\frac{z_1}{z_2} = \frac{r_1 e^{j\theta 1}}{r_2 e^{j\theta 2}} = \frac{r_1}{r_2} e^{j(\theta 1 - \theta 2)} \tag{6.39}$$

We can obtain the above equations also from standard trigonometric addition (or subtraction) formulae as follows:

$$\begin{aligned} e^{j\theta 1} e^{j\theta 2} &= (\cos\theta_1 + j\sin\theta_1)(\cos\theta_2 + j\sin\theta_2) \\ &= (\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + j(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2) \\ &= \cos(\theta_1 + \theta_2) + j\sin(\theta_1 + \theta_2) = e^{j(\theta 1 + \theta 2)} \\ \frac{e^{j\theta 1}}{e^{j\theta 2}} &= \frac{\cos\theta_1 + j\sin\theta_1}{\cos\theta_2 + j\sin\theta_2} = \frac{(\cos\theta_1 + j\sin\theta_1)(\cos\theta_2 - j\sin\theta_2)}{(\cos\theta_2 + j\sin\theta_2)(\cos\theta_2 - j\sin\theta_2)} \\ &= (\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2) + j(\sin\theta_1\cos\theta_2 - \cos\theta_1\sin\theta_2) \\ &= \cos(\theta_1 - \theta_2) + j\sin(\theta_1 - \theta_2) = e^{j(\theta 1 - \theta 2)} \end{aligned}$$

Similarly, we can obtain Equation (6.40) and Equation (6.41) as follows:

$$e^{z1+z2} = e^{z1} e^{z2} \tag{6.40}$$

$$e^z = e^{x+jy} = e^x e^{jy} \tag{6.41}$$

$$\begin{aligned} \because\ e^{z1} e^{z2} &= e^{x1+jy1} e^{x2+jy2} = e^{x1}(\cos y_1 + j\sin y_1) e^{x2}(\cos y_2 + j\sin y_2) \\ &= e^{x1} e^{x2}\{(\cos y_1\cos y_2 - \sin y_1\sin y_2) + j(\sin y_1\cos y_2 + \cos y_1\sin y_2)\} \\ &= e^{x1+x2}\{\cos(y_1 + y_2) + j\sin(y_1 + y_2)\} = e^{x1+x2} e^{j(y1+y2)} = e^{x1+jy1+x2+jy1} = e^{z1+z2} \end{aligned}$$

Equations (6.33)–(6.41) are very useful for calculating wave motion.
Now,

$$zz^* = (x + jy)(x - jy) = x^2 + y^2 = |z|^2 \tag{6.42}$$

Thus the modulus of the complex quantity $|z|$ will be:

$$|z| \equiv \sqrt{zz^*} = \sqrt{x^2 + y^2} \tag{6.43}$$

Since $\cos 2\pi = 1$, $\sin 2\pi = 0$, $\cos \pi = -1$, $\sin \pi = 0$, $\cos(\pi/2) = 0$, $\sin(\pi/2) = 1$, it follows that:

$$e^{j2\pi} = 1 \tag{6.44}$$

$$e^{j\pi} = e^{-j\pi} = -1, \quad e^{\pm j\pi/2} = \pm j \tag{6.45}$$

The function e^z is periodic; it repeats itself every $j\,2\pi$ rad.

$$e^{z+j2\pi} = e^z e^{j2\pi} = e^z \tag{6.46}$$

Any complex number can be expressed as the sum of a real component $\mathrm{Re}(z)$ and an imaginary component $\mathrm{Im}(z)$.

$$z = \mathrm{Re}(z) + j\mathrm{Im}(z) \tag{6.47}$$

Thus

$$\mathrm{Re}(z) = \frac{1}{2}(z + z^*) \tag{6.48a}$$

$$\mathrm{Im}(z) = \frac{1}{2j}(z - z^*) \tag{6.48b}$$

Using polar coordinates, we can say that

$$\mathrm{Re}(z) = r\cos\theta \tag{6.49a}$$

$$\mathrm{Im}(z) = r\sin\theta \tag{6.49b}$$

We could choose either the real or the imaginary component, in order to describe a harmonic wave. In practice, however, the real component is normally used.

$$U(x,t) = \mathrm{Re}(Ae^{j(\omega t - kx + \varepsilon)}) \tag{6.50}$$

Equation (6.50) is equivalent to:

$$U(x,t) = A\cos(\omega t - kx + \varepsilon) \tag{6.51}$$

Henceforth, we will use Equation (6.52) to describe wave functions, whenever it proves convenient to do so.

$$U(x,t) = Ae^{j(\omega t - kx + \varepsilon)} = Ae^{j\varphi} \tag{6.52}$$

Proof of the Euler Formula Equation (6.33)

$$e^{j\theta} = \cos\theta + j\sin\theta \tag{6.32}$$

We define z as follows:

$$z = \cos\theta + j\sin\theta$$

Differentiating both sides with respect to θ, we get:

$$dz/d\theta = -\sin\theta + j\cos\theta \tag{1}$$

Multiplying both the right and left sides of the equation by *j,* we obtain:

$$jdz/d\theta = -j\sin\theta - \cos\theta = -z \tag{2}$$

Thus

$$dz/z = jd\theta \tag{3}$$

If we integrate the left and the right sides of the equation we get:

$$\int 1/z\, dz = j\int d\theta \tag{4}$$

In other words,

$$\log z = j\theta \tag{5}$$

Therefore,

$$z = e^{j\theta} \tag{6}$$

An Alternative Proof of the Euler Formula Equation (6.33)

$$e^{j\theta} = \cos\theta + j\sin\theta \tag{6.32}$$

e^x, $\cos x$, and $\sin x$ can be represented by the following series, all of which are Maclaurin expansions.

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \tag{7}$$

$$\cos x = \sum_{n=0}^{\infty} \frac{-1^n}{(2n)!} x^{2n} \tag{8}$$

$$\sin x = \sum_{n=0}^{\infty} \frac{-1^n}{(2n+1)!} x^{2n+1} \tag{9}$$

That is,

$$e^x = 1 + x + x^2/2! + x^3/3! + \cdot\cdot\cdot\cdot\cdot + x^n/n! + \cdot\cdot\cdot \tag{7'}$$

$$\cos x = 1 - x^2/2! + x^4/4! - x^6/6! + \cdot\cdot\cdot\cdot\cdot (-1)^n x^{2n}/(2n)! + \cdot\cdot\cdot \tag{8'}$$

$$\sin x = x - x^3/3! + x^5/5! - x^7/7! + \cdot\cdot\cdot\cdot\cdot + (-1)^n x^{2n+1}/(2n+1)! + \cdot\cdot\cdot \tag{9'}$$

From Equation (7′):

$$e^{jx} = 1 + jx - x^2/2! - jx^3/3! + x^4/4! + jx^5/5! - x^6/6! - jx^7/7! + \cdot\cdot\cdot \tag{10}$$

From Equation (8′), Equation (9′), and Equation (10):

$$e^{jx} = \cos x + j\sin x \tag{6.32'}$$

Here, we have derived the Euler formula in Equation (6.32), where $re^{j(\omega t - kz)}$ denotes a vector with length r which rotates counterclockwise (or clockwise) around the origin, as shown in the Argand diagram.

Readers wishing to study the properties of light waves in further depth may refer to Appendix C, which provides detailed explanations of plane waves, spherical waves, and Gaussian beams, and lists their respective equations.

6.3 Problems Relating to Wave Motion

Please download the Excel® files from the companion web site, which show calculation details for all problems in this book as listed in Appendix G

Problem 6.1 Superposition of light waves (1)

Graph the superposition of two light waves having the same frequency, the same amplitude, and a phase difference of $\pi/2$ rad, with the time t on the horizontal axis and the electric field U on the vertical axis. Calculate the irradiance (intensity) of the superposition. Assume that the velocity of light $c = 3 \times 10^8$m/s and the wavelength $\lambda = 600$ nm.

Solution

The values of the frequency *f,* angular velocity ω*,* propagation number *k,* and other parameters are as follows:

Wavelength $\lambda = 6.00 \times 10^{-7}$ m Velocity of light $c = 3.00 \times 10^8$ m/s

Frequency $f = c/\lambda = 5.00 \times 10^{14}$ Hz Period $\tau = 1/f = 2 \times 10^{-15}$ s

Angular velocity $\omega = 2\pi f = 3.14 \times 10^{15}$ rad/s

Propagation number $k = 2\pi/\lambda = 1.047197551 \times 10^7$ rad/m

The wave function U for a light wave can be represented as follows:

$$U = \cos(\omega t - kz + \varepsilon)$$

If we keep the values of z and ε constant at $z = 0, \varepsilon = 0,$ or $\pi/2$, we can represent the two waves U_A and U_B and their superposition U_{A+B} as follows:

$$U_A = \cos(\omega t), \quad U_B = \cos(\omega t + \pi/2), \quad U_{A+B} = U_A + U_B = \cos(\omega t) + \cos(\omega t + \pi/2)$$

We can create calculation tables for U_A, U_B, U_{A+B} for t. We can then graph these functions over the time interval from $t = 0$ to $t = 2\tau$ as in Figure 6.5a,b. We can do the same for ${U_A}^2, {U_{A+B}}^2,$ and ${U_{A+A}}^2$.

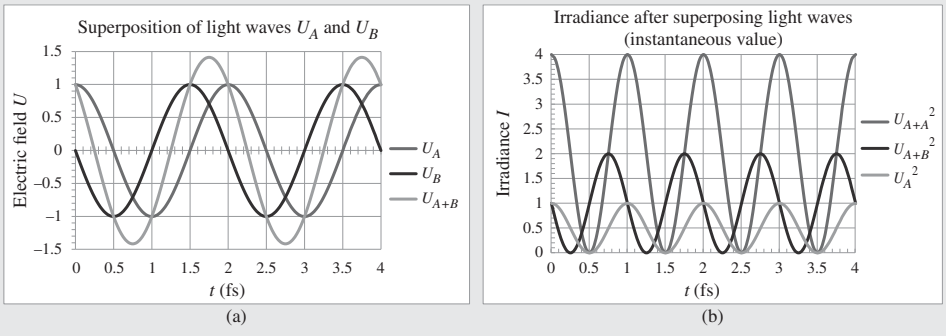


Figure 6.5 (a) Superposition of light waves. (b) Irradiance after superposition

Figure 6.5a shows that the amplitude of the superposition U_{A+B} is about 1.4 times greater than that of the constituent parts. The irradiance I can be expressed as $I =< U^2 >$. (< > means the

time-averaged value, which will be half the maximum value U^2). Figure 6.5b shows the instantaneous values of the irradiance of the original wave I_A, the irradiance of the superposition I_{A+B} (where the phase difference equals $\pi/2$), and the irradiance I_{A+A} after superposing two waves in-phase. The values of the irradiances are:

$$I_A = < U_A{}^2 > = 0.5 \qquad I_{A+B} = < U_{A+B}{}^2 > = 1 \qquad I_{A+A} = < U_{A+A}{}^2 > = 2$$

Notes

1. If the two waves have the same amplitude and a phase difference of $\pi/2$, the amplitude of their superposition will be about 1.4 times higher, while the irradiance will be exactly double.
2. If the waves (having the same amplitude) are superposed in phase, the amplitude will double, while the irradiance will be four times greater than the value for each wave. However, if the phase difference between the waves is π, both the amplitude and the irradiance of the superposition will be zero. (These are examples of constructive and destructive interference.)

Problem 6.2 Superposition of light waves (2)

Calculate the irradiance of a superposition of two light waves as a function of their phase difference $\alpha - \beta$, where the waves have the same frequency *f,* non-identical amplitudes whose respective values are *a* and *b,* and non-identical phases whose values are α and β.

Solution

The two light waves U_a, U_b can be represented by the following equations:

$$U_a = a\cos(\omega t - kz + \alpha), \quad U_b = b\cos(\omega t - kz + \beta), \quad \omega = 2\pi f, k = 2\pi/\lambda$$

Holding z constant ($z = 0$) we obtain:

$$U_a = a\cos(\omega t + \alpha) = a\exp[j(\omega t + \alpha)] = a\exp[j\alpha]\exp[j\omega t]$$
$$U_b = b\cos(\omega t + \beta) = b\exp[j(\omega t + \beta)] = b\exp[j\beta]\exp[j\omega t]$$

The electric field U and the irradiance I of the superposition of the two waves U_a and U_b will be:

$$U = U_a + U_b = \{a\exp(j\alpha) + b\exp(j\beta)\}\exp(j\omega t)$$
$$I = < UU^* > \quad (< > \text{ shows the time-averaged value})$$

Here, we simplify UU^* into a time independent form by omitting the term $\exp(j\omega t)$ in the above equation:

$$\begin{aligned} UU^* &= \{a\exp(j\alpha) + b\exp(j\beta)\}\{a\exp(j\alpha) + b\exp(j\beta)\}^* \\ &= \{a\cos\alpha + b\cos\beta + j(a\sin\alpha + b\sin\beta)\}\{a\cos\alpha + b\cos\beta - j(a\sin\alpha + b\sin\beta)\} \\ &= a^2 + b^2 + 2ab(\cos\alpha\cos\beta + \sin\alpha\sin\beta) = a^2 + b^2 + 2ab\cos(\alpha - \beta) \end{aligned}$$

The same results can be obtained applying the cosine rule to the triangle OAC in the Argand diagram in Figure 6.6a.

$$U^2 = U_a{}^2 + U_b{}^2 + 2U_aU_b\cos(\alpha - \beta)$$

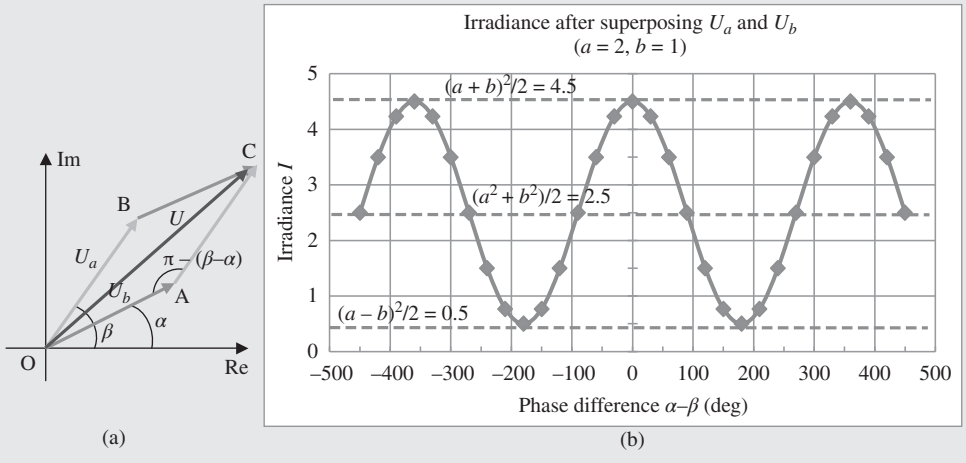


Figure 6.6 (a) Superposing waves by the method of vector addition. (b) Irradiance after superposition

An example of the calculation results (with $a = 2$, $b = 1$) is shown in Figure 6.6b.

As the graph shows, the irradiance I reaches its maximum value, $I = (a + b)^2/2 = 4.5$, when the phase difference is zero. It attains its minimum value of $I = 0.5$ when the phase difference equals $\pm\pi$, and it exhibits its average value, $I = 2.5$, when the phase difference is $\pm\pi/2$.

Note

The above calculations show how two waves behave when superposed. If the two waves were in the same phase, their superposition would display its maximum irradiance, and if they had a phase difference of $\pm\pi$, their superposition would display minimum irradiance.

Problem 6.3 Superposition of light waves (3)

Draw a graph of the light wave in Problem 6.2, with the time t along the horizontal axis and the electric field U along the vertical axis, as per Problem 6.1. Draw a graph of the irradiance after superposing the two waves. Assume that the velocity of light $c = 3 \times 10^8$ m/s, the wavelength $\lambda = 600$ nm, and the phase difference is $\pi/2$ or π.

Solution

The values of the frequency f, angular velocity ω, propagation number k, and other parameters are as follows:

Wavelength $\lambda = 6.00 \times 10^{-7}$ m Period $\tau = 1/f = 2 \times 10^{-15}$ s

Velocity of light $c = 3.00 \times 10^8$ m/s Angular velocity $\omega = 2\pi f = 3.14 \times 10^{15}$ rad/s

Frequency $f = c/\lambda = 5.00 \times 10^{14}$Hz

Propagation number $k = 2\pi/\lambda = 1.047197551 \times 10^{-7}$ rad/m

The light wave U can be expressed by the equation:

$$U = u\cos(\omega t - kz + \varepsilon)$$

If we keep $z = 0$, we obtain:

$$U_A = a\cos(\omega t),\ \ U_{B1} = b\cos(\omega t - \pi/2),\ \ U_A + U_{B1} = U_{A+B1} = a\cos(\omega t) + b\cos(\omega t - \pi/2)$$

$$U_A = a\cos(\omega t),\ \ U_{B2} = b\cos(\omega t - \pi),\ \ U_A + U_{B2} = U_{A+B2} = a\cos(\omega t) + b\cos(\omega t - \pi)$$

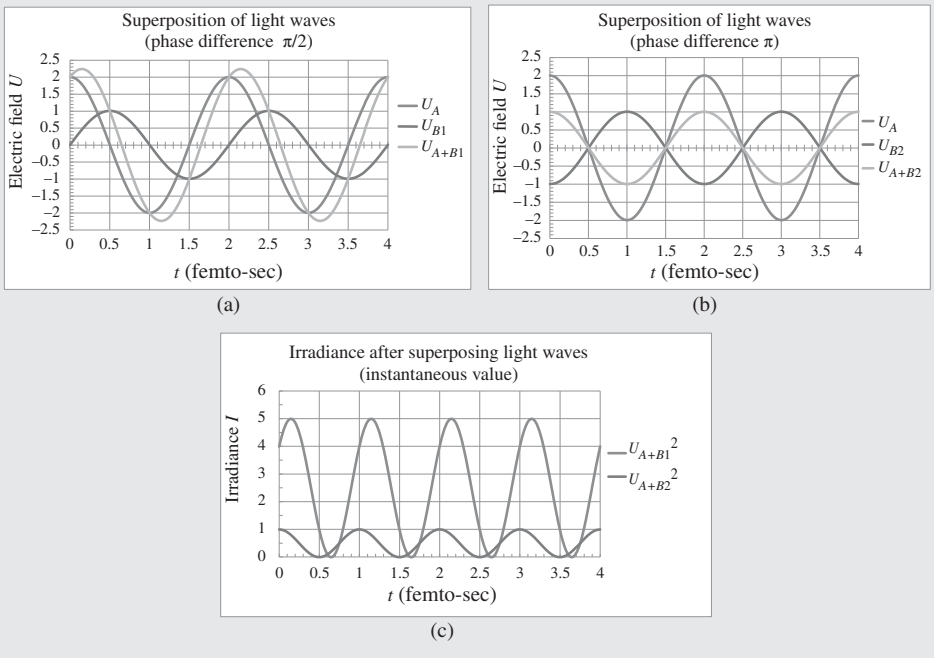


Figure 6.7 (a) Superposition of light waves (phase difference $\pi/2$). (b) Superposition of light waves (phase difference π). (c) Irradiance after superposition

As shown in Figure 6.7a–c, we can graph the above equations, with the time t on the horizontal axis and the electric field U on the vertical axis. We can also graph the irradiance of ${U_{A+B1}}^2$ and ${U_{A+B2}}^2$. Here, we shall assume that $a = 2$ and $b = 1$.

Note

The irradiance of the superposition will have a peak value of 5 when the phase difference equals $\pi/2$ and 1 when the phase difference equals π. Usually, however, we express the irradiance not in terms of its peak value, but in terms of its time-averaged value. Thus the irradiance will be half its peak value, that is, 2.5 or 0.5. Here, we have obtained the same result as in Problem 6.2.

Reference

[1] Hect, E. and Zajac, A. (1973) *Optics*, Addison-Wesley, USA.

7

Calculations for Focusing Gaussian Beams

Since light beams (including lasers) are a kind of electromagnetic waves, they need to satisfy the electromagnetic wave equation[1] derived from Maxwell's equations. A Gaussian beam is the most basic type of laser beam whose electric field equation can be derived from a solution to the electromagnetic wave equation. In this chapter, we will study the key features of Gaussian beams, as well as some simple methods for performing calculations on them.

7.1 What is a Gaussian Beam?

A Gaussian beam is a light beam whose transverse electric field and irradiance distributions are described by Gaussian functions [1–3]. This beam is the basic type of beam found in lasers. As shown in Figure 7.1, a Gaussian beam has a beam waist. This is the narrowest point of the beam, where its diameter reaches a minimum value. The beam diameter gradually increases on either side of the waist. The dotted lines in the figure depict various wave-front surfaces, over which the light beam will be in phase. The beam radius w_z can be defined as the radius at which the electric field and irradiance drop to $1/e$ and $1/e^2$ of their respective values on the beam axis.

Equation (7.1), Equation (7.2), Equation (7.3) and Equation (7.4), which describe a Gaussian beam, are mathematically derived solutions to the electromagnetic wave equation. (The reader may refer to Appendix C for details.) The electric field U at the point (z, ρ) can be represented as follows:

$$U(z,\rho) = A\frac{w_0}{w}\exp[j(kz-\varphi)]\exp\left[-\left(\frac{\rho}{w}\right)^2\right]\exp\left[-j\frac{k\rho^2}{2R}\right] \tag{7.1}$$

Energy conservation | Phase | Electric field distribution | Spherical wave front

[1] $\frac{\partial^2 E}{\partial x^2}+\frac{\partial^2 E}{\partial y^2}+\frac{\partial^2 E}{\partial z^2}=\varepsilon_0\mu_0\frac{\partial^2 E}{\partial t^2}$: Electromagnetic wave equation derived from Maxwell's equations. This equation represents the electromagnetic wave equation derived from Maxwell's equations (where $\boldsymbol{E}$ is the electric field, ε_0 is the dielectric index in a vacuum, and μ_0 is the magnetic permeability in a vacuum), which has the same form as the three-dimensional wave equation Equation (C.18) in Appendix C. Here, $\varepsilon_0\mu_0 = 1/c^2$ (where c is the velocity of light in a vacuum).

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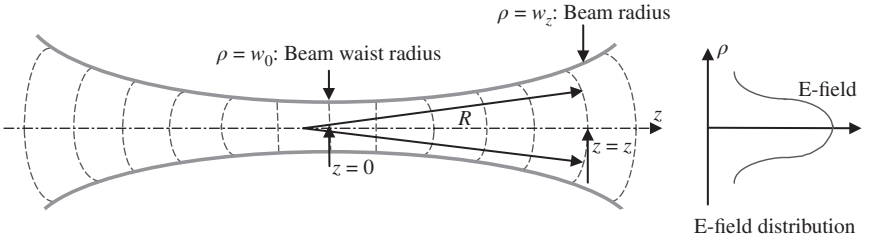


Figure 7.1 Gaussian beam

where the parameters *w, R, φ* are described by the following formulae:

$$w^2 = {w_0}^2 \left[1 + \left(\frac{\lambda z}{\pi {w_0}^2}\right)^2\right] \quad w = \text{Beam radius at distance } z \tag{7.2}$$

$$R = z\left[1 + \left(\frac{\pi {w_0}^2}{\lambda z}\right)^2\right] \quad R = \text{Radius of curvature of wave front at distance } z \tag{7.3}$$

$$\varphi = \tan^{-1}\left(\frac{\lambda z}{\pi {w_0}^2}\right) \quad \varphi = \text{Phase} \tag{7.4}$$

where

λ = Wavelength
$k \equiv 2\pi/\lambda$ = Propagation number
z = Distance
ρ = Radius of the beam (distance from the axis)
w_0 = Beam radius at the beam waist
w = Beam radius at distance z
A = Coefficient
j = Imaginary unit

We can derive Equation (7.5) and Equation (7.6) from Equation (7.2) and Equation (7.3). These equations allow us to calculate the beam waist radius w_0 and the beam waist distance z_0, once we know the values of the beam radius w and the radius of curvature R of the spherical wave front.

$${w_0}^2 = \frac{w^2}{1 + \left(\frac{\pi w^2}{\lambda R}\right)^2} \tag{7.5}$$

$$z_0 = \frac{R}{1 + \left(\frac{\lambda R}{\pi w^2}\right)^2} \tag{7.6}$$

7.1.1 First Term in Equation (7.1): Law of Energy Conservation Multiplier

The term $A\ w_0/w$ is entailed by the law of energy conservation. The amplitude of the electric field at distance z will be a multiple, w_0/w, of the value at the beam waist. It follows that the amplitude of the

electric field is inversely proportional to the beam diameter, which entails that the irradiance is inversely proportional to the square of the beam diameter.

7.1.2 Second Term in Equation (7.1): Phase

The term $\exp[j(kz - \varphi)]$ represents the phase at a distance z from the beam waist. The significance of this term is that the phase will vary with $k\,z$ (which is proportional to z) and φ (which is a function of z), as shown in Equation (7.4).

7.1.3 Third Term in Equation (7.1): Electric Field Distribution

The term $\exp[-(\rho/w)^2]$ represents the electric field distribution across the cross section of the beam. It attains a maximum value on the z-axis ($\rho = 0$), and gradually tapers off as ρ (the distance from the axis) increases. This distribution has the form of a Gaussian function, where the irradiance (or intensity) is equal to the square of the amplitude of the electric field. Thus the irradiance distribution $I(\rho)$ can be expressed by Equation (7.8).

$$U(\rho) = U_0 \exp\left[-\frac{\rho^2}{w^2}\right] : \text{ Electric field (Amplitude)} \tag{7.7}$$

$$I(\rho) = I_0 \exp\left[-\frac{2\rho^2}{w^2}\right] : \text{ Irradiance (Intensity)} \tag{7.8}$$

Figure 7.2 is a graph of Equation (7.7) and Equation (7.8), depicting the normalized distribution of the electric field amplitude and the irradiance, over a cross section of a Gaussian beam.

In the graph, the values at $\rho = w$ (i.e., $\rho/w = 1$) are:

$$U(\rho)|_{\rho=w} = 0.367879U_0(= 36.8\%) \quad I(\rho)|_{\rho=w} = 0.135335I_0(= 13.5\%)$$

ρ/w	E-field U	Irradiance I
–1.4	0.1408584	0.0198411
–1.2	0.2369278	0.0561348
–1	0.3678794	0.1353353
–0.8	0.5272924	0.2780373
–0.6	0.6976763	0.4867523
–0.4	0.8521438	0.726149
–0.2	0.9607894	0.9231163
0	1	1
0.2	0.9607894	0.9231163
0.4	0.8521438	0.726149
0.6	0.6976763	0.4867523
0.8	0.5272924	0.2780373
1	0.3678794	0.1353353
1.2	0.2369278	0.0561348
1.4	0.1408584	0.0198411

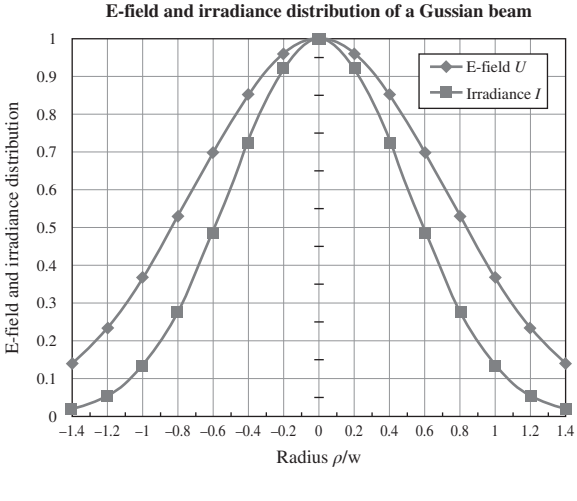


Figure 7.2 Electric field and irradiance distribution of a Gaussian beam

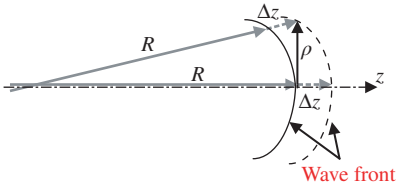


Figure 7.3 Wave-front sphere

The beam radius of a Gaussian beam is defined as that radius where the electric field strength is 36.8% (=1/*e*) of its value on the axis, or the irradiance is 13.5% (=1/e^2) of its value on the axis.

7.1.4 Fourth Term in Equation (7.1): Spherical Wave Front

The term $\exp[-jk\rho^2/(2R)]$ represents the phase. A point at a distance of ρ from the axis will be $k\rho^2/(2R)$ behind the phase at the z-axis. For a spherical wave with a radius of curvature R, as illustrated in Figure 7.3, after a time period of Δt has elapsed, the wave will have traveled forward by a distance of Δz and the radius of curvature is now $R + \Delta z$. In other words,

$$k(R + \Delta z) = k\sqrt{R^2 + \rho^2}$$

Squaring both sides, we get:

$$R^2 + 2R\Delta z + \Delta z^2 = R^2 + \rho^2$$

If we assume that $\Delta z \ll R$, it follows that:

$$2R\,\Delta z = \rho^2 \;\therefore\; \Delta z = \rho^2/(2R)$$

The above equation entails that the phase at a point at distance ρ from the axis will come into contact with a wave front with a radius of curvature $R + \Delta z$, after a time period of Δt has elapsed. Thus the term $\exp[-jk\rho^2/(2R)]$ shows that the surface of the wave front is a sphere with a curvature radius of R.

7.1.5 Equation (7.2): Beam Radius

The equation ${w_z}^2 = {w_0}^2[1 + \{\lambda z/(\pi {w_0}^2)\}^2]$ describes the beam radius w_z at a distance z from the beam waist. The greater the distance from the beam waist, the larger the radius of the beam will be. For very large values of z, $\lambda z \gg (\pi {w_0}^2)$, so w_z can be approximated by $w_z \cong \lambda z/(\pi w_0)$. This implies that w_z will increase in proportion to the distance z from the origin, with a divergence slope of $\tan\theta$ as shown in Equation (7.9) (Figure 7.4).

$$\tan\theta = \frac{w_z}{z} = \frac{\lambda}{\pi w_0} \tag{7.9}$$

7.1.6 Equation (7.3): Radius of Curvature of the Spherical Wave Front

The equation $R = z[1 + \{\pi {w_0}^2/(\lambda z)\}^2]$ describes the radius of curvature of the spherical wave front. In the equation, as z tends to zero ($z \to 0$), R tends to infinity ($R \to \infty$). This entails that the wave front will

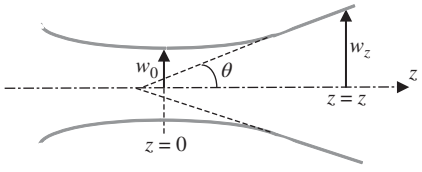


Figure 7.4 Gaussian beam divergence

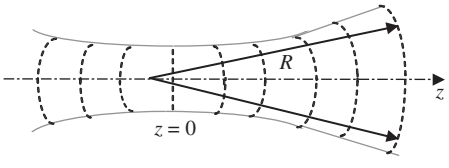


Figure 7.5 Wave-front curvature radius of a Gaussian beam

be a flat plane at the beam waist. As we move away from the beam waist, R gets smaller and it reaches a minimum value at a distance of $z_R = \pi {w_0}^2/\lambda$ (the Rayleigh range). At a distance larger than the Rayleigh range, R gets larger. For very large values of z, where $(\lambda z)^2 \gg (\pi {w_0}^2)^2$, R will take a value approximately equal to the distance z (Figure 7.5).

$$R \cong z \tag{7.10}$$

7.1.7 Equation (7.5): Beam Waist Radius

The equation ${w_0}^2 = w^2/[1 + \{\pi w^2/(\lambda R)\}^2]$ is the equation for calculating the beam waist radius w_0 from the beam radius w and the radius of curvature R of the spherical wave front. If R satisfies $\lambda R \ll \pi w^2$, that is, if we focus a collimated beam by a lens with a short focal length, then Equation (7.5) can be approximated by:

$$w_0 \cong \frac{\lambda R}{\pi w} = \frac{\lambda f}{\pi w} \tag{7.11}$$

In Equation (7.11), w_0 represents the beam waist radius when a collimated beam w is focused by a lens with a short focal length.

7.1.8 Equation (7.6): Distance to Beam Waist

The equation $z_0 = R/[1 + \{\lambda R/(\pi w^2)\}^2]$ is the equation for calculating the beam waist distance z_0 from the beam radius w and the radius of curvature R of the spherical wave front. If R satisfies $\lambda R \ll \pi w^2$, that is, if we focus a collimated beam w by a lens with a short focal length, then Equation (7.6) can be approximated by:

$$z_0 \cong R = f \tag{7.12}$$

In Equation (7.12), z_0 represents the focusing distance when a collimated beam w is focused by a lens with a short focal length as shown in Figure 7.6.

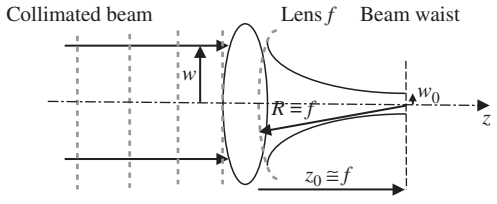


Figure 7.6 Beam waist distance

We have discussed the characteristics of Gaussian beams at considerable length here. The key differences between Gaussian beam theory and geometrical optics are as follows:

1. Geometrical optics postulates an infinitesimal focal point. In practice, however, beams cannot be focused onto a spot with an infinitesimal radius; they have a finite, minimum size at the beam waist, as shown in the preceding discussion of Gaussian beam theory.
2. In geometrical optics, a collimated beam can maintain a constant diameter over an infinite range. In practice, all beams – including collimated beams – have a beam waist. As we move away from the waist, the diameter of the beam gradually increases.
3. We can precisely specify a Gaussian beam, simply by specifying its wavelength and the diameter of the beam waist.
4. Alternatively, we can obtain the beam waist radius w_0 and the beam waist distance z_0 from the beam radius w and the radius of curvature R of the spherical wave front at an arbitrary position (with a fixed wavelength).

7.2 Equations for Focusing a Gaussian Beam

In this section, we will examine the equations for focusing a Gaussian beam. Figure 7.7 illustrates a convex lens focusing a Gaussian beam [4, 5]. From the perspective of geometrical optics, the lens converts an incident beam with a spherical wave front having a negative radius of curvature R into an output beam with a spherical wave front having a positive radius of curvature R'. The two rays have the same height at the lens. In this example, the Gaussian lens formula Equation (1.24) can be rewritten as follows:

$$\frac{1}{R'} = \frac{1}{R} + \frac{1}{f} \quad \text{where} \quad \frac{1}{f} = \text{Refractive power of the lens} \tag{7.13}$$

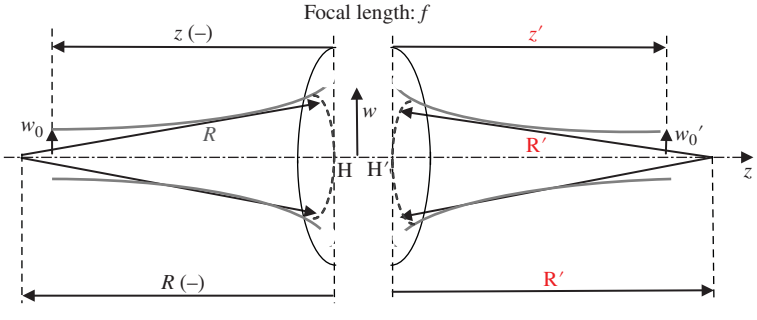


Figure 7.7 Gaussian beam focusing using a lens

Using Gaussian beam theory and Geometrical optics, we can derive the image point (w_0', z') from the object point (w_0, z). The calculation procedure is as follows:

① First, we substitute the values of w_0 and z into Equation (7.2) and Equation (7.3), in order to obtain the beam radius w and the radius of curvature R of the wave front at the lens.
② Next, we substitute R and f into Equation (7.13), in order to obtain the radius of curvature R' of the wave front after exiting the lens.
③ Finally, we substitute w (from ①) and R' (from ②) into Equation (7.5) and Equation (7.6), in order to obtain the beam waist radius w_0' and the beam waist position z_0' as an image point.

Using the above method, we can calculate the beam waist radius w_0' and position z_0' of a given Gaussian beam, after exiting the lens. The equations for calculating z_0' and w_0' will be as follows. (Here, we shall rewrite the symbols z and z_0' as s and s', respectively, as these symbols are used in the Gaussian lens formula.)

$$\frac{1}{s'} = \frac{1}{s + \dfrac{{z_R}^2}{s+f}} + \frac{1}{f} \tag{7.14}$$

$$w_0' = \frac{w_0}{\sqrt{\left(1 + \dfrac{s}{f}\right)^2 + \left(\dfrac{z_R}{f}\right)^2}} \tag{7.15}$$

$$\text{where } z_R = \frac{\pi {w_0}^2}{\lambda} \ (z_R = \text{Rayleigh range}) \tag{7.16}$$

Equation (7.14) can be rewritten as:

$$s' = \left(s + \frac{{z_R}^2}{s+f}\right) \frac{f}{s + f + \dfrac{{z_R}^2}{s+f}} \tag{7.14′}$$

Equation (7.14), Equation (7.15) and Equation (7.16) are the equations used for Gaussian beam focusing calculations. These equations are very useful and convenient for calculating the beam waist size and its position in a Gaussian beam focusing optical system. The problems at the end of this chapter are examples of real-life situations involving the focusing of a Gaussian beam within an optical system. When applying the equations for focusing a Gaussian beam, the reader is advised to keep the following facts in mind.

1. The Gaussian beam focusing equations are based on the case of a pure Gaussian beam whose transverse mode is TEM_{00} (where the irradiance distribution exactly matches that of an ideal Gaussian distribution). However pure Gaussian laser beams rarely exist in the real world. In practice, even a laser beam of TEM_{00} will include a few other higher order modes and its irradiance distribution will diverge from that of an ideal Gaussian distribution. The greater the divergence, the greater the difference will be between the actual (focused) beam waist size and its calculated theoretical value. In the next section, we will examine an approximate calculation method for a real life Gaussian beam, using the M^2 factor.

   For example, let us suppose that a beam truncated by the lens aperture is focused on a larger spot than that of the Gaussian beam. Here, it is desirable to make the aperture size about three times greater

than the beam size, so as not to deform the shape of the beam from that of an ideal Gaussian beam. (For further details on truncated Gaussian beam focusing, the reader may refer to Chapter 9.)
2. If we are using a concave lens in an optical system, we can perform Gaussian beam focusing calculations by making the sign of the focal length of the lens negative, in the same way as we did in our earlier discussion of paraxial theory.
3. If we want to obtain a parallel (collimated) beam, we need to set the optical configuration, so as to maximize the beam waist size w_0' in Equation (7.15).
4. If we are calculating the beam waist size and position of an elliptical Gaussian beam, where the irradiance distributions for the *x*-axis and *y*-axis are different from each other, we can calculate them for the *x*- and *y*- axes independently of one another.

7.3 The M^2 (M Squared) Factor

Until now, our study of Gaussian beam theory has been based on the assumption that the beam is an ideal Gaussian beam with a transverse mode of TEM_{00} (which means that the beam's irradiance distribution over its cross-section exactly matches that of an ideal Gaussian distribution) . However in practice, even a laser beam of TEM_{00} will include a few other higher order modes and it will differ to some degree from an ideal beam, so the focused beam waist size will be a little greater than that of the ideal Gaussian beam. Here, we can calculate the approximate waist size of the focused beam, by introducing an M^2 factor, which corresponds to the beam quality [5]. (The reader may refer to ISO 11146 for further details.)

7.3.1 Definition of the M^2 Factor

The beam divergence angle (actually, half-angle) θ for high values of z can be expressed as a function of the wavelength λ and the beam waist radius w_0:

$$\tan\theta = \frac{\lambda}{\pi w_0} \tag{7.9}$$

Assuming that θ is small, we can approximate Equation (7.9) by the following equations.

$$\theta = \frac{\lambda}{\pi w_0} \tag{7.17a}$$

$$\text{or } w_0 = \frac{\lambda}{\pi\theta} \tag{7.17b}$$

$$\text{or } \theta w_0 = \frac{\lambda}{\pi} \tag{7.17c}$$

Equation (7.17c) entails that the product of the beam waist radius w_0 and the beam divergence angle θ is a constant, whose value depends on the wavelength λ. Hence, if the beam waist is small, then the beam divergence angle will be large; if the waist is large, the divergence angle will be small. This relationship can also be applied to the actual beam, whose focused beam waist size is greater than that of an ideal Gaussian beam. Here, we shall define the M^2 factor as the ratio of the focused beam waist size to that of the ideal Gaussian beam. (It is M^2 times greater.) This M^2 factor can be used to gauge the beam's quality. If we insert an M^2 factor, Equation (7.17a–c) can be rewritten as follows:

$$\boldsymbol{\Theta} = \frac{M^2\lambda}{\pi W_0} \tag{7.18a}$$

$$\text{or } W_0 = \frac{M^2\lambda}{\pi\boldsymbol{\Theta}} \tag{7.18b}$$

$$\text{or } \boldsymbol{\Theta} W_0 = \frac{M^2\lambda}{\pi} \tag{7.18c}$$

where $\boldsymbol{\Theta}$ and W_0 are the beam divergence angle (half-angle) and the beam waist radius for the actual beam, respectively.

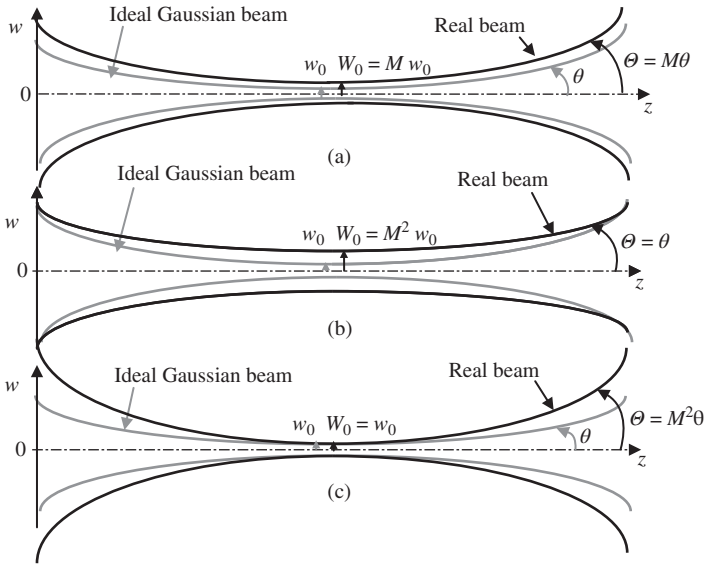


Figure 7.8 Conceptual relationship between real and ideal beams. (a) If the real beam is everywhere larger than the ideal beam by a factor of *M,* then the divergence angle will also be larger by the same factor. (b) If both beams have the same divergence angle, but the real beam has a larger waist, then it will be larger by a factor of M^2. (c) If both beams have the same size waist, but the real beam diverges faster, then it will diverge faster by a factor of M^2

When the beam profile matches that of an ideal Gaussian beam, $M^2 = 1$. The value of M^2 will be greater than 1 when the beam profile diverges from the ideal case. The poorer the beam quality, the greater the value of M^2 will be. Figure 7.8 illustrates the conceptual relationship between real and ideal beams.

7.3.2 Beam Propagation Equations after Taking M^2 Factor into Account

In Appendix C, we derived the Gaussian beam equations from the three-dimensional wave equation. $Q(z)$ in Equation (C.38) is a complex beam parameter for obtaining the Gaussian beam equations. (See Appendix C for details.)

$$Q(z) = z + q_0 \tag{C.38a}$$

$$q_0 = \frac{j\pi {w_0}^2}{\lambda} \tag{C.39a}$$

$$Q(z) = z + \frac{j\pi {w_0}^2}{\lambda} \tag{C.38a′}$$

We can calculate $1/Q\,(z)$ from Equation (C.38a′) as follows:

$$\frac{1}{Q(z)} = \frac{1}{z + j\dfrac{\pi {w_0}^2}{\lambda}} = \frac{z}{z^2 + \left\{\dfrac{\pi {w_0}^2}{\lambda}\right\}^2} - j\frac{\dfrac{\pi {w_0}^2}{\lambda}}{z^2 + \left\{\dfrac{\pi {w_0}^2}{\lambda}\right\}^2}$$

$$= \frac{1}{z\left[1 + \left\{\dfrac{\pi {w_0}^2}{\lambda z}\right\}^2\right]} - j\frac{\lambda}{\pi {w_0}^2\left[1 + \left\{\dfrac{\lambda z}{\pi {w_0}^2}\right\}^2\right]}$$

Inspecting the terms in the above equation, we can simplify the equation as follows:

$$\frac{1}{Q(z)} = \frac{1}{R(z)} - j\frac{\lambda}{\pi w(z)^2} \tag{7.19}$$

$$R(z) = z\left[1 + \left\{\frac{\pi {w_0}^2}{\lambda z}\right\}^2\right] \tag{7.3}$$

$$w(z)^2 = {w_0}^2\left[1 + \left\{\frac{\lambda z}{\pi {w_0}^2}\right\}^2\right] \tag{7.2}$$

Equation (7.19) shows that the real part of $1/Q\,(z)$ represents the inverse of the radius of curvature of the wave front $1/R$, while the imaginary part represents a function of the beam radius w.

Here, we shall assume that the real beam radius W is larger than the ideal beam radius w by a factor of M everywhere, as shown in Figure 7.8a.

$$W(z) = Mw(z) \tag{7.20}$$

Substituting Equation (7.20) into Equation (C.38a′) we obtain:

$$Q(z) = z + j\frac{\pi {W_0}^2}{M^2\lambda} \tag{7.21}$$

We can calculate $1/Q\,(z)$ from Equation (7.21) as follows:

$$\frac{1}{Q(z)} = \frac{1}{z + j\dfrac{\pi {W_0}^2}{M^2\lambda}} = \frac{z}{z^2 + \left\{\dfrac{\pi {W_0}^2}{M^2\lambda}\right\}^2} - j\frac{\dfrac{\pi {W_0}^2}{M^2\lambda}}{z^2 + \left\{\dfrac{\pi {W_0}^2}{M^2\lambda}\right\}^2}$$

$$= \frac{1}{z\left[1 + \left\{\dfrac{\pi {W_0}^2}{M^2\lambda z}\right\}^2\right]} - j\frac{M^2\lambda}{\pi {W_0}^2\left[1 + \left\{\dfrac{M^2\lambda z}{\pi {W_0}^2}\right\}^2\right]}$$

Inspecting the terms in the above equation, we can simplify the equation as follows:

$$\frac{1}{Q(z)} = \frac{1}{R(z)} - j\frac{M^2\lambda}{\pi W(z)^2} \tag{7.22}$$

$$R(z) = z\left[1 + \left\{\frac{\pi {W_0}^2}{M^2\lambda z}\right\}^2\right] \tag{7.23}$$

$$W(z)^2 = {W_0}^2\left[1 + \left\{\frac{M^2\lambda z}{\pi {W_0}^2}\right\}^2\right] \tag{7.24}$$

Additionally, we can derive Equation (7.25) and Equation (7.26) from Equation (7.23) and Equation (7.24):

$$Z_0 = \frac{R}{1 + \left(\dfrac{M^2\lambda R}{\pi W^2}\right)^2} \tag{7.25}$$

$${W_0}^2 = \frac{W^2}{1 + \left\{\dfrac{\pi W^2}{M^2\lambda R}\right\}^2} \tag{7.26}$$

where Z_0 is the distance to the beam waist and W_0 is the beam waist radius for the actual beam. Equation (7.23), Equation (7.24), Equation (7.25), and Equation (7.26) are very similar to Equation (7.2), Equation (7.3), Equation (7.5), and Equation (7.6), which we studied in the previous section. The only difference is that λ has been replaced by $M^2\lambda$ in Equation (7.23), Equation (7.24), Equation (7.25), and Equation (7.26). Thus we can take account of the M^2 factor while performing Gaussian beam focusing calculations, simply by changing λ to $M^2\lambda$.

Here, we will examine the Rayleigh range Z_R for the actual beam. Replacing w_0 with W_0 and λ with $M^2\lambda$, respectively, in Equation (7.16), we obtain:

$$Z_R = \frac{\pi {W_0}^2}{M^2\lambda} = \frac{\pi(Mw_0)^2}{M^2\lambda} = \frac{\pi {w_0}^2}{\lambda} \tag{7.27}$$

Equation (7.27) is identical to Equation (7.16) which is for the ideal Gaussian beam. Thus the Rayleigh ranges z_R for the ideal Gaussian beam and for the actual beam are identical to one another. Using the same methods, we find that the radius of curvature R for the wave front and the distance to beam waist z_0 for the ideal Gaussian beam and for the actual beam are also identical.

7.4 Sample Gaussian Beam Focusing Problems

Please download the Excel® files from the companion web site, which show calculation details for all problems in this book as listed in Appendix G

Problem 7.1 Focusing a Parallel Beam

Calculate the focused beam waist size ${w_0}'$ for a lens satisfying the following specifications.

Wavelength λ	0.000532 mm
Beam waist radius of the incident beam w_0	0.5 mm at $1/e^2$
Beam waist position of the incident beam s	−40 mm
Focal length of the focusing lens f	2, 4, 10, 20, or 40 mm

See Figure 7.9.

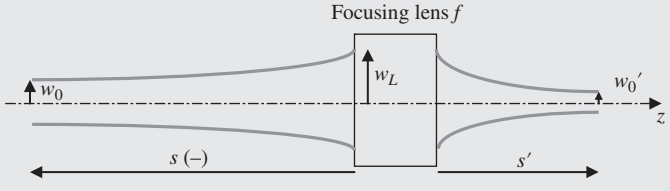


Figure 7.9 Focusing a parallel beam

Solution

(1) Equations: The focused beam waist radius ${w_0}'$, position s', and beam radius at the lens w_L will be:

$$s' = \frac{\left(s + \frac{{z_R}^2}{s+f}\right) f}{s + f + \frac{{z_R}^2}{s+f}} \tag{7.14'}$$

$${w_0}' = \frac{w_0}{\sqrt{\left(1 + \frac{s}{f}\right)^2 + \left(\frac{z_R}{f}\right)^2}} \tag{7.15}$$

$$z_R = \frac{\pi {w_0}^2}{\lambda} \tag{7.16}$$

$$w_L = w_0 \sqrt{1 + \left(\frac{\lambda s}{\pi {w_0}^2}\right)^2} \tag{7.2}$$

(2) Calculation results (Figure 7.10a,b)

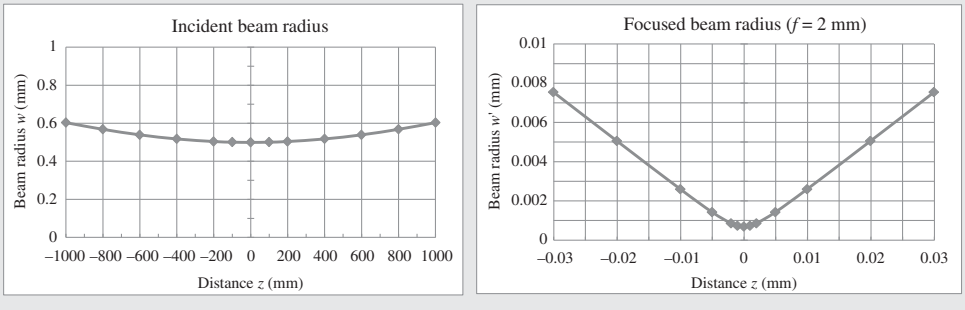


Figure 7.10 (a) Incident beam radius and (b) focused beam radius

f (mm)	2	4	10	20	40
s' (mm)	2.00007	4.00026	10.00138	20.00367	40
${w_0}'$ (mm)	0.00068	0.00135	0.00339	0.00677	0.01355
w_L (mm)	0.50018	same as on the left	"	"	"

Notes

1. We have obtained a focused beam whose waist is at a point very close to the second focal point.
2. The beam waist size will get smaller as the focal length of the lens decreases.
3. If $f = 40$ mm, $s = -40$ mm, then $s' = 40$ mm. This means that if we set the incident beam waist at the first focal point, the exiting beam waist will also be at the second focal point.

Problem 7.2 Beam expanders

In Figure 7.11a,b, calculate the collimated beam radii of Gaussian beams after exiting the beam expanders. (All units are in millimeters, and the wavelength $\lambda = 0.000532$ mm or 532 nm.)

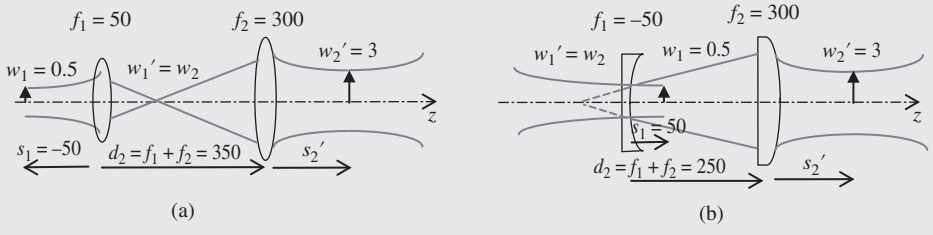


Figure 7.11 (a) Configuration A and (b) configuration B

Solution

(1) Equations: For each lens ($n = 1, 2$), we can calculate the output beam waist radius and its position w_n' and s_n' from the corresponding parameters for the incident beam (w_n and s_n). Finally, we can derive the beam waist radius w_2' and its position s_2' of the exiting beam from each expander.

$$s_n = -(d_n - s_{n-1}')$$

$$s_n' = \frac{\left(s_n + \frac{z_{Rn}^2}{s_n + f_n}\right) f_n}{s_n + f_n + \frac{z_{Rn}^2}{s_n + f_n}} \tag{7.14′}$$

$$w_n = w_{n-1}'$$

$$w_n' = \frac{w_n}{\sqrt{\left(1 + \frac{s_n}{f_n}\right)^2 + \left(\frac{z_{Rn}}{f_n}\right)^2}} \tag{7.15}$$

$$z_{Rn} = \frac{\pi w_n^2}{\lambda} \text{ (Rayleigh range)} \tag{7.16}$$

$$w_z = w_0 \sqrt{1 + \left(\frac{\lambda z}{\pi w_0^2}\right)^2} \tag{7.2}$$

(2) Calculation results: Using the above equations, we can calculate the exiting beam waist radius $w_{2'}$ and its position $s_{2'}$ from the incident beam waist w_1 and its position s_1 as follows:
Configuration A ($\lambda = 0.000532$ mm, $w_1 = 0.5$ mm, $s_1 = -50$ mm). Units: mm

	f	d	s	w	z_R	s'	w'
$n = 1$	50	50	−50	0.5	1476.3123	50	0.01693
$n = 2$	300	350	−300	0.01693	1.69341	300	3

Configuration B ($\lambda = 0.000532$ mm, $w_1 = 0.5$ mm, $s_1 = 50$ mm). Units: mm

	f	d	s	w	z_R	s'	w'
$n = 1$	−50	−50	50	0.5	1476.3123	−50	0.01693
$n = 2$	300	350	−300	0.01693	1.69341	300	3

We obtained the same results for s_2' and w_2' for both Configuration A and Configuration B.

The exiting beam radius is shown in Figure 7.12, which was calculated by using Equation (7.2), where the wavelength $\lambda = 0.000532$ mm and the beam waist radius $w_0 = 3$ mm. The exiting beam maintains its radius of 3–4.12 mm over a range of 50 m. This means that the beam is almost perfectly collimated.

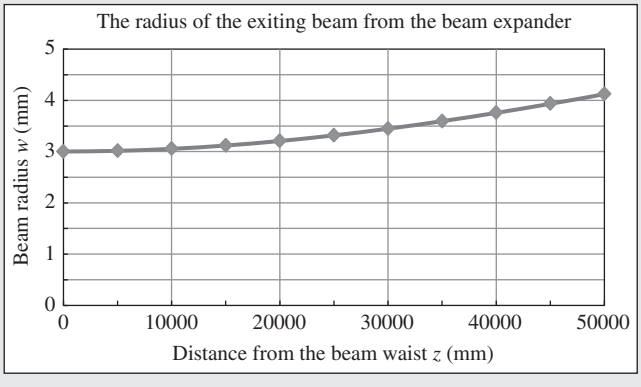


Figure 7.12 Radius of the exiting beam

Beam waist radius w_0 which minimizes the beam radius w over a range z

The beam radius w is expressed as follows:

$$w = w_0\sqrt{1 + \left(\frac{\lambda z}{\pi {w_0}^2}\right)^2} \tag{7.2}$$

Differentiating w with respect to w_0, $dw/dw_0 = 0$ is the condition which minimizes the value of w:

$$\frac{dw}{dw_0} = \frac{1 - \{\lambda z/(\pi {w_0}^2)\}^2}{\sqrt{1 + \{\lambda z/(\pi {w_0}^2)\}^2}} = 0$$

From the above equation we obtain:

$$w_0 = \sqrt{\frac{\lambda z}{\pi}}$$

If we set $\lambda = 0.000532$ mm and $z = 50\,000$ mm, we get a similar value for w_0 to those obtained above:

$$w_0 = 2.9098184 \text{ mm}, \qquad w = 4.1151046 \text{ mm } (= \sqrt{2}w_0)$$

The above value of w_0 minimizes the beam radius over a range of $z = 50\,000$ mm (Figure 7.13a). We refer to this range as the Rayleigh range and we can express it as follows, using the equation for w_0 above:

$$z_R = \frac{\pi {w_0}^2}{\lambda} \qquad \text{(Rayleigh range)} \tag{7.16}$$

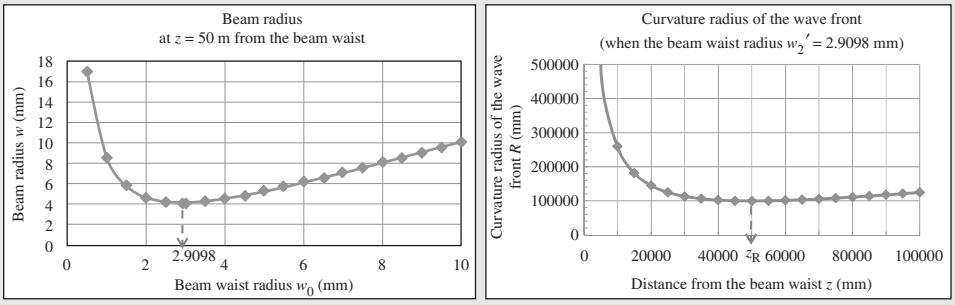


Figure 7.13 (a) Beam radius and (b) wave-front curvature radius of the exiting beam

We can also express z_R as the range z which minimizes the curvature radius R of the wave front.

$$R_z = z\left[1 + \left\{\frac{\pi {w_0}^2}{\lambda z}\right\}^2\right] = z + \frac{(\pi {w_0}^2)^2}{\lambda^2 z} \qquad \text{(Curvature radius of the wave front)} \tag{7.3}$$

Differentiating R with respect to z, $dR/dz = 0$ is the condition which minimizes the value of R_z:

$$\frac{dR}{dz} = 1 - \frac{(\pi {w_0}^2)^2}{(\lambda z)^2} = 0$$

From the above equations, we obtained

$$z_R = \frac{\pi {w_0}^2}{\lambda} \ (= 50\,000 \text{ mm}) \qquad \text{(Rayleigh range)} \tag{7.16}$$

$$R_z = 2z_R = 100\,000 \text{ mm}$$

The Rayleigh range z_R also means the range where the curvature radius R of the wave front displays its minimum value as shown in Figure 7.13b.

Note

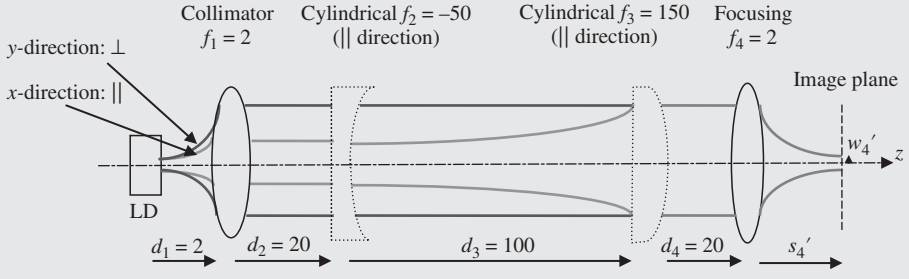
Beam expanders A and B will generate identical beams which are almost perfectly collimated, maintaining a radius of 3–4.12 mm over a range of 50 m.

Problem 7.3 A laser system: laser diode, collimator, beam expander, and focusing lens

Calculate the focused spot size w_4' for the laser system configured in Figure 7.14: laser diode (LD), collimator, unidirectional beam expander (to generate a circular beam) and focusing lens.

Please download the Excel® files from the companion web site, which show calculation details for all problems in this book as listed in Appendix G

(All units are in mm.)



LD wavelength λ (mm)	0.00067		
LD radiation angle $\theta_\perp$° (HAHM)	14.5	LD radiation angle $\theta_\parallel$° (HAHM)	4.8
Collimator f_1 (mm)	2	d_1 (mm)	2
Concave cylindrical lens f_2 (mm)	–50	d_2 (mm)	20
Convex cylindrical lens f_3 (mm)	150	d_3 (mm)	100
Focusing lens f_4 (mm)	2	d_4 (mm)	20
HAHM: Half angle at half-maximum ⊥: The direction perpendicular to the LD junction plane. ∥: The direction parallel to the LD junction plane.			

Figure 7.14 Configuration of the laser system

Solution

(1) Obtain the Gaussian beam radius (where the irradiance is $1/e^2$ of the value at the center) of the beam after exiting the collimator lens

In Figure 7.15, the irradiance I' after exiting the collimator lens (on the secondary principal plane $\sum'$) is $1/\cos\theta$ of that of the incident beam I (on the primary principal surface $\sum$), due to the concentration of the beam by the collimator lens.

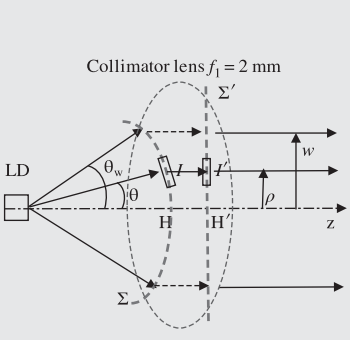
$$I' = I/\cos\theta$$

The ray height ρ after exiting the collimator will be:

$$\rho = f\sin\theta$$

We can calculate the ray height ρ and the irradiance I' of the ray after exiting the collimator, which corresponds to an incident ray with a radiation angle of HAHM. Assuming that the beam

after exiting the collimator has the irradiance pattern of a Gaussian distribution $I = \exp[-2(\rho/w)^2]$, we can calculate the Gaussian beam radius, where the value of the irradiance equals $1/e^2$ times the value at the center. Collimated Gaussian beam radii (at $1/e^2$) are obtained as $w_{||} = 0.2850008$ mm, and $w_{\perp} = 0.871199$ mm as shown in Figure 7.15.



	∥	⊥	
LD radiation angle θ_{HAHM} =	4.8	14.5°	
Irradiance right before the lens I =	0.5	0.5	
Irradiance after the lens I' =	0.5017597	0.5164502	$I' = I/\cos\theta_{HAHM}$ mm
Focal length of the lens f_1 =	2	2	
Ray height after the lens ρ =	0.1673557	0.50076	$\rho = f_1 \sin\theta_{HAHM}$ mm

∥ direction

ρ ∥ mm	ρ/w	Irradiance I'
0	0	1
	0.2	0.9231163
	0.4	0.726149
0.1673557	0.5872112	0.5017597
	0.8	0.2780373
0.2850008	1	0.1353353

⊥ direction

ρ ⊥ mm	ρ/w	Irradiance I'
0	0	1
	0.2	0.9231163
	0.4	0.726149
0.50076	0.5747941	0.5164502
	0.8	0.2780373
0.871199	1	0.1353353

at $1/e^2$

Figure 7.15 Calculation for the Gaussian beam radius exiting the collimator

(2) Obtain the beam waist size (radius) of the virtual beam emitted from the LD

Assuming that the LD emits an ideal Gaussian beam, we can calculate the virtual beam waist size (radius) w_0 at the LD emission point, by using a reverse chain of argumentation, taking as our starting point the size (radius) w of the beam after exiting the collimator (whose focal length is f). Equation (7.11) enables us to derive the beam waist size (radius) w_0 when we focus a collimated beam (of radius w) using a short focal length lens, and Equation (7.12) allows us to calculate the beam waist distance z_0:

$$w_0 \cong \frac{\lambda R}{\pi w} = \frac{\lambda f}{\pi w} \tag{7.11}$$

$$z_0 \cong R = f \tag{7.12}$$

We can regard this focused beam waist size as the waist size $w_{0||}$ and $w_{0\perp}$ of a virtual beam emitted from the LD.

$$w_{0||} = 0.0014966 \text{ mm}, \quad w_{0\perp} = 0.0004896 \text{ mm}, \quad z_0 = 2 \text{ mm}$$

(3) Calculate the focused spot size w_4' using Gaussian beam focusing equations [Equation (7.14′), Equation (7.15), and Equation (7.16)].

$$s_n = -(d_n - s_{n-1}') \qquad w_n = w_{n-1}'$$

$$s_n' = \{s_n + z_{Rn}^2/(s_n + f_n)\} f_n / \{s_n + f_n + z_{Rn}^2/(s_n + f_n)\} \tag{7.14′}$$

$$w_n' = w_n / \sqrt{\{1 + (s_n/f_n)\}^2 + (z_{Rn}/f_n)^2} \tag{7.15}$$

$$z_{Rn} = \pi w_n^2/\lambda \tag{7.16}$$

In the laser system shown in Figure 7.16, applying Equation (7.14′), Equation (7.15), and Equation (7.16), we can calculate the beam waist size (radius) w_n' and its position s_n' step by

step, until we finally obtain the beam waist size w_4' and its position s_4'. w_4' and s_4' are calculated independently in the || and ⊥ directions. The calculated results are as follows (units are in millimeters):

|| direction beam waist position s_4' (mm)	2.00016
|| direction beam waist radius w_4' (mm)	0.000494
⊥ direction beam waist position s_4' (mm)	2.000043
⊥ direction beam waist radius w_4' (mm)	0.000489

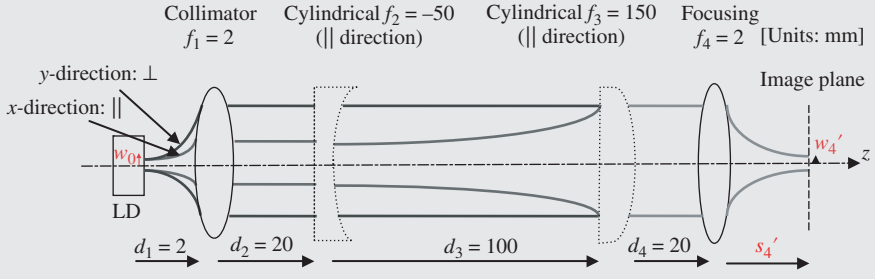


Figure 7.16 Configuration of the laser system

The radius and position of the beam waist in the || direction and the ⊥ direction are almost identical, which means that the beam waist is almost perfectly round.

Notes

1. In this problem we did not take into account the truncation of the beam, caused by a lens. If the beam is truncated by a lens aperture, the focused beam waist size will be larger than the calculated value (see Problem 9.4 in Chapter 9).

Problem 7.4 Generating a line spot

Configure an optical system which generates a line spot 0.01 mm in diameter ($1/e^2$ irradiance) in the x-direction and 0.2 mm in diameter in the y-direction from a round incident Gaussian beam of 0.68 mm in diameter. (Assume that $\lambda = 532$ nm.)

Solution 1: Beam expander

We can configure an optical system with a beam expander in the x-direction (two cylindrical lenses with focal lengths f_1 and f_2) and a focusing lens f_3, as illustrated in Figure 7.17.

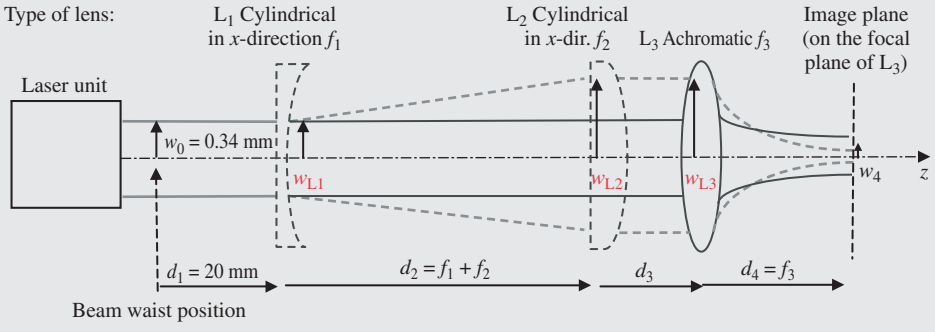


Figure 7.17 Generating a line spot using a beam expander

Assuming that $w_{L3y} = w_{0y}$, we can roughly estimate w_{4y} using Equation (7.11), where $f_3 = 200$ mm.

$$w_{4y} \cong \lambda R/(\pi w_{L3y}) = \lambda f_3/(\pi w_{L3y}) = 0.099612 \text{ mm}$$

$w_{4y} \fallingdotseq 0.1$ mm satisfies the requirements for this problem. If we set the x-expander to 20×, the beam radius at L_3 becomes $w_{L3x} \fallingdotseq 20 w_{0x} = 20 \times 0.34$ mm $= 6.8$ mm, and the radius of the focused spot in the x-direction can be calculated using Equation (7.11), as follows:

$$w_{4x} \cong \lambda R/(\pi w_{L3x}) = \lambda f_3/(\pi w_{L3x}) = 0.004981 \text{ mm}$$

We obtained $w_{4x} \approx 0.005$mm, which satisfies the requirement. The optical configuration will be:

$$f_1 = -15 \text{ mm}, \quad f_2 = 300 \text{ mm}, \quad f_3 = 200 \text{ mm}$$
$$d_1 = 20 \text{ mm}, \quad d_2 = 285 \text{ mm}, \quad d_3 = 10 \text{ mm}, \quad d_4 = 200 \text{ mm}$$

Next, we can verify the output spot size w_{4x} and w_{4y} by recalculating the spot size using the Gaussian beam focusing Equation (7.14′), Equation (7.15), and Equation (7.16), as well as Equation (7.2) as follows:

$$s_n = -(d_n - s_{n-1}') \qquad w_n = w_{n-1}'$$

$$s_n' = \{s_n + z_{Rn}^2/(s_n + f_n)\} f_n / \{s_n + f_n + z_{Rn}^2/(s_n + f_n)\} \tag{7.14′}$$

(Beam waist position after the nth lens)

$$w_n' = w_n / \sqrt{\{1 + (s_n/f_n)\}^2 + (z_{Rn}/f_n)^2} \quad \text{(Beam waist radius after the } n\text{th lens)} \tag{7.15}$$

$$z_{Rn} = \pi w_n^{\,2}/\lambda \quad \text{(Rayleigh range)} \tag{7.16}$$

$$w_{Ln} = w_n \sqrt{1 + \{\lambda s_n/(\pi w_n^{\,2})\}^2} \quad \text{(Beam radius at the } n\text{th lens)} \tag{7.2}$$

The calculation results in the x-direction are $s_{3x}' = 200.00723$, $w_{3x}' = 0.0049745$ mm. This means that the beam waist position lies precisely on the image plane, and the spot size on the image plane will be $w_{4x} \approx w_{3x}' = 0.00497$ mm. The calculation results in the y-direction are $s_{3y}' = 209.599, w_{3y}' = 0.0982$ mm. This means that the beam waist position is $(z = 209.599 - 200 =)$ 9.599 mm behind the image plane, and the beam radius on the image plane will be $w_{4y} = w_{3y}'\sqrt{[1 + (\lambda z/(\pi w_{3y}'^2))^2]} = 0.0996$ mm. Here, we have obtained almost the same results as we did in the previous calculation. Additionally, the values we obtained satisfy the requirements for this problem.

Solution 2: Cylindrical lenses, without using a high magnification ratio expander

Here, we shall configure an alternative optical system, which satisfies the requirements of this problem, without using a high magnification ratio expander.

We can configure an optical system with a 5 × and 6.5 × magnification beam expander in the x- and y-directions, respectively, and a focusing lens with focal length f_4, as shown in Figure 7.18. Here we shall deliberately move the focusing point of y-direction a little way along the z-axis, in order to obtain a spot which is elongated in the y-direction on the image plane. We have selected a value of 50 mm for the focal length of f_4, in order to obtain a spot with the same width in the x-direction as the spot described in Solution 1.

$$w_{5x} \cong \lambda R/(\pi w_{L4x}) = \lambda f_4/(\pi w_{L4x}) = 0.004981 \text{ mm}$$

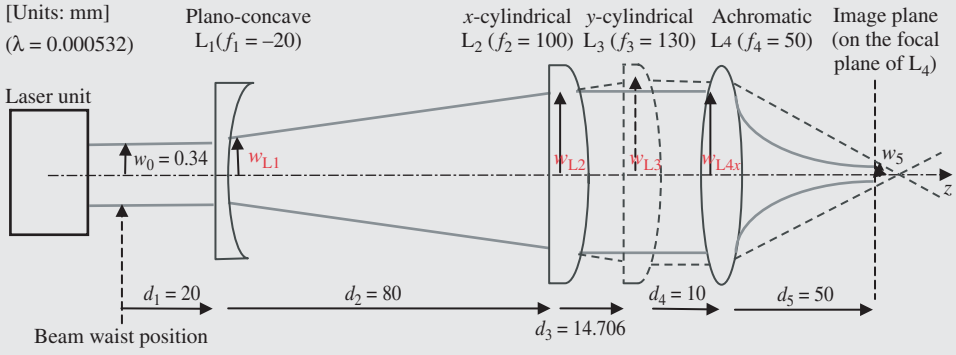


Figure 7.18 Generating a line spot without using a high magnification ratio expander

Ray tracing calculation in the y-direction, using paraxial theory to obtain the value of d_3

$$w_n = w_{n-1} + d_n\theta_{n-1} \tag{3.1}$$

$$\theta_n = \theta_{n-1} - w_n/f_n \tag{3.2}$$

We can perform ray tracing calculations, using paraxial theory [Equation (3.1) and Equation (3.2)]. If we let $d_3 = 14.706$ mm, then w_{5y} will be 0.1 mm. In practice, d_3 needs to be slightly

greater than 14.706 mm, due to the aberration generated by the thickness of the lens f_2 (see Problem 2.4 in Chapter 2).

Gaussian beam focusing calculation

We can verify the output spot size w_{5x} and w_{5y} by recalculating the spot size using the Gaussian beam focusing equations [Equation (7.14′), Equation (7.15), Equation (7.16), and Equation (7.2)]. We obtain values in the x-direction of $s_{4x}{}' = 50.00749$, $w_{4x}{}' = 0.004974$ mm. This means that the beam waist position is just at the image plane, and the spot size on the image plane will be $w_{5x} \approx w_{4x}{}' = 0.004974$ mm, which satisfies the requirement that $w_{5x} = 0.005$ mm. We obtain values in the y-direction of $s_{4y}{}' = 52.68001$, $w_{4y}{}' = 0.004974$ mm, and w_{5y} can be derived by applying Equation (7.2), as follows:

$$w_{5y} = w_{4y}{}'\sqrt{1 + [\lambda z/(\pi w_{4y}{}'^2)]^2} = 0.100468 \text{ mm} \quad \text{where } z = s_4{}' - d_5 = 2.68001 \text{ mm}$$

This value of w_{5y} also satisfies the specifications for the problem. The converging point in the y-direction lies on the plane $z = 2.68$ mm, after exiting the image plane.

Notes

1. If we use two or more cylindrical lenses in an optical system, we need to align each lens's rotation angle around the optical axis very precisely, or else we will be unable to generate a micro spot at the focus (see Problem 5.4 in Chapter 5).
2. We need to calculate the aberration value of the optical system we designed, to ensure that the aberration falls within a permissible range (see Problem 5.6 in Chapter 5).

Problem 7.5 A comparison of two Gaussian beams with M^2 factors of 1 and 1.3 respectively

Compare two Gaussian beams whose M^2 factors are 1 and 1.3, respectively, whose wavelength is 405 nm and whose beam waist radius is 10 μm (11.40175 μm) for the beam where $M^2 = 1$ ($M^2 = 1.3$).

Solution

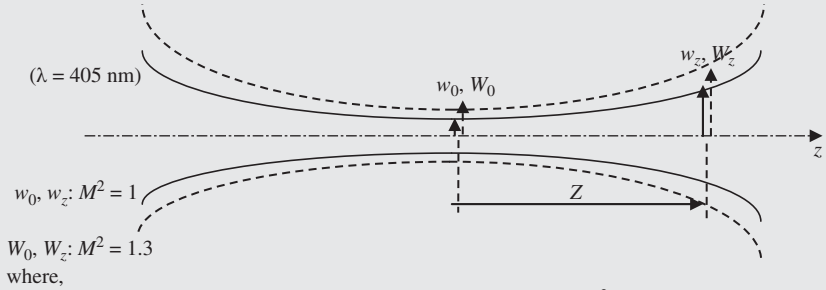
When we compare two Gaussian beams whose M^2 factors are 1 and 1.3, respectively, we can see that the beam width of the later is $\sqrt{1.3}$ times wider than that of the former, everywhere along the z-axis as shown in Figure 7.19.

The beam radius W_z, the curvature radius of the wave front R_z and the Rayleigh range Z_R can be expressed as follows:

$$W_z{}^2 = W_0{}^2\left[1 + \left\{\frac{M^2\lambda z}{\pi W_0{}^2}\right\}^2\right] \tag{7.24}$$

$$R_z = z\left[1 + \left\{\frac{\pi W_0{}^2}{M^2\lambda z}\right\}^2\right] \tag{7.23}$$

$$Z_R = \frac{\pi W_0{}^2}{M^2\lambda} \tag{7.27}$$



where,

w_0 (=10 μm) = Beam waist radius for a Gaussian beam whose M^2 factor =1
W_0 (Mw_0 = 11.40175 μm) = Beam waist radius for a Gaussian beam whose M^2 factor =1.3
w_z = Beam radius at a distance Z from the beam waist whose M^2 factor =1
W_z (=Mw_z) = Beam radius at a distance Z from the beam waist whose M^2 factor =1.3

Figure 7.19 Two Gaussian beams whose M^2 factors are 1 and 1.3

The calculated values of W_z and R_z for the above beams are shown on the graphs in Figure 7.20a,b.

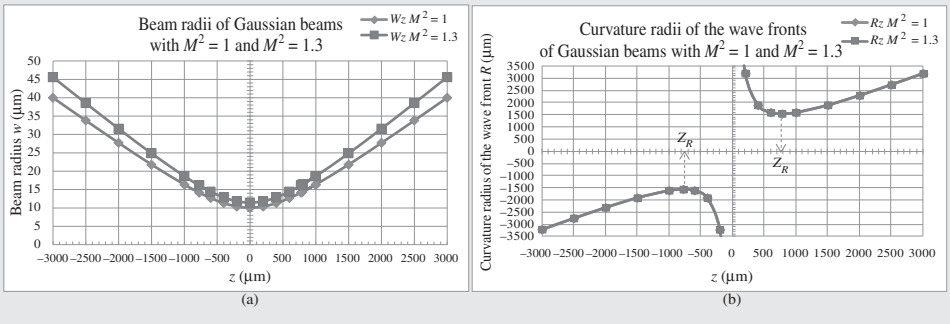


Figure 7.20 (a) Beam radii and (b) wave-front curvature radii of two Gaussian beams with $M^2 = 1$ and $M^2 = 1.3$

Notes

1. A beam with an M^2 factor of 1.3 is $\sqrt{1.3}$ times wider than a beam with M^2 factor of 1, everywhere along the z-axis. However, the curvature radii of the wave fronts of these two beams will have identical values, all the way along the z-axis.
2. In the Gaussian beam focusing equations, we can take the M^2 factor into account simply by replacing λ with $M^2\ \lambda$. The beam waist distance s' after exiting the lens and the beam waist radius W_0 after the lens can be expressed as follows. (For more details, see the Excel calculation files on the companion web site.)

$$\frac{1}{s'} = \frac{1}{s + \dfrac{Z_R^2}{s+f}} + \frac{1}{f} \tag{7.14}$$

$$W_0{}' = \frac{W_0}{\sqrt{\left(1 + \dfrac{s}{f}\right)^2 + \left(\dfrac{Z_R}{f}\right)^2}} \tag{7.15}$$

$$Z_R = \frac{\pi W_0^2}{M^2 \lambda} \tag{7.27}$$

References

[1] Kudo, K. and Uehara, F. (1990) *Kiso-kogaku*, Gendaikogakusya Co., Ltd, Japan (in Japanese).
[2] Suhara, T. (1998) *Koha-kogaku*, Corona Publishing Co., Ltd, Japan (in Japanese).
[3] Hayamizu, Y. (1989) *Hikari-kiki no kogaku* II, Optomechatronics Association, Japan (in Japanese).
[4] Matsui, Y. (1988) *Kogaku-kei toriatsukai no kiso*, Keigaku Publishing Co., Ltd, Japan (in Japanese).
[5] Melles Griot (2003) *Optical Parts Catalogue*, Melles Griot Co., Ltd, Japan.

8

Diffraction: Theory and Calculations

In the previous chapter, we discussed calculation methods that can be used to focus a Gaussian beam. However, the same results as those listed in the previous chapter can be obtained using diffraction theory. In this chapter, we will examine the concepts and the basic principles of diffraction phenomena. An opaque body with an aperture of any shape (e.g., a slit or a pinhole) placed midway between a point (light) source and an observation point will cause light to deviate from its customary mode of rectilinear propagation, which it displays when unimpeded. The term "diffraction theory" includes both Fresnel diffraction and Fraunhofer diffraction. The former can be defined as the theoretical treatment of diffraction for cases in which the source and/or the observation point are not very distant from the aperture, while the latter theory deals with the special case in which the source of light and the observation point are both very distant from the aperture. In this chapter, we will confine ourselves to a discussion of Fraunhofer diffraction, which has a multitude of practical, everyday applications. We shall use the examples of a slit, a rectangular aperture and a circular aperture to illustrate our discussion.

8.1 The Concept of Diffraction

Diffraction is a common phenomenon which can occur in any kind of wave. Waves, after passing through a small aperture, will spread out in all directions. We refer to this phenomenon as diffraction. To illustrate with a simple example: if a pebble is dropped into still water, it generates a wave spreading out from the point where it hit the surface of the water (Figure 8.1). The wave will spread out in all directions. If we repeat the experiment after placing a small slit on the surface of the water, the wave, after passing through the slit, will generate a secondary circular wavelet (diffraction wave) spreading out from the aperture.

If this experiment is performed in a three-dimensional space, the wave generated will be a spherical wave.

8.1.1 Huygens' Explanation of Wave Propagation Phenomena

In Figure 8.2, the wave from the point of origin O advances, spreading outwards with a spherical wave front. When the wave front reaches the sphere S, each tiny segment of the wave front S will generate new wavelets (secondary spherical waves). These secondary waves, when superposed (after taking into

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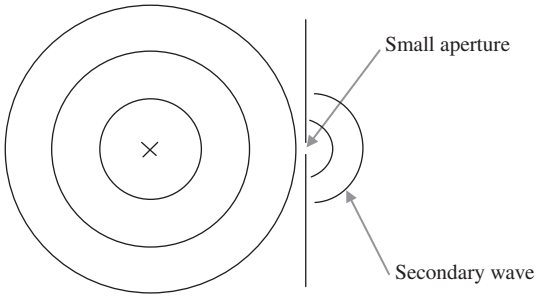


Figure 8.1 Diffraction wave generated by a small aperture on the surface of a body of water

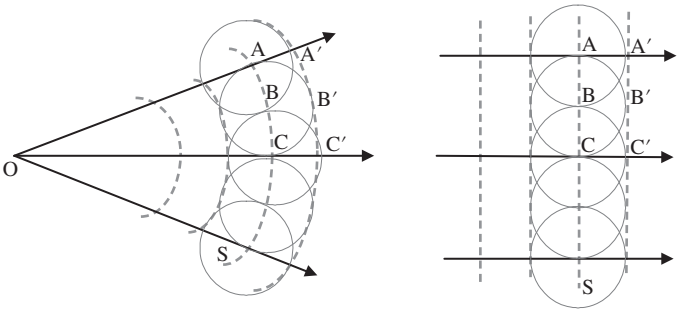


Figure 8.2 Huygens–Fresnel principle

account their amplitudes and relative phases), yield a spherical wave A′ B′ C′ as an envelope of the wavelets. The spherical wave A′ B′ C′ will in turn generate new wavelets, which form the next spherical wave. If these processes are continually repeated, they generate a light wave which propagates forward over time. At a point very distant from the origin, the curvature of the wave front will appear almost flat, like a plane wave (see Figure 8.2, right-hand diagram).

We can explain these diffraction phenomena using the Huygens–Fresnel principle. Think of a plane wave (a wave having a plane wave front) which is irradiated onto a slit at a perpendicular angle as in Figure 8.3. When the plane wave reaches the slit, it generates secondary circular wavelets at each infinitesimally small segment across the aperture. Near the center of the slit, these secondary waves, when superposed, result in a plane wave. However, near the edge of the slit, the secondary wavelets cannot generate a plane wave, because one half of the incident wave is blocked by the wall and the contributions of the different wavelets cannot generate a plane wave, as they do in the center of the slit. When the wave front moves toward the wall, it spreads out from the edge of the slit. We refer to these phenomena as "diffraction." The diffraction angle increases as the slit aperture decreases, or alternatively, as the wavelength increases.

In Figure 8.3, the amplitude of the wave which reaches the observation point P can be calculated by adding up all the contributions from the wave in the plane of the aperture, which generates secondary spherical wavelets originating from each tiny segment across the aperture. The (wave) amplitude at point P can thus be calculated by superimposing all the secondary wavelets from the plane of the aperture which reach point P. In this chapter, we will only be discussing Fraunhofer diffraction, so both the light source and the observation point can be regarded as being very distant from the aperture. In this case,

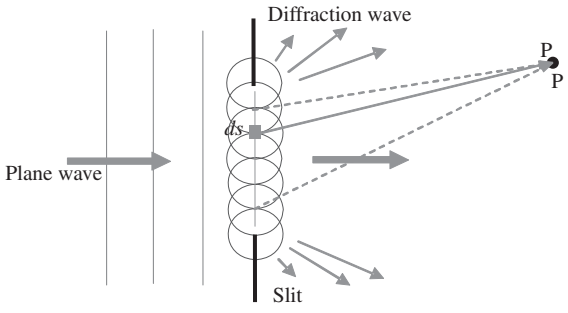


Figure 8.3 Diffraction at a slit

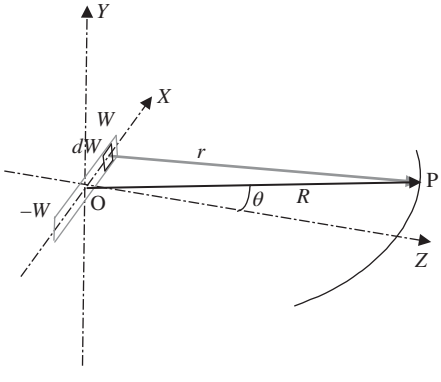


Figure 8.4 Diffraction from a slit aperture

light which is incident onto the aperture will be a plane wave, and the phase of the secondary wavelets will be the same. Below, we will discuss how the diffraction wave can be calculated for various different shapes of the aperture [1, 2].

8.2 Diffraction at a Slit Aperture

Figure 8.4 shows a slit being irradiated with a plane wave, where the X-axis corresponds to the length of the slit and the Z-axis to the direction of light propagation . The interval $\pm W$ represents the length of the slit (from $-W$ to $+W$). The point P is located on the X–Z plane, at a distance R from the center of the aperture. As we showed above, the electric field dU at P, associated with a differential segment dW of the wave light source will be:

$$dU = \frac{\varepsilon_L}{r}\sin(\omega t - kr)\, dW \cong \frac{\varepsilon_L}{R}\sin(\omega t - kr)\, dW \tag{8.1}$$

where

r is the distance from the differential segment dW on the slit to point P,
R is the distance from the center of the aperture O to point P,
k is the propagation number, $k = 2\pi/\lambda$,
λ is the wavelength, and
ε_L is the source strength per unit length.

As we showed in Chapter 6, the field at P can be expressed as a wave equation of the form $\sin(\omega t - kr)$, whose coefficient ε_L/r is inversely proportional to the distance r. As the point P is very distant from the aperture, r is almost the same as R, and ε_L/r can be replaced with ε_L/R. Now let θ be the angle between OP and the Z-axis. If $r \gg W$, we can make a further approximation:

$$r \cong R - x\sin\theta \tag{8.2}$$

We can then derive the electric field U at P by integrating dU over the entire aperture length, from $-W$ to $+W$, as follows:

$$U = \frac{\varepsilon_L}{R}\int_{-W}^{W} \sin[\omega t - k(R - x\sin\theta)]dx \tag{8.3}$$

If the function described in Equation (8.3) is integrated over the entire length of the aperture [see "Step-by-step calculation of the integral in Equation (8.3)" below for supporting calculations], then we obtain the result:

$$U = \frac{\varepsilon_L}{R}2W\frac{\sin(kW\sin\theta)}{kW\sin\theta}\sin(\omega t - kR) \tag{8.4}$$

If we let

$$\beta \equiv kW\sin\theta \tag{8.5}$$

then Equation (8.4) can be rewritten as follows:

$$U = 2\frac{W\varepsilon_L}{R}\frac{\sin\beta}{\beta}\sin(\omega t - kR) \tag{8.6}$$

The irradiance I can be expressed as

$$I = <|U|^2>$$

where $<>$ denotes the average value over time. It then follows that

$$I = |U|^2/2$$

$$\because <\sin^2(\omega t - kR)> = 1/2 \tag{8.7}$$

The irradiance $I(\theta)$ at P will therefore be:

$$I(\theta) = \frac{1}{2}\left(\frac{2\varepsilon_L W}{R}\right)^2\frac{\sin^2\beta}{\beta^2} = I(0)\frac{\sin^2\beta}{\beta^2} \tag{8.8}$$

where I (0) is the irradiance when P lies on the Z-axis.

When $\theta = 0$, $\sin^2\beta/\beta^2$ reaches its maximum value of 1.

Equation (8.8) thus shows that the irradiance is inversely proportional to the square of the distance, and that it is a function of $\sin^2\beta/\beta^2$.

Step-by-step Calculation of the Integral in Equation (8.3)

$$\begin{aligned}\int_{-W}^{W} \sin[\omega t - k(R - x\sin\theta)]dx &= \int_{-W}^{W} \sin[kx\sin\theta + \omega t - kR]dx \\ &= -\frac{1}{k\sin\theta}[\cos(kx\sin\theta + \omega t - kR)]_{-W}^{W} \\ &= -\frac{1}{k\sin\theta}[\cos(kW\sin\theta + \omega t - kR) - \cos(-kW\sin\theta + \omega t - kR)] \\ &= -\frac{1}{k\sin\theta}[-2\sin(\omega t - kR)\sin(kW\sin\theta)] \\ &= 2W\frac{\sin(kW\sin\theta)}{kW\sin\theta}\sin(\omega t - kR)\end{aligned}$$

8.3 Diffraction Calculations Using Numerical Integration

Even if the integral in Equation (8.3) had an algebraic form which could not be straightforwardly integrated, we could still calculate the integral, using the mathematical technique of numerical integration. In this section, we will obtain an approximation of Equation (8.3), using numerical integration.

$$U = \frac{\varepsilon_L}{R}\int_{-W}^{W} \sin[\omega t - k(R - x\sin\theta)]dx \tag{8.3}$$

Rewriting Equation (8.3) in the form $e^{j\psi}$, we obtain:

$$U = \frac{\varepsilon_L}{R}\int_{-W}^{W} \exp[j\{\omega t - k(R - x\sin\theta)\}]dx = \frac{\varepsilon_L}{R}\exp[j(\omega t - kR)]\int_{-W}^{W} \exp[j(kx\sin\theta)]dx \tag{8.9}$$

Here we have moved $\exp[j(\omega t - kR)]$ outside the integral, because it is independent of the derivative dx. Rewriting the integral in Equation (8.9), and replacing the exponential function with cos and sin functions, we get:

$$\int_{-W}^{W} \exp[j(kx\sin\theta)]dx = \int_{-W}^{W} [\cos(kx\sin\theta) + j\sin(kx\sin\theta)]dx \tag{8.9′}$$

We can use numerical integration to perform an integral calculation on the complex function described in Equation (8.9′). The sum of two complex numbers can be calculated by adding their real and imaginary components independently. In the Argand diagram (Figure 8.5):

$$\mathrm{OP}_1 = \cos\psi_1 + j\sin\psi_1 \quad \mathrm{OP}_2 = \cos\psi_2 + j\sin\psi_2$$

The addition of the two complex numbers above can be performed using the same method as for vector addition, that is:

$$\mathrm{OP} = \mathrm{OP}_1 + \mathrm{OP}_2 = \cos\psi_1 + \cos\psi_2 + j(\sin\psi_1 + \sin\psi_2)$$

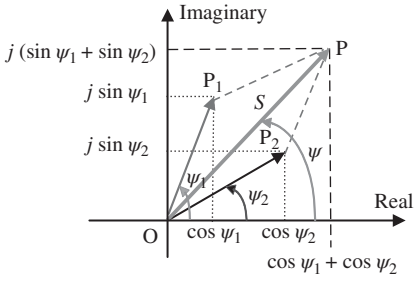


Figure 8.5 Vector addition in an Argand diagram

The amplitude and the phase of the sum will be:

$$\text{Amplitude} = \sqrt{(\cos\psi_1 + \cos\psi_2)^2 + (\sin\psi_1 + \sin\psi_2)^2}$$

$$\text{Phase} = \tan^{-1}\left[\frac{\sin\psi_1 + \sin\psi_2}{\cos\psi_1 + \cos\psi_2}\right]$$

Likewise, we can calculate the integral in Equation (8.9′) as the sum of the vectors in the Argand diagram. Adding the real parts and the imaginary parts independently, the integral in Equation (8.9′) can be calculated as:

$$\text{Total amplitude } S = \sqrt{(\Sigma_i \cos\psi_i)^2 + (\Sigma_i \sin\psi_i)^2}\Delta x \tag{8.10}$$

$$\text{Total phase } \psi = \tan^{-1}\left[\frac{\Sigma_i \sin\psi_i}{\Sigma_i \cos\psi_i}\right] \tag{8.11}$$

Substituting the numerical integration results obtained above into Equation (8.9), we get:

$$U = \frac{\varepsilon_L}{R} S \exp[j(\omega t - kR + \psi)] \tag{8.12}$$

We can rewrite Equation (8.12) using the cosine and sine functions:

$$U = \frac{\varepsilon_L}{R} S[\cos(\omega t - kR + \psi) + j\sin(\omega t - kR + \psi)] \tag{8.12′}$$

For representing a wave, we can choose either the real component (a cosine function) or the imaginary component (a sine function) of Equation (8.12′). Here, we will represent this wave by using the imaginary component (a sine function or sine wave) which has the same form as the result we obtained in Section 8.2.

$$U = \frac{\varepsilon_L}{R} S\ \sin(\omega t - kR + \psi) \tag{8.13}$$

In the case of Equation (8.9′), calculations for the phase ψ yield the result: $\psi = 0$ ($\because \Sigma_i \sin\psi_i = 0$). See Problem 8.2 for an actual calculation example using numerical integration.

8.4 Diffraction at a Rectangular Aperture

In Figure 8.6, a plane wave is irradiated onto a $(2a \times 2b)$ rectangular aperture from a source on the left. Point P is situated on a screen located at a distance Z from the aperture. The electric field dU at P which

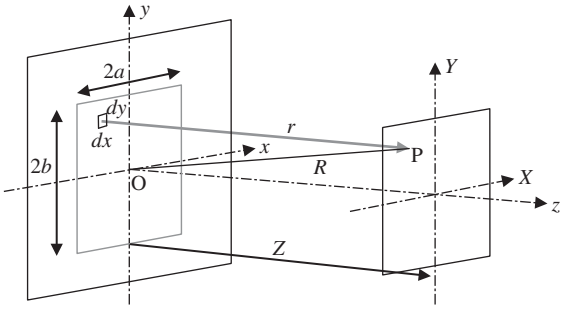


Figure 8.6 Diffraction at a rectangular aperture

is associated with the differential component of the field at the aperture ($dxdy = dS$) can be expressed as follows:

$$dU = \frac{\varepsilon_A}{r} \exp[j(\omega t - kr)]dS \tag{8.14}$$

$$r = \sqrt{(X - x)^2 + (Y - y)^2 + Z^2} \tag{8.15}$$

where

ε_A = Source strength at dS, if we treat the aperture as the source of the secondary wave
r = Distance from dS to P

The distance R from the center of the aperture to P is:

$$R = \sqrt{X^2 + Y^2 + Z^2} \tag{8.16}$$

Combining Equation (8.15) and Equation (8.16), the distance r can be expressed as follows:

$$r = R\sqrt{1 + \frac{x^2 + y^2}{R^2} - \frac{2(Xx + Yy)}{R^2}} \tag{8.17}$$

In the case when $R \gg x$, $R \gg y$, in Equation (8.17) we can approximate r by the following formula:

$$r \cong R\sqrt{1 - \frac{2(Xx + Yy)}{R^2}} \tag{8.18}$$

This case is commonly referred to as Fraunhofer diffraction. If we retain only the first two terms in the binomial expansion, we get:

$$r \cong R\left[1 - \frac{2\,(Xx + Yy)}{R^2}\right]^{1/2} \cong R\left[1 - \frac{Xx + Yy}{R^2}\right] = R - (Xx + Yy)/R \tag{8.19}$$

The electric field U at P can be calculated by integrating dU over the entire area of the aperture. Using Equation (8.14) and Equation (8.19), U can be derived:

$$U = \iint_{Aperture} \frac{\varepsilon_A}{r} e^{j(\omega t - kr)} dS = \frac{\varepsilon_A}{R} e^{j(\omega t - kR)} \iint_{Aperture} e^{jk(Xx + Yy)/R} dS \tag{8.20}$$

If we now define $dS = dxdy$, then we can rewrite Equation (8.20) as follows:

$$U = \frac{\varepsilon_A}{R} e^{j(\omega t - kR)} \int_{-a}^{a} e^{jkXx/R} dx \int_{-b}^{b} e^{jkYy/R} dy \tag{8.21}$$

If we define

$$\alpha' \equiv kaX/R \tag{8.22a}$$

$$\beta' \equiv kbY/R \tag{8.22b}$$

then

$$\int_{-a}^{a} e^{jkXx/R} dx = a\frac{e^{j\alpha'} - e^{-j\alpha'}}{j\alpha'} = 2a\frac{\sin\alpha'}{\alpha'} \tag{8.23}$$

Likewise,

$$\int_{-b}^{b} e^{jkYy/R} dy = b\frac{e^{j\beta'} - e^{-j\beta'}}{j\beta'} = 2b\frac{\sin\beta'}{\beta'} \tag{8.24}$$

$$\because e^{jx} - e^{-jx} = \cos x + j\sin x - [\cos(-x) + j\sin(-x)] = 2j\sin x$$

From the above, it follows that:

$$U = \frac{A\varepsilon_A}{R} e^{j(\omega t - kR)} \frac{\sin\alpha'}{\alpha'} \frac{\sin\beta'}{\beta'} \tag{8.25}$$

where A is the area of the aperture, $A = 4ab$. The irradiance I will be $I = < |U|^2 >$. Hence,

$$I(X,Y) = I(0,0)\left[\frac{\sin\alpha'}{\alpha'}\right]^2 \left[\frac{\sin\beta'}{\beta'}\right]^2 \tag{8.26a}$$

$$I(0,0) = \frac{A^2 {\varepsilon_A}^2}{2R^2} \tag{8.26b}$$

where $I(0,0)$ is the irradiance at the point where $X = 0, Y = 0$.

Let us now examine the relationship between the diffraction patterns generated by a slit and those generated by a rectangular aperture. In Equation (8.26a):

$$\lim_{x\to 0} \frac{\sin x}{x} = 1$$

When $X = 0$, then $\alpha' = 0$. Likewise, when $Y = 0$, then $\beta' = 0$. Thus $I(0,Y)$ and $I(X,0)$ will have the same form as the equation for a slit, Equation (8.8).

8.5 Diffraction at a Circular Aperture

As we saw in the previous section, the electric field which is generated at an aperture can be expressed by Equation (8.20), in the case when a plane wave is irradiated onto the aperture and the observation point is located at a great distance from the aperture:

$$U = \frac{\varepsilon_A}{R} e^{j(\omega t - kR)} \iint_{Aperture} e^{jk(Xx+Yy)/R} dS \tag{8.20}$$

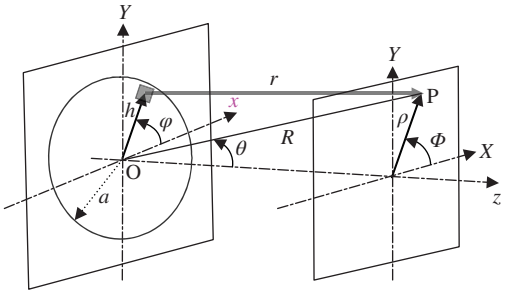


Figure 8.7 Diffraction at a circular aperture

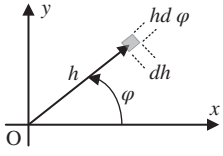


Figure 8.8 Polar coordinates on the aperture plane

Since the aperture is circular, we will use polar coordinates for both the aperture plane and the observation plane as shown in Figure 8.7 and Figure 8.8.

$$x = h\ \cos\varphi, \qquad y = h\ \sin\varphi, \qquad dS = hdhd\varphi \quad \text{on the aperture plane}$$

$$X = \rho\ \cos\Phi, \qquad Y = \rho\ \sin\Phi \quad \text{on the observation plane}$$

If we rewrite Equation (8.20), applying the above equations, we get:

$$U = \frac{\varepsilon_A}{R} e^{j(\omega t - kR)} \int_{h=0}^{a} \int_{\varphi=0}^{2\pi} e^{j(kh\rho/R)\cos(\varphi-\Phi)} hdhd\varphi \tag{8.27}$$

In Equation (8.27), the aperture is symmetrical about the z-axis, so the solution U must also be symmetrical about the z-axis. Thus U must be independent of Φ. In order to simplify the equation, we will arbitrarily let $\Phi = 0$. The portion of the double integral associated with the variable φ is:

$$\int_{\varphi=0}^{2\pi} e^{j(kh\rho/R)\cos\varphi} d\varphi \tag{8.28}$$

Inspecting the above integral, we observe that it has the form of a Bessel function,

$$J_0(p) = \frac{1}{2\pi} \int_0^{2\pi} e^{jp\cos q} dq \tag{8.29}$$

which is a Bessel function of order zero [3]. Generalizing, we can write:

$$J_m(p) = \frac{j^{-m}}{2\pi} \int_0^{2\pi} e^{j(mq + p\cos q)} dq \tag{8.30}$$

which represents a Bessel function of order m. Rewriting Equation (8.27) using the Bessel function Equation (8.29), we get:

$$U = \frac{\varepsilon_A}{R} e^{j(\omega t - kR)} 2\pi \int_0^a J_0\left(\frac{kh\rho}{R}\right) h dh \tag{8.31}$$

Bessel functions have the interesting property that the order of their derivatives is one less than that of the original function, as can be seen in the following equation:

$$d/dx[x^m J_m(x)] = x^m J_{m-1}(x) \tag{8.32}$$

When $m = 1$, it is immediately apparent that

$$\int_0^x x' J_0(x') dx' = x J_1(x) \text{ (where } x' \text{ is a dummy variable)} \tag{8.33}$$

Rewriting the integral in Equation (8.31) using the variable $\zeta = kh\rho/R$, that is, $dh = [R/(k\rho)]d\zeta$, we get:

$$\int_{h=0}^a J_0\left(\frac{kh\rho}{R}\right) h dh = \left(\frac{R}{k\rho}\right)^2 \int_{\xi=0}^{ka\rho/R} J_0(\zeta)\zeta d\zeta \tag{8.34}$$

$$\because h dh = R/(k\rho)\zeta R/(k\rho) d\zeta = [R/(k\rho)]^2 \zeta d\zeta \tag{8.35}$$

Rewriting Equation (8.31) using Equation (8.33) and Equation (8.34), we obtain:

$$U(t) = \frac{\varepsilon_A}{R} e^{j(\omega t - kR)} 2\pi a^2 \frac{R}{ka\rho} J_1\left(\frac{ka\rho}{R}\right) = \frac{\varepsilon_A \pi a^2}{R} \frac{2J_1\left(\frac{ka\rho}{R}\right)}{\frac{ka\rho}{R}} e^{j(\omega t - kR)} \tag{8.36}$$

$$\because \left(\frac{R}{k\rho}\right)^2 \frac{ka\rho}{R} = a^2 \frac{R}{ka\rho} \tag{8.37}$$

We have obtained an electric field generated at a circular aperture, as in Equation (8.36). The irradiance I can be calculated as follows, using $I = < |U|^2 >$ or $I = 1/2UU^*$.

$$I = \frac{{\varepsilon_A}^2 A^2}{2R^2} \left[\frac{2J_1\left(\frac{ka\rho}{R}\right)}{\frac{ka\rho}{R}}\right]^2 \qquad A = \pi a^2 = \text{the area of the aperture} \tag{8.38}$$

Since it can be proved that $\lim_{x\to 0} 2J_1(x)/x = 1$ (see 'Proof that $\lim_{x\to 0} 2J_1(x)/x = 1$'), when $\rho = 0$, the value of the right-hand term in Equation (8.38) will be:

$$\left[\frac{2J_1(0)}{0}\right]^2 = 1$$

Thus when $\rho = 0$, I can be expressed as follows:

$$I(0) = \frac{{\varepsilon_A}^2 A^2}{2R^2}$$

Hence Equation (8.38) can be rewritten as:

$$I(\rho) = I(0)\left[\frac{2J_1\left(\frac{ka\rho}{R}\right)}{\frac{ka\rho}{R}}\right]^2 \tag{8.39}$$

$$\text{or, } I(\theta) = I(0)\left[\frac{2J_1(ka\sin\theta)}{ka\sin\theta}\right]^2 \tag{8.40}$$

$$\because \sin\theta = \rho/R \tag{8.41}$$

Proof that $\lim_{x\to 0} 2J_1(x)/x = 1$

When $m = 1$, Equation (8.32) can be written as:

$$d/dx[xJ_1(x)] = xJ_0(x)$$

Thus

$$J_0(x) = \frac{d}{dx}J_1(x) + \frac{J_1(x)}{x}$$

We can see that $J_0(0) = 1$ and $J_1(0) = 0$ (see Problem 8.4). In the above equation, the ratio of $J_1(x)/x$ as x approaches zero has the same limit as the ratio of the separate derivatives of its numerator and denominator (L'Hôpital's rule). Here, the derivative of the numerator is identical to the first term of the above equation and the derivative of the denominator is equal to 1. Hence,

$$\lim_{x\to 0}\frac{J_1(x)}{x} = \frac{1}{2} \qquad \lim_{x\to 0}\frac{2J_1(x)}{x} = 1$$

Bessel Function

A Bessel function of order n can be represented as an infinite series [5]:

$$J_n(x) = \sum_{k=0}^{\infty}\frac{(-1)^k x^{n+2k}}{2^{n+2k}k!(n+k)!} \qquad n = 0, 1, 2, \cdots \tag{8.42}$$

For instance, a Bessel function $J_0(x)$ of order 0 and a Bessel function $J_1(x)$ of order 1 have the following expanded forms:

$$J_0(x) = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^4(2!)^2} - \frac{x^6}{2^6(3!)^2} + \cdots + (-1)^k\frac{x^{2k}}{2^{2k}(k!)^2} + \cdots \tag{8.43}$$

$$J_1(x) = \frac{x}{2} - \frac{x^3}{2^3 2!} + \frac{x^5}{2^5 2!3!} - \frac{x^7}{2^7 3!4!} + \cdots + (-1)^k\frac{x^{2k+1}}{2^{2k+1}k!(k+1)!} + \cdots \tag{8.44}$$

8.6 Diffraction Wave Generated after the Incident Wave Exits a Focusing Lens

In this section, we shall examine a diffraction wave generated after the incident wave exits a focusing lens. As we have learned in this chapter, the irradiance pattern of diffraction waves on the observation

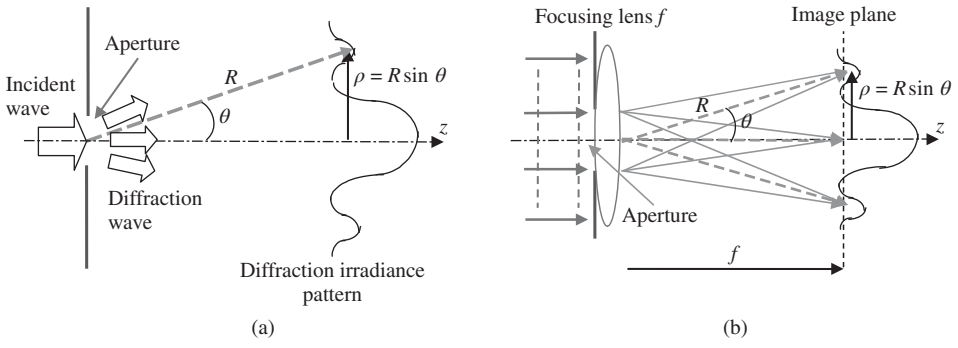


Figure 8.9 (a) Diffraction by an aperture with no lens. (b) Diffraction by an aperture with a focusing lens

plane can be expressed as a function of the angle θ between the diffraction ray and the z-axis. This means that the farther away we situate the observation plane, the wider the diffraction pattern will be. If we place a focusing lens f on the aperture plane as in Figure 8.9b, the incident rays will be focused at the focal point, and the diffraction pattern will appear on the focal plane as a function of the angle θ, whose size ρ satisfies the equation $\rho = f \sin\theta$.

$$\rho = R\sin\theta \cong f\sin\theta \tag{8.45}$$

Previously, we obtained a Fraunhofer diffraction pattern at far field (the exact pattern is shown at an infinite distance). However, if we insert a lens at the aperture plane, we can obtain exactly the same diffraction pattern on the focal plane of the lens (see Chapter 9 and Appendix F for a more detailed explanation).

8.7 Diffraction Calculation Problems

Please download the Excel® files from the companion web site, which show calculation details for all problems in this book as listed in Appendix G

Problem 8.1 Diffraction caused by a slit

Calculate the far-field irradiance distribution of the diffraction wave when a plane wave ($\lambda = 532$ nm) is irradiated onto a slit of length $W = \pm 0.05$ mm as shown in Figure 8.10. Repeat the calculations for a slit of width $W = \pm 0.1$ mm.

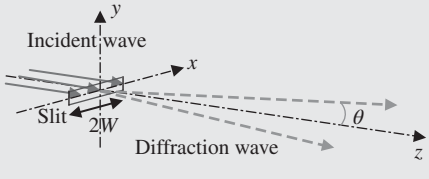


Figure 8.10 Diffraction caused by a slit

Equations

$$U = \frac{\varepsilon_L}{R}\int_{-W}^{W} \sin[\omega t - k(R - x\sin\theta)]dx \tag{8.3}$$

$$U = 2\frac{W\varepsilon_L}{R}\frac{\sin\beta}{\beta}\sin(\omega t - kR) \tag{8.6}$$

$$\beta \equiv kW\ \sin\theta \quad k \equiv \frac{2\pi}{\lambda}$$

$$I(\theta) = \frac{1}{2}\left(\frac{2W\varepsilon_L}{R}\right)^2\frac{\sin^2\beta}{\beta^2} = I(0)\frac{\sin^2\beta}{\beta^2} \tag{8.7′}$$

$I(0)$: Irradiance at the axis

$\beta \equiv kW\ \sin\theta$

Solution

The calculation results for Equation (8.7′) with $I(0) = 1$ are shown in the graph in Figure 8.11.

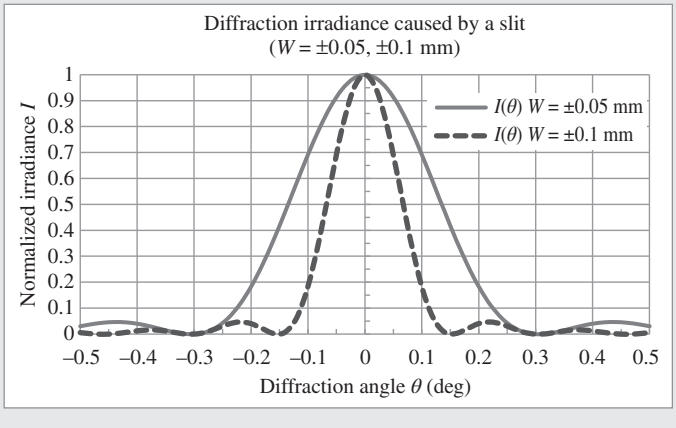


Figure 8.11 Diffraction irradiance caused by a slit

Settings

λ	0.000532 mm
W	±0.05 mm, ±0.1 mm
θ	0 – 0.5°

Notes

1. The irradiance I of the diffracted wave from the slit (length: ±0.05 mm) reaches its maximum value when θ equals zero, and decreases rapidly as the absolute value θ gets larger. When θ reaches ±0.31°, the irradiance I falls to zero. For higher values, an attenuated oscillation pattern recurs.
2. If the length of the slit is doubled, the diffraction angle will be half the original value.

Problem 8.2 Slit diffraction calculation using numerical integration

Calculate the diffraction caused by the slit ($\lambda = 0.532$ μm, $W = \pm 0.05$ mm) in Problem 8.1 as shown in Figure 8.10, using the mathematical technique of numerical integration.

Equations

$$U = \frac{\varepsilon_L}{R}\int_{-W}^{W} \sin[\omega t\text{-}k(R\text{-}x\sin\theta)]dx \tag{8.3}$$

$$U = \frac{\varepsilon_L}{R}\int_{-W}^{W} \exp[j\{\omega t - k(R - x\sin\theta)\}]dx$$

$$= \frac{\varepsilon_L}{R}\exp[j(\omega t - kR)]\int_{-W}^{W} \exp[j(kx\sin\theta)]dx \tag{8.9}$$

$$\int_{-W}^{W} \exp[j(kx\sin\theta)]dx = \int_{-W}^{W} [\cos(kx\sin\theta) + j\sin(kx\sin\theta)]dx \tag{8.9'}$$

$$\text{Total amplitude} \quad S = \sqrt{\left(\sum_i \cos\psi_i\right)^2 + \left(\sum_i \sin\psi_i\right)^2}\,\Delta x \quad \psi_i = kx_i\sin\theta \tag{8.10}$$

$$\text{Total phase} \quad \psi = \tan^{-1}\frac{\sum_i \sin\psi_i}{\sum_i \cos\psi_i} \tag{8.11}$$

$$\text{Electric field} \quad U = \varepsilon_L/R\ S\ \exp[j(\omega t - kR + \psi)] \tag{8.12}$$

$$\text{Irradiance} \quad I = <|U|^2> = |U|^2/2 \tag{8.6}$$

Solution

We can calculate Equation (8.9′), from the numerical integration of Equation (8.10) and Equation (8.11), as follows. First, we sum the values in the integral Equation (8.9′), keeping the real part and the imaginary part separate from one another. Next, we perform vector addition of the real sum and the imaginary sum, to obtain the total amplitude of the diffraction waves. In the case of Equation (8.9′), the imaginary component of the integral has the form of an odd function, and its integral (for the range from $-W$ to $+W$) will be zero, so we will just calculate the real part. The calculation results are shown in Figure 8.12.

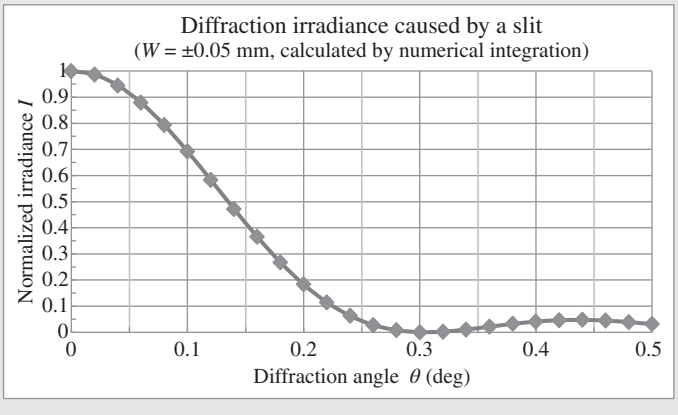


Figure 8.12 Diffraction irradiance by a slit (calculated by numerical integration)

Settings

Wavelength λ	0.000532 mm
Length $\pm W$	$\pm$0.05 mm
Diffraction angle θ	0–0.5°
Interval Δx	0.001 mm

Notes

1. We have obtained the same results as those calculated for Equation (8.7) in Problem 8.1.
2. The term in the integral in Equation (8.3) represents a diffraction wave and its phase. In Equation (8.9), we changed its form to that of a complex number. This enabled us to calculate the sum of the amplitudes of the waves (which are in different phases) using numerical integration.

Problem 8.3 Diffraction caused by a rectangular aperture

Calculate the far field irradiance pattern for the diffraction wave generated when a plane wave ($\lambda = 685$ nm) is irradiated onto a rectangular aperture ($2b = 0.04$ mm, $2a = 0.02$ mm) as shown in Figure 8.13.

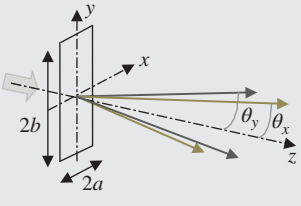


Figure 8.13 Diffraction caused by a rectangular aperture

Equations

$$U = \frac{\varepsilon_A}{R} e^{j(\omega t - kR)} \int_{-a}^{a} e^{jkXx/R} dx \int_{-b}^{b} e^{jkYy/R} dy \tag{8.21}$$

$$U = \frac{A\varepsilon_A}{R} e^{j(\omega t - kR)} \frac{\sin \alpha'}{\alpha'} \frac{\sin \beta'}{\beta'} \tag{8.25}$$

$$I(X, Y) = I(0, 0) \left[\frac{\sin \alpha'}{\alpha'} \right]^2 \left[\frac{\sin \beta'}{\beta'} \right]^2 \tag{8.26a}$$

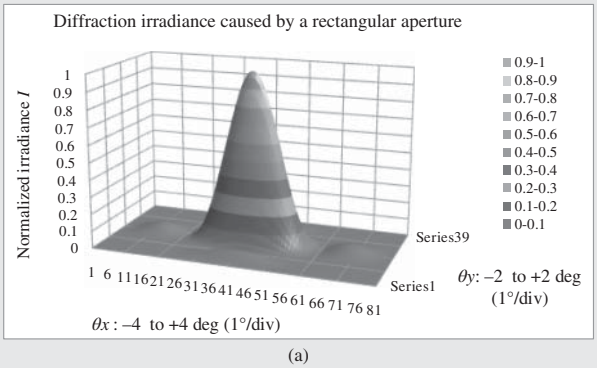
$$I(0, 0) = A^2 \varepsilon_A^2 / (2R^2) \tag{8.26b}$$

$$\alpha' \equiv kaX/R \tag{8.22a}$$

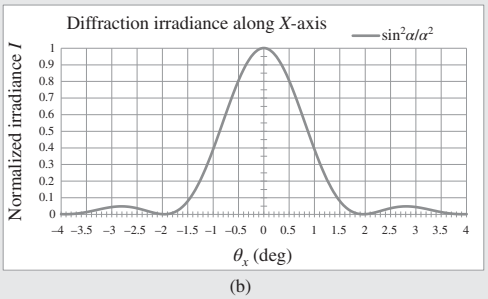
$$\beta' \equiv kbY/R \tag{8.22b}$$

Solution

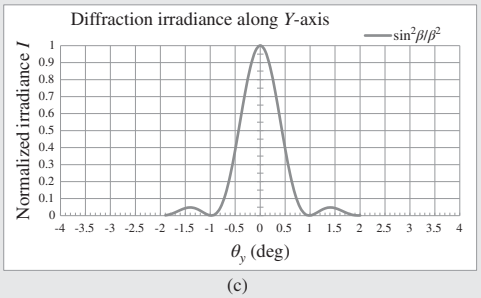
The calculation results for Equation. (8.26a) in the case when $I(0, 0) = 1$ are shown in the graphs in Figure 8.14a–c.



(a)



(b)



(c)

Figure 8.14 Diffraction irradiance caused by a rectangular aperture. (a) Three-dimensional irradiance. (b) Irradiance along *X*-axis. (c) Irradiance along *Y*-axis

Settings

λ	0.000685 mm
$2a$	0.04 mm
$2b$	0.02 mm

Notes

1. The irradiance of a diffraction wave generated at a rectangular aperture will have the same distribution pattern as the irradiance generated at a slit when the *x*- and *y*-components are multiplied.
2. Thus the irradiance generated by a rectangular aperture in the *x*-cross section (or *y*-cross section) will have the same distribution pattern as that generated at a slit.

Problem 8.4 Calculating Bessel functions

Each Bessel function can be expressed as an infinite series of polynomials. Bessel functions can be arranged in an order from zero to infinity: $J_0, J_1, J_2, J_3, \cdots$ [5].

$$J_n(x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{n+2k}}{2^{n+2k} k!(n+k)!} \qquad n = 0, 1, 2 \cdots \tag{8.42}$$

$$\text{For } n = 0, J_0(x) = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^4(2!)^2} - \frac{x^6}{2^6(3!)^2} + \cdots + (-1)^k \frac{x^{2k}}{2^{2k}(k!)^2} + \cdots \tag{8.43}$$

$$\text{For } n = 1, J_1(x) = \frac{x}{2} - \frac{x^3}{2^3 2!} + \frac{x^5}{2^5 2! 3!} - \frac{x^7}{2^7 3! 4!} + \cdots + (-1)^k \frac{x^{2k+1}}{2^{2k+1} k!(k+1)!} + \cdots \tag{8.44}$$

Calculate the Bessel functions J_n whose order $n = 0,\ 1,\ 2,\ 3,\ 4,$ and 5.

Solution

By setting the objective cell to "=BESSEL *J* (*n*,*x*)," you will be able to obtain values for the Bessel functions. The diagrams in Figure 8.15 illustrate the calculation results for Bessel functions.

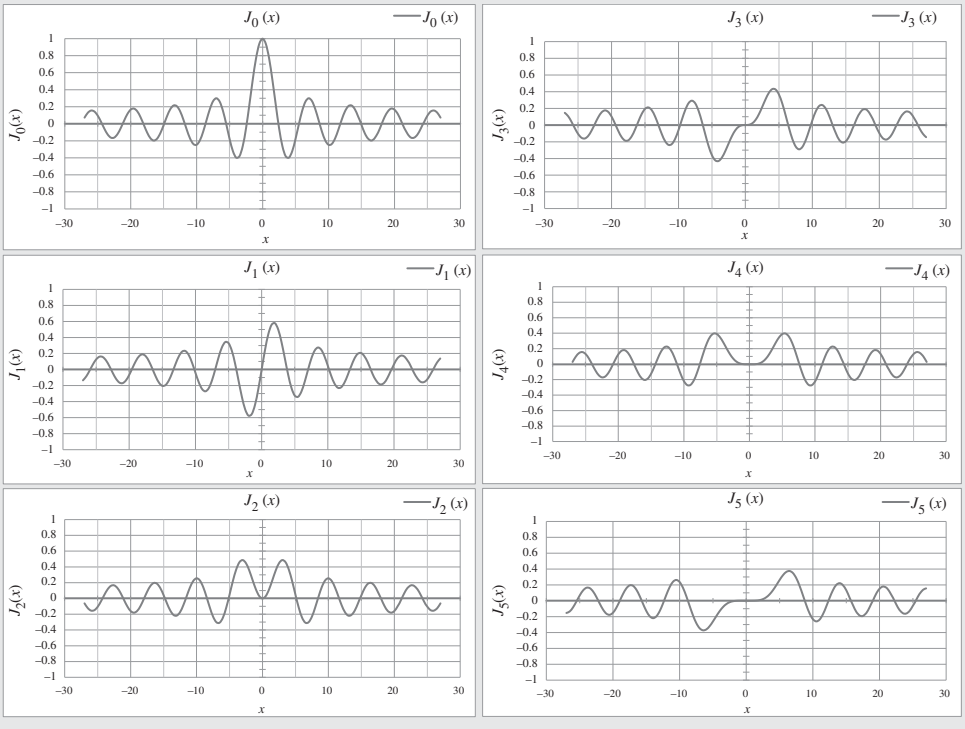


Figure 8.15 Bessel functions

Notes

1. Bessel functions have an oscillation pattern which gradually attenuates as the absolute value of x gets larger, as shown in Figure 8.15.
2. Bessel functions have the interesting property that the order of their derivatives is one less than that of the original function, as can be seen in Figure 8.15.

Problem 8.5 Diffraction caused by a circular aperture

Calculate the far field irradiance pattern of a diffraction wave when a plane wave ($\lambda = 685$ nm) is irradiated onto a circular aperture of diameter 2 mm as shown in Figure 8.16a. Also calculate the irradiance pattern on the focal plane when a focusing lens ($f = 10$ mm) is placed at the aperture plane as shown in Figure 8.16b.

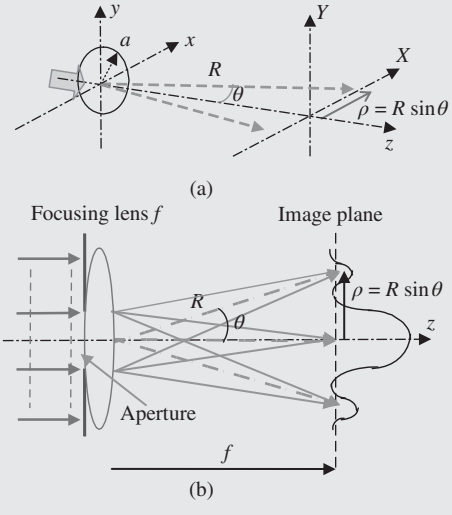


Figure 8.16 (a) Diffraction caused by a circular aperture and (b) diffraction pattern after exiting a focusing lens

Equations

$$I(\theta) = I(0)\left[\frac{2J_1(ka\sin\theta)}{ka\sin\theta}\right]^2 \tag{8.40}$$

Diffraction pattern after exiting a focusing lens *f*

We can calculate the diffraction pattern described by Equation (8.40) on the focal plane of a lens, with radius ρ, where

$$\rho = R\sin\theta \cong f\sin\theta \tag{8.45}$$

Solution

The calculation results obtained by using Equation (8.40) are as in Figure 8.17a,b.

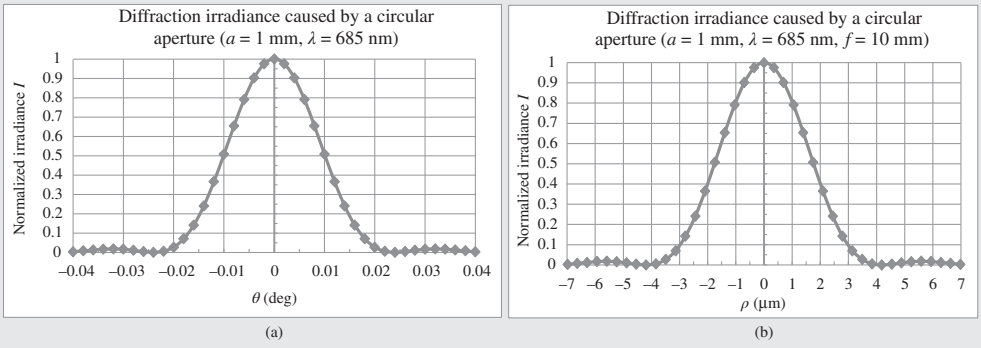


Figure 8.17 Diffraction irradiance caused by a circular aperture. (a) With no lens. (b) With a focusing lens

Settings

Wavelength λ	= 0.000685 mm
Aperture radius a	= 1 mm
Focal length f	= 10 mm

Note

1. We can make the diffraction pattern smaller by using a larger aperture, or light with a shorter wavelength, or a lens with a shorter focal length.

References

[1] Sagai, J. (1997) *Kogaku no Kiso*, Corona Publishing Co., Ltd, Japan (in Japanese).
[2] Hect, E. and Zajac, A. (1973) *Optics*, Addison-Wesley, USA.
[3] Born, M. and Wolf, E. (1974) *Principles of Optics I (Tokai-daigaku syuppan-kai)*, Tokai University Press, Japan (in Japanese).
[4] Melles Griot (2003) *Optical Parts Catalogue*, Melles Griot Co., Ltd, Japan.
[5] Reddick, H.W. and Miller, F.H. (1959) *Advanced Mathematics for Engineers*, Charles E. Tuttle Company, Japan.

9

Calculations for Gaussian Beam Diffraction

In the previous chapter, we looked at methods for calculating the diffraction waves generated by an aperture, when the irradiance distribution is uniform. However, light beams usually have a non-uniform distribution – for example, a Gaussian distribution. In this chapter, we will investigate the properties of various diffraction waves whose irradiance distribution at the aperture is non-uniform, such as a Gaussian beam.

9.1 The Power and the Central Irradiance of a Gaussian Beam

We shall begin by examining the relationship between the power P and the central irradiance I_0 of a Gaussian beam.

9.1.1 The Power of a Truncated Gaussian Beam

The irradiance I at radius ρ in a Gaussian beam of diameter $2w$ can be described by the following equation, in which I_0 denotes the central irradiance.

$$I = I_0 \exp\left(-\frac{2\rho^2}{w^2}\right) \tag{7.8}$$

The power P_a of a beam after passing through a circular aperture of radius a can be obtained by integrating the irradiance I over the entire area of the aperture. This is equivalent to calculating the volume of the solid by rotating the Gaussian curve around the I-axis, as in Figure 9.1. We can express this idea mathematically, as follows:

$$P_a = \int_0^a I 2\pi\rho d\rho = I_0 \int_0^a 2\pi \exp\left[-\frac{2\rho^2}{w^2}\right] \rho d\rho \tag{9.1}$$

Equation (9.1) enables us to calculate the power of the truncated Gaussian beam. Here,

$$\frac{d[\exp(-2\rho^2/w^2)]}{d\rho} = -\frac{4\rho}{w^2} \exp\left[-\frac{2\rho^2}{w^2}\right] \tag{9.2}$$

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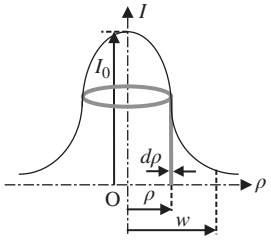


Figure 9.1 Power of a Gaussian beam

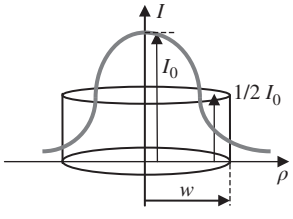


Figure 9.2 Power of a Gaussian beam vs. power of a circular beam with uniform irradiance

Hence,

$$\exp\left[-\frac{2\rho^2}{w^2}\right]\rho = -\frac{w^2}{4}\frac{d[\exp(-2\rho^2/w^2)]}{d\rho}$$

$$\int \exp\left(-\frac{2\rho^2}{w^2}\right)\rho d\rho = -\frac{w^2}{4}\exp(-2\rho^2/w^2) \tag{9.3}$$

The integral in Equation (9.1) can be solved as follows, using Equation (9.3).

$$P_a = -\pi\frac{w^2}{2}I_0\left[\exp\left(-\frac{2\rho^2}{w^2}\right)\right]_0^a = \frac{\pi w^2}{2}I_0\left\{1-\exp\left(-\frac{2a^2}{w^2}\right)\right\} \tag{9.4}$$

Equation (9.4) represents the power of a Gaussian beam after passing through an aperture of radius a. The total power P of a Gaussian beam can be calculated for the limiting case when $a \to \infty$:

$$P = \lim_{a\to\infty} P_a = \frac{\pi w^2}{2}I_0 \tag{9.5}$$

Inspecting Equation (9.5), we see that the power of a Gaussian beam (with a central irradiance I_0 and a beam radius of w) is equal to the power of a beam having a uniform irradiance of $I_0/2$ and a radius w which is equal to that of the Gaussian beam (Figure 9.2).

From Equation (9.5), the central irradiance of Gaussian beam can be calculated as:

$$I_0 = \frac{2P}{\pi w^2} \tag{9.6}$$

The irradiance can be expressed as:

$$I = < u^2 > = |U|^2/2$$

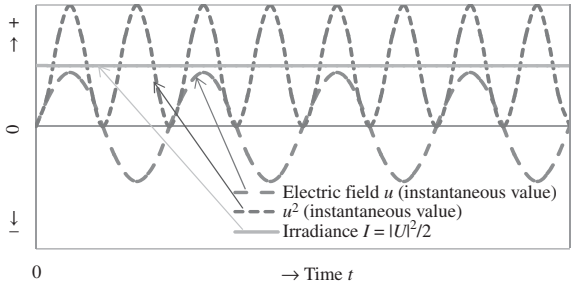


Figure 9.3 Electric field and irradiance

where

u = Electric field
U = Amplitude of the electric field
$< >$ = Average value over time

Figure 9.3 shows the relationship between an electric field and its irradiance.
The electric field U_0 at the center will be:

$$U_0 = \sqrt{2I_0} = \sqrt{\frac{4P}{\pi w^2}} \tag{9.7}$$

Truncated Power of Gaussian Beam
From Equation (9.4) and Equation (9.5), the power of a beam which is truncated by an aperture of radius a can be calculated as follows:

$$P_t = P - P_a = \frac{\pi w^2}{2} I_0 \exp\left(-\frac{2a^2}{w^2}\right) \tag{9.8}$$

The truncated power of a Gaussian beam also possesses the form of a Gaussian function.

9.1.2 The Power and the Central Irradiance of an Elliptical Gaussian Beam

The central irradiance of a circular Gaussian beam can be described by Equation (9.6), which we examined above:

$$I_0 = \frac{2P}{\pi w^2} \tag{9.6}$$

The central irradiance I_0 of an elliptical Gaussian beam will therefore be:

$$I_0 = \frac{2P}{\pi w_a w_b} \tag{9.9}$$

where

w_a = Radius in the x-direction
w_b = Radius in the y-direction

as shown in Figure 9.4.

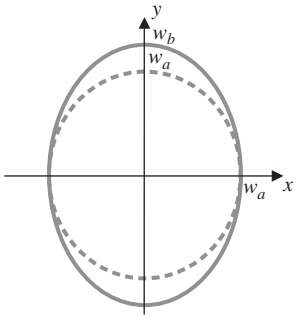


Figure 9.4 Circular beam and elliptical beam

Likewise, the amplitude of the electric field at the center U_0 will be:

$$U_0 = \sqrt{\frac{4P}{\pi w_a w_b}} \tag{9.10}$$

9.1.3 Irradiance Distribution of an Elliptical Gaussian Beam

The irradiance $I(\rho)$ and the electric field $U(\rho)$ of a circular Gaussian beam can be expressed as follows:

$$I(\rho) = I_0 \exp\left[-2\left(\frac{\rho}{w}\right)^2\right] \tag{7.8}$$

$$U(\rho) = U_0 \exp\left[-\left(\frac{\rho}{w}\right)^2\right] \tag{7.7}$$

The irradiance of an elliptical Gaussian beam on the x-axis $I(x,0)$ and on the y-axis $I(0,y)$ can be represented by the following two equations (Figure 9.5a):

$$I(x,0) = I_0 \exp\left[-2\left(\frac{x}{w_a}\right)^2\right] \tag{9.11}$$

$$I(0,y) = I_0 \exp\left[-2\left(\frac{y}{w_b}\right)^2\right] \tag{9.12}$$

The irradiance $I(x,y)$ of an elliptical Gaussian beam at the point (x,y) can be obtained by multiplying Equation (9.11) by Equation (9.12).

$$I(x,y) = I_0 \exp\left[-2\left(\frac{x}{w_a}\right)^2\right] \exp\left[-2\left(\frac{y}{w_b}\right)^2\right] = I_0 \exp\left[-2\left\{\left(\frac{x}{w_a}\right)^2 + \left(\frac{y}{w_b}\right)^2\right\}\right] \tag{9.13}$$

Hence the irradiance $I(x,y)$ of an elliptical Gaussian beam can be described by the equation:

$$I(x,y) = I_0 \exp\left[-2\left\{\left(\frac{x}{w_a}\right)^2 + \left(\frac{y}{w_b}\right)^2\right\}\right] \tag{9.14}$$

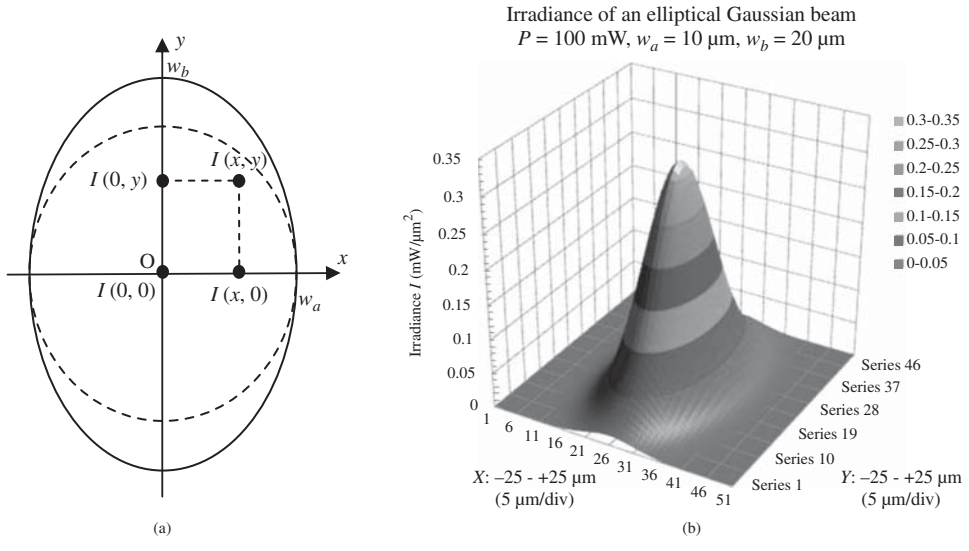


Figure 9.5 (a) Elliptical Gaussian beam. (b) Irradiance distribution of an elliptical Gaussian beam

Likewise, the electric field $U(x,y)$ can be described by the equation:

$$U(x,y) = U_0 \exp\left[-\left\{\left(\frac{x}{w_a}\right)^2 + \left(\frac{y}{w_b}\right)^2\right\}\right] \tag{9.15}$$

Figure 9.5b shows an example of the irradiance distribution of an elliptical beam whose power $P = 100$ mW, and whose x- and y-radii are $w_a = 10$ μm and $w_b = 20$ μm, respectively.

9.1.4 *The Power and the Central Irradiance of a One-Dimensional Gaussian Beam*

The equation for a one-dimensional Gaussian beam can be expressed by the following equation, obtained by replacing the coefficients w_0/w_z in Equation (7.1) with $\sqrt{(w_0/w_z)}$, and the variable ρ with x.

$$U(z,x) = A\sqrt{\frac{w_0}{w_z}}\exp[-j(kz-\psi)]\exp\left[-\left(\frac{x}{w_z}\right)^2\right]\exp\left(\frac{-jkx^2}{2R}\right) \tag{7.1′}$$

A one-dimensional Gaussian beam along the z-axis has a Gaussian electric field distribution pattern in the x-direction and maintains a constant value in the y-direction as y tends to infinity, as shown in Figure 9.6a.

The irradiance distribution of a one-dimensional Gaussian beam can be expressed as the square of the third term in Eq. (7.1′), as shown in Equation (9.16).

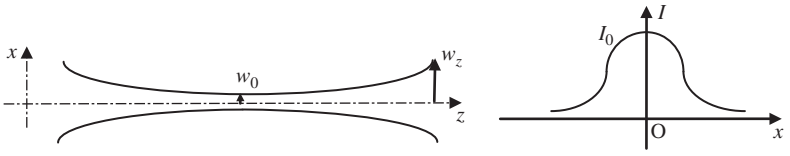


Figure 9.6 (a) A one-dimensional Gaussian beam. (The beam maintains a constant value in the *y*-direction.) (b) Irradiance of a one-dimensional Gaussian beam

$$I(x) = I_0 \exp\left[-2\left(\frac{x}{w}\right)^2\right] \tag{9.16}$$

Figure 9.6b is a graph of Equation (9.16). Here, we define the power P of a one-dimensional Gaussian beam as the area lying between the curve and the x-axis. [In reality, the height of the curve in the figure will remain the same, even as y (whose direction is vertical to the page) tends to infinity. Hence the above definition refers to the power of a unit length in the y-direction.]

The power of a one-dimensional Gaussian beam can be obtained by integrating Equation (9.16).

$$P = \int_{-\infty}^{+\infty} I(x)dx = \int_{-\infty}^{+\infty} I_0 \exp\left[-2\left(\frac{x}{w}\right)^2\right]dx = \sqrt{\frac{\pi}{2}}wI_0 \quad \text{(see Calculation of the integral below)} \tag{9.17}$$

The central irradiance I_0 can be expressed as:

$$I_0 = \sqrt{\frac{2}{\pi}}\frac{P}{w} \tag{9.18}$$

Likewise, the central electric field U_0 and electric field distribution $U(x)$ can be represented as:

$$U_0 = \sqrt{2I_0} = \sqrt{\frac{2\sqrt{2}P}{\sqrt{\pi}w}} \tag{9.19}$$

$$U(x) = U_0 \exp\left[-\left(\frac{x}{w}\right)^2\right] \tag{9.20}$$

Calculation of the Integral in Equation (9.17) $\int_{-\infty}^{+\infty} \exp[-ax^2]dx$

If $G = \int_{-\infty}^{+\infty} \exp(-ax^2)dx$, G^2 can be expressed in terms of the variables x and y:

$$G^2 = \int_{-\infty}^{+\infty} \exp(-ax^2)dx \int_{-\infty}^{+\infty} \exp(-ay^2)dy = \int_{-\infty}^{+\infty} \exp[-a(x^2+y^2)]dxdy \tag{9.21}$$

Rewriting Equation (9.21) with polar coordinates, we obtain:

$$G^2 = \int_0^{+\infty} \exp(-a\rho^2)\rho d\rho \int_0^{2\pi} d\psi = 2\pi \int_0^{+\infty} \rho \exp(-a\rho^2)d\rho \tag{9.22}$$

Applying integration by substitution to Equation (9.22) defining $v = \rho^2 (\rho = v^{1/2})$, we get:

$$G^2 = 2\pi \int_0^\infty \frac{1}{2} \exp[-av] dv = \pi \left[-\frac{1}{a} \exp(-av) \right]_0^\infty = \frac{\pi}{a} \tag{9.23}$$

Hence,

$$G = \int_{-\infty}^{+\infty} \exp(-ax^2) dx = \sqrt{\frac{\pi}{a}} \tag{9.24}$$

9.2 General Equations for Waves Diffracted by an Aperture

Figure 9.7 shows a diffraction wave generated at an aperture, when we irradiate a plane wave onto the aperture. The electric field U at the point Q(X,Y) on the observation plane, which is at a distance Z from the aperture, can be obtained by integrating the (secondary wave) electric field dU at Q(X,Y), generated by the electric field u at the differential segment P(x,y) on the aperture, over the entire aperture [1, 2].

$$U(X, Y) \cong \frac{j}{\lambda} \iint_{aperture} u(x, y) \frac{\exp[-jkr]}{r} K(\theta) dxdy \tag{9.25}$$

Equation (9.25) is derived from the Fresnel–Kirchhoff diffraction formula and represents the diffraction field at Q(X,Y), where

Coefficient j	=	Imaginary unit. Since $\exp(j\pi/2) = j$, diffraction waves change their phase by $\pi/2$
Coefficient $1/\lambda$, $1/r$ (in the integral)	=	Proportionality constants. Diffraction field is inversely proportional to the wavelength λ and the distance r
$u(x,y)$	=	Electric field of the plane wave irradiated onto the aperture
$\exp(-j\,kr)$	=	Phase delay caused by the distance r
$K(\theta)$	=	Inclination factor. θ is the angle between the secondary wavelet and the direction perpendicular to the wave front of the primary wave. $K(\theta) = (1 + \cos\theta)/2$ (in the case of a scalar wave)

When θ is small, the coefficient $1/r$ in the integral in Equation (9.25) can be approximated by $1/r \approx 1/Z$, and the inclination factor $K(\theta)$ can also be approximated by $K(\theta) \approx 1$. This enables us to derive the following equation:

$$U(X, Y) \cong \frac{j}{\lambda Z} \iint_{aperture} u(x, y) \exp(-jkr) dxdy \quad \text{General diffraction equation} \tag{9.26}$$

In Figure 9.7, r can be expressed as:

$$r = \sqrt{Z^2 + (x - X)^2 + (y - Y)^2} = Z\sqrt{1 + \frac{(x - X)^2 + (y - Y)^2}{Z^2}}$$

The above equation can be expanded[1] as follows:

$$r \cong Z + \frac{1}{2Z}[(x - X)^2 + (y - Y)^2] - \frac{1}{8Z^3}[(x - X)^2 + (y - Y)^2]^2 + \cdots \tag{9.27}$$

[1] When $n = 1/2, -1 < x < 1, \quad (1 + x)^n = 1 + x/2 - x^2/8 + \cdots + [n(n-1)(n-2)\cdots(n-r+1)]x^r/r! + \cdots$

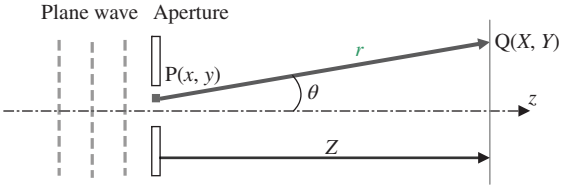


Figure 9.7 Diffraction wave generated at an aperture

Dropping the third and subsequent terms, we obtain the following approximation.

$$U(X,Y) \cong \frac{j}{\lambda Z}\exp(-jkZ)\exp\left(-jk\frac{X^2+Y^2}{2Z}\right)\iint u(x,y)\exp\left(jk\frac{Xx+Yy}{Z}\right) \times \exp\left(-jk\frac{x^2+y^2}{2Z}\right)dxdy \tag{9.28}$$

Equation (9.28) is referred to as the **Fresnel diffraction equation.** It describes the case where the observation distance Z is such that $k[(x-X)^2+(y-Y)^2]^2/(8Z^3) \ll \pi/2$, that is, $Z^3 \gg 4[(x-X)^2+(y-Y)^2]^2/(8\lambda)$ (Fresnel region), so that Equation (9.28) is satisfied. [The third and the subsequent terms in Equation (9.27) can therefore be dropped.]

Additionally, if the condition $k(x^2+y^2)/(2Z) \ll \pi/2$ is satisfied, that is, if $Z \gg 2(x^2+y^2)/\lambda$, then we can drop the final term in the integral in Equation (9.28), to obtain the following approximation:

$$U(X,Y) \cong \frac{j}{\lambda Z}\exp(-jkZ)\exp\left(-jk\frac{X^2+Y^2}{2Z}\right)\iint u(x,y)\exp\left(jk\frac{Xx+Yy}{Z}\right)dxdy \tag{9.29}$$

Equation (9.29) is commonly referred to as the **Fraunhofer diffraction equation.** It describes the special case where the observation distance Z is very remote, that is, $Z \gg 2(x^2+y^2)/\lambda$.

The diffraction calculations we studied in Chapter 8 apply to this kind of diffraction.

In Sections 9.3 and 9.4, we will look at some real-life applications of diffraction calculations to focusing optical systems and collimating optical systems, respectively. (See Appendix E for further study where Fresnel diffraction and Fraunhofer diffraction are explained in detail.)

9.3 Diffraction Wave Equations for a Focused Beam

9.3.1 Diffraction Wave Equation for a Beam Focused on the Focal Plane

Figure 9.8 depicts a diffraction wave on the focal plane of a focusing lens (focal length = f) when a parallel (collimated) beam is irradiated onto the aperture of the lens. Here, we suppose that the plane

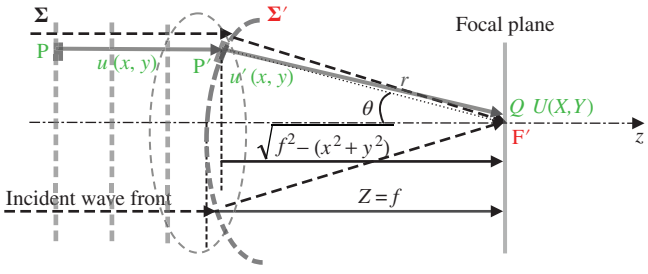


Figure 9.8 Diffraction wave for focusing a beam

wave front of the incident beam **Σ** will become a spherical wave front **Σ′** (which lies on the secondary principal surface) after passing through the lens, and the secondary wavelets generated on **Σ′** will generate an electric field $U(X,Y)$ at point Q on the focal plane of the lens. Thus the electric field $U(X,Y)$ can be obtained by integrating the electric field $u'(x,y)$ at P′ on **Σ′** over the entire aperture. In a similar fashion, we can calculate the electric field $U(X,Y)$ by applying Equation (9.26) where we integrate the electric field (amplitude) $u(x,y)$ at P on **Σ** with its phase assuming it as emitted from P′ on **Σ′** (the phase varies in direct proportion to the distance $P'Q = r$).

$$U(X,Y) \cong \frac{j}{\lambda Z}\iint_{aperture} u(x,y)\exp(-jkr)dxdy \quad \text{General diffraction wave equation} \tag{9.26}$$

Here, we will examine the inclination factor in Equation (9.26). The irradiance at point P′ on the wave front **Σ′** will be $\cos\theta$ times the value at P on **Σ** (where θ is the angle between P′F′ and the optical axis), due to the rays being slanted at an angle to the surface **Σ′** at P′ (which is because of the slanted wave-front surface **Σ′** at P′). In addition, the irradiance at Q on the focal plane generated by an incident angle θ will also be $\cos\theta$ times[1] the value generated when the incident angle is zero, due to the rays being slanted at an angle to the focal plane. On the other hand, the size of the differential segment P′ on **Σ′** will be $1/\cos\theta$ times greater than that of P on **Σ**. This means that the electric field contribution to Q generated by integrating the electric fields at P′ on **Σ′** will be $(1/\cos\theta)$ times the value obtained by integrating the electric fields at P on **Σ**, and the irradiance contribution to Q will be $(1/\cos\theta)^2$. Thus the overall value of the inclination factor will be 1 [as the factors $\cos\theta$, $\cos\theta$, and $(1/\cos\theta)^2$ cancel each other out], and we can treat Equation (9.26) as being a precise formula for scalar waves. From inspecting Figure 9.8, it can be seen that r can be expressed by the formula:

$$r = \sqrt{f^2 - (x^2+y^2) + (X-x)^2 + (Y-y)^2} = \sqrt{f^2 + X^2 + Y^2 - 2(Xx+Yy)} \tag{9.30}$$

Using the same method as we used in Equation (9.27), we can approximate the value of r by Equation (9.31), if $k[X^2+Y^2-2(Xx+Yy)]^2/(8f^3) \ll \pi/2$, that is, if $f^3 \gg 4[X^2+Y^2-2(Xx+Yy)]^2/(8\lambda)$:

$$r = \sqrt{f^2 + X^2 + Y^2 - 2(Xx+Yy)} = f\sqrt{1 + \frac{X^2+Y^2-2(Xx+Yy)}{f^2}} \cong f + \frac{X^2+Y^2}{2f} - \frac{Xx+Yy}{f} \tag{9.31}$$

[1] The angle lying between P′Q and the axis can be approximated as θ as far as the focused spot size is small.

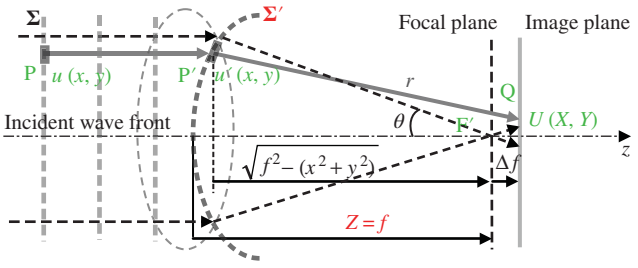


Figure 9.9 Diffraction wave for focusing a beam on a defocused plane

Substituting Equation (9.31) into Equation (9.26), we obtain a diffraction wave $U(X,Y)$ at the focal plane:

$$U(X,Y) \cong \frac{j}{\lambda f}\exp(-jkf)\exp\left(-jk\frac{X^2+Y^2}{2f}\right)\iint u(x,y)\exp\left[jk\frac{Xx+Yy}{f}\right]dxdy \qquad (9.32)$$

Diffraction wave equation for a focused beam

Equation (9.32) is identical with the Fraunhofer diffraction equation, Equation (9.29). This shows that the diffraction wave equation for a focused beam is equivalent to that of the Fraunhofer diffraction, which describes observations made from a far field point. Equation (9.32) can also be derived by using the method of "Wave-front conversion by a lens" (see Appendix F).

9.3.2 *Diffraction Wave Equation for a Focused Beam on a Defocused Plane*

Figure 9.9 represents the diffraction wave $U(X,Y)$ at a point Q on the plane at a distance Δf from the focal plane when a beam is focused on the focal point of the lens. In the figure, the distance r from a point $P'(x,y)$ on the secondary principal surface of the lens to the observation point $Q(X,Y)$ will be:

$$\begin{aligned} r &= \sqrt{\left\{\sqrt{f^2-(x^2+y^2)}+\Delta f\right\}^2+(X-x)^2+(Y-y)^2} \\ &= \sqrt{f^2+\Delta f^2+2\Delta f\sqrt{f^2-(x^2+y^2)}+X^2+Y^2-2(Xx+Yy)} \\ &= f\sqrt{1+\Delta f^2/f^2+2\Delta f\sqrt{f^2-(x^2+y^2)}/f^2+(X^2+Y^2)/f^2-2(Xx+Yy)/f^2} \end{aligned} \qquad (9.33)$$

Using the same method as we used in Equation (9.27), we can approximate Equation (9.33) by the following series, if $k[\Delta f^2+2\Delta f\sqrt{\{f^2-(x^2+y^2)\}}+X^2+Y^2-2(Xx+Yy)]^2/(8f^3) \ll \pi/2$, that is, if $f^3 \gg 4[\Delta f^2+2\Delta f\sqrt{\{f^2-(x^2+y^2)\}}+X^2+Y^2-2(Xx+Yy)]^2/(8\lambda)$:

$$r \cong f+\Delta f^2/(2f)+\Delta f\sqrt{f^2-(x^2+y^2)}/f+(X^2+Y^2)/(2f)-(Xx+Yy)/f \qquad (9.34)$$

Substituting Equation (9.34) into Equation (9.26), we obtain the following equation for the diffraction wave $U(X,Y)$:

$$U(X,Y) \cong \frac{j}{\lambda f}\exp\left(-jk\left(f+\frac{\Delta f^2}{2f}\right)\right)\exp\left(jk\frac{X^2+Y^2}{2f}\right)$$
$$\iint u(x,y)\exp\left(jk\frac{Xx+Yy}{f}\right)\exp\left[-jk\Delta f\sqrt{1-(x^2+y^2)/f^2}\right]dxdy \tag{9.35}$$

In the calculations that follow, we shall drop the following terms from Equation (9.35):

1. the coefficient j and the term $\exp[-jk(f + \Delta f^2/2f)]$, which represents a constant phase; and
2. the term $\exp[-jk(X^2 + Y^2)/(2f)]$, since the spot size is very small ($X^2 + Y^2 \ll 2f$) and we can treat this term as effectively equal to 1.

We then obtain Equation (9.36), which approximates real-life cases very well.

$$U(X,Y) \cong \frac{1}{\lambda f}\iint u(x,y)\exp\left(jk\frac{Xx+Yy}{f}\right)\exp\left[-jk\Delta f\sqrt{1-(x^2+y^2)/f^2}\right]dxdy \tag{9.36}$$

Diffraction wave equation for a focused beam on a defocused plane

The root in the last term in Equation (9.36) can be extended into a series. However, we will not be using this method here, so as to avoid any errors that may be caused by this approximation.

9.3.3 *Diffraction Equation for a Focused Beam, in Polar Coordinates*

In this section, we will express Equation (9.36) using the polar coordinates $h\angle\psi$ (for the aperture plane) and $\rho\angle\varphi$ (for the observation plane), where the aperture is defined to be a circular aperture and the shape of the beam is also circular. In this case, the diffraction wave pattern will be symmetrical about the axis, so we can fix the observation point Q(X,Y) on the X-axis as Q(ρ,0), as shown in Figure 9.10a,b. Then we can express the terms in Equation (9.36) using the polar coordinates (h, ψ, ρ) instead of (x,y) and (X,Y) as shown in Equation (9.37).

$$\iint_{aperture} => \int_0^a \int_0^{2\pi}$$
$$\exp[jk(Xx+Yy)/f] => \exp[jk\rho h\cos\psi/f]$$
$$\exp[-jk\Delta f\sqrt{\{1-(x^2+y^2)/f^2\}}] => \exp[-jk\Delta f\sqrt{(1\text{-}h^2/f^2)}]$$
$$dxdy => hd\psi dh \tag{9.37}$$

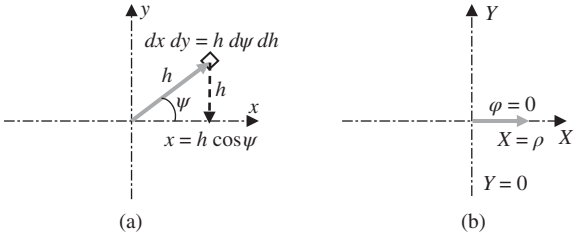


Figure 9.10 Coordinate conversion to polar coordinates. (a) Aperture plane $(x,y) => h\angle\psi$. (b) Observation plane $(X,Y) => \rho\angle\varphi$

Applying the above coordinate conversion in Equation (9.37), we can rewrite the diffraction wave equation (9.36) as follows:

$$U(\rho) \cong \frac{1}{\lambda f}\int_0^a \int_0^{2\pi} u(h)\exp\left[jk\frac{\rho h\cos\psi}{f}\right]\exp\left[-jk\Delta f\sqrt{1-h^2/f^2}\right]hdhd\psi \tag{9.38}$$

Diffraction wave equation for a focused beam on a defocused plane (in polar coordinates)

9.3.4 *One-Dimensional Expression of the Diffraction Wave for Focusing a Beam*

If we limit both the aperture and the observation plane to a single dimension, we can represent the diffraction wave by the following equation [converting Equation (9.38) into a single dimension]:

$$U(X) \cong \frac{1}{\sqrt{\lambda f}}\int u(x)\exp\left[jk\frac{xX}{f}\right]\exp\left[-jk\Delta f\sqrt{1-x^2/f^2}\right]dx \tag{9.39}$$

Diffraction wave equation for a focused beam on a defocused plane (one-dimensional case)

9.4 Diffraction Wave Equations for a Collimated Beam

9.4.1 *Diffraction Wave Equation for a Collimated Beam*

Figure 9.11 illustrates a beam which diverges after passing through its beam waist, and a convex lens which collimates the diverging beam. The first focal point F of the lens is situated at the beam waist. In the figure, if we imagine an aperture located on the beam waist (which has a plane wave front), the secondary wavelets from the point P(x,y) on the aperture will generate an electric field $U'(X,Y)$ at point Q′ on the primary principal surface $\mathbf{\Sigma}'$ of the lens, and subsequently give rise to an electric field $U(X,Y)$ at point Q on the wave front $\mathbf{\Sigma}$ after exiting the lens. Here, we will examine the inclination factor. The irradiance at Q′ will be approximately $\cos\theta$ times the value generated on the axis, due to the beam being inclined at an angle θ (where θ is the angle between FQ′ and the optical axis). On the other hand, the

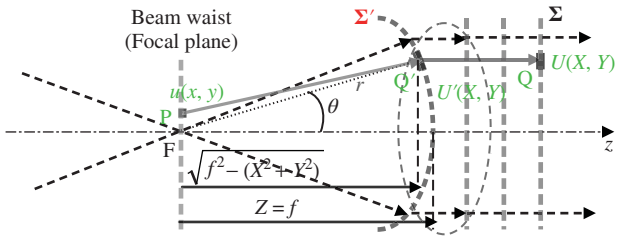


Figure 9.11 Diffraction wave for collimating a beam

irradiance at Q will be approximately $(1/\cos\theta)$ times that of at Q′, due to the collimator lens. Thus the overall inclination factor will be 1 (as the two factors cancel each other out). Hence the electric field $U(X,Y)$ at Q will be:

$$U(X,Y) \cong \frac{j}{\lambda Z}\iint_{aperture} u(x,y)\exp(-jkr)dxdy \quad \text{General diffraction wave equation} \tag{9.26}$$

From Figure 9.11, r in Equation (9.26) can be expressed as follows:

$$r = \sqrt{f^2-(X^2+Y^2)+(X-x)^2+(Y-y)^2} = \sqrt{f^2+x^2+y^2-2(Xx+Yy)} \tag{9.40}$$

Using the same method as we used in Equation (9.27), we can approximate Equation (9.40) as follows, if $k[x^2+y^2-2(Xx+Yy)]^2/(8f^3) \ll \pi/2$, that is, if $f^3 \gg 4[x^2+y^2-2(Xx+Yy)]^2/(8\lambda)$:

$$\begin{aligned} r &= \sqrt{f^2-2(Xx+Yy)+(x^2+y^2)} = f\sqrt{1-2(xX+Yy)/f^2+(x^2+y^2)/f^2} \\ &\cong f-(Xx+Yy)/f+(x^2+y^2)/(2f) \end{aligned} \tag{9.41}$$

If we substitute Equation (9.41) into Equation (9.26), we obtain the diffraction wave equation for collimating a beam:

$$U(X,Y) \cong \frac{j}{\lambda f}\exp(-jkf)\iint_{aperture} u(x,y)\exp\left[jk\frac{Xx+Yy}{f}\right]\exp\left[-jk\frac{x^2+y^2}{2f}\right]dxdy \tag{9.42}$$

Diffraction wave equation for a collimated beam

The diffraction wave equation for collimating a beam, Equation (9.42), can be converted into the same form as the Fraunhofer diffraction equation, simply by dropping the term $\exp[-jk(x^2+y^2)/(2f)]$, whose value is negligible, if $k(x^2+y^2)/(2f) \ll \pi/2$, that is, $f \gg 2(x^2+y^2)/\lambda$.

9.4.2 *Diffraction Wave Equation for a Collimated Beam with a Defocused Setting*

If the lens is shifted by Δf from its normal position (where $\Delta f \ll f$) as shown in Figure 9.12, the distance r from the point P(x,y) on the beam waist to the point Q′(X,Y) on the primary principal surface of the

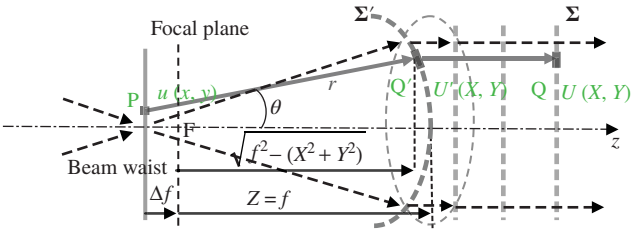


Figure 9.12 Diffraction wave for a collimated beam with a defocused setting

lens will be:

$$r = \sqrt{\left\{\sqrt{f^2-(X^2+Y^2)}+\Delta f\right\}^2+(X-x)^2+(Y-y)^2}$$

$$= \sqrt{f^2+\Delta f^2+2\Delta f\sqrt{f^2-(X^2+Y^2)}+x^2+y^2-2(Xx+Yy)} \tag{9.43}$$

Using the same method as in Equation (9.27), we can approximate Equation (9.43) as follows if $k[\Delta f^2+2\Delta f\sqrt{\{f^2-(X^2+Y^2)\}}+x^2+y^2-2(Xx+Yy)]^2/(8f^3) \ll \pi/2$, that is, if $f^3 \gg 4[\Delta f^2+2\Delta f\sqrt{\{f^2-(X^2+Y^2)\}}+x^2+y^2-2(Xx+Yy)]^2/(8\lambda)$:

$$r \cong f+\Delta f^2/(2f)+\Delta f\sqrt{f^2-(X^2+Y^2)}/f+(x^2+y^2)/(2f)-(Xx+Yy)/f \tag{9.44}$$

If we substitute Equation (9.44) into Equation (9.26), then we obtain the diffraction wave equation for collimating a beam with a defocused setting.

$$U(X,Y) \cong \frac{j}{\lambda f}\exp\left[-jk\left(f+\frac{\Delta f^2}{2f}\right)\right]\exp\left[-jk\Delta f\sqrt{1-(X^2+Y^2)/f^2}\right]$$
$$\iint u(x,y)\exp\left(jk\frac{Xx+Yy}{f}\right)\exp\left(-jk\frac{x^2+y^2}{2f}\right)dxdy \tag{9.45}$$

In the calculations that follow, we shall drop the following terms from Equation (9.45):

1. the coefficient j and the term $\exp[-jk(f+\Delta f^2/(2f))]$, which represents a constant phase; and
2. the term $\exp[-jk(x^2+y^2)/(2f)]$, as the beam waist size is very small ($x^2+y^2 \ll 2f$) and we treat this term as effectively equal to 1.

We then obtain Equation (9.46), which approximates real-life cases very well.

$$U(X,Y) \cong \frac{1}{\lambda f}\exp\left[-jk\Delta f\sqrt{1-(X^2+Y^2)/f^2}\right]\iint u(x,y)\exp\left(-jk\frac{Xx+Yy}{f}\right)dxdy \tag{9.46}$$

Diffraction wave equation for a collimated beam with a defocused setting

Equation (9.46) and Equation (9.36) both have a similar form, and if $\Delta f = 0$, then the two equations are equivalent. In other words, the diffraction equations for focusing and for collimating a beam are equivalent when $\Delta f = 0$.

9.4.3 Diffraction Wave Equation for a Collimated Beam, in Polar Coordinates

Here, we will express Equation (9.46) using the polar coordinates $h\angle\psi$ (aperture plane) and $\rho\angle\varphi$ (observation plane), where the aperture is defined to be a circular aperture and the shape of the beam shape is also circular. In this case, the diffraction wave pattern will be symmetrical about the axis, so we can fix the observation point Q(X,Y) on the X-axis as Q(ρ,0), as shown in Figure 9.10. Then we can express the terms in Equation (9.46) using the polar coordinates (h, ψ, ρ) instead of (x,y) and (X,Y), as shown in Equation (9.37). Applying the coordinate conversion in Equation (9.37), we can rewrite the diffraction wave equation (9.46) as follows:

$$U(\rho) \cong \frac{1}{\lambda f} \exp\left[-jk\Delta f\sqrt{1-\rho^2/f^2}\right] \int_0^a \int_0^{2\pi} u(h) \exp\left(jk\frac{\rho h \cos\psi}{f}\right) h d\psi dh \tag{9.47}$$

Diffraction wave equation for a collimated beam with a defocused setting (in polar coordinates).

9.4.4 One-Dimensional Expression of a Diffraction Wave for a Collimated Beam

If we limit both the aperture and the observation plane to a single dimension, we can express the diffraction wave for collimating a beam as follows:

$$U(X) \cong \frac{1}{\sqrt{\lambda f}} \exp[-jk\Delta f\sqrt{1-X^2/f^2}] \int u(x) \exp\left(jk\frac{xX}{f}\right) dx \tag{9.48}$$

Diffraction wave equation for a collimated beam with a defocused setting (one-dimensional case)

Equation (9.48) can be derived from the two-dimensional wave equation Equation (9.47), by converting it into a single dimension.

9.5 Diffraction Calculation Program

In this book, we have used Visual Basic for Application (VBA) programs to perform calculations of the diffraction waves for focusing and collimating a beam. The calculation examples can be found in the Excel file on the companion web site.

9.5.1 Insertion of Complex Amplitude Transmission Coefficients

As shown in Table 9.1, we can input the amplitude $u_t(x,y)$ and the phase $\phi_t(x,y)$ of the complex amplitude transmission coefficients that are combined with the electric field at the aperture. Using these transmission coefficients, we can express the size of the aperture by setting the amplitude $u_t(x,y)$ to zero outside the aperture. Alternatively, we can express the phase shift factor at the aperture by setting $\phi_t(x,y)$ combined with the phase of incident wave (e.g., the wave-front aberration occurring in the optical system) by setting $\phi_i(x,y)$.

Table 9.1 Electric field and transmission coefficients at the aperture

Electric field at aperture	Transmission coefficients	Diffraction	Electric field at observation point
Amplitude $u_i(x,y)$ Phase $\phi_i(x,y)$	Amplitude $u_t(x,y)$ Phase $\phi_t(x,y)$	⇨	Amplitude $U_o(X,Y)$ Phase $\phi_o(X,Y)$
		⇩	
Overall electric field at aperture		**Diffraction**	**Electric field at observation point**
Amplitude $u(x,y) = u_i(x,y)u_t(x,y)$ Phase $\phi(x,y) = \phi_i(x,y) + \phi_t(x,y)$		⇨	Amplitude $U_o(X,Y)$ Phase $\phi_o(X,Y)$

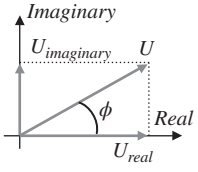


Figure 9.13 Complex number representation

9.5.2 *Diffraction Wave Calculations Using the Numerical Integration Method*

In this program, we will perform diffraction wave calculations using the numerical integration method.

$$U_o = \iint u(x,y)\exp[j\phi(x,y)]dxdy \tag{9.49}$$

We can rewrite Equation (9.49), replacing $\exp(j\phi)$ with $\cos\phi + j\sin\phi$ as in Figure 9.13.

$$U_o = \iint u(x,y)[\cos\phi(x,y) + j\sin\phi(x,y)]dxdy \tag{9.50}$$

The above integration can be performed by adding the real and imaginary values separately.

$$U_{real} = \Sigma[u(x,y)\cos\phi(x,y)]\Delta x\Delta y \tag{9.51}$$

$$U_{imaginary} = \Sigma[u(x,y)\sin\phi(x,y)]\Delta x\Delta y \tag{9.52}$$

In this book, the author applied Simpson's rule of numerical integration for calculating Equation (9.51) and Equation (9.52) (see Appendix D for details).

We can then obtain the combined amplitude U by adding the real number U_{real} to the imaginary number $U_{imaginary}$ using the method of vector addition. The combined amplitude U and phase ϕ will be:

$$U_o = \sqrt{{U_{real}}^2 + {U_{imaginary}}^2} \tag{9.53}$$

$$\phi_o = \arctan\left(\frac{U_{imaginary}}{U_{real}}\right) \tag{9.54}$$

9.5.3 Calculation Programs

The companion web site lists the following VBA programs for calculating diffraction fields:

① Two-dimensional *X*-*Y* diffraction calculation (Focus, Collimate) Equation (9.36) and Equation (9.46)
② Two-dimensional R-θ diffraction calculation (Focus, Collimate) Equation (9.38) and Equation (9.47)
③ One-dimensional diffraction calculation (SlitFocus, SlitCollimate) Equation (9.39) and Equation (9.48).

The two-dimensional R-θ diffraction calculation applies only to beams which are symmetrical about the axis. Both ② and ③ are performed by the same program.[3]

9.6 Operating Instructions for the Diffraction Calculation Programs

9.6.1 The Two-Dimensional R-θ Diffraction Calculation Program, "Focus/Collimate "

Specify the following items on the calculation table as shown in Figure 9.14.

① **Row and column header address(es):** Input the address (row and column) of the header cell for the calculation results table. Additional header addresses may be input to the right of the first address, if multiple calculations are desired. The cells to the right of the last row and column number input should be left blank, so that the program can recognize the end of the row and column number settings.
② **Optical system settings**: Input the optical configuration parameters in the following order:

Parameter	Units	Notes
Wavelength λ	(μm)	–
Focal length of the lens f	(μm)	–
Differential segment[a] $\Delta\psi$	(°)	Should be a divisor[b] of 180.0°
Defocus Δf	(μm)	–
Type of calculation	–	"Focus" or "Collimate"

[a]To keep the calculation as accurate as possible, it is better not to set large values for $\Delta\psi$.
[b]Divisors of 180.0° : 0.1°, 0.2°, 0.3°, 0.4°, 0.5°, 0.6°, 0.8°, 0.9°, 1.0°, 1.2°, 1.5°, and so on.

③ **Incident field settings:** Input the incident field parameters – that is, the amplitude and phase – at each specified radius in the aperture. The pitch between successive values of h should be constant.

Parameter	Units	Notes
Radius[a] h	(μm)	Should increase as the row number increases
Incident field amplitude u_i (h)	$\sqrt{(\text{mW}/\mu\text{m}^2)}$	–
Incident field phase[3] ϕ_i (h)	(rad)	–

[a]To keep the calculation as accurate as possible, it is better not to set a large pitch between successive values of h.

[3] The diffraction calculations assume that the incident wave has a plane wave front. The incident wave phases need to be assigned small values which are not very different from the plane wave front, so the calculated results will be reasonably accurate.

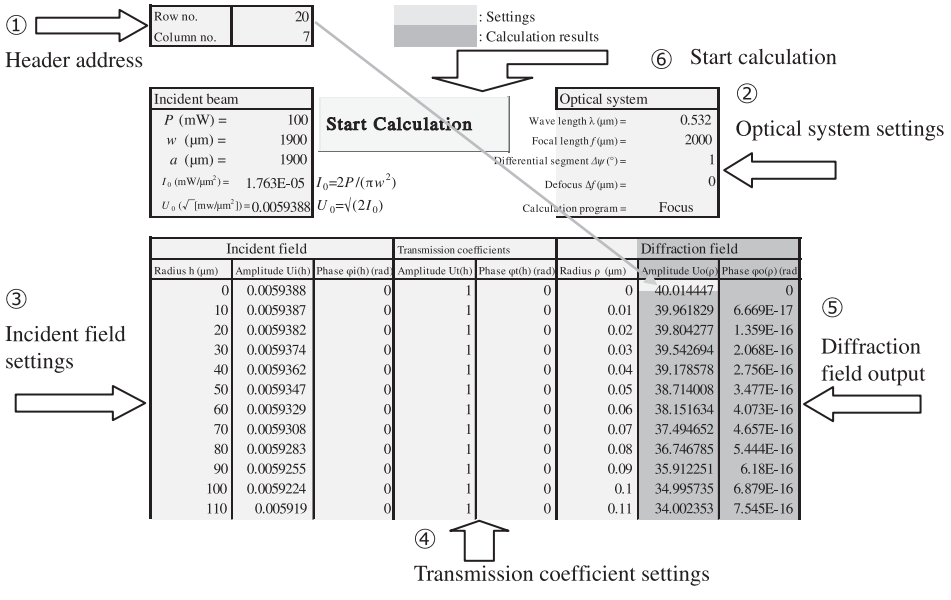


Figure 9.14 A calculation table for the two-dimensional R-θ diffraction calculation program

The cell below the specified radius should be left blank, so that the program can recognize the end of the incident field settings. The number of radius settings should not exceed 32 000.

④ **Transmission coefficient settings:** Input the amplitude coefficient $u_t(h)$ and the phase $\phi_t(h)$ of the complex amplitude transmission coefficients at the aperture.

Transmission coefficients	Units	Notes
Amplitude coefficient $u_t\ (h)$	–	–
Phase $\phi_t\ (h)$	(rad)	–

The amplitude $u(h)$ and the phase $\phi(h)$ of the field obtained from combining these transmission coefficients can be defined as follows:

$$u(h) = u_i(h)u_t(h) \qquad \phi(h) = \phi_i(h) + \phi_t(h)$$

⑤ **Diffraction field output:** Calculation results for the diffraction field, showing the amplitude $U_o(\rho)$ and phase $\phi_o(\rho)$ at each radius ρ, will be output in the table specified.

Parameter	Units	Notes
Radius ρ	(μm)	Input
Diffraction field amplitude $U_o(\rho)$	$\sqrt{(\text{mW}/\mu\text{m}^2)}$	Output
Diffraction field phase $\phi_o(\rho)$	(rad)	Output

⑥ **Start calculation:** Clicking on this button will initiate execution of the calculation program.

9.6.2 The One-Dimensional Diffraction Calculation Program, "SlitFocus"/"SlitCollimate"

This calculation will run on the same program as the two-dimensional R-θ diffraction calculation program.

Specify the following items on the calculation table as shown in Figure 9.15.

① **Row number and column number:** Input the address (row and column) of the header cell for the calculation results table. If additional header addresses are input to the right of the first address, multiple calculations will be performed. The cells to the right of the last row and column number should be left blank, so that the program can recognize the end of the row and column number settings.

② **Optical system settings:** Input the optical configuration parameters in the following order.

Parameter	Units	Notes
Wavelength λ	(μm)	–
Focal length of the lens f	(μm)	–
Defocus Δf	(μm)	–
Type of calculation	–	"SlitFocus" or "SlitCollimate"

③ **Incident field settings:** Input the values of the incident field (amplitude and phase), corresponding to each input position in the aperture. The pitch between successive values of x should be constant.

Parameter	Units	Notes
Position x[a]	(μm)	Should increase as the row number increases
Incident field amplitude $u_i(x)$	$\sqrt{}$(mW/μm)	–
Incident field phase $\phi_i(x)$	(rad)	–

[a]To keep the calculation as accurate as possible, it is better not to set a large pitch between successive values of x.

The cell just below the input values of the position should be left blank, so that the program can recognize the end of the incident field settings. The number of position settings should not exceed 32 000.

④ **Complex amplitude transmission coefficient settings:** Input the amplitude $u_t(x)$ and phase $\phi_t(x)$ of the complex amplitude transmission coefficients for the aperture.

Transmission coefficient	Units	Notes
Amplitude $u_t(x)$	–	–
Phase $\phi_t(x)$	(rad)	–

By combining these transmission coefficients, the amplitude $u(h)$ and the phase $\phi(h)$ of the field at the aperture can be defined as follows:

$$u(x) = u_i(x)u_t(x) \qquad \phi(x) = \phi_i(x) + \phi_t(x)$$

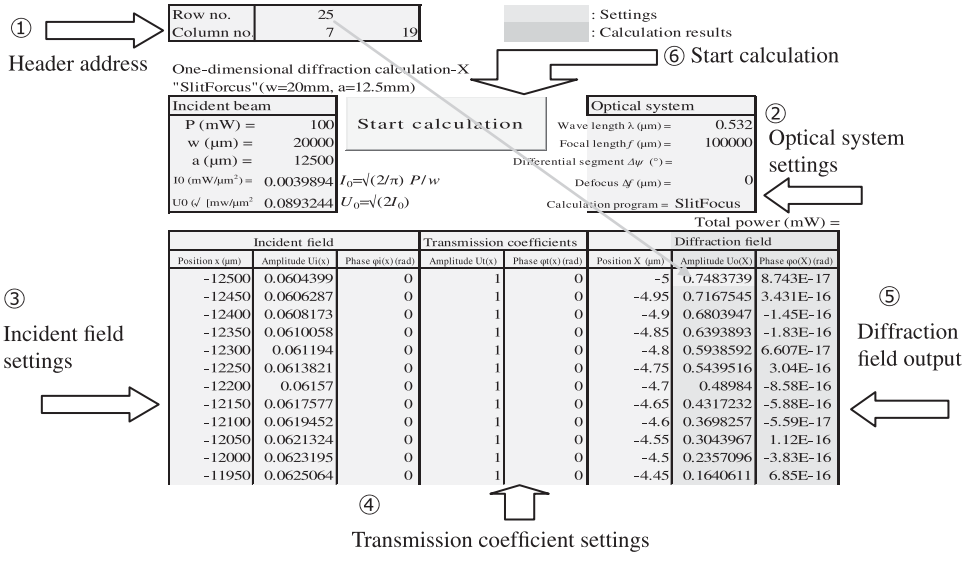


Figure 9.15 A calculation table for the one-dimensional diffraction calculation program

⑤ **Diffraction field output:** Calculated results for the diffraction field, showing the amplitude $U_o(X)$ and phase $\phi_o(X)$ at each position X on the observation plane, will be output in the table specified.

Parameter	Units	Notes
Position X	(μm)	Input
Diffraction field amplitude $U_o(X)$	$\sqrt{}$(mW/μm)	Output
Diffraction field phase $\phi_o(X)$	(rad)	Output

⑥ **Start calculation:** Clicking on this button will initiate execution of the calculation program.

Note

The one-dimensional diffraction calculation program and the two-dimensional R-θ diffraction calculation program actually operate according to the same underlying program. The user can select the desired program by selecting an appropriate value for the calculation program cells in ②**:**

One-dimensional calculation program: “SlitFocus” or “SlitCollimate”
Two-dimensional R-θ diffraction calculation program: “Focus” or “Collimate.”

9.6.3 The Two-Dimensional X-Y Diffraction Calculation Program, “Focus”/“Collimate”

The two-dimensional X-Y diffraction calculation program uses seven Excel sheets (five sheets for input settings and two sheets for outputs).

Sheets for input settings	Sheets for outputs
(1) "Sheet1": Main sheet	(6) "U_o": Diffraction field amplitude
(2) "U_i": Incident field amplitude	(7) "ϕ_o": Diffraction field phase
(3) "ϕ_i": Incident field phase	
(4) "U_t": Transmission coefficient amplitude	
(5) "ϕ_t": Transmission coefficient phase	

Specify the following input items on the calculation table as shown in Figure 9.16.

(1) "Sheet1": The Main Sheet

① **Settings for the optical configuration:** Input the optical configuration, as specified in the following table.

Parameter	Units	Notes
Type of calculation	–	"Focus" or "Collimate"
Wavelength λ	(μm)	–
Focal length f	(μm)	–
Defocus Δf	(μm)	–

② **Settings for the sheet name and cell address (row number and column number):** Input the sheet name and cell address of the incident field amplitude U_i, the incident field phase ϕ_i, the transmission coefficient amplitude U_t, the transmission coefficient phase ϕ_t, the diffraction field amplitude U_o (output), and the diffraction field phase ϕ_o (output), as shown in Table 9.2.

Notes

1. The incident field phase, the transmission coefficient amplitude and the transmission coefficient phase can be input directly as (uniform) fixed values, instead of being tabulated on the other spreadsheets. In this case, it will be unnecessary to input the sheet name and cell address.
2. The incident field amplitude, the incident field phase, the transmission coefficient amplitude, and the transmission coefficient phase should all be on separate sheets.
3. If the x-value of the input data exceeds the maximum column number allowed by Excel (only for old versions of Excel), the data table can be divided into multiple data tables, all in different rows. In this case, the addresses of the multiple tables (the sheet name, the row number, and the column number) will all need to be entered in the same row of the main sheet, in consecutive order. The cells on the right of the last table address should be left blank, so that the program will recognize the end of the table address parameters. (See Problem 9.10 for a worked example.)

③ **Start calculation:** Clicking on this button will initiate execution of the calculation program.

(2) Sheet "U_i": Incident Field Amplitude Settings

Input the incident field amplitude $U_i[\sqrt{(\mathrm{mW}/\mu\mathrm{m}^2)}]$ for each specified x–y position within the aperture. If the number of x values exceeds the maximum column number allowed by Excel [only for old versions

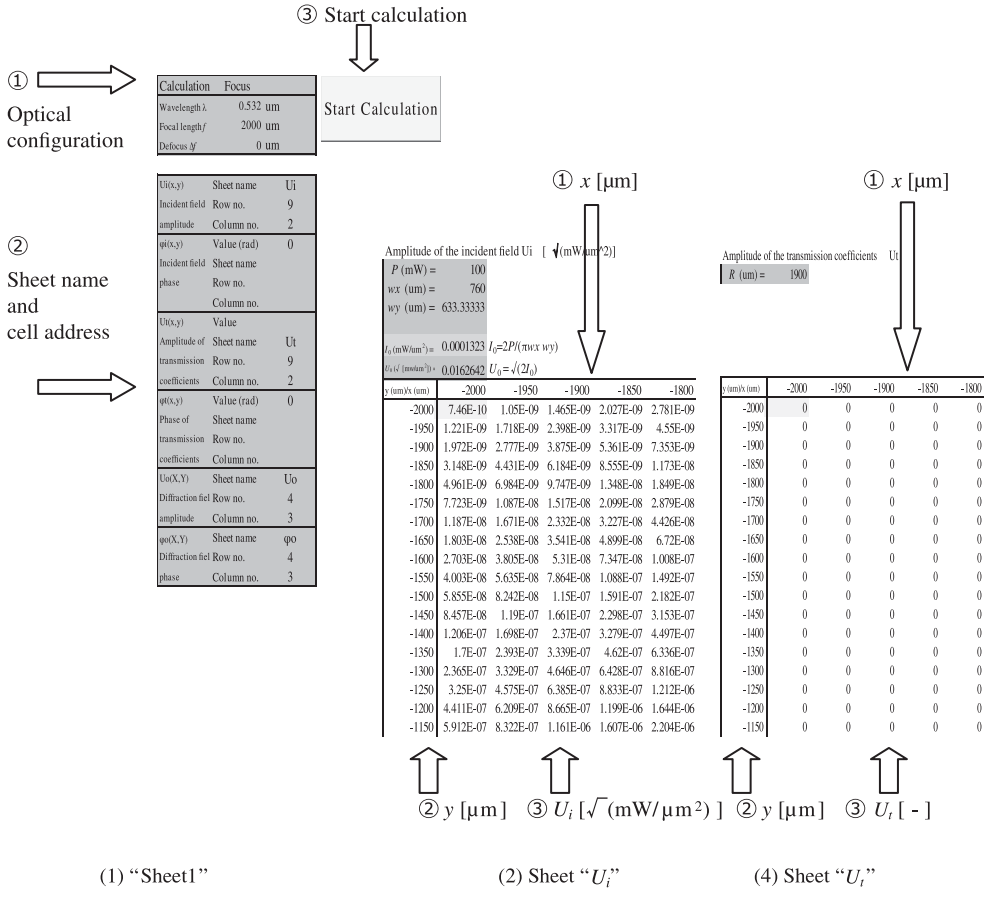


Figure 9.16 An input table for the two-dimensional X-Y diffraction calculation program

Table 9.2 Settings for the sheet name and cell address (row number and column number)

Items		Fixed value	Sheet name	Row number	Column number
Input settings	Incident field amplitude	–	○ U_i	○	○
	Incident field phase	Δ	Δ ϕ_i	Δ	Δ
	Transmission coefficient amplitude (additional field amplitude)	Δ	Δ U_t	Δ	Δ
	Transmission coefficient phase (additional field phase)	Δ	Δ ϕ_t	Δ	Δ
Output settings	Diffraction field amplitude	–	○ U_o	○	○
	Diffraction field phase	–	○ ϕ_o	–	–

○means necessary settings, and Δ means that there is no need to set the values on each sheet if we have already input the values on Sheet 1.

of Excel (1997–2003 Excel)], the data table can be divided into multiple data tables, all in different rows. (See Problem 9.10 for a worked example.)

① x (μm): Input the x-values for various points in the aperture into a single row, in increasing order from left to right, keeping the pitch constant.
② y (μm): Input the y-values for various points in the aperture into a single column, in increasing order from top to bottom, keeping the pitch constant.
③ $U_i[\sqrt{(\mathrm{mW}/\mu\mathrm{m}^2)}]$: Input the incident field amplitude values into the corresponding cells in the x–y table.

Note

The number of cells used for inputting the x-values and y-values, respectively, should not exceed 2000.

(3) Sheet "φ_i": Incident Field Phase Settings

Input the incident field phase ϕ_i (rad) for each x-y position in the aperture into the ϕ_i table on sheet "φ_i," using the same x-y data as for the U_i table above.

① x (μm): Input the x-values for various points in the aperture into a single row, in increasing order from left to right, keeping the pitch constant.
② y (μm): Input the y-values for various points in the aperture into a single column, in increasing order from top to bottom, keeping the pitch constant.
③ ϕ_i (rad): Input the incident field phase values into the corresponding cells in the x–y table.

(4) Sheet "U_t": Amplitude Settings of the Complex Amplitude Transmission Coefficients

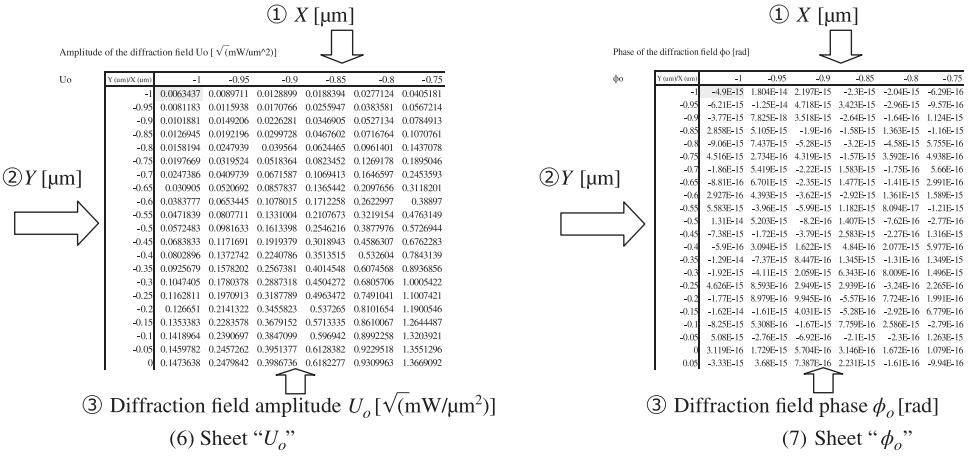
Input the amplitude coefficient U_t of the complex amplitude transmission coefficients for each x–y position in the aperture into the U_t table, using the same x–y data as for the U_i table above.

① x (μm): Input the x-values for various points in the aperture into a single row, in increasing order from left to right, keeping the pitch constant.
② y (μm): Input the y-values for various points in the aperture into a single column, in increasing order from top to bottom, keeping the pitch constant.
③ U_t (–): Input the amplitude coefficient of the complex amplitude transmission coefficients into the corresponding cells in the x–y table.

(5) Sheet "φ_t": Phase Settings for the Complex Amplitude Transmission Coefficients

Input the phase shift ϕ_t for the complex amplitude transmission coefficients for each x–y position in the aperture into the ϕ_t table, using the same x–y data as for the U_i table above.

① x (μm): Input the x-values for various points in the aperture into a single row, in increasing order from left to right, keeping the pitch constant.
② y (μm): Input the y-values for various points in the aperture into a single column, in increasing order from top to bottom, keeping the pitch constant.
③ ϕ_t (rad): Input the phase of the complex amplitude transmission coefficients into the corresponding cells in the x–y table.



Y (um)/X (um)	-1	-0.95	-0.9	-0.85	-0.8	-0.75
-1	0.0063437	0.0089711	0.0128899	0.0188394	0.0277124	0.0405181
-0.95	0.0081183	0.0115938	0.0170766	0.0255947	0.0383581	0.0567214
-0.9	0.0101881	0.0149206	0.0226281	0.0346905	0.0527134	0.0784913
-0.85	0.0126945	0.0192196	0.0299728	0.0467602	0.0716764	0.1070761
-0.8	0.0158194	0.0247939	0.039564	0.0624465	0.0961401	0.1437078
-0.75	0.0197669	0.0319524	0.0518364	0.0823452	0.1269178	0.1895046
-0.7	0.0247386	0.0409739	0.0671587	0.1069413	0.1646597	0.2453593
-0.65	0.030905	0.0520692	0.0857837	0.1365442	0.2097656	0.3118201
-0.6	0.0383777	0.0653445	0.1078015	0.1712258	0.2622997	0.38897
-0.55	0.0471839	0.0807711	0.1331004	0.2107673	0.3219154	0.4763149
-0.5	0.0572483	0.0981633	0.1613398	0.2546216	0.3877976	0.5726944
-0.45	0.0683833	0.1171691	0.1919379	0.3018943	0.4586307	0.6762283
-0.4	0.0802896	0.1372742	0.2240786	0.3513515	0.532604	0.7843139
-0.35	0.0925679	0.1578202	0.2567381	0.4014548	0.6074568	0.8936856
-0.3	0.1047405	0.1780378	0.2887318	0.4504272	0.6805706	1.0005422
-0.25	0.1162811	0.1970913	0.3187789	0.4963472	0.7491041	1.1007421
-0.2	0.126651	0.2141322	0.3455823	0.537265	0.8101654	1.1900546
-0.15	0.1353383	0.2283578	0.3679152	0.5713335	0.8610067	1.2644487
-0.1	0.1418964	0.2390697	0.3847099	0.596942	0.8992258	1.3203921
-0.05	0.1459782	0.2457262	0.3951377	0.6128382	0.9229518	1.3551296
0	0.1473638	0.2479842	0.3986736	0.6182277	0.9309963	1.3669092

Y (um)/X (um)	-1	-0.95	-0.9	-0.85	-0.8	-0.75
-1	-4.9E-15	1.804E-14	2.197E-15	-2.3E-15	-2.04E-15	-6.29E-16
-0.95	-6.21E-15	-1.25E-14	4.718E-15	3.423E-15	-2.96E-15	-9.57E-16
-0.9	-3.77E-15	7.825E-18	3.518E-15	-2.64E-15	-1.64E-16	1.124E-15
-0.85	2.858E-15	5.105E-15	-1.9E-16	-1.58E-15	1.363E-15	-1.16E-15
-0.8	-9.06E-15	7.437E-15	-5.28E-15	-3.2E-15	-4.58E-15	5.755E-16
-0.75	4.516E-15	2.734E-16	4.319E-15	-1.57E-15	3.592E-16	4.938E-16
-0.7	-1.86E-15	5.419E-15	-2.22E-15	1.583E-15	-1.75E-16	5.66E-16
-0.65	-8.81E-16	6.701E-15	-2.35E-15	1.477E-15	-1.41E-15	2.991E-16
-0.6	2.927E-16	4.393E-15	-3.62E-15	-2.92E-15	1.361E-15	1.589E-15
-0.55	5.583E-15	-3.96E-15	-5.99E-15	1.182E-15	8.094E-17	-1.21E-15
-0.5	1.31E-14	5.203E-15	-8.2E-16	1.407E-15	-7.62E-16	-2.77E-16
-0.45	-7.38E-15	-1.72E-15	-3.79E-15	2.583E-15	-2.27E-16	1.316E-15
-0.4	-5.9E-16	3.094E-15	1.622E-15	4.84E-16	2.077E-15	5.977E-16
-0.35	-1.29E-14	-7.37E-15	8.447E-16	1.345E-15	-1.31E-16	1.349E-15
-0.3	-1.92E-15	-4.11E-15	2.059E-15	6.343E-16	8.009E-16	1.496E-15
-0.25	4.626E-15	8.593E-16	2.949E-15	2.939E-16	-3.24E-16	2.265E-16
-0.2	-1.77E-15	8.979E-16	9.945E-16	-5.57E-16	7.724E-16	1.991E-16
-0.15	-1.62E-14	-1.61E-15	4.031E-15	-5.28E-16	-2.92E-16	6.779E-16
-0.1	-8.25E-15	5.308E-16	-1.67E-15	7.759E-16	2.586E-15	-2.79E-16
-0.05	5.08E-15	-2.79E-15	-6.92E-16	-2.1E-15	-2.3E-16	1.263E-15
0	3.119E-16	1.729E-15	5.704E-16	3.146E-16	1.672E-16	1.079E-16
0.05	-3.33E-15	3.68E-15	7.387E-16	2.231E-15	-1.61E-16	-9.94E-16

Figure 9.17 An output table for the two-dimensional X-Y diffraction calculation program

After combining these transmission coefficients, the amplitude $U(x,y)$ and the phase $\phi(x,y)$ will be:

$$U(x,y) = U_i(x,y) \times U_t(x,y) \qquad \phi(x,y) = \phi_i(x,y) + \phi_t(x,y)$$

Specify the following output items on the calculation table as shown in Figure 9.17.

(6) Sheet "U_o": Diffraction Field Amplitude Output

The calculation results for the diffraction field amplitude $U_o[\sqrt{(\mathrm{mW}/\mu\mathrm{m}^2)}]$ will be displayed on sheet "U_o" for each $X-Y$ position input.

① X (μm): Input the x-coordinate of the diffraction field in the row, as per Figure 9.17.
② Y (μm): Input the y-coordinate of the diffraction field in the column, as per Figure 9.17.
③ $U_o[\sqrt{(\mathrm{mW}/\mu\mathrm{m}^2)}]$: The calculation results for the diffraction field amplitude will be displayed in the corresponding cells in the $X-Y$ table.

Note

The number of cells used for outputting the x-values and y-values, respectively, should not exceed 2000.

(7) Sheet "φ_o": Diffraction Field Phase Output

The calculated results for the diffraction field phase ϕ_o (rad) will be displayed on the sheet "φ_o" for each X-Y position input.

① X (μm): The x-coordinate of the diffraction field is automatically displayed in the row, as per the "U_o" table.
② Y (μm): The y-coordinate of the diffraction field is automatically displayed in the column, as per the "U_o" table.
③ ϕ_o (rad): The calculation results for the diffraction field phase will be displayed in the corresponding cells in the $X-Y$ table.

Notes

1. This diffraction calculation program requires a large amount of computer memory. Other applications should be closed down before running this program.
2. This program uses the numerical integration calculation method. In order to obtain accurate results, numerical integration should generally have a small integration pitch, and an integration range which is sufficiently large (for decreasing functions). In this book, we calculate the total power of the diffraction field. If the total power does not satisfy the law of conservation of energy, the calculation results may be in error. Users are advised to verify that the total power satisfies the conservation law of energy after obtaining their calculation results.

9.7 Gaussian Beam Diffraction Calculation Problems

Please download the Excel files from the companion web site, which show calculation details for all problems in this book as listed in Appendix G

Problem 9.1 Central irradiance for a Gaussian beam

Calculate the central irradiance for a Gaussian beam whose power $P = 10\,\text{mW}$ and whose radius $w = 1\,\text{mm}$.

Solution

For a Gaussian beam whose radius is w and whose central irradiance is I_0, the power P_a within radius a will be:

$$P_a = \int_0^a 2\pi\rho I d\rho = I_0 \int_0^a 2\pi \exp\left(-\frac{2\rho^2}{w^2}\right)\rho d\rho \tag{9.1}$$

$$= -\pi\frac{w^2}{2} I_0 \left[\exp\left(-\frac{2\rho^2}{w^2}\right)\right]_0^a = \frac{\pi w^2}{2} I_0 \left\{1 - \exp\left(-\frac{2a^2}{w^2}\right)\right\} \tag{9.4}$$

The power P_∞ of the whole beam can be obtained by substituting $a = \infty$ in Equation (9.4).

$$P_\infty = \lim_{a\to\infty} P_a = \frac{\pi w^2}{2} I_0 \tag{9.5}$$

Thus the central irradiance I_0 and the central electric field U_0 will be:

$$I_0 = \frac{2P}{\pi w^2} \tag{9.6}$$

$$U_0 = \sqrt{2I_0} = \sqrt{\frac{4P}{\pi w^2}} \tag{9.7}$$

The central irradiance I_0 and the central electric field U_0 can be obtained by substituting $P = 10\,\text{mW}$ and $w = 1000\,\mu\text{m}$ into Equation (9.6) and Equation (9.7).

$$I_0 = 6.3662 \times 10^{-6}(\text{mW}/\mu\text{m}^2) \qquad U_0 = 0.0035681\ (\sqrt{(\text{mW}/\mu\text{m}^2)})^4$$

[4] The unit for the electric field is defined as [V/m] or [N/C] by the International System of Units. However in this book we use the unit of $[\sqrt{(\text{mW}/\mu\text{m}^2)}]$ for the sake of conbinience.

Note

The electric field distribution $U(\rho)$ (or$U(x,y)$) and the irradiance distribution $I(\rho)$ (or $I(x,y)$) of the Gaussian beam can be expressed as follows:

$$U(\rho) = U_0 \exp\left(-\frac{\rho^2}{w^2}\right) \quad \text{or} \quad U(x,y) = U_0 \exp\left(-\frac{x^2+y^2}{w^2}\right) \tag{7.7′}$$

$$I(\rho) = I_0 \exp\left(-\frac{2\rho^2}{w^2}\right) \quad \text{or} \quad I(x,y) = I_0 \exp\left(-2\frac{x^2+y^2}{w^2}\right) \tag{7.8′}$$

The three-dimensional graph in Figure 9.18 shows the irradiance distribution for this Gaussian beam.

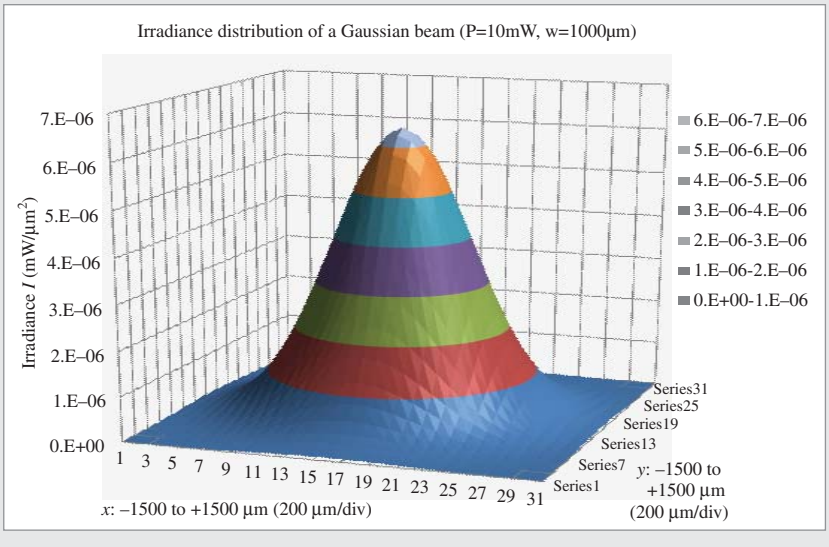


Figure 9.18 Irradiance distribution of a Gaussian beam (P = 10 mW, w = 1000 μm)

Problem 9.2 Diffraction irradiance for a focused Gaussian beam

In Figure 9.19, calculate the diffraction spot irradiance and the spot size for a collimated Gaussian beam [λ = 532 nm, P = 100 mW, w(radius) = 1.9 mm] focused by a lens (aperture radius a = 1.9 mm, f = 2 mm) using the two-dimensional R-θ diffraction calculation program[5], "Focus."

Solution

Settings: As shown in Figure 9.20, input the incident field and the transmission coefficients with their radius h in 10 μm increments, ranging from 0 to 1900 μm. [The incident field can be calculated using Equation (9.7) and Equation (7.7′) as in Problem 9.1.] The diffraction field (output) radius ρ will range from 0 to 1.0 μm.

[5] Diffraction calculation in this book has been successfully tested using Microsoft Excel 2003, Excel 2007 and Excel 2010.

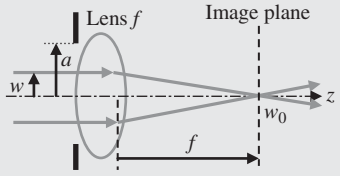


Figure 9.19 Focusing a Gaussian beam

Calculations: Click "Start calculation," then the calculation results for the diffraction field $U_o(\rho)$ and the phase $\phi_o(\rho)$ will be displayed on the right of the setting table. We can calculate the irradiance $I_o(\rho)$ using: $I_o(\rho) = U_o(\rho)^2/2$.

Row no.	20	: Settings	
Column no.	7	: Calculation results	

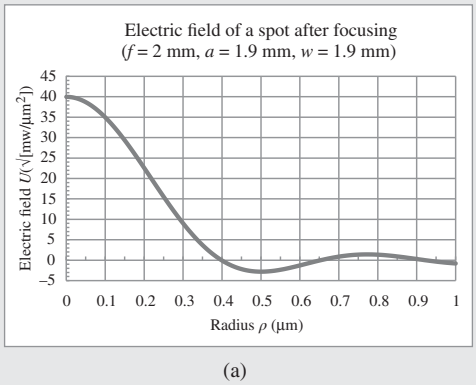
Incident beam			Optical system	
P (mW) =	100	Start Calculation	Wave length λ (um) =	0.532
w (um) =	1900		Focal length f (um) =	2000
a (um) =	1900		Differential segment $\Delta\psi$ (°) =	1
I_o (mW/um²) =	1.763E-05	$I_0=2P/(\pi w^2)$	Defocus Δf(um) =	0
U_o (√[mw/um²	0.0059388	$U_0=\sqrt{(2I_0)}$	Calculation program =	Focus

Total power (mW)
86.119409

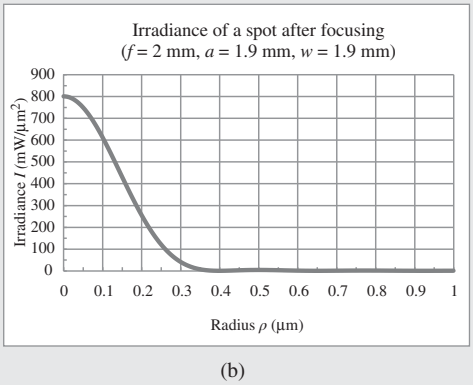
Incident field			Transmission coefficients		Diffraction field			$I=U^2/2$		
Radius h (um)	Amplitude U(h)	Phase φ(h) (rad)	Amplitude U(h)	Phase φ(h) (rad)	Radius ρ (um)	Amplitude Uo(ρ)	Phase φo(ρ) (rad)	Irradiance Io(ρ)	Uo cosφo	Beam Power (mW)
0	0.0059388	0	1	0	0	40.014447	0	800.57799	40.014447	0
10	0.0059387	0	1	0	0.01	39.961829	6.669E-17	798.4739	39.961829	0.5016959
20	0.0059382	0	1	0	0.02	39.804277	1.359E-16	792.19025	39.804277	0.9954956
30	0.0059374	0	1	0	0.03	39.542694	2.068E-16	781.81234	39.542694	1.4736815
40	0.0059362	0	1	0	0.04	39.178578	2.756E-16	767.48047	39.178578	1.9288888
50	0.0059347	0	1	0	0.05	38.714008	3.477E-16	749.3872	38.714008	2.3542693
60	0.0059329	0	1	0	0.06	38.151634	4.073E-16	727.77357	38.151634	2.7436417
70	0.0059308	0	1	0	0.07	37.494652	4.657E-16	702.92447	37.494652	3.0916233
80	0.0059283	0	1	0	0.08	36.746785	5.444E-16	675.1631	36.746785	3.3937399
90	0.0059255	0	1	0	0.09	35.912251	6.18E-16	644.84488	35.912251	3.6465119
100	0.0059224	0	1	0	0.1	34.995735	6.879E-16	612.35072	34.995735	3.8475131
110	0.005919	0	1	0	0.11	34.002353	7.545E-16	578.08001	34.002353	3.9954022
120	0.0059152	0	1	0	0.12	32.937617	8.412E-16	542.44332	32.937617	4.0899263
130	0.0059111	0	1	0	0.13	31.807393	9.032E-16	505.85511	31.807393	4.1318958
140	0.0059067	0	1	0	0.14	30.617856	9.781E-16	468.72655	30.617856	4.1231341
150	0.0059019	0	1	0	0.15	29.375451	1.071E-15	431.45856	29.375451	4.0664011
160	0.0058969	0	1	0	0.16	28.086843	1.172E-15	394.43538	28.086843	3.9652969
170	0.0058915	0	1	0	0.17	26.758871	1.202E-15	358.01858	26.758871	3.824145
180	0.0058858	0	1	0	0.18	25.398497	1.334E-15	322.54182	25.398497	3.6478621
190	0.0058797	0	1	0	0.19	24.012761	1.41E-15	288.30635	24.012761	3.4418163
200	0.0058734	0	1	0	0.2	22.608731	1.529E-15	255.57735	22.608731	3.2116797

Figure 9.20 Settings on the two-dimensional R-θ diffraction calculation table

Calculation results (Figure 9.21a,b)



(a)



(b)

Figure 9.21 (a) Diffraction field and (b) irradiance of a spot after focusing

Notes

1. We have obtained a spot central irradiance of 800.6 mW/μm^2 for a 100 mW Gaussian beam.
2. A very small spot size (0.512 μm in diameter) is obtained when we use a lens with a short focal length and a large numerical aperture.

Problem 9.3 Size of the incident beam that will maximize the focused spot irradiance

In Problem 9.2, we calculated the central irradiance I_0 of the focused spot, using the two-dimensional R-θ diffraction calculation program. In the same manner, we will calculate the focused spot irradiance, by varying the incident beam radius incrementally under a fixed aperture size, and obtain the incident beam radius which maximizes the spot irradiance (Figure 9.22).

Laser power $P = 100$ mW
Wavelength $\lambda = 0.532$ μm
Focal length $f = 2$ mm
Aperture radius $a = 1.9$ mm
Incident beam radius w = 7.6 to 0.76 mm. (Aperture/beam size ratio $c = a/w = 0.25$ to 2.5.)

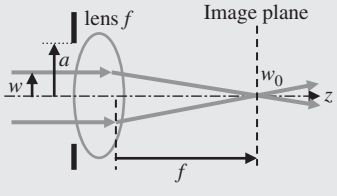
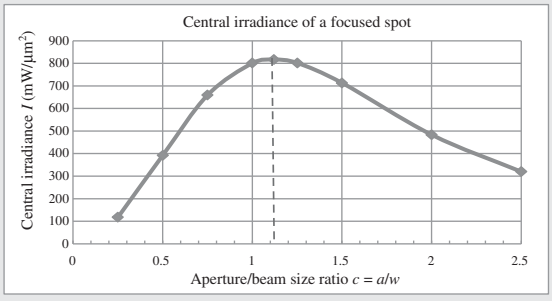


Figure 9.22 Focusing a Gaussian beam (by varying the incident beam radius)

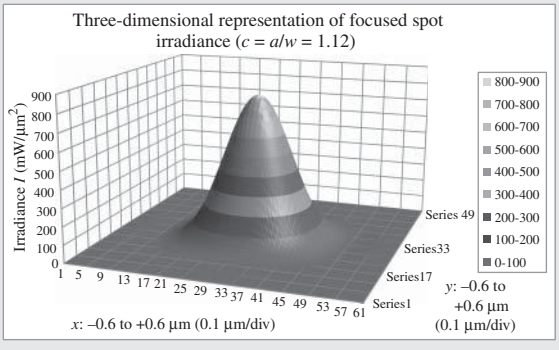
Solution

First, we will make a new calculation sheet for this problem by copying the Excel book for Problem 9.2. As with the calculation sheet for Problem 9.2, we can calculate the diffraction field for the cases when $c = a/w = 0.25$, 0.5, 0.75, 1, 1.12, 1.25, 1.5, 2, 2.5, and obtain the central irradiance. (Here we shall use the two-dimensional R-θ diffraction calculation program, "Focus.")

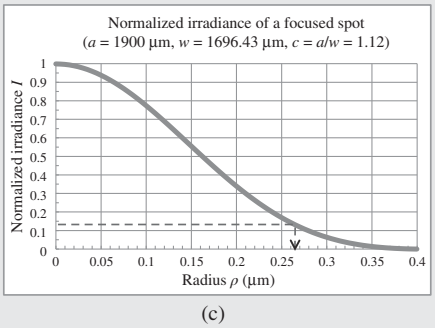
Calculation results: The maximum irradiance is obtained when $c = a/w = 1.12$, as shown in Figure 9.23a. The maximum irradiance is: $I_0 = 815.98$ mW/μm^2 for a spot radius of $w_0 = 0.264$ μm as in Figure 9.23b,c.



(a)



(b)



(c)

Figure 9.23 Focused spot irradiance which is maximized when $c = a/w = 1.12$. (a) Aperture/beam size ratio vs. central irradiance. (b) Focused spot irradiance when a/w=1.12. (c) Normalized irradiance of the focused spot when a/w=1.12

Note

The central irradiance is maximized when the aperture/beam size ratio $c = a/w$ is 1.12.

Problem 9.4 Diffraction calculation for a truncated/untruncated Gaussian beam

In Figure 9.24, calculate the irradiance of the diffraction spot in Problem 9.2, letting the aperture/beam size ratio $c = a/w = 2,\ 2.5,\ 3$. Compare the results with the Gaussian beam focusing calculation in Chapter 7 where the beam is not truncated.

Laser power $P = 100$ mW
Wavelength $\lambda = 0.532$ μm
Focal length $f = 2$ mm
Aperture radius $a = 1.9$ mm
Incident beam radius $w = 0.95, 0.76,$ and 0.633 mm.
(Aperture/beam size ratio $c = a/w = 2,\ 2.5,\ 3$)

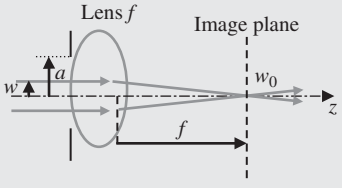


Figure 9.24 Focusing a truncated/untruncated Gaussian beam

Gaussian beam focusing equations

Focused spot radius

$$w_0 \approx \lambda R/(\pi w) = \lambda f/(\pi w) \tag{7.11′}$$

Central irradiance

$$I_0 = 2P/(\pi {w_0}^2) \tag{9.6}$$

Irradiance distribution

$$I(\rho) = I_0 \exp(-2\rho^2/{w_0}^2) \tag{7.8}$$

Solution

Calculate the diffraction field $U(\rho)$ using the two-dimensional R-θ diffraction calculation program, "Focus." The diffraction irradiance $I(\rho)$ can be calculated as follows:

$$I(\rho) = U(\rho)^2/2$$

Calculate the focused spot irradiance $I(\rho)$ using the Gaussian beam focusing equations Equation (7.8), Equation (7.11′), and Equation (9.6).

Calculation results (Figure 9.25a,b)

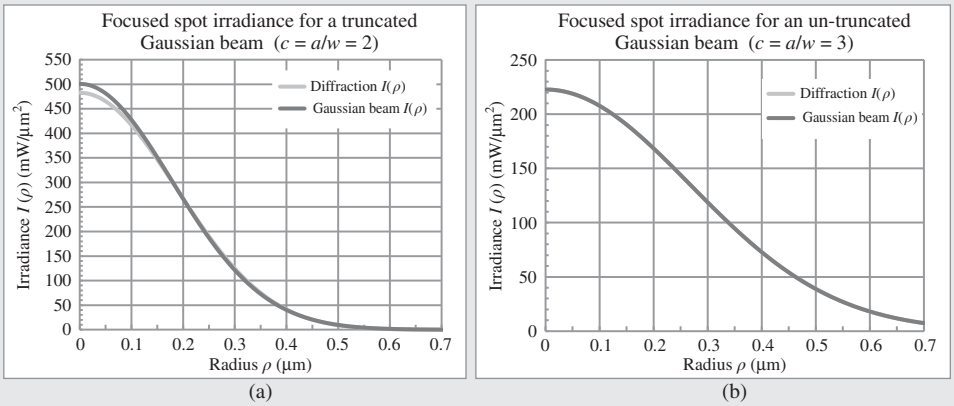


Figure 9.25 Focused spot irradiance for (a) a truncated and (b) an untruncated Gaussian beam

Notes

1. The diffraction calculation for the truncated Gaussian beam yields the same results as the Gaussian beam focusing calculation when the aperture is sufficiently large ($c = a/w \geqq 3$). This means that the Gaussian beam focusing calculation is valid for an untruncated beam.
2. If the aperture size is at least three times the radius of the Gaussian beam, then the aperture will have no effect on the Gaussian beam.

Problem 9.5 Diffraction calculation for an elliptical Gaussian beam after passing through a circular aperture

In Figure 9.26, calculate the diffraction spot irradiance for a collimated elliptical Gaussian beam ($\lambda = 532$ nm, $P = 100$ mW) focused by a lens ($f = 2$ mm, aperture radius $a = 1.9$ mm), using the two-dimensional X-Y diffraction calculation program "Focus," where the incident beam radii w_a (in the x-direction), w_b (in the y-direction) are specified as follows:

① $w_a = 0.76$ mm, $w_b = 0.6333$ mm
② $w_a = 1.9$ mm, $w_b = 0.6333$ mm.

Solution

The central field amplitude U_0, and field distribution $U(x,y)$ for an elliptical Gaussian beam are:

$$U_0 = \sqrt{\frac{4P}{\pi w_a w_b}} \tag{9.10}$$

$$U(x,y) = U_0 \exp\left[-\left\{\left(\frac{x}{w_a}\right)^2 + \left(\frac{y}{w_b}\right)^2\right\}\right] \tag{9.15}$$

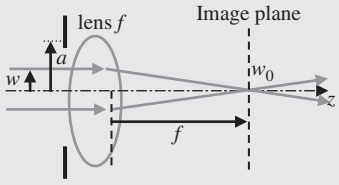


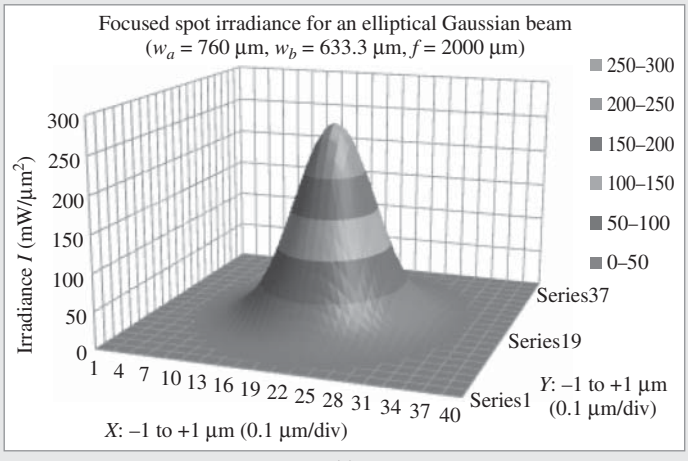
Figure 9.26 Focusing an elliptical Gaussian beam

Substituting the above ① and ② into Equation (9.10), we obtain the following values of U_0:

① $U_0 = 0.016264$ ($\sqrt{(\text{mW}/\mu\text{m}^2)}$),
② $U_0 = 0.010286$ ($\sqrt{(\text{mW}/\mu\text{m}^2)}$)

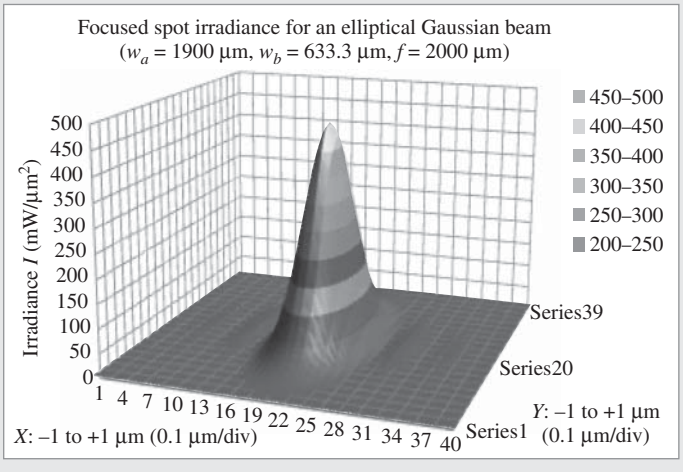
Next, we calculate the diffraction field $U_o(X,Y)$ using the two-dimensional X-Y diffraction calculation program "Focus," by setting the incident field $U(x,y)$ as in Equation (9.15). The calculation results for the diffraction irradiance $I_o(= U_o{}^2/2)$ are shown in Figure 9.27a,b.

① $w_a = 0.76, w_b = 0.6333$ mm
Diffraction spot central irradiance:
$I_{o0} = 266.68$ (mW/μm^2)
Diffraction spot radii are as follows:
$w_{oa} = 0.45$ μm
$w_{ob} = 0.54$ μm
② $w_a = 1.9, w_b = 0.6333$ mm
Diffraction spot central irradiance
$I_{o0} = 458.15$ (mW/μm^2)
Diffraction spot radii are as follows:
$w_{oa} = 0.24$ μm
$w_{ob} = 0.54$ μm



(a)

Figure 9.27 Focused spot irradiance for an elliptical Gaussian beam. (a) Focused spot irradiance ① when $w_a = 0.76$ mm, $w_b = 0.6333$ mm. (b) Focused spot irradiance ② when $w_a = 1.9$ mm, $w_b = 0.6333$ mm



(b)

Figure 9.27 *(continued)*

Problem 9.6 Diffraction calculation for an elliptical Gaussian beam after passing through a semicircular aperture

In Figure 9.28, calculate the diffraction spot irradiance for a collimated elliptical Gaussian beam ($\lambda = 532$ nm, $P = 100$ mW) focused by a lens ($f = 2$ mm, aperture radius $a = 1.9$ mm), using the two-dimensional X-Y diffraction calculation program "Focus," where the incident beam radii w_a (in the x-direction), w_b (in the y-direction) are given as:

$$w_a = 1.9 \text{ mm}, w_b = 0.6333 \text{ mm}$$

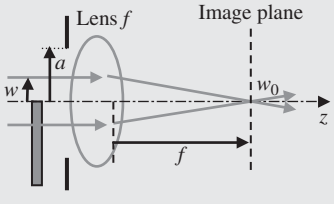


Figure 9.28 Focusing an elliptical Gaussian beam (with a semicircular aperture)

Cover one half of the lens aperture with a barrier lying in the same direction as w_a, as shown in Figure 9.28.

Solution

The central field amplitude U_0, and field distribution $U(x,y)$ for an elliptical Gaussian beam are as follows:

$$U_0 = \sqrt{\frac{4P}{\pi w_a w_b}} \tag{9.10}$$

$$U(x,y) = U_0 \exp\left[-\left\{\left(\frac{x}{w_a}\right)^2 + \left(\frac{y}{w_b}\right)^2\right\}\right] \tag{9.15}$$

Substituting the values of w_a and w_b into Equation (9.10), we obtain the following value for U_0:

$$U_0 = 0.010286\ (\sqrt{(\mathrm{mW}/\mu\mathrm{m}^2)}) \tag{9.15}$$

We can calculate the diffraction field using the two-dimensional X-Y diffraction calculation program, "Focus." The calculation results for the diffraction irradiance I_o $(= U_o{}^2/2)$ are shown in Figure 9.29.

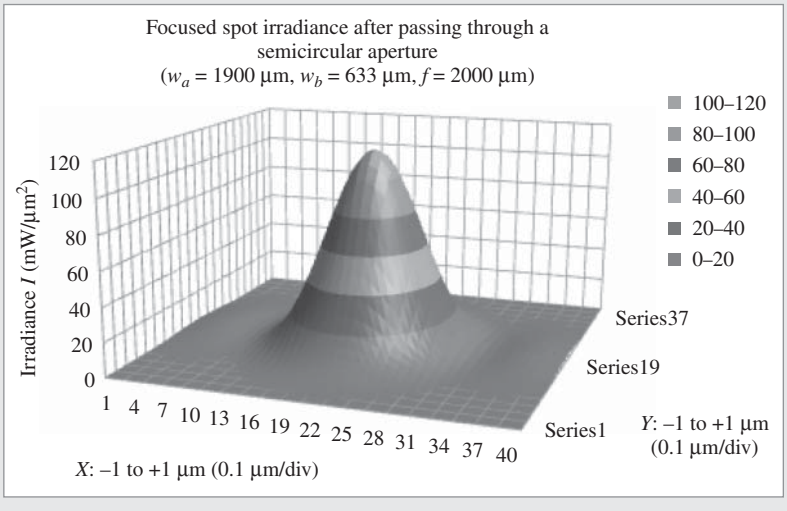


Figure 9.29 Focused spot irradiance after passing through a semicircular aperture.
Central irradiance of diffraction spot $I_{o0} = 115.67\ (\mathrm{mW}/\mu\mathrm{m}^2)$.
Radii of diffraction spot $w_{oa} = 0.44\ \mu\mathrm{m}$ $w_{ob} = 0.54\ \mu\mathrm{m}$

Notes

1. We have obtained a similarly shaped spot using the semicircular aperture as we did when using the circular aperture.
2. For the semicircular aperture, the radius of the spot in the masked direction is double that of the full-circle aperture, while the radius in the other direction is the same as that of the full-circle aperture. The central irradiance of the diffraction spot will be one-fourth (1/4) that of the full-circle aperture.

Problem 9.7 Diffraction calculation for a uniform irradiance beam after passing through a ring aperture

In Figure 9.30, calculate the diffraction spot irradiance for a collimated uniform irradiance beam ($\lambda = 532$ nm, $P = 100$ mW, with diameter = 3.8 mm) focused by a lens ($f = 2$ mm) with a ring aperture (outer/inner radius $a = 1.9$ mm/$b = 1.52$ mm), using the two-dimensional X-Y diffraction calculation program, "Focus." Compare your results with those obtained using a full-circle aperture (radius $a = 1.9$ mm).

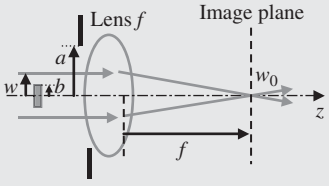
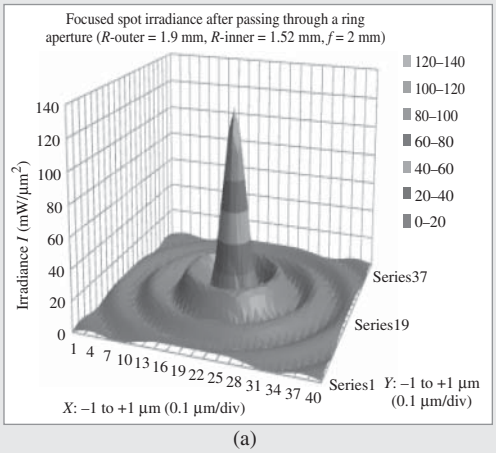


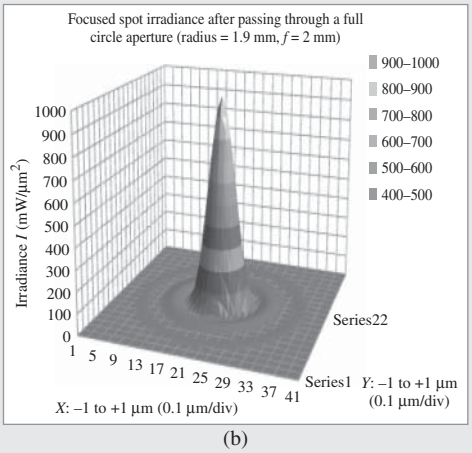
Figure 9.30 Focusing a uniform irradiance beam (with a ring aperture)

Solution

Calculate the diffraction field for the optical configuration described above, using the two-dimensional X-Y diffraction calculation program, "Focus." The calculation results for the diffraction irradiance I ($= U^2/2$) are shown in Figure 9.31a–c.

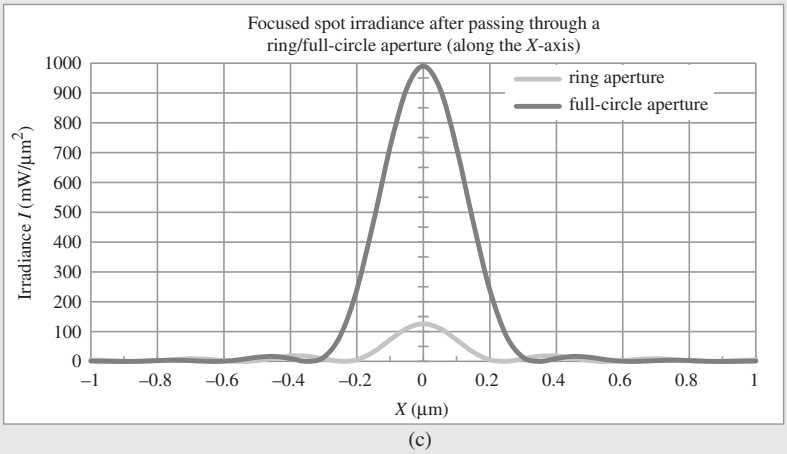


(a)



(b)

Figure 9.31 Focused spot irradiance after passing through a ring/full-circle aperture. (a) Ring aperture. (b) Full-circle aperture. (c) Ring/full-circle aperture (along X-axis)



(c)

Figure 9.31 *(continued)*

Note

As can be seen from Figure 9.31c, the ring aperture generates a smaller diffraction spot size than that generated by the original, full-circle aperture.

Problem 9.8 Double diffraction calculation for a truncated Gaussian beam

In Figure 9.32, calculate the diffraction spot field focused on plane B (f_1 = 100 mm) from the collimated, Gaussian, incident beam (wavelength λ = 0.532 μm, power P = 100 mW, beam radius w = 5000 μm, aperture radius a = 5000 μm). Derive the diffraction field collimated on plane C (f_2 = 100 mm) from the calculated field on plane B, using the two-dimensional R-θ diffraction calculation programs, "Focus" and "Collimate."

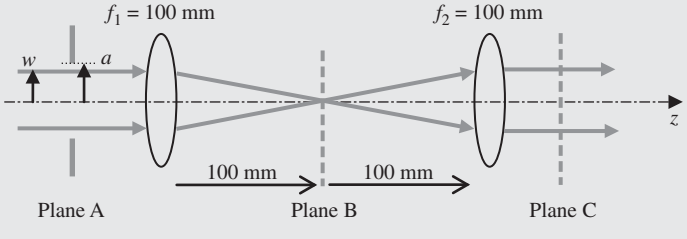
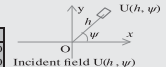


Figure 9.32 Double diffraction for a truncated Gaussian beam

Solution

Settings on the calculation table (Figure 9.33)

Two-dimensional R-θ diffraction calculation



Incident beam	
P (mW) =	100
w (um) =	5000
a (um) =	5000
I_0 (mW/um^2) =	2.546E-06
U_0 ($\sqrt{}$ [mw/um^2]) =	0.0022568

$I_0 = 2P/(\pi w^2)$

$U_0 = \sqrt{(2I_0)}$

"Focus"

Optical system	
Wave length λ (um) =	0.532
Focal length f (um) =	100000
Differential segment $\Delta\psi$ (°) =	0.5
Defocus Δf (um) =	0
Calculation program =	Focus

"Collimate"

Optical system	
Wave length λ (um) =	0.532
Focal length f (um) =	100000
Differential segment $\Delta\psi$ (°) =	0.2
Defocus Δf (um) =	0
Calculation program =	Collimate

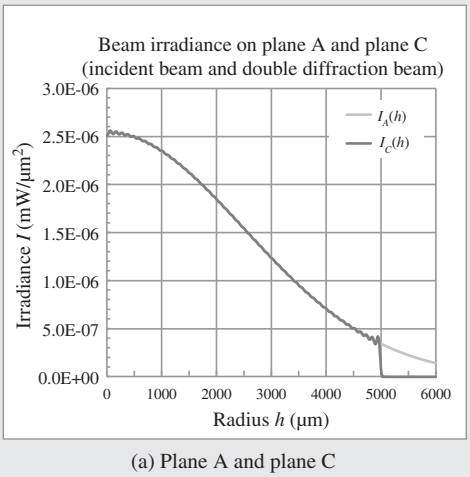
Incident field (Plane A)			Transmission coefficients		Diffraction field B			Transmission coefficients		Diffraction field C		
Radius h (um)	Amplitude U(h)	Phase φ(h) (rad)	Amplitude Ut(h)	Phase φt(h) (rad)	Radius ρ (um)	Amplitude Uo(ρ)	Phase φo(ρ) (rad)	Amplitude Ut(ρ)	Phase φt(ρ) (rad)	Radius h (um)	Amplitude Uo(h)	Phase φo(h) (rad)
0	0.0022568	0	1	0	0	2.1060235	0	1	0	0	0.0022413	−2.7E-13
20	0.0022567	0	1	0	0.2	2.1029551	7.384E-17	1	0	20	0.0022473	−2.4E-13
40	0.0022566	0	1	0	0.4	2.0937695	1.451E-16	1	0	40	0.0022591	−1.6E-13
60	0.0022564	0	1	0	0.6	2.0785249	2.182E-16	1	0	60	0.0022648	−5.9E-14
80	0.0022562	0	1	0	0.8	2.0573179	2.898E-16	1	0	80	0.0022598	2.36E-14
100	0.0022559	0	1	0	1	2.0302829	3.617E-16	1	0	100	0.002251	6.21E-14
120	0.0022555	0	1	0	1.2	1.9975905	4.152E-16	1	0	120	0.0022484	5.21E-14
140	0.002255	0	1	0	1.4	1.9594465	4.956E-16	1	0	140	0.002254	1.47E-14
160	0.0022544	0	1	0	1.6	1.9160897	5.833E-16	1	0	160	0.00226	−1.8E-14
180	0.0022538	0	1	0	1.8	1.8677909	6.643E-16	1	0	180	0.0022589	−2.7E-14
200	0.0022532	0	1	0	2	1.8148496	7.159E-16	1	0	200	0.0022519	−1.3E-14

Figure 9.33 Settings on the two-dimensional R-θ diffraction calculation table

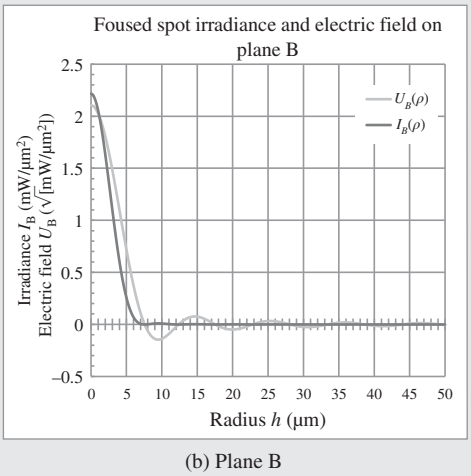
Radii for plane A: h = 0–5000 μm, plane B: ρ = 0–500 μm, plane C: h = 0–6000 μm

Calculation results

Calculated irradiances for plane B, as well as planes A and C, are shown in Figure 9.34a,b.[6]



(a) Plane A and plane C



(b) Plane B

Figure 9.34 Calculated irradiances at planes A, B, and C. (a) Planes A and B. (b) Plane C

The irradiance for plane C turns out to be the same as for plane A.

Notes

1. A broad, decreasing oscillation[6] can be seen on plane B (generated by the truncated beam).
2. After a double diffraction, the original image on plane A reappears on plane C.

[6] The extensive oscillation of the electric field on plane B requires a range large enough to integrate the secondary waves from plane B.

Problem 9.9 Diffraction calculation for a Gaussian beam after passing through a slit (one-dimensional diffraction calculation)

In Figure 9.35, calculate the diffraction spot irradiance for a collimated one-dimensional Gaussian beam ($\lambda = 0.532$ μm, $P = 100$ mW) from a slit after being focused by a cylindrical lens $f = 100$ mm, where the slit width $2a$ and the incident beam width $2w$ are as follows:

① Slit width $2a = 25$ mm, incident beam width $2w_a = 40$ mm
② Slit width $2b = 20$ mm, incident beam width $2w_b = 80$ mm.

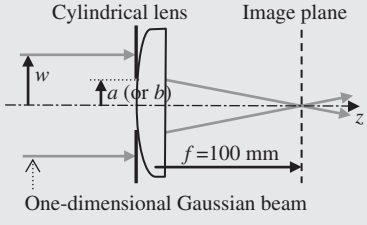


Figure 9.35 Focusing a one-dimensional Gaussian beam

Solution

The central irradiance I_0, peak amplitude U_0, and electric field U of a one-dimensional Gaussian beam (width: $2w$) are as follows:

$$I_0 = \sqrt{\frac{2}{\pi}}\frac{P}{w} \tag{9.18}$$

$$U_0 = \sqrt{2I_0} \tag{9.19}$$

$$U = U_0 \exp\left[-\left(\frac{x}{w}\right)^2\right] \tag{9.20}$$

The two solutions for U_0 are:

① $U_0 = 0.0893$ ($\sqrt{(\text{mW}/\mu\text{m})}$),
② $U_0 = 0.0632$ ($\sqrt{(\text{mW}/\mu\text{m})}$).

We can calculate the diffraction field $U(X)$ and the irradiance $I(X)$ using the one-dimensional diffraction calculation program, "SlitFocus," by setting the incident field $U(x,y)$, described in Equation (9.20), as shown in Figure 9.36.

One-dimensional diffraction calculation-X
"SlitForcus"(w=20mm, a=12.5mm)

Incident beam		
P (mW) =	100	
w (um) =	20000	
a (um) =	12500	
I0 (mW/um²) =	0.0039894	$I_0=\sqrt{(2/\pi)}\ P/w$
U0 (√[mw/um2]	0.0893244	$U_0=\sqrt{(2I_0)}$

Optical system	
Wave length λ (um) =	0.532
Focal length f (um) =	100000
Differential segment Δψ (°) =	
Defocus Δf (um) =	0
Calculation program =	SlitFocus

Total power (mW) = 76.770177

Incident field			Transmission coefficients		Diffraction field			Focued spot irradiance
Position x (um)	Amplitude Ui(x)	Phase φi(x) (rad)	Amplitude Ut(x)	Phase φt(x) (rad)	Position X (um)	Amplitude Uo(X)	Phase φo(X) (rad)	I(X) 2wa=40mm, 2aa=25mm
-12500	0.0604399	0	1	0	-5	0.7483739	8.743E-17	0.2800317
-12450	0.0606287	0	1	0	-4.95	0.7167545	3.431E-16	0.2568685
-12400	0.0608173	0	1	0	-4.9	0.6803947	-1.45E-16	0.2314685
-12350	0.0610058	0	1	0	-4.85	0.6393893	-1.83E-16	0.2044093
-12300	0.061194	0	1	0	-4.8	0.5938592	6.607E-17	0.1763344
-12250	0.0613821	0	1	0	-4.75	0.5439516	3.04E-16	0.1479417
-12200	0.06157	0	1	0	-4.7	0.48984	-8.58E-16	0.1199716

Figure 9.36 Settings on the one-dimensional diffraction calculation table

Calculation results

The diffraction irradiances are shown in Figure 9.37.

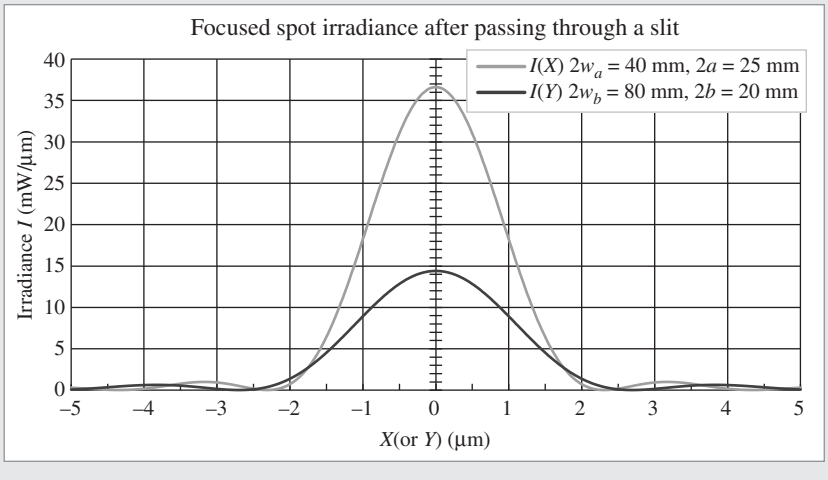


Figure 9.37 Diffraction irradiance for a one-dimensional Gaussian beam

The central irradiances I_0 for the diffraction field are as follows:

① $I_{0a} = 36.60$ mW/μm
② $I_{0b} = 14.39$ mW/μm.

Note

For practical purposes, the one-dimensional beam can be regarded as a flat beam of infinite width, in a direction perpendicular to the paper, as shown in Figure 9.37. The cylindrical lens and slit can also be regarded as having an infinite width, in a direction perpendicular to the paper. Therefore the power of the one-dimensional beam mentioned above is defined for a unit width in the direction perpendicular to the paper.

Problem 9.10 Two-dimensional diffraction calculation combining one-dimensional diffractions

Formulate a method for generalizing the one-dimensional slit diffraction calculations in Problem 9.9 to the case of a two-dimensional rectangular aperture diffraction, as shown in Figure 9.38, where the aperture size $2a \times 2b = 25 \times 20$ mm, incident beam radius $w_a \times w_b = 20 \times 40$ mm, and the focal length of the lens $f = 100$ mm.

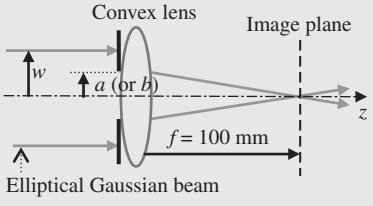


Figure 9.38 Focusing a two-dimensional Gaussian beam

Solution

Generalizing from one to two dimensions entails replacing a one-dimensional Gaussian beam with an elliptical Gaussian beam whose radii are w_a and w_b [see Equation (9.11), Equation (9.12) and Equation (9.13)], replacing the slit with a rectangular aperture of length and width $2a \times 2b$, and replacing the cylindrical lens with a spherical lens. As shown by Equation (8.8) and Equation (8.26a) in Chapter 8, the diffraction irradiance $I(X,Y)$ from a rectangular aperture has the form of a product of the two diffraction irradiances from the slits, $I_a(X)$ and $I_b(Y)$. Thus we can express the diffraction irradiance $I(X,Y)$ from a rectangular aperture using the diffraction irradiances $I_a(X)$ and $I_b(Y)$ from the slits (one for the length and the other for the width) as follows:

$$I(X,Y) = P[I_a(X)/P]\,[I_b(Y)/P] = P[I_a(X,0)/P]\,[I_b(0,Y)/P] \tag{9.55}$$

$I_a(X)$ = Irradiance component from slit, parallel to X-axis
[$I_a(X)/P$ = Irradiance per unit power]
$I_b(Y)$ = Irradiance component from slit, parallel to Y-axis
[$I_b(Y)/P$ = Irradiance per unit power]
P = Incident beam power

Calculation results (Figure 9.39a–c).

① Figure 9.39a shows the two-dimensional diffraction calculated by combining[7] the slit diffractions in Problem 9.9, using Equation (9.55).
② Figure 9.39b shows the results directly calculated by the two-dimensional X-Y diffraction calculation program "Focus".[8]
③ Figure 9.39c shows that the difference between ① and ② is zero, that is, ① = ②.

[7] The two-dimensional data table can be generated from the one-dimensional data tables, using the (Excel) Offset function.
[8] In the two-dimensional x-y diffraction calculation above, the number of x-values exceeds the maximum column number allowed by Excel 2003 (or older), so we shall divide the data table into multiple data tables with different rows.

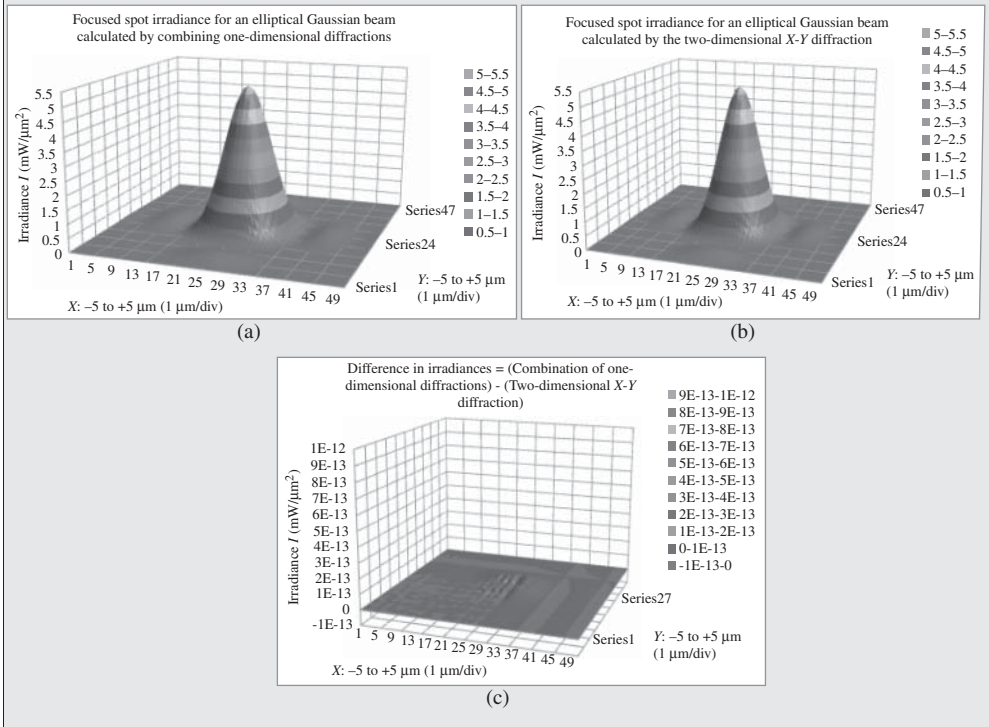


Figure 9.39 Two-dimensional diffraction irradiance combining *x*- and *y*- one-dimensional diffraction. (a) Combining *x*- and *y*- one-dimensional diffraction. (b) Calculated by *X*-*Y* two-dimensional diffraction. (c) (a)−(b)

Note

The diffraction irradiance $I(X,Y)$ of a Gaussian beam from a rectangular aperture can be expressed as a product of the two diffraction irradiances $I(X)$ and $I(Y)$ from the slit [see Equation (9.55)].

Problem 9.11 Calculation of focused spot irradiance, after taking account of wave-front aberration

Calculate the wave-front aberration for the optical system (consisting of a beam expander and a focusing lens) in Figure 9.40, using the two-dimensional VBA ray tracing program described in Chapter 4. Next, calculate the electric field at the front plane of the focusing lens L_3 using the Gaussian beam focusing equations in Chapter 7. Then calculate the diffraction irradiance on the image plane using the two-dimensional R-θ diffraction calculation program "Focus," after taking account of wave-front aberration, where the wavelength $\lambda = 0.532$ μm, power $P = 100$ mW, and the incident beam radius $w = 0.9$ mm. (The beam is a collimated Gaussian beam.) Graph the focused spot irradiance distribution for the distance where the central irradiance is maximized.

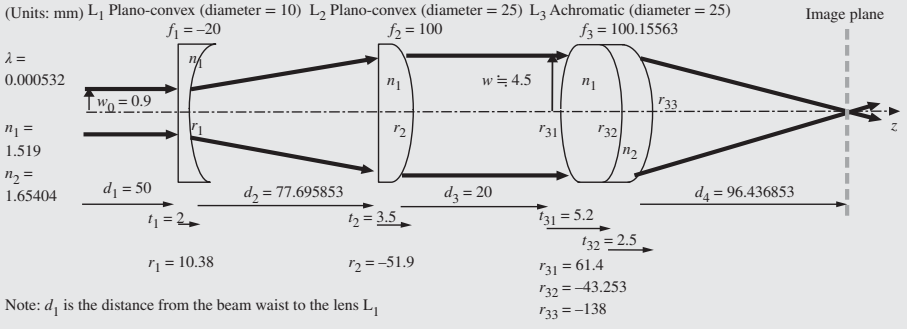


Figure 9.40 Laser optical system consisting of a beam expander and a focusing lens

Solution

Calculation of the wave-front aberration by performing a ray tracing

Input the optical configurations above into the two-dimensional VBA ray tracing program described in Chapter 4 as in Figure 9.41 and calculate the wave-front aberration after exiting the lens L_3. (We shall place the reference wave front at the vertex of the back surface of the lens L_3.)

λ (mm) = 0.000532	Surface no. =	1	2	3	4	5	6	7	8	Aberration	10	Image Plane	12		
	Surface type =	Plane	Sphere	Plane	Sphere	Plane	Sphere	Sphere	Sphere	Plane	Plane	Plane	Plane		
n_0 = 1	Refractive index n =	1.519	1	1.519	1	1	1.519	1.65404	1	1	1	1	1	zImageSet =	
	Distance d (mm) =	50	2	77.695853	3.5	20	0	5.2	2.5	0	95	1.4368533	1.4368533	yImageSet =	
	r or f (mm) =		10.38		-51.9		61.4	-43.253	-138					zImage =	96.436853
	Tilt angle α (degrees) =	0		0		0				0	0	0	0	yImage =	0
2w0=0.9*2= 1.8	Offset (mm) =	0	0	0	0	0	0	0	0	0	0	0	0	rImage =	96.436853

00 (degrees)	tan000	w00	w01	w02	w03	w04	w05	w06	w07	w08	w09	w10	w11	w12	OPL	WF aberration
Distance za (mm) =		0	50	52	129.69585	133.19585	153.19585	153.19585	158.39585	160.89585	160.89585	255.89585	257.33271	258.76958	OPL (mm)	W (lambdas)
Ray01	0	-1.8	-1.8	-1.8	-8.974411	-9.137825	-9.133772	-9.133639	-8.949601	-8.855933	-8.829819	-0.10848	0.0234282	0.1553365	167.68031	1.0423514
Ray02	0	-1.65	-1.65	-1.65	-8.198927	-8.355525	-8.352508	-8.352425	-8.171803	-8.089027	-8.069128	-0.102226	0.018272	0.1387696	167.74687	0.7377222
Ray03	0	-1.5	-1.5	-1.5	-7.430959	-7.578853	-7.576656	-7.576605	-7.402743	-7.330001	-7.315196	-0.09533	0.0138685	0.1230673	167.80694	0.5046762
Ray04	0	-1.35	-1.35	-1.35	-6.669629	-6.807203	-6.805646	-6.805617	-6.641442	-6.577979	-6.56728	+0.087806	0.0101946	0.1081951	167.86072	0.3314699
Ray05	0	-1.2	-1.2	-1.2	-5.9143	-6.040001	-6.038935	-6.038919	-5.886079	-5.832135	-5.824679	-0.079683	0.0072086	0.0941004	167.9084	0.207073
Ray06	0	-1.05	-1.05	-1.05	-5.164076	-5.276696	-5.275997	-5.275989	-5.138482	-5.091685	-5.086724	-0.071006	0.0048558	0.0807174	167.95013	0.1214283
Ray07	0	-0.9	-0.9	-0.9	-4.418296	-4.516756	-4.516324	-4.516321	-4.39512	-4.355882	-4.352776	-0.061828	0.0030714	0.067971	167.98603	0.0655555
Ray08	0	-0.75	-0.75	-0.75	-3.676274	-3.75967	-3.759424	-3.759423	-3.656697	-3.624009	-3.62222	-0.052212	0.0017836	0.0557792	168.01623	0.0316157
Ray09	0	-0.6	-0.6	-0.6	-2.93735	-3.00494	-3.004816	-3.004815	-2.920649	-2.895373	-2.894461	-0.042224	0.0009157	0.0440551	168.04082	0.0129494
Ray10	0	-0.45	-0.45	-0.45	-2.200882	-2.25208	-2.252028	-2.252028	-2.188032	-2.169305	-2.168921	-0.031934	0.0003871	0.0327085	168.05986	0.004497
Ray11	0	-0.3	-0.3	-0.3	-1.466244	-1.500616	-1.5006	-1.5006	-1.457524	-1.445147	-1.445034	-0.021417	0.0001148	0.0216467	168.07343	0.00088892
Ray12	0	-0.15	-0.15	-0.15	-0.73282	-0.750077	-0.750075	-0.750075	-0.728414	-0.722258	-0.722244	-0.010747	1.437E-05	0.0107756	168.08155	5.057E-05
Ray13	0	0	0	0	0	0	0	0	0	0	0	0	0	0	168.08425	0

Figure 9.41 Settings on the two-dimensional VBA ray tracing calculation table

Ray tracing results (Figure 9.42a,b).

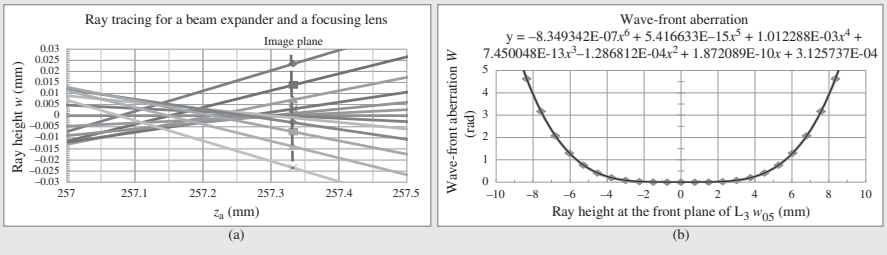


Figure 9.42 Ray tracing results with wave-front aberration. (a) Ray tracing results. (b) Wave-front aberration

If we graph the wave-front aberration y(in rad) against the ray height (in mm) at the front plane of L_3 on the horizontal x-axis, we will obtain the following approximation from the graph.

Approximate equation: $y = -8.349342\text{E}-07x^6 + 1.012288\text{E}-03x^4 - 1.286812\text{E}-04x^2$(rad).

Next, calculate the electric field of the beam at the front plane of L_3 using the Gaussian beam focusing equations. (We approximated the amplitude of the electric field at the front plane of the lens L_3 using the Gaussian beam focusing equations.)

Then, we calculate the diffraction irradiance of the focused spot using the two-dimensional R-θ diffraction calculation "Focus" as shown in Figure 9.43, where the aperture plane is defined as lying on the front plane of L_3, after inputting the electric field amplitude and phase which correspond with the Gaussian beam electric field amplitude and the wave-front aberration we calculated above.

Two-dimensional R-θ diffraction calculation

Incident beam		
P (mW) =	100	
w (μm) =	4500.457	
a (μm) =	12500	
I0 (mW/μm²) =	3.143E-06	$I_0=2P/(\pi w^2)$
U0 (√(mW/μm²)) =	0.0025073	$U_0=\sqrt{(2I_0)}$

Defocus= -91 μm

Optical system	
Wave length λ (μm) =	0.532
Focal length f (μm) =	100155.63
Differential segment Δw (°) =	1
Defocus Δf (μm) =	-91
Calculation program =	Focus

Gaussian beam calculation

w_0= 3.7686478

$I_0=2P/(\pi w_0^2)$= 4.8423835

$I=I0\ EXP(-2*(\rho/w0)^2)$

Total power (mW)	Incident power
99.899917	99.983523

Incident field			Transmission coefficients		Diffraction field			I=U²/2		
Radius h (μm)	Amplitude U(h)	Phase φi(h) (rad)	Amplitude U(h)	Phase φt(h) (rad)	Radius ρ (μm)	Amplitude Uo(ρ)	Phase φo(ρ) (rad)	Irradiance Io(ρ)	Beam power (mW)	Incident power P(h) (mW)
0	0.0025073	0	1	0	0	2.7673507	-0.070452	3.8291149	0	0
100	0.002506	-1.2E-06	1	0	0.1	2.765732	-0.070453	3.8246366	0.240309	0.1972958
200	0.0025023	-3.5E-06	1	0	0.2	2.7608805	-0.070453	3.8112304	0.4789333	0.3934245
300	0.0024961	-3.4E-06	1	0	0.3	2.7528102	-0.070453	3.788982	0.7142063	0.5872302
400	0.0024875	5.32E-06	1	0	0.4	2.7415446	-0.070451	3.7580335	0.9444968	0.7775803
500	0.0024765	3.11E-05	1	0	0.5	2.7271162	-0.070443	3.7185815	1.1682268	0.9633756
600	0.0024631	8.48E-05	1	0	0.6	2.7095667	-0.070428	3.670876	1.3838876	1.1435616
700	0.0024473	0.00018	1	0	0.7	2.6889467	-0.070402	3.6152173	1.5900556	1.3171383
800	0.0024293	0.000332	1	0	0.8	2.6653156	-0.070359	3.5519535	1.7854066	1.483169

Figure 9.43 Settings on the two-dimensional R-θ diffraction calculation table

Two-dimensional R-θ diffraction calculation "Focus"

Electric field at the aperture:

$U(h) = U_0 \exp(-(h/w)^2)$ (from Gaussian beam focusing calculation) where

$U_0 = \sqrt{(4P/(\pi w^2))} = 0.002507255$ ($\sqrt{(\text{mW}/\mu\text{m}^2)}$)($P = 100$ mW, $w = 4500.456955$ μm).

Phase (wave-front aberration): $\phi_i(h) = -8.349342\text{E}-07h^6 + 1.012288\text{E}-03h^4 - 1.286813\text{E}-04h^2$(rad)

Calculation results

Diffraction irradiance is obtained by shifting the image plane over the range 0 to −140 μm, as shown in Figure 9.44a,b. Figure 9.44a shows that the central irradiance is maximized when "Image plane shift" $\Delta z = -91$ μm, while Figure 9.44b depicts the diffraction irradiance distribution (at $\Delta z = -91$ μm) as a function of the radius, and also the irradiance obtained by the Gaussian beam calculation which takes no account of aberration and beam truncation.

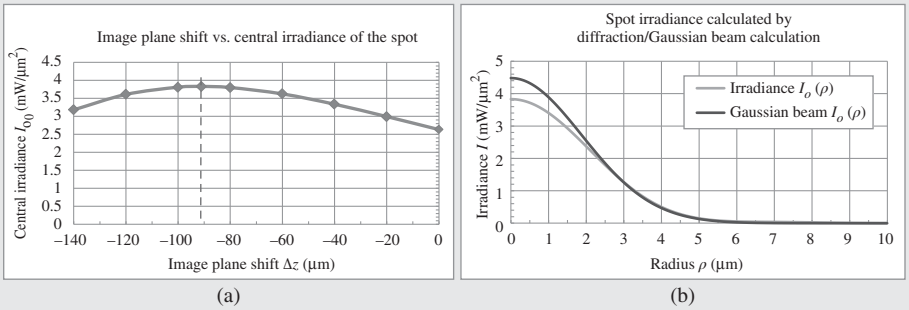


Figure 9.44 Diffraction irradiance is obtained by shifting the image plane. (a) Image plane shift vs. central irradiance. (b) Irradiance calculated by diffraction and Gaussian beam calculation

Notes

1. The diffraction irradiance, after taking wave-front aberration into account, was obtained by performing the two-dimensional R-θ diffraction calculation "Focus," where the aperture plane was placed at the front plane of L_3 with its electric field by the Gaussian beam calculation and its phase by the wave-front aberration, and by shifting the image plane.
2. We obtained the maximum spot irradiance $I_0 = 3.83\ \text{mW}/\mu\text{m}^2$ at a defocus $\Delta z = -91\ \mu\text{m}$.
3. The central irradiance fell from 4.5 to 3.83 $\text{mW}/\mu\text{m}^2$ due to aberration.

Problem 9.12 Diffraction calculation for a collimated Gaussian beam

In Figure 9.45, calculate the diffraction irradiance of the collimated Gaussian beam exiting the lens ($f_1 = 2$ mm) which is generated from the beam waist at the object plane, using the two-dimensional X-Y diffraction calculation, "Collimate." Here the wavelength λ, the beam power P, and the incident beam waist radii w_{ix} and w_{iy} (elliptical) at the object plane are as follows: $\lambda = 0.406\ \mu\text{m}$, $P = 100$ mW, $w_{ix} = 0.7\ \mu\text{m}$, $w_{iy} = 0.5\ \mu\text{m}$.

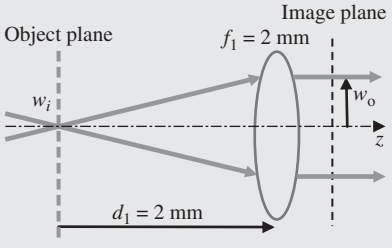


Figure 9.45 Collimating an elliptical Gaussian beam

Solution

1. Calculate the collimated beam radius w_o and the central irradiance I_{o0} after exiting the lens, using the Gaussian beam equations Equation (7.11′) and Equation (9.9), respectively.

$$w_o = \lambda f/(\pi w_i) \tag{7.11′}$$

$$I_{o0} = 2P/(\pi w_{ox} w_{oy}) \tag{9.9}$$

The calculation results are as follows:

$$w_{ox} = 369.2395\ \mu\text{m},\quad w_{oy} = 516.9353\ \mu\text{m},\quad I_{o0} = 0.0003335\ \text{mW}/\mu\text{m}^2.$$

2. Calculate the diffraction field $U_o(X,Y)$ and the diffraction irradiance $I_o(X,Y)$ for the collimated beam exiting the lens, using the two-dimensional X-Y diffraction calculation program "Collimate." The diffraction irradiance $I_o(X,Y)$ of the collimated beam is calculated as in Figure 9.46a–c.

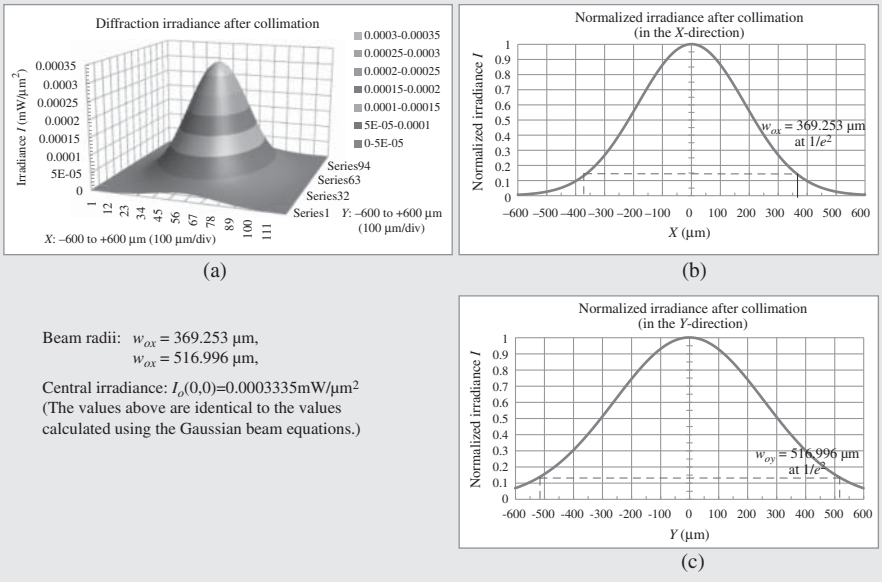


Figure 9.46 Diffraction irradiance after collimation. (a) Irradiance depicted in three dimensions. (b) Normalized irradiance along the *X*-axis. (c) Normalized irradiance along the *Y*-axis

Notes

1. We obtained the same irradiance data for the values calculated using the Gaussian beam equations and those calculated by the two-dimensional X-Y diffraction calculation program.
2. In the diffraction calculation program, the range and the increment value of the position for setting the incident fields should be chosen carefully[9] and with appropriate values, in order to maintain a reasonable degree of accuracy in the calculation.

References

[1] Sagai, J. (1997) *Kogaku no Kiso*, Corona Publishing Co., Ltd, Japan (in Japanese).
[2] Suhara, T. (1998) *Koha-kogaku*, Corona Publishing Co., Ltd, Japan (in Japanese).

[9] Here, we input the *x*-*y* coordinates of the electric field of the incident beam in 0.02μm increments, ranging from −2μm to 2μm for both *x* and *y*. (To maintain a reasonable degree of accuracy in the numerical integration calculation, the electric field should be input with a range three times greater than the Gaussian beam size.)

Appendix A

Paraxial Theory: A Detailed Account

Here, we will derive equations for obtaining focal lengths and principal points for a combination of two lenses and for a simple lens [1].

A.1 Derivation of Equation (1.39), Equation (1.40), Equation (1.41), and Equation (1.42) for Obtaining Focal Lengths and Principal Points for a Combination of Two Lenses

In Figure A.1, f_1 and f_1' represent the first and second focal lengths, while $\mathbf{H_1}$ and $\mathbf{H_1}'$, respectively, represent the primary and the secondary principal planes of the first lens. Similarly, f_2, f_2', $\mathbf{H_2}$ and $\mathbf{H_2}'$ represent the corresponding lengths and planes for the second lens, while δ represents the distance between the focal points F_1' and F_2. ($\delta = F_1'F_2$, $\delta > 0$ in the example in the figure.) The incident ray **R** (parallel to the axis of height w), passes through the primary principal plane $\mathbf{H_1}$, is refracted at the secondary principal plane $\mathbf{H_1}'$, passes through the second focal point F_1', is refracted at the primary principal plane $\mathbf{H_2}$, travels parallel to the axis, is refracted at the secondary principal plane $\mathbf{H_2}'$ and finally exits the second lens. The ray $\mathbf{R}'$ exiting the lens will intersect the axis at F'. Thus F' must be the second focal point of the combination lens, as the incident ray **R** is stipulated to be parallel to the axis. F_1' and F' are the conjugate points for the second lens, so we can apply Newton's formula, Equation (1.28), to the second lens as follows:

$$F_2'F' = -\frac{f_2 f_2'}{\delta} \tag{A.1}$$

Next, if we extend the incident ray **R** to the right in the figure, the point where the (extended) incident ray **R** intersects with ray $\mathbf{R}'$ will lie on the secondary principal plane $\mathbf{H}'$ of the combination lens. This is because the ray height at the primary principal plane of the combination lens is w, so the ray height at the secondary principal plane $\mathbf{H}'$ must also be w. The focal length f' of the combination lens will therefore be the distance from $\mathbf{H}'$ to F'. Now let us consider the ray which diverges from ray **R** at the first focal plane $\mathbf{F_2}$ of the second lens and then travels parallel to the axis. This ray will pass through the primary principal plane $\mathbf{H_2}$, be refracted at the secondary principal plane $\mathbf{H_2}'$ and pass through the second focal point F_2' of the second lens. In the figure, the ray $H_{2A}'F_2'$ and the ray $\mathbf{R}'$ must be parallel to each other,

Optical Design Using Excel®: Practical Calculations for Laser Optical Systems, First Edition. Hiroshi Nakajima.
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Companion Website: www.wiley.com/go/nakajima/excel

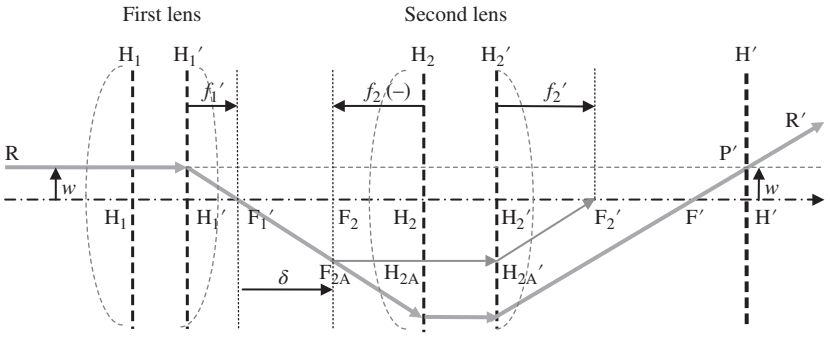


Figure A.1 Combination of two lenses

because rays starting from a point on the first focal plane $\mathbf{F_2}$ must exit the lens parallel to each other. In Figure A.1,

$$\frac{H'P'}{H'F'} = \frac{H_2'H_{2A}'}{H_2'F_2'} \qquad \frac{H_1'F_1'}{w} = \frac{F_2F_1'}{F_2F_{2A}}$$

After multiplying the above equations, we obtain the following equation:

$$\frac{H_1'F_1'}{H'F'} = \frac{F_2F_1'}{H_2'F_2'} \quad \therefore \quad \frac{f_1'}{f'} = -\frac{\delta}{f_2'}$$

since $H'P' = w$, and $F_2F_{2A} = H_2H_{2A} = H_2'\,H_{2A}'$.

Thus,

$$f' = -\frac{f_1'f_2'}{\delta} \tag{A.2}$$

Using the same reasoning as above, the first focal length f for the combination lens will be (this time, the incident ray is from the right):

$$f = \frac{f_1f_2}{\delta} \tag{A.3}$$

The distance F_1F (between the first focal point of the first lens and that of the combination lens) and the distance $F_2'F'$ (between the second focal point of the second lens and that of the combination lens) will be obtained applying Newton's formula, Equation (1.28):

$$F_1F = \frac{f_1f_1'}{\delta} \tag{A.4a}$$

$$F_2'F' = -\frac{f_2f_2'}{\delta} \tag{A.4b}$$

The distance H_1H (between the primary principal points of the first lens and the combination lens) and the distance $H_2'H'$ (between the secondary principal points of the second lens and the combination lens) can therefore be expressed as follows:

$$H_1H = \frac{f_1d}{\delta} \tag{A.5a}$$

$$H_2'H' = \frac{f_2'd}{\delta} \tag{A.5b}$$

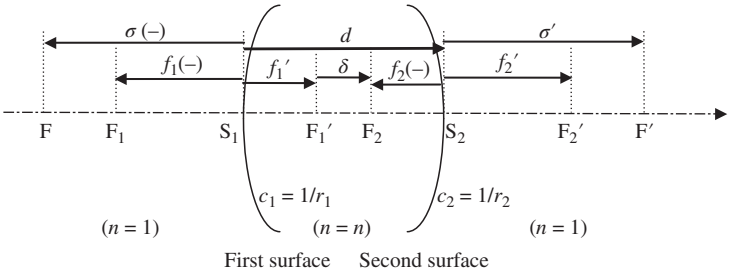


Figure A.2 A simple lens, viewed as a combination of two surfaces

where

$$d = \mathrm{H_1'H_2} = f_1' + \delta - f_2$$

Equation (A.2) and Equation (A.3) are equivalent to Equation (1.40) and Equation (1.39), and Equation (A.5a,b) is equivalent to Equation (1.41) and Equation (1.42), respectively.

A.2 Derivation of Equation (1.34), Equation (1.35), and Equation (1.36) for Obtaining Focal Lengths and Principal Points of a Simple Lens

We can regard a simple lens as a combination of a front and a back surface imaging system. In Figure A.2, f_1, f_1' and f_2, f_2' represent the focal lengths of the front surface and back surface, respectively, while δ is the distance between the two imaging systems. It can be seen that $\delta = \mathrm{F_1'F_2}$. From Figure A.2,

$$\delta = -f_1' + d + f_2 \tag{A.6}$$

Using Equation (A.2) and Equation (A.3), the focal length of the combination lens f, f' can be expressed as:

$$\frac{1}{f} = \frac{\delta}{f_1 f_2} = \frac{1}{f_1} - \frac{f_1'}{f_1}\frac{1}{f_2} + \frac{d}{f_1 f_2} \tag{A.7a}$$

$$\frac{1}{f'} = -\frac{\delta}{f_1' f_2'} = -\frac{f_2}{f_2'}\frac{1}{f_1'} + \frac{1}{f_2'} - \frac{d}{f_1' f_2'} \tag{A.7b}$$

Using the curvature radii r_1, r_2 of the surfaces, f_1, f_1', f_2, f_2' can be expressed as follows. [The reader is invited to refer to Equation (A.21a,b) below for a more detailed derivation.]

$$f_1 = \frac{1}{1-n} r_1 \tag{A.8a}$$

$$f_1' = \frac{n}{n-1} r_1 \tag{A.8b}$$

$$f_2 = \frac{n}{n-1} r_2 \tag{A.8c}$$

$$f_2' = \frac{1}{1-n} r_2 \tag{A.8d}$$

Substituting Equation (A.8a–d) into Equation (A.7a,b), we obtain:

$$\frac{1}{f'} = -\frac{1}{f} = (n-1)\left[\left(\frac{1}{r_1} - \frac{1}{r_2}\right) + \frac{(n-1)d}{n r_1 r_2}\right] \tag{1.34}$$

The distances from the vertices S, S′ to the focal points $\sigma = S_1F$, $\sigma' = S_2F'$ will be:

$$\sigma = f_1 + F_1F = f_1 + \frac{f_1 f_1'}{\delta} \qquad \sigma' = f_2' + F_2'F' = f_2' - \frac{f_2 f_2'}{\delta}$$

We can re-express the above equation in terms of the curvature radii r_1, r_2 and the refractive index n:

$$\sigma = -\frac{r_1}{n-1}\frac{(n-1)d + nr_2}{(n-1)d + n(r_2 - r_1)} \tag{A.9a}$$

$$\sigma' = -\frac{r_2}{n-1}\frac{(n-1)d - nr_1}{(n-1)d + n(r_2 - r_1)} \tag{A.9b}$$

Likewise, the distances from the vertices S_1, S_2 to the principal points $p = S_1H$, $p' = S_2H'$ can be obtained from Equation (A.5a,b), Equation (A.2), and Equation (A.3), as follows:

$$p = \frac{f_1 d}{\delta} = \frac{fd}{f_2} \qquad p' = \frac{f_2' d}{\delta} = -\frac{f' d}{f_1'}$$

We can re-express the above equations in terms of the curvature radii r_1, r_2, and the refractive index n.

$$p = -\frac{r_1 d}{n(r_2 - r_1) + (n-1)d} \tag{1.35}$$

$$p' = -\frac{r_2 d}{n(r_2 - r_1) + (n-1)d} \tag{1.36}$$

Derivation of Equation (A.8a–d)

Relationship between the Focal Lengths f and f'

In this section, we shall derive the relationship between the two focal lengths f and f' where the refractive index of the medium surrounding the object is n and that of the medium surrounding the image point is n'.

In Figure A.3, both ray a and ray b pass through the principal planes of the lenses **H** and **H′**, respectively, traveling at the same height h. Ray a passes through the first focal point F, and ray b passes through the second focal point F′. Ray a can be viewed as a part of the light beam emitted from F, so its wave-front surface after exiting the lens will lie precisely on the secondary principal plane $H'H_1'$. Thus the optical path length $[FH_1H_1']$ will be equal to $[FHH']$.

$$[FH] + [HH'] = [FH_1] + [H_1H_1'] \tag{A.10}$$

FH_1 can be approximated as follows:

$$FH_1 = (f^2 + h^2)^{1/2} \cong -f\left(1 + \frac{1}{2}\frac{h^2}{f^2}\right) \tag{A.11a}$$

$$\therefore [FH_1] = -nf\left(1 + \frac{1}{2}\frac{h^2}{f^2}\right) \tag{A.11b}$$

[FH] can be expressed very simply, as follows:

$$[FH] = -nf \tag{A.12}$$

Substituting Equation (A.11a,b) and Equation (A.12) into Equation (A.10), we obtain:

$$[HH'] - [H_1H_1'] = -\frac{1}{2}n\frac{h^2}{f} \tag{A.13}$$

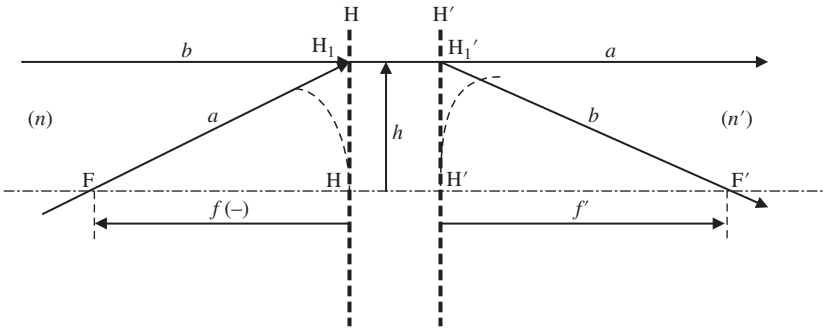


Figure A.3 Relationship between the two focal lengths

If we perform the same transformations for ray b (on the image side), we get:

$$[\mathrm{HH}'] - [\mathrm{H_1H_1}'] = \frac{1}{2}n'\frac{h^2}{f'} \tag{A.14}$$

From Equation (A.13) and Equation (A.14), we can derive the relationship between f and f':

$$\frac{n'}{f'} = -\frac{n}{f} \tag{A15}$$

If $n' = n$, then $f = -f'$. Substituting Equation (A15) into Equation (1.33), we obtain:

$$\frac{n'}{s'} - \frac{n}{s} = \frac{n'}{f'} = -\frac{n}{f} \tag{A.16}$$

Equation (A.16) represents the lens formula, where the refractive indices are n (on the object side) and n' (on the image side).

Imaging by a Single Surface Sphere

In this appendix, we shall derive the imaging equation for a single surface sphere. In Figure A.4, O and O′ are the conjugate points on the axis, s (=SO) and s' (=SO′) are their distances from the vertex S, and Σ and Σ′ are the wave fronts whose centers of curvature are situated at O and O′, respectively. The optical path length of the ray a from O to O′ must be equal to the ray traveling along the axis. Hence,

$$[\mathrm{QP}] = [\mathrm{ST}] \tag{A.17}$$

The z-coordinate of P can be approximated by $(1/2)h^2/r$ with its origin on S (h is the ray height on the surface of the sphere) and the z-coordinate of Q can likewise be approximated by $(1/2)h^2/s$. Thus,

$$[\mathrm{QP}] = \frac{1}{2}h^2 n\left(\frac{1}{r} - \frac{1}{s}\right) \tag{A.18}$$

(Here, we have assumed that the ray maintains the same height h from Q to P.) Similarly,

$$[\mathrm{ST}] = \frac{1}{2}h^2 n'\left(\frac{1}{r} - \frac{1}{s'}\right) \tag{A.19}$$

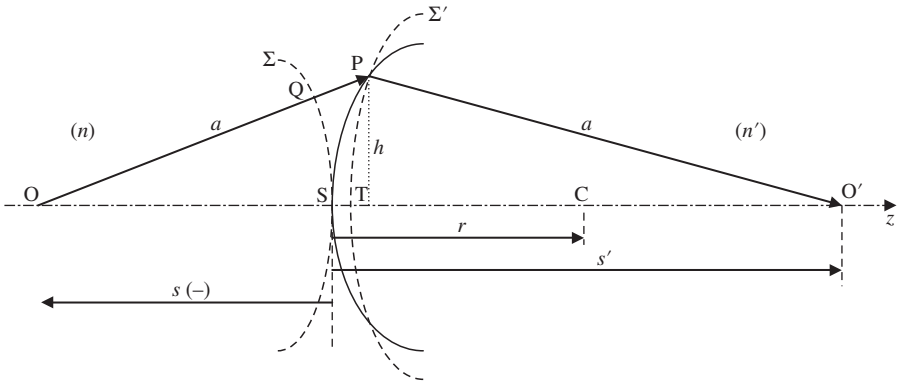


Figure A.4 Imaging by a single surface sphere

Substituting Equation (A.18) and Equation (A.19) into Equation (A.17), we can derive the imaging equation for a single surface sphere:

$$\frac{n'}{s'} - \frac{n}{s} = \frac{n' - n}{r} \tag{A.20}$$

Comparing Equation (A.20) with Equation (A.16), we can obtain the focal lengths f and f' for a single surface sphere:

$$f = \frac{n}{n - n'} r \tag{A.21a}$$

$$f' = \frac{n'}{n' - n} r \tag{A.21b}$$

Thus we can finally derive f_1, f_1', f_2, f_2' in Equation (A.8), by using Equation (A.21a,b).

Reference

[1] Miyake, K. (1979) *Kika-kogaku*, Kyoritsu Publishing Co., Ltd, Japan (in Japanese).

Appendix B

Table of Refractive Indices for BK7

Table B.1 A table of refractive indices for BK7 [1]

Wavelength (nm)	Refractive index
313.2	1.548
325	1.545
334.2	1.543
337.1	1.541
351.1	1.539
355	1.539
363.8	1.536
365	1.536
398.8	1.531
404.7	1.530
435.8	1.527
441.6	1.526
457.9	1.525
465.8	1.524
472.7	1.523
476.5	1.523
480	1.523
486.1	1.522
488	1.522
496.5	1.522
501.7	1.521
514.5	1.520
532	1.519
546.1	1.519
587.6	1.517
589.3	1.517
632.8	1.515
643.9	1.515
656.3	1.514

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Table B.1 (*continued*)

Wavelength (nm)	Refractive index
694.3	1.513
706.5	1.513
830	1.510
852.1	1.510
904	1.509
1014	1.508
1064	1.507
1100	1.507
1200	1.505
1300	1.504
1350	1.504
1380	1.503
1400	1.503
1500	1.501
1600	1.500
1700	1.499
1800	1.497
1900	1.497
2000	1.495
2100	1.493
2200	1.492
2220	1.492
2250	1.491
2300	1.490
2400	1.487
2500	1.485
2600	1.484
2650	1.483
2700	1.483
2720	1.482
2750	1.481

Reference

[1] Sigma-Koki (2003) *Optical Parts Catalogue*, Sigma-Koki Co., Ltd, Japan.

Appendix C

Equations for Plane Waves, Spherical Waves, and Gaussian Beams

In this appendix, we shall derive the equations for plane waves, spherical waves, and Gaussian beams which can be used to represent laser beams.

C.1 Equations for Plane Waves

Wave fronts are surfaces over which a wave has the same phase. By definition, the wave fronts of plane waves are planes. All of these wave fronts are perpendicular to the direction of wave propagation [1]. Figure C.1 depicts a plane wave which propagates from the origin O in the direction of $\boldsymbol{k}$ (where $\boldsymbol{k}$ is the propagation vector). The mathematical expression for all of these wave fronts is:

$$\boldsymbol{r} \cdot \boldsymbol{k} = \text{constant} \qquad \text{(Inner product)} \tag{C.1}$$

that is,

$$\boldsymbol{r} \cos\theta = \text{constant} \qquad \text{where } \theta = \text{the angle between vector } \boldsymbol{r} \text{ and vector } \boldsymbol{k} \tag{C.2}$$

Figure C.1a depicts wave fronts traveling along the z-axis in the x–z plane, while Figure C.1b depicts the wave fronts traveling in an arbitrary direction in x–y–z space.

We can now construct a set of planes over which the value $U(\boldsymbol{r})$ varies in a sinusoidal fashion in space, namely:

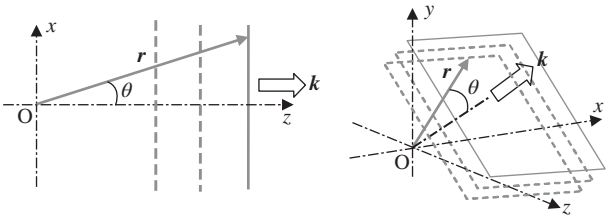
$$U(\boldsymbol{r}) = A \sin(\boldsymbol{r} \cdot \boldsymbol{k}) \tag{C.3}$$

$$U(\boldsymbol{r}) = A \cos(\boldsymbol{r} \cdot \boldsymbol{k}) \tag{C.4}$$

$$U(\boldsymbol{r}) = A \exp(j\boldsymbol{k} \cdot \boldsymbol{r}) \tag{C.5}$$

For each of these expressions, $U(\boldsymbol{r})$ remains constant over every plane defined by the equation $\boldsymbol{k}\cdot\boldsymbol{r}$ = constant. Since we are discussing harmonic functions here, we would expect these functions to

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(a) (b)

Figure C.1 A plane wave propagating along the ***k***-axis. (a) Propagating along the z-axis. (b) Propagating along an arbitrary direction in three-dimensional space

repeat themselves in space after traveling an interval of λ in the direction of vector ***k.*** The spatially repetitive nature of these harmonic functions can be expressed by the formula:

$$U(\boldsymbol{r}) = U(\boldsymbol{r} + \lambda \boldsymbol{k}/k) \tag{C.6}$$

where k is the magnitude of ***k*** and ***k***/k is a unit vector parallel to it. In exponential form, this is equivalent to:

$$Ae^{j\boldsymbol{k}\cdot\boldsymbol{r}} = Ae^{j\boldsymbol{k}(\boldsymbol{r}+\lambda\boldsymbol{k}/k)} = Ae^{j\boldsymbol{k}\boldsymbol{r}}e^{j\lambda\boldsymbol{k}}$$

Hence it follows that

$$e^{j\lambda k} = 1 = e^{j2\pi}$$

and

$$\lambda k = 2\pi \qquad k = 2\pi/\lambda \tag{6.10}$$

where k is the propagation number. At any fixed point in space where $\boldsymbol{k}\cdot\boldsymbol{r}$ is constant, the phase is constant, and hence $U(\boldsymbol{r})$ is constant. This means that the planes will not move. To get things moving, $U(\boldsymbol{r})$ must be made to vary over time. As we saw in Chapter 6, we can introduce a time-dependent variable ωt as follows:

$$U(\boldsymbol{r}, t) = A\ \exp[j(\boldsymbol{k}\cdot\boldsymbol{r} \mp \omega t)] \qquad \text{Plane wave equation} \tag{C.7}$$

where A, ω, ***k*** are constant. At any time, the wave has the same phase over each wave front. The wave function $U(\boldsymbol{r}, t)$ has the same value over a wave front, as A is constant over the wave front. In the case when the direction of the vector ***k*** lies along the x-axis, Equation (C.7) can be re-expressed as Equation (C.8):

$$\boldsymbol{k}\cdot\boldsymbol{r} = kr_k = kx$$

$$U(\boldsymbol{r}, t) = A\exp[j(kx \mp \omega t)] \tag{C.8}$$

Equation (C.8) is the same as Equation (6.52), which we studied in Chapter 6.

$$U(x, t) = A\exp[j(kx - \omega t + \varepsilon)] = A\exp[j\varphi] \tag{6.52}$$

We can now describe the plane wave traveling in x–y–z space as follows:

$$U(x, y, z) = A\exp[j(k_x x + k_y y + k_z z \mp \omega t)] \tag{C.9}$$

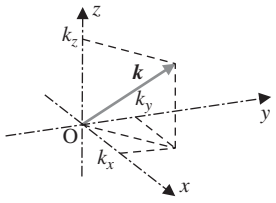


Figure C.2 Vector $\boldsymbol{k}$ in $x-y-z$ space

or

$$U(x, y, z) = A \exp[j(k(\alpha x + \beta y + \gamma z) \mp \omega t)] \tag{C.10}$$

where α, β, γ are the direction cosines of $\boldsymbol{k}$.

The magnitude of the propagation vector $\boldsymbol{k}$ is given by

$$|k| = k = (k_x{}^2 + k_y{}^2 + k_z{}^2)^{1/2} \tag{C.11}$$

$$\alpha^2 + \beta^2 + \gamma^2 = 1 \tag{C.12}$$

Figure C.2 depicts the vector $\boldsymbol{k}$ in $x-y-z$ space.

C.2 Three-dimensional Differential Equation of Wave Motion

In this appendix, we shall attempt to generalize the one-dimensional formula for wave motion which was discussed in Chapter 6, to three dimensions [1].

$$\frac{\partial^2 U}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 U}{\partial t^2} \tag{6.26}$$

Clearly, the three-dimensional plane wave described by Equation (C.10) represents the solution to the three-dimensional wave equation which we are trying to derive. If we partially differentiate Equation (C.10) by *x, y, z,* and *t* twice, we get:

$$\frac{\partial^2 U}{\partial x^2} = -\alpha^2 k^2 U \tag{C.13}$$

$$\frac{\partial^2 U}{\partial y^2} = -\beta^2 k^2 U \tag{C.14}$$

$$\frac{\partial^2 U}{\partial z^2} = -\gamma^2 k^2 U \tag{C.15}$$

$$\frac{\partial^2 U}{\partial t^2} = -\omega^2 U \tag{C.16}$$

Adding Equation (C.13), Equation (C.14), and Equation (C.15), and considering that $\alpha^2 + \beta^2 + \gamma^2 = 1$, we get:

$$\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2} = -k^2 U \tag{C.17}$$

Combining Equation (C.17) and the time-differential equation (C.16), considering $v = \omega/k$. we obtain the following equation:

$$\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 U}{\partial t^2} \tag{C.18}$$

Equation (C.18) has a symmetrical form regarding *x, y,* and *z,* and it also has the same form as the one-dimensional wave equation (6.26). It turns out that this is the three-dimensional wave equation which we were trying to obtain. Using the Laplacian operator,

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \tag{C.19}$$

Equation (C.18) can be expressed in a simpler form:

$$\nabla^2 U = \frac{1}{v^2}\frac{\partial^2 U}{\partial t^2} \tag{C.20}$$

The plane wave equation Equation (C.10) can also be expressed as Equation (C.10′). Clearly, it is a solution to the wave equation Equation (C.18).

$$U(x, y, z) = A \ \exp[j(k(\alpha x + \beta y + \gamma z) \mp \omega t)] \tag{C.10}$$

$$= A \ \exp[jk(\alpha x + \beta y + \gamma z \mp vt)] \tag{C.10′}$$

C.3 Equations for Spherical Waves

If a pebble is dropped into still water, it generates a wave which spreads out in all directions from the point where the pebble hit the surface of the water. In this appendix, we shall discuss what happens when the number of dimensions of the wave increases from two (an x–y plane) to three (an x–y–z space). Let us consider an idealized point source of light, where radiation is emitted uniformly from the source, in all directions. The source is isotropic and the resulting wave fronts are concentric spheres, which increase in diameter as they spread out into the surrounding space. For the sake of convenience, we shall use spherical polar coordinates (Figure C.3) to mathematically represent the behavior of symmetrical wave fronts [1].

Using spherical coordinates, the Laplacian operator can be expressed as follows:

$$\begin{aligned}\nabla^2 &\equiv \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2} \\ &= \frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial\theta^2} + \frac{1}{r^2}\cot\theta\frac{\partial}{\partial\theta} + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2} \\ x &= r\sin\theta\cos\varphi, \qquad y = r\sin\theta\sin\varphi, \qquad z = r\cos\theta\end{aligned} \tag{C.21}$$

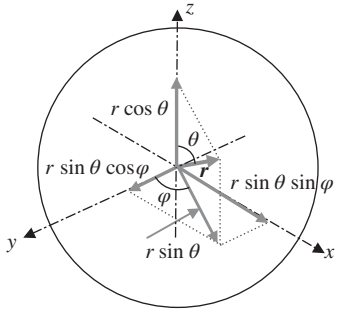


Figure C.3 The geometry of spherical coordinates

A more detailed derivation of Equation (C.21) is provided in Section C.6. We now need to find a suitable description for waves which are spherically symmetrical. These waves are characterized by the fact that they do not depend on θ and φ; hence we can drop the terms θ and φ.

$$U(\boldsymbol{r}) = U(r, \theta, \varphi) = U(r) \tag{C.22}$$

The Laplacian of $U(r)$ is then simply:

$$\nabla^2 U(r) = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial U}{\partial r}\right) = \frac{\partial^2 U}{\partial r^2} + \frac{2}{r}\frac{\partial U}{\partial r} \tag{C.23}$$

Equation (C.23) can also be expressed in a slightly different form, as follows:

$$\nabla^2 U = \frac{1}{r}\frac{\partial^2}{\partial r^2}(rU) \tag{C.24}$$

$$\because \frac{1}{r}\frac{\partial^2}{\partial r^2}(rU) = \frac{1}{r}\frac{\partial}{\partial r}\left(\frac{\partial}{\partial r}(rU)\right) = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial U}{\partial r} + U\right) = \frac{1}{r}\left(r\frac{\partial^2 U}{\partial r^2} + 2\frac{\partial}{\partial r}U\right)$$

$$= \frac{\partial^2 U}{\partial r^2} + \frac{2}{r}\frac{\partial U}{\partial r}$$

Hence the differential wave equation (C.20) can be written as:

$$\frac{1}{r}\frac{\partial^2}{\partial r^2}(rU) = \frac{1}{v^2}\frac{\partial^2 U}{\partial t^2} \tag{C.25}$$

Multiplying both sides by *r,* we obtain:

$$\frac{\partial^2}{\partial r^2}(rU) = \frac{1}{v^2}\frac{\partial^2}{\partial t^2}(rU) \tag{C.26}$$

Equation (C.26) has the same form as the one-dimensional wave equation (6.26), if we replace U with rU. Hence the solution to Equation (C.26) will simply be:

$$rU(r,t) = f(r - vt) \tag{C.27a}$$

or

$$U(r,t) = \frac{f(r - vt)}{r} \tag{C.27b}$$

Equation (C.27a,b) represents a spherical wave propagating radially outward from the origin, at a constant speed *v,* and having an arbitrary function form *f*. Another solution is given by:

$$U(r,t) = \frac{g(r + vt)}{r}$$

The above represents a wave converging toward the origin. The general solution will be:

$$U(r,t) = C_1\frac{f(r - vt)}{r} + C_2\frac{g(r + vt)}{r} \tag{C.28}$$

Here, if we regard the function *f* as a harmonic wave, Equation (C.28) will become:

$$U(r,t) = \frac{A}{r}\cos[k(r \mp vt)] \tag{C.29}$$

or

$$U(r,t) = \frac{A}{r}\exp[k(r \mp vt)] \tag{C.30}$$

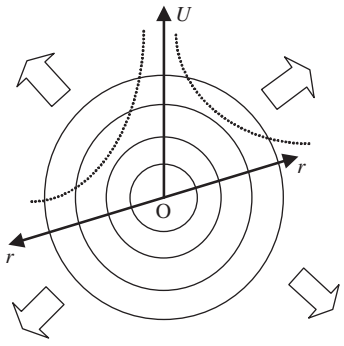


Figure C.4 Amplitude of a spherical wave

The above equation represents a spherical wave, in which the constant A is called the source strength. At any fixed point in time, this represents a cluster of concentric spheres filling the entire space as shown in Figure C.4. Each wave front (or surface of constant phase) is given by the equation:

$$kr = \text{constant}$$

The coefficient $1/r$ in the above equation signifies that a spherical wave decreases in amplitude as it travels outwards, unlike a plane wave.

C.4 Helmholtz's Wave Equation

The equation for a plane wave, Equation (C.7), and the equation of a spherical wave, Equation (C.30), which we studied previously, can be expressed as follows:

$$U(\boldsymbol{r}, t) = \boldsymbol{A} \exp[j(\omega t \mp \boldsymbol{k} \cdot \boldsymbol{r} + \varphi_0)] \tag{C.7′}$$

$$U(\boldsymbol{r}, t) = (\boldsymbol{A}/\boldsymbol{r}) \exp[j(\omega t \mp \boldsymbol{k} \cdot \boldsymbol{r} + \varphi_0)] \tag{C.30′}$$

Inspecting the above equations, we find that both of them have the form of the product of a space-dependent term and a time-dependent term. Generalizing, we can express these waves as follows:

$$U(r, t) = U_0(r)e^{j\omega t} = U_0(x, y, z)e^{j\omega t} \tag{C.31}$$

Additionally, the three-dimensional wave equation can be expressed as follows:

$$\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 U}{\partial t^2} \tag{C.18}$$

$$\nabla^2 U = \frac{1}{v^2}\frac{\partial^2 U}{\partial t^2} \tag{C.20}$$

Rewriting Equation (C.20), we obtain:

$$\nabla^2 U - \frac{1}{v^2}\frac{\partial^2 U}{\partial t^2} = 0 \tag{C.20′}$$

Substituting Equation (C.31) into Equation (C.20′), we get:

$$\nabla^2 U_0 - \frac{1}{v^2}(-\omega^2)U_0 = 0$$

If we substitute $v = f\lambda$ and $\omega = 2\pi f$ into the above equation, then we obtain:

$$\nabla^2 U_0 - \frac{1}{(f\lambda)^2}[-(2\pi f)^2]U_0 = 0 \qquad \nabla^2 U_0 + \left(\frac{2\pi}{\lambda}\right)^2 U_0 = 0$$

Rewriting the above, we get:

$$\nabla^2 U_0 + k^2 U_0 = 0 \tag{C.32a}$$

That is,

$$\frac{\partial^2 U_0}{\partial x^2} + \frac{\partial^2 U_0}{\partial y^2} + \frac{\partial^2 U_0}{\partial z^2} + k^2 U_0 = 0 \tag{C.32b}$$

We have thus obtained a wave equation which is independent of time t. This equation is commonly referred to as Helmholz's wave equation [2].

C.5 Equations for Gaussian beams

If a light source is small but not point-sized, then the wave front emitted from the source can no longer be regarded as a sphere. If the light emitted from a source exhibits sharp directivity, like a laser beam, then the wave front near the axis on which the source lies will behave like a superimposed form of a plane wave and a spherical wave. We can express the wave as:

$$U = E_0 u(x, y, z) e^{-jkz} \quad \text{where} \quad k = 2\pi/\lambda$$

As shown in Figure C.5, if we take the z-axis to be in the direction in which the wave exhibits sharp directivity, then the wave is symmetrical around the z-axis, and we can express the wave using a cylindrical coordinate system (ρ, φ, z). We can drop the term φ because of this symmetry.

$$U = E_0 u(r) e^{-jkz} = E_0 u(x, y, z) e^{-jkz} = E_0 u(\rho, \varphi, z) e^{-jkz} = E_0 u(\rho, z) e^{-jkz} \tag{C.33}$$

$$k = 2\pi/\lambda \qquad \rho^2 = x^2 + y^2 \qquad \varphi = \tan^{-1}(y/x)$$

Here, e^{-jkz} represents the phase factor of the wave propagating along the z-axis, just as it does for plane waves. $u(\rho,z)$ is a function which represents the difference between the wave and a plane wave. Let us assume that $u(\rho,z)$ varies gradually with regard to ρ and z. Substituting Equation (C.33) into wave equation (C.32), we can obtain a differential equation which $u(\rho,z)$ should satisfy. (See Section C.6.2 below for a more detailed explanation.)

$$\left(\frac{1}{\rho} \frac{\partial}{\partial \rho} + \frac{\partial^2}{\partial \rho^2} \right) u - 2jk \frac{\partial u}{\partial z} = 0 \tag{C.34}$$

where $|d^2u/(k^2dz^2)| \ll |du/(kdz)|$

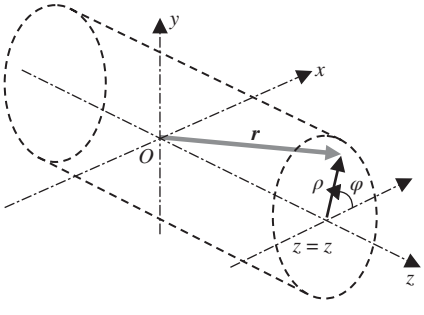


Figure C.5 Cylindrical coordinate system

The first term of Equation (C.34) is the expression we get when we convert the orthogonal coordinate system (x,y) to a polar coordinate system (ρ, φ). That is,

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \frac{1}{\rho}\frac{\partial}{\partial \rho} + \frac{\partial^2}{\partial \rho^2} \tag{C.35}$$

To solve Equation (C.34), we define $P(z)$ and $Q(z)$ as functions of a complex variable.

$$u(\rho, z) = \exp\left[-j\left\{P(z) + \frac{k\rho^2}{2Q(z)}\right\}\right] \tag{C.36}$$

Substituting Equation (C.36) into Equation (C.34), we obtain the equation which $P(z)$ and $Q(z)$ should satisfy.

$$\frac{dP(z)}{dz} + \frac{j}{\mathrm{Q}(z)} = 0 \tag{C.37a}$$

$$\frac{dQ(z)}{dz} - 1 = 0 \tag{C.37b}$$

The solution to Equation (C.37a,b) will then be as shown in the following. (For a more detailed explanation, the reader may refer to Section C.6.3.)

$$Q(z) = z + q_0 \tag{C.38a}$$

$$P(z) = -j\ln\left(1 + \frac{z}{q_0}\right) + p_0 \tag{C.38b}$$

Let us define the constants q_0, p_0 as follows:

$$q_0 = j\frac{\pi {w_0}^2}{\lambda} \tag{C.39a}$$

$$p_0 = 0 \tag{C.39b}$$

Substituting Equation (C.38a,b) and Equation (C.39a,b) into Equation (C.36), we obtain the following equations. (For a more detailed explanation, the reader may refer to Section C.6.4.)

$$u(\rho, z) = \frac{w_0}{w}\exp\left[j\varphi - \rho^2\left(\frac{1}{w^2} + j\frac{k}{2R}\right)\right] \tag{C.53}$$

$$w^2 \equiv {w_0}^2\left[1 + \left(\frac{\lambda z}{\pi {w_0}^2}\right)^2\right] \tag{7.2}$$

$$R \equiv z\left[1 + \left(\frac{\pi {w_0}^2}{\lambda z}\right)^2\right] \tag{7.3}$$

$$\varphi \equiv \tan^{-1}\left[\frac{\lambda z}{\pi {w_0}^2}\right] \tag{7.4}$$

Substituting Equation (C.53) into Equation (C.33), we obtain Equation (7.1′), which is identical to Equation (7.1):

$$\boldsymbol{U} = \boldsymbol{E}_0\frac{w_0}{w}\exp[-j(kz - \varphi)]\exp\left[-\left(\frac{\rho}{w}\right)^2\right]\exp\left[\frac{-jk\rho^2}{2R}\right] \tag{7.1′}$$

We have thus obtained the Gaussian beam equations which were discussed in Chapter 7 [2, 3].

C.6 Detailed Derivations of Equation (C.21), Equation (C.34), and Equation (C.38) and Gaussian beam equations [Equation (7.1), Equation (7.2), Equation (7.3), and Equation (7.4)]

C.6.1 Detailed Derivation of Equation (C.21)

$$\frac{\partial^2 f}{\partial x^2}+\frac{\partial^2 f}{\partial y^2}+\frac{\partial^2 f}{\partial z^2}=\frac{\partial^2 f}{\partial r^2}+\frac{2}{r}\frac{\partial f}{\partial r}+\frac{1}{r^2}\frac{\partial^2 f}{\partial \theta^2}+\frac{1}{r^2}\cot\theta\frac{\partial f}{\partial \theta}+\frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial \varphi^2} \tag{C.21}$$

where $x=r\sin\theta\cos\varphi$, $y=r\sin\theta\sin\varphi$, $z=r\cos\theta$

In Figure C.6, $r=\sqrt{x^2+y^2+z^2}$, $\theta=\arctan\frac{\sqrt{x^2+y^2}}{z}$, $\varphi=\arctan\frac{y}{x}$

Applying $\frac{d}{da}(\sqrt{a})=\frac{1}{2\sqrt{a}}$ $\quad\frac{d}{da}(\arctan a)=\frac{1}{1+a^2}$

$$\frac{\partial r}{\partial x}=\frac{x}{r}=\sin\theta\cos\varphi$$

$$\frac{\partial r}{\partial y}=\frac{y}{r}=\sin\theta\sin\varphi$$

$$\frac{\partial r}{\partial z}=\frac{z}{r}=\cos\theta$$

$$\frac{\partial\theta}{\partial x}=\frac{1}{1+\frac{x^2+y^2}{z^2}}\frac{x}{z\sqrt{x^2+y^2}}=\frac{1}{\sqrt{x^2+y^2+z^2}}\frac{z}{\sqrt{x^2+y^2+z^2}}\frac{x}{\sqrt{x^2+y^2}}=\frac{1}{r}\cos\theta\cos\varphi$$

$$\frac{\partial\theta}{\partial y}=\frac{1}{1+\frac{x^2+y^2}{z^2}}\frac{y}{z\sqrt{x^2+y^2}}=\frac{1}{\sqrt{x^2+y^2+z^2}}\frac{z}{\sqrt{x^2+y^2+z^2}}\frac{y}{\sqrt{x^2+y^2}}=\frac{1}{r}\cos\theta\sin\varphi$$

$$\frac{\partial\theta}{\partial z}=\frac{1}{1+\frac{x^2+y^2}{z^2}}\sqrt{x^2+y^2}\left(-\frac{1}{z^2}\right)=-\frac{1}{\sqrt{x^2+y^2+z^2}}\frac{\sqrt{x^2+y^2}}{\sqrt{x^2+y^2+z^2}}=-\frac{1}{r}\sin\theta$$

$$\frac{\partial\varphi}{\partial x}=\frac{1}{1+\frac{y^2}{x^2}}\left(-\frac{y}{x^2}\right)=-\frac{y}{x^2+y^2}=-\frac{y}{\sqrt{x^2+y^2}}\frac{1}{\sqrt{x^2+y^2}}=-\frac{\sin\varphi}{r\sin\theta}$$

$$\frac{\partial\varphi}{\partial y}=\frac{1}{1+\frac{y^2}{x^2}}\left(\frac{1}{x}\right)=\frac{x}{x^2+y^2}=\frac{x}{\sqrt{x^2+y^2}}\frac{1}{\sqrt{x^2+y^2}}=\frac{\cos\varphi}{r\sin\theta}\qquad\frac{\partial\varphi}{\partial z}=0$$

Substituting the above equations into the differential equations:

$$\frac{\partial f}{\partial x}=\frac{\partial f}{\partial r}\frac{\partial r}{\partial x}+\frac{\partial f}{\partial\theta}\frac{\partial\theta}{\partial x}+\frac{\partial f}{\partial\varphi}\frac{\partial\varphi}{\partial x},$$

$$\frac{\partial^2 f}{\partial x^2}=\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)$$

$$=\frac{\partial}{\partial r}\left(\frac{\partial f}{\partial x}\right)\cdot\frac{\partial r}{\partial x}+\frac{\partial}{\partial\theta}\left(\frac{\partial f}{\partial x}\right)\cdot\frac{\partial\theta}{\partial x}+\frac{\partial}{\partial\varphi}\left(\frac{\partial f}{\partial x}\right)\cdot\frac{\partial\varphi}{\partial x},$$

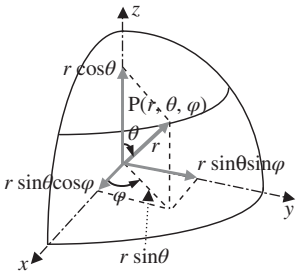


Figure C.6 Coordinates of a sphere

we obtain $\partial^2 f/\partial x^2$. In a similar manner, we can derive $\partial^2 f/\partial y^2$ and $\partial^2 f/\partial z^2$. Finally, we can obtain Equation (C.21) as follows [4]:

$$\frac{\partial^2 f}{\partial x^2}+\frac{\partial^2 f}{\partial y^2}+\frac{\partial^2 f}{\partial z^2}=\frac{\partial^2 f}{\partial r^2}+\frac{2}{r}\frac{\partial f}{\partial r}+\frac{1}{r^2}\frac{\partial^2 f}{\partial \theta^2}+\frac{1}{r^2}\cot\theta\frac{\partial f}{\partial \theta}+\frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial \varphi^2} \tag{C.21}$$

C.6.2 Detailed Derivation of Equation (C.34)

$$\left(\frac{1}{\rho}\frac{\partial}{\partial \rho}+\frac{\partial^2}{\partial \rho^2}\right)u-2jk\frac{\partial u}{\partial z}=0 \tag{C.34}$$

Substituting Equation (C.33) into the wave equation (C.32b) and Equation (C.35), we obtain Equation (C.34).

$$U_0=E_0u(x,y,z)e^{-jkz}=E_0u(\rho.\varphi,z)e^{-jkz}=E_0u(\rho,z)e^{-jkz} \tag{C.33}$$

$$\frac{\partial^2 U_0}{\partial x^2}+\frac{\partial^2 U_0}{\partial y^2}+\frac{\partial^2 U_0}{\partial z^2}+k^2U_0=0 \tag{C.32b}$$

$$\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}=\frac{1}{\rho}\frac{\partial}{\partial \rho}+\frac{\partial^2}{\partial \rho^2} \tag{C.35}$$

Partially differentiating Equation (C.33) twice by z,

$$\frac{\partial U_0}{\partial z}=E_0\left[e^{-jkz}\frac{\partial u}{\partial z}-jkue^{-jkz}\right]$$

$$\frac{\partial^2 U_0}{\partial z^2}=E_0\frac{\partial}{\partial z}\left[e^{-jkz}\frac{\partial u}{\partial z}-jkue^{-jkz}\right]=E_0\left[e^{-jkz}\frac{\partial^2 u}{\partial z^2}-2jke^{-jkz}\frac{\partial u}{\partial z}-k^2e^{-jkz}u\right] \tag{C.53}$$

From Equation (C.35),

$$\frac{\partial^2 U_0}{\partial x^2}+\frac{\partial^2 U_0}{\partial y^2}=E_0\left(\frac{1}{\rho}\frac{\partial u}{\partial \rho}+\frac{\partial^2 u}{\partial \rho^2}\right)e^{-jkz} \tag{C.41}$$

From Equation (C.53) and Equation (C.41)

$$\frac{\partial^2 U_0}{\partial x^2}+\frac{\partial^2 U_0}{\partial y^2}+\frac{\partial^2 U_0}{\partial z^2}=E_0\left(\frac{1}{\rho}\frac{\partial u}{\partial \rho}+\frac{\partial^2 u}{\partial \rho^2}+\frac{\partial^2 u}{\partial z^2}-2jk\frac{\partial u}{\partial z}-k^2u\right)e^{-jkz} \tag{C.42}$$

Substituting Equation (C.42) into Equation (C.32b), we get:

$$\frac{1}{\rho}\frac{\partial u}{\partial \rho}+\frac{\partial^2 u}{\partial \rho^2}+\frac{\partial^2 u}{\partial z^2}-2jk\frac{\partial u}{\partial z}=0 \tag{C.43}$$

Here we can drop the term $\partial^2 u/\partial z^2$, assuming that $\partial u^2/\partial z^2 \ll k\partial u/\partial z$, since u varies gradually with respect to z. Thus we obtain Equation (C.34):

$$\left(\frac{1}{\rho}\frac{\partial}{\partial \rho}+\frac{\partial^2}{\partial \rho^2}\right)u-2jk\frac{\partial u}{\partial z}=0 \tag{C.34}$$

C.6.3 Detailed Derivation of Equation (C.38a,b)

$$Q(z)=z+q_0 \tag{C.38a}$$

$$P(z)=-j\ln\left(1+\frac{z}{q_0}\right)+p_0 \tag{C.38b}$$

If the solution to the differential equation (C.34) has the form of Equation (C.36), then we can derive the conditions Equation (C.38a,b) under which $P(z)$ and $Q(z)$ satisfy the differential equation Equation (C.34), for all values of ρ.

$$\left(\frac{1}{\rho}\frac{\partial}{\partial \rho}+\frac{\partial^2}{\partial \rho^2}\right)u-2jk\frac{\partial u}{\partial z}=0 \tag{C.34}$$

$$u(\rho,z)=\exp\left[-j\left\{P(z)+\frac{k\rho^2}{2Q(z)}\right\}\right] \tag{C.36}$$

Partially differentiating Equation (C.36) by ρ and then by z, we get:

$$\frac{\partial u}{\partial \rho}=-j\frac{k\rho}{Q(z)}\exp\left[-j\left\{P(z)+\frac{k\rho^2}{2Q(z)}\right\}\right] \tag{C.44}$$

$$\frac{\partial^2 u}{\partial \rho^2}=-j\frac{k}{Q(z)}\exp\left[-j\left\{P(z)+\frac{k\rho^2}{2Q(z)}\right\}\right]-\frac{k^2\rho^2}{Q^2(z)}\exp\left[-j\left\{P(z)+\frac{k\rho^2}{2Q(z)}\right\}\right] \tag{C.45}$$

$$\frac{\partial u}{\partial z}=-j\frac{dP(z)}{dz}\exp[-jP(z)]\exp\left[-\frac{j}{2}\frac{k\rho^2}{Q(z)}\right]+\frac{j}{2}\frac{k\rho^2}{Q^2(z)}\frac{dQ(z)}{dz}\exp[-jP(z)]\exp\left[-\frac{j}{2}\frac{k\rho^2}{Q(z)}\right] \tag{C.46}$$

The terms in Equation (C.34) can be derived as follows:

$$\text{The first term}=\frac{1}{\rho}\frac{\partial u}{\partial \rho}=-j\frac{k}{Q(z)}\exp\left[-j\left\{P(z)+\frac{k\rho^2}{2Q(z)}\right\}\right] \tag{C.47}$$

$$\text{The second term}=\frac{\partial^2 u}{\partial \rho^2}=\left[-j\frac{k}{Q(z)}-\frac{k^2\rho^2}{Q^2(z)}\right]\exp\left[-j\left\{P(z)+\frac{k\rho^2}{2Q(z)}\right\}\right] \tag{C.48}$$

$$\text{The third term}=-2jk\frac{\partial u}{\partial z}=\left[-2k\frac{dP(z)}{dz}+\frac{k^2\rho^2}{Q^2(z)}\frac{dQ(z)}{dz}\right]\exp\left[-j\left\{P(z)+\frac{k\rho^2}{2Q(z)}\right\}\right] \tag{C.49}$$

Substituting Equation (C.47), Equation (C.48) and Equation (C.49) into Equation (C.34), we get:

$$\left[-2j\frac{k}{\mathrm{Q}(z)} - 2k\frac{dP(z)}{dz} - \frac{k^2\rho^2}{\mathrm{Q}^2(z)} + \frac{k^2\rho^2}{\mathrm{Q}^2(z)}\frac{d\mathrm{Q}(z)}{dz}\right]\exp\left[-j\left\{P(z) + \frac{k\rho^2}{2Q(z)}\right\}\right] = 0 \tag{C.50}$$

The conditions under which Equation (C.50) is satisfied for all values of ρ are as follows:
(for terms **not** containing $\rho^2 = 0$)

$$\frac{dP(z)}{dz} + \frac{j}{Q(z)} = 0 \tag{C.37a}$$

(for terms containing $\rho^2 = 0$)

$$\frac{dQ(z)}{dz} - 1 = 0 \tag{C.37b}$$

From Equation (C.37b),

$$Q(z) = z + q_0 \tag{C.38a}$$

Substituting Equation (C.38a) into Equation (C.37a), we obtain Equation (C.38b).

$$\frac{dP(z)}{dz} = -j\frac{1}{z+q_0}$$

$$P(z) = -j\ln\left(1 + \frac{z}{q_0}\right) + p_0 \tag{C.38b}$$

C.6.4 Gaussian Beam equations [*Equation (7.1), Equation (7.2), Equation (7.3), and Equation (7.4)*]

$$u(\rho, z) = \frac{w_0}{w}\exp\left[j\varphi - \rho^2\left(\frac{1}{w^2} + j\frac{k}{2R}\right)\right] \tag{C.53}$$

$$w^2 \equiv {w_0}^2\left[1 + \left(\frac{\lambda z}{\pi {w_0}^2}\right)^2\right] \tag{7.2}$$

$$R \equiv z\left[1 + \left(\frac{\pi {w_0}^2}{\lambda z}\right)^2\right] \tag{7.3}$$

$$\varphi \equiv t\mathrm{an}^{-1}\left[\frac{\lambda z}{\pi {w_0}^2}\right] \tag{7.4}$$

$$Q(z) = z + q_0 \tag{C.38a}$$

$$P(z) = -j\ln\left(1 + \frac{z}{q_0}\right) + p_0 \tag{C.38b}$$

$$u(\rho, z) = \exp\left[-j\left\{P(z) + \frac{k\rho^2}{2Q(z)}\right\}\right] = \exp[-jP(z)]\exp\left[-j\frac{k\rho^2}{2Q(z)}\right] \tag{C.36}$$

Substituting Equation (C.38a,b) into Equation (C.36), defining $q_0 = j\pi {w_0}^2/\lambda$, $p_0 = 0$, we get:

$$\exp[-jP(z)] = \exp\left[-\ln\left\{1+\frac{\lambda z}{j\pi {w_0}^2}\right\}\right] = \frac{1}{1+\dfrac{\lambda z}{j\pi w_0}} = \frac{1}{\sqrt{1+\dfrac{\lambda^2 z^2}{\pi^2 {w_0}^4}}}\frac{\pi {w_0}^2 + j\lambda z}{\sqrt{(\pi {w_0}^2)^2+(\lambda z)^2}} \tag{C.51}$$

$$\downarrow \qquad\qquad \downarrow$$

$$w_0/w \qquad\qquad \exp(j\varphi)$$

$$\exp\left[-j\frac{k\rho^2}{2Q(z)}\right] = \exp\left[\frac{k\rho^2}{2}\frac{-j}{z+\dfrac{j\pi {w_0}^2}{\lambda}}\right] = \exp\left[\frac{-jk\rho^2\left(1-\dfrac{j\pi {w_0}^2}{\lambda z}\right)}{2z\left(1+\dfrac{j\pi {w_0}^2}{\lambda z}\right)\left(1-\dfrac{j\pi {w_0}^2}{\lambda z}\right)}\right]$$

$$= \exp\left[\frac{-jk\rho^2}{2z\left(1+\dfrac{\pi^2 {w_0}^4}{\lambda^2 z^2}\right)}\right]\exp\left[\frac{-\rho^2\pi^2 {w_0}^2}{\lambda^2 z^2+\pi^2 {w_0}^4}\right] = \exp\left[\frac{-jk\rho^2}{2z\left[1+\left(\dfrac{\pi {w_0}^2}{\lambda z}\right)^2\right]}\right]\exp\left[\frac{-\rho^2}{{w_0}^2\left(1+\dfrac{\lambda^2 z^2}{\pi^2 {w_0}^4}\right)}\right] \tag{C.52}$$

$$\downarrow \qquad\qquad\qquad \downarrow$$

$$\exp[-jk\rho^2/(2R)] \qquad\qquad \exp[-\rho^2/w^2]$$

Observing Equation (C.51) and (C.52) and comparing with Equation (7.2), Equation (7.3), and Equation (7.4), we find that $\exp[-jP(z)]$ is equal to $w_0/w\ \exp(j\,\varphi)$, and $\exp[-jk\rho^2/\{2Q(z)\}]$ is equal to $\exp[-jk\rho^2/(2R)]\ \exp[-\rho^2/w^2]$. Hence Equation (C.36) can be rewritten as follows.

$$u(\rho, z) = \frac{w_0}{w}\exp\left[j\varphi - \rho^2\left(\frac{1}{w^2}+j\frac{k}{2R}\right)\right] \tag{C.53}$$

Substituting Equation (C.53) into Equation (C.33), we obtain Equation (7.1′), which is identical to Equation (7.1).

$$\boldsymbol{U} = \boldsymbol{E}_0\frac{w_0}{w}\exp[-j(kz-\varphi)]\exp\left[-\left(\frac{\rho}{w}\right)^2\right]\exp\left[\frac{-jk\rho^2}{2R}\right] \tag{7.1′}$$

We have thus obtained the Gaussian beam equations which we discussed in Chapter 7.

Using Equation (7.2), Equation (7.3), and Equation (7.4), we can calculate w_0/w, $\exp[j\varphi]$, $\exp[-jk\rho^2/(2R)]$ and $\exp[-(\rho/w)^2]$ [in Equation (7.1′)] as follows:

$$\frac{w_0}{w} = \frac{w_0}{w_0\sqrt{1+\left(\dfrac{\lambda z}{\pi {w_0}^2}\right)^2}} = \frac{1}{\sqrt{1+\left(\dfrac{\lambda z}{\pi {w_0}^2}\right)^2}} \tag{C.54}$$

$$\exp[j\varphi] = \exp\left[j\tan^{-1}\left(\frac{\lambda z}{\pi {w_0}^2}\right)\right] = \frac{\pi {w_0}^2 + j\lambda z}{\sqrt{(\lambda z)^2+(\pi {w_0}^2)^2}} \tag{C.55}$$

$$\exp\left[-\frac{jk\rho^2}{2R}\right] = \exp\left[\frac{-jk\rho^2}{2z\left[1+\left(\frac{\pi {w_0}^2}{\lambda z}\right)^2\right]}\right] \tag{C.56}$$

$$\exp\left[-\frac{\rho^2}{w^2}\right] = \exp\left[\frac{-\rho^2}{{w_0}^2\left[1+\left(\frac{\lambda z}{\pi {w_0}^2}\right)^2\right]}\right] \tag{C.57}$$

References

[1] Hect, E. and Zajac, A. (1973) *Optics*, Addison-Wesley, USA.
[2] Kudo, K. and Uehara, F. (1990) *Kiso-kogaku*, Gendaikogakusya Co., Ltd, Japan (in Japanese).
[3] Suhara, T. (1998) *Koha-kogaku*, Corona Publishing Co., Ltd, Japan (in Japanese).
[4] Wada, M. (1988) *Bibun sekibun*, Iwanami Publishing Co., Ltd, Japan (in Japanese).

Appendix D

Numerical Integration Methods

In calculations of the definite integral $\int_a^c f(x)dx$, we can use numerical integration when it is difficult to find the anti-derivative of the integrand $f(x)$. We can obtain an approximation of the integral by breaking up the interval $[a, c]$ into a finite number of subintervals, computing an approximation for each subinterval, and then adding up the calculated areas [1].

D.1 Trapezoidal Rule

In order to approximate the area enclosed by the curve $f(x)$, $x = a$, $x = c$, and the x-axis in Figure D.1, we can break the interval $[a, c]$ up into n subintervals, each of length d, and sum the areas S_1, S_2, S_3, S_n. Each partitioned area has an arc at the top. However, we can replace this with a line connecting the two end-points, thereby making a trapezoid. Doing this for the entire interval generates a set of n trapezoids, which can be used to approximate the area under the curve. The heights needed to calculate the areas of the trapezoids are y_0, y_1, y_2, y_n. Thus the area to be calculated, ΣS, will be:

$$\begin{aligned}\sum S &= S_1 + S_2 + S_3 + \cdots + S_n \\ &= \frac{(y_0 + y_1)d}{2} + \frac{(y_1 + y_2)d}{2} + \frac{(y_2 + y_3)d}{2} + \cdots + \frac{(y_{n-1} + y_n)d}{2} \\ &= \frac{(y_0 + y_n)d}{2} + (y_1 + y_2 + y_3 + \cdots + y_{n-1})d \end{aligned} \tag{D.1}$$

D.2 Simpson's Rule (a Better Approximation)

As shown in Figure D.2, we can break up the interval $[a, c]$ into $2m$ subintervals of width d and then create m partitioned areas $S_1, S_2, \cdots S_m$ by joining each pair of adjacent subareas. Here, we shall treat the arc at the top of each divided area as a parabola $y = Ax^2 + Bx + C$ which passes through the three points on the arc. The partitioned area S_i can be approximated as follows:

$$S_i \cong \frac{d}{3}[y_{2i-2} + 4y_{2i-1} + y_{2i}] \tag{D.2}$$

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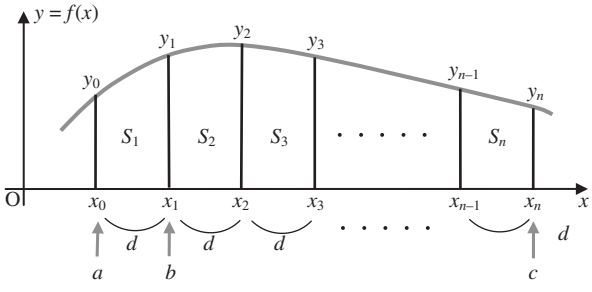


Figure D.1 Trapezoidal rule

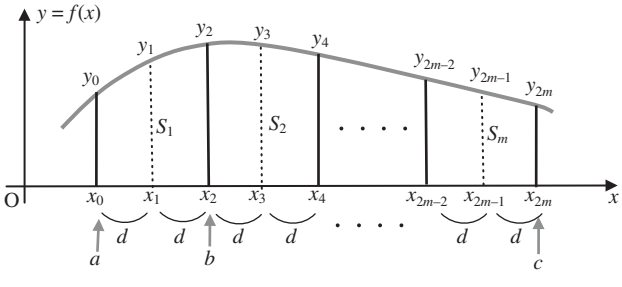


Figure D.2 Simpson's rule

where $d = (c - a)/(2\text{ m})$, and where y_{2i-2}, $y_{2i-1,}$ and y_{2i} represent the vertical heights of the arc at each boundary. The total area ΣS will therefore be:

$$\begin{aligned}\sum S &= S_1 + S_2 + S_3 + \cdots + S_m \\ &\cong \frac{d}{3}[y_0 + 4y_1 + y_2 + y_2 + 4y_3 + y_4 + \cdots + y_{2m-2} + 4y_{2m-1} + y_{2m}] \\ &= \frac{d}{3}[y_0 + 4y_1 + 2y_2 + 4y_3 + 2y_4 + 4y_5 + 2y_6 + \cdots + 2y_{2m-2} + 4y_{2m-1} + y_{2m}] \end{aligned} \tag{D.3}$$

Equation (D.3) is a good approximation of $\int_a^c f(x)dx$.

Proof of Equation (D.2)

Here we shall prove that

$$\int_a^b y dx = \frac{(b-a)}{6}(y_0 + 4y_1 + y_2) \tag{D.4}$$

where $y = f(x) = Ax^2 + Bx + C$ and $\mathrm{P}_0\,(a, y_0)$, $\mathrm{P}_1\,((a+b)/2, y_1)$, $\mathrm{P}_2\,(b, y_2)$ are three points through which the curve $y = f(x)$ passes.

The definite integral Equation (C.4) can be expressed as follows:

$$\int_a^b y dx = \int_a^b (Ax^2 + Bx + C)dx = \left[\frac{A}{3}x^3 + \frac{B}{2}x^2 + Cx\right]_a^b = \frac{A}{3}(b^3 - a^3) + \frac{B}{2}(b^2 - a^2) + C(b - a)$$
$$= \frac{(b-a)}{3}\left[A\left(a^2 + ab + b^2\right) + \frac{3}{2}B(a+b) + 3C\right] \tag{D.5}$$

The vertical heights y_0, y_1, y_2 of the three points P_0, P_1, P_2 can be computed as follows:

$$y_0 = Aa^2 + Ba + C \tag{D.6}$$

$$y_1 = A\left(\frac{a+b}{2}\right)^2 + B\left(\frac{a+b}{2}\right) + C = \frac{1}{4}[(a^2 + 2ab + b^2)A + (2a + 2b)B + 4C] \tag{D.7}$$

$$y_2 = Ab^2 + Bb + C \tag{D.8}$$

Hence

$$y_0 + 4y_1 + y_2 = 2\left[A\left(a^2 + ab + b^2\right) + \frac{3}{2}B(a+b) + 3C\right] \tag{D.9}$$

Substituting Equation (D.9) into Equation (D.5), we obtain Equation (D.4).

$$\int_a^b y dx = \frac{b-a}{6}(y_0 + 4y_1 + y_2) \tag{D.4}$$

Reference

[1] Wada, M. (1988) *Bibun sekibun*, Iwanami Publishing Co., Ltd, Japan (in Japanese).

Appendix E

Fresnel Diffraction and Fraunhofer Diffraction

Here we will study Fresnel diffraction and Fraunhofer diffraction in detail [1].

E.1 How Fresnel and Fraunhofer Diffraction Patterns Are Generated

The diffraction irradiance pattern will very much depend on the distance from the aperture to the observation point. Figure E.1 is a rough illustration of the various diffraction irradiance patterns that can be observed at different distances from the aperture, while it is being irradiated with collimated light.

① *Near-field pattern*:
Irradiance pattern observed very close to the aperture ($z <$ a few wavelengths).
② *Fresnel diffraction pattern*:
Irradiance pattern observed at an intermediate distance from the aperture.
③ *Far-field pattern (Fraunhofer diffraction pattern)*:
Irradiance pattern observed at a very great distance from the aperture ($z > D^2/\lambda$).

Just past the aperture (near-field), the irradiance pattern observed is very simple: it reveals nothing more than the shape of the aperture. However, at a distance more than a few wavelengths from the aperture (Fresnel region), the irradiance pattern changes: it now consists of a combination of the shape of the aperture and the diffraction pattern generated by the edge of the aperture. Finally, at a distance which is sufficiently far from the aperture (far-field), the shape of the aperture disappears entirely from the irradiance pattern, which now only displays the diffraction pattern.

E.2 Fundamentals of Diffraction Theory

In Figure E.2, a light wave from a point source O is diffracted at point P in the aperture and generates a diffraction field at an observation point Q on the screen. The field (which has a complex

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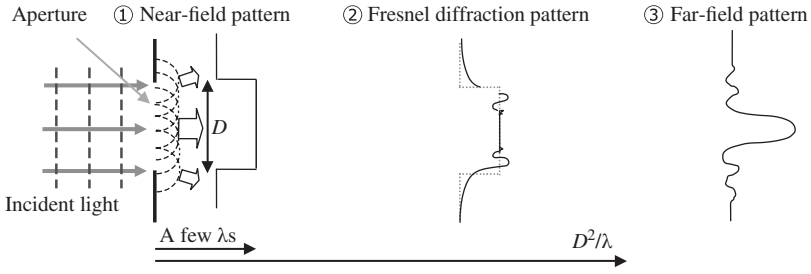


Figure E.1 Diffraction irradiance patterns observed at various distances from the aperture

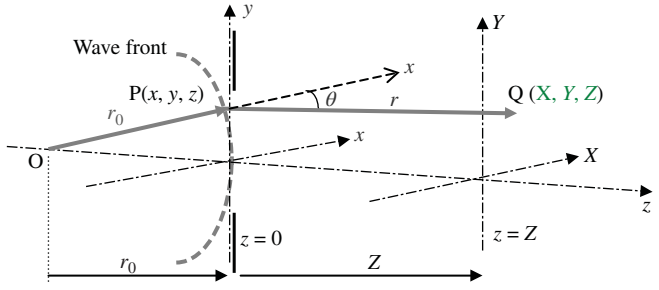


Figure E.2 Diffraction from an aperture

amplitude) at Q will be of the form shown in Equation (E.1), provided that the following conditions are satisfied:

1. The aperture size is sufficiently large when compared with the wavelength.
2. The distances from the source O to the aperture P, and from the aperture P to the observation point Q are sufficiently long, when compared with the size of the aperture.

$$U(X,Y) \cong \frac{j\varepsilon_0}{\lambda}\iint \exp(-jkr_0)\exp(-jkr)\frac{K(\theta)}{r_0 r}dS \tag{E.1}$$

where

$$K(\theta) \equiv \frac{1+\cos\theta}{2} \tag{E.2}$$

In Equation (E.1), ε_0 represents the amplitude of the wave at a unit distance from the source, dS is a tiny sector of the area of the aperture, and $K(\theta)$ is the inclination factor, which can be approximated by the equation $K(\theta) \approx 1$ for diffraction waves situated near the axis. Equation (E.1) is commonly referred to as the Kirchhoff approximation. It applies to all diffraction-related phenomena. The following remarks illustrate the significance of the terms used in Equation (E.1).

1. If coherent light is used, then the secondary waves from the aperture will interfere with each other on the observation plane.

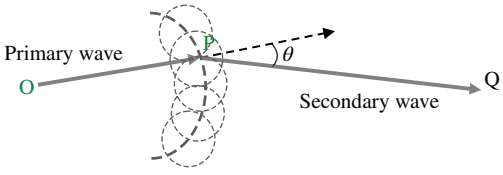


Figure E.3 Inclination factor $K(\theta) = (1 + \cos\theta)/2$

2. The term $1/(r_0 r)$ in the integral signifies that the amplitude of the spherical wave will decrease in inverse proportion to the distance traveled, while the irradiance will decrease in inverse proportion to the square of the distance traveled.
3. The product $\exp[-jkr_0]\exp(-jkr)$ in the integral signifies that the phase delay will be proportional to the distance traveled from the source, that is, $l = r_0 + r$.

$$\exp[-jkl] = \exp[-jk(r_0 + r)] = \exp[-jkr_0]\exp[-jkr]$$

4. The term j/λ signifies that the secondary waves from the aperture will have an amplitude proportional to $1/\lambda$, and a phase shift of $\pi/2$ rad [$\because j = \exp(j\pi/2)$].
5. As shown in Figure E.3, the term $K(\theta)$ in the integral means that the amplitude of the secondary wave from the aperture will have a maximum value of 1 in the direction of propagation and will have a value of zero when the wave is traveling in the reverse direction.

$$(\because (1 + \cos 0)/2 = 1, \text{ and } (1 + \cos\pi)/2 = 0).$$

We can assume that $K(\theta) \approx 1$ for waves traveling forward near the axis.

E.3 Fresnel Diffraction

In Figure E.2, P(x, y, z) is a point on the wave front at the aperture, whose coordinates have their origin at the center of the aperture, and Q(X,Y) represents the observation point on a screen which is placed at a distance Z from the aperture. The distance r from P (the wave front at the aperture) to Q (the observation point) can be expressed as follows:

$$r = \sqrt{(X - x)^2 + (Y - y)^2 + (Z - z)^2} \tag{E.3}$$

Since P(x, y, z) is situated on the wave front, which is a sphere whose center is at O, the values x, y, z can be expressed as:

$$x^2 + y^2 + (z + r_0)^2 = {r_0}^2 \tag{E.4}$$

If $|z| << r_0$, then z in Equation (E.4) can be approximated as follows:

$$z = -\frac{x^2 + y^2}{2r_0} \tag{E.5}$$

We can substitute Equation (E.5) into Equation (E.3) and expand the results into the following series:

$$r = Z\left(1 + \frac{X^2 + Y^2}{2Z^2}\right) - \left(\frac{xX + yY}{Z} - \frac{x^2 + y^2}{2Z_h}\right) + \cdots \tag{E.6}$$

Z_h is the harmonic mean ($1/Z_h = 1/r_0 + 1/Z$) between the two distances r_0 (from the object to the aperture) and Z (from the aperture to the image). If we irradiate a plane wave perpendicularly onto the

aperture, r_0 tends towards ∞, and then Z_h tends towards Z. When the aperture size is small (i.e., when x, y are small comparing with Z, r_0), we can drop the third and subsequent terms in Equation (E.6). Substituting Equation (E.6) into Equation (E.1), we obtain:

$$U(X,Y) = C\iint u_0(x,y)\exp\left[-jk\frac{x^2+y^2}{2Z}\right]\exp\left[jk\frac{xX+yY}{Z}\right]dxdy \tag{E.7a}$$

$$C \equiv \frac{j\varepsilon_0\exp(-jkr_0)}{\lambda r_0 Z}\exp\left[-jkZ\left(1+\frac{X^2+Y^2}{2Z^2}\right)\right] \tag{E.7b}$$

Here, we placed a body which has an amplitude transmission coefficient of $u_0(x,y)$ at the aperture as in Equation (E.7a), and C is a constant which is independent of the aperture coordinates x and y. Equation (E.7a) is commonly referred to as the Fresnel diffraction equation [1,2].

E.4 Fraunhofer Diffraction

When both the source and the observation point are situated far enough from the aperture (i.e., $Z_h >> k(x^2+y^2)/2$), the term $\exp[-jk(x^2+y^2)/(2Z)]$ can be dropped from Equation (E.7a), as it is identical to unity, for all practical purposes. This gives us the following equations:

$$U(X,Y) = C\iint u_0(x,y)\exp\left[jk\frac{xX+yY}{Z}\right]dxdy \tag{E.8a}$$

$$C \equiv \frac{j\varepsilon_0\exp(-jkr_0)}{\lambda r_0 Z}\exp[-jkZ]\exp\left[-jk\frac{X^2+Y^2}{2Z}\right] \tag{E.8b}$$

Combining Equation (E.8a) and Equation (E.8b), we obtain:

$$U(X,Y) = \frac{j\varepsilon_0\exp(-jkr_0)\exp(-jkZ)}{\lambda r_0 Z}\exp\left[-jk\frac{X^2+Y^2}{2Z}\right]\iint u_0(x,y)\exp\left[jk\frac{xX+yY}{Z}\right]dxdy \tag{E.8c}$$

Equation (E.8c) is commonly referred to as the Fraunhofer diffraction equation [1, 2]. We can compare Equation (E.8c) with Equation (9.29), which is the Fraunhofer diffraction equation that we discussed in Chapter 9, but differently expressed.

$$U(X,Y) \cong \frac{j}{\lambda Z}\exp[-jkZ]\exp\left[-jk\frac{X^2+Y^2}{2Z}\right]\iint u(x,y)\exp\left[jk\frac{Xx+Yy}{Z}\right]dxdy \tag{9.29}$$

In Equation (E.8c), if we drop the term $\exp[-jkr_0]$ (which represents a constant phase shift), and define $u(x, y)$ in Equation (9.29) as follows:

$$u(x,y) = \frac{\varepsilon_0}{r_0}u_0(x,y)$$

then we can see that Equation (9.29) is indeed equivalent to Equation (E.8c). The term $\exp[-jk(X^2+Y^2)/(2Z)]$ which is just before the integral, represents a spherical wave front propagating along the z-axis.

Moreover, Equation (8.20), which we discussed in Chapter 8, turns out to be equivalent to Equation (E.8c).

$$U = \frac{\varepsilon_A}{R}e^{j(\omega t-kR)}\iint_{Aperture} e^{jk(xX+yY)/R}dS \tag{8.20}$$

Here in Equation (E.8c), we replaced Z by R, dropped the term $j\exp[-jkr_0]$ (which represents a constant phase shift), the term $\exp[-jk(X^2+Y^2)/(2Z)]$ (assuming $k(X^2+Y^2) << 2Z$) and the term $u_0(x,y)$ (an amplitude transmission coefficient of a body placed at the aperture), and defined ε_A (the amplitude of the wave at the aperture) as follows:

$$\varepsilon_A = \frac{\varepsilon_0}{\lambda r_0}$$

References

[1] Sagai, J. (1997) *Kogaku no Kiso*, Corona Publishing, Co., Ltd, Japan (in Japanese).
[2] Suhara, T. (1998) *Koha-kogaku*, Corona Publishing Co., Ltd, Japan (in Japanese).

Appendix F

Wave-Front Conversion by a Lens

F.1 Wave-Front Conversion by a Lens

As we saw in Section 1.5.2, if we irradiate a plane wave onto a convex lens parallel to the axis, the wave front after exiting the lens will be a sphere converging at the second focal point F′ (Figure F.1).

The phase of a spherical wave diverging from the point F′ can be expressed as:

$$u_p = \exp(-jkr) \tag{F.1}$$

In Figure F.1, the wave travels in the opposite direction to Equation (F.1). Hence we can replace the minus sign by a positive sign:

$$u_p = \exp(jkr) \tag{F.2}$$

In Figure F.2, the distance r from P(x,y) (just after exiting the thin lens) to the focal point F′ is:

$$r = \sqrt{x^2 + y^2 + f^2} \tag{F.3}$$

where f is the focal length of the lens and x, y are the ray height at the lens. When $x,y \ll f$ (for near-axis rays), r can be approximated as:

$$r \cong f + \frac{x^2 + y^2}{2f} \tag{F.4}$$

Here we omit the constant term f from Equation (F.4) then the complex amplitude transmission coefficient u_{L} of a convex lens (which shows the phase exiting the lens) can be represented as follows:

$$u_{\mathrm{L}} = \exp\left[jk\frac{x^2 + y^2}{2f}\right] \tag{F.5}$$

Equation (F.5) tells us that convex lenses shift the phase of the incident wave forward by an angle which is in proportion to the square of the ray height. We can also apply Equation (F.5) to concave lenses, by changing the value of f from positive to negative [1,2].

F.2 Diffraction Field Equation for a Plane Wave after it Exits a Convex Lens

Here, we shall examine the diffraction field for a plane wave (collimated beam) after exiting the convex lens in Figure F.1 [2]. We can multiply the Fresnel diffraction equations, Equation (E.7a,b), by the

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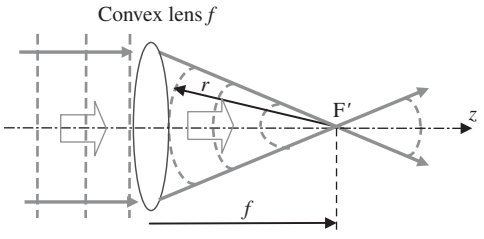


Figure F.1 Wave-front conversion by a convex lens

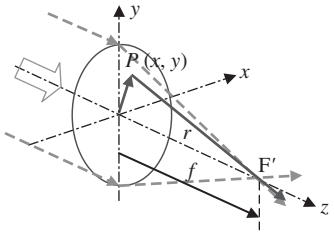


Figure F.2 Ray height (*x*,*y*)

complex amplitude transmission coefficient in the lens equation Equation (F.5).

$$U(X,Y) = C\iint u_0(x,y)\exp\left[-jk\frac{x^2+y^2}{2Z}\right]\exp\left[jk\frac{xX+yY}{Z}\right]dxdy \tag{E.7a}$$

$$C \equiv \frac{j\varepsilon_0\exp(-jkr_0)}{\lambda r_0 Z}\exp\left[-jkZ\left(1+\frac{X^2+Y^2}{2Z^2}\right)\right] \tag{E.7b}$$

$$u_{\mathrm{L}}(x,y) = \exp\left[jk\frac{x^2+y^2}{2f}\right] \tag{F.5}$$

In Equation (E.7a), the wave incident onto the lens is a plane wave, so $Z = f$, and hence:

$$u_L(x,y)\exp\left[-jk\frac{x^2+y^2}{2Z}\right] = \exp\left[jk\frac{x^2+y^2}{2f}\right]\exp\left[-jk\frac{x^2+y^2}{2f}\right] = 1$$

Equation (E.7a) will then be:

$$U(X,Y) = C\iint u_0(x,y)\exp\left[jk\frac{xX+yY}{Z}\right]dxdy \tag{E.7a′}$$

Equation (E.7b) is:

$$C = \frac{j\varepsilon_0\exp(-jkr_0)}{\lambda r_0 Z}\exp(-jkZ)\exp\left[-jk\frac{X^2+Y^2}{2Z}\right] \tag{E.7b′}$$

Combining Equations (E.7a′) and Equation (E.7b′), we obtain the following equation:

$$U(X,Y) = \frac{j\varepsilon_0\exp(-jkr_0)\exp(-jkZ)}{\lambda r_0 Z}\exp\left[-jk\frac{X^2+Y^2}{2Z}\right]\iint u_0(x,y)\exp\left[jk\frac{xX+yY}{Z}\right]dxdy \tag{E.8c}$$

We have finally obtained the above equation. After inspecting the equation, we realize that the equation is identical to the Fraunhofer diffraction equation Equation (E.8c). In our original discussion of Fraunhofer diffraction, a Fraunhofer diffraction image could only be obtained at a distance far from the aperture. However, we can now see that if we insert a convex lens at the aperture, it is possible to obtain a Fraunhofer diffraction image on the second focal plane of the lens.

References

[1] Sagai, J. (1997) *Kogaku no Kiso*, Corona Publishing Co., Ltd, Japan (in Japanese).
[2] Suhara, T. (1998) *Koha-kogaku*, Corona Publishing Co., Ltd, Japan (in Japanese).

Appendix G

List of Excel Calculation Files on the Companion Web Site

File Name	Contents
1 Reflectance and transmittance	Reflectance and transmittance calculation
3.1–3.2 Ray tracing using paraxial theory	Problems for paraxial theory ray tracing
4.1–4.16, 4.20 Two-dimensional ray tracing	Problems for two-dimensional ray tracing
4.1–4.16, 4.19–4.20 Two-dimensional VBA ray tracing	Problems for two-dimensional VBA ray tracing
4.17–4.18 Ray tracing for an aspheric lens	Two-dimensional ray tracing for an aspheric lens
4.19 A laser beam scanning system	Two-dimensional ray tracing for a laser beam scanning system
5.1–5.7 Three-dimensional ray tracing	Problems for three-dimensional ray tracing
5.1–5.7 Three-dimensional VBA ray tracing	Problems for three-dimensional VBA ray tracing
5.6 A laser beam expander and focusing	Two-dimensional VBA ray tracing for a laser beam expander and focusing
6.1–6.3 Superposition of light waves	Problems of superposition of light waves
7.1–7.5 Gaussian beam focusing	Problems for Gaussian beam focusing
8.1–8.5 Diffraction calculation	Problems of diffraction calculation
9.1 Central irradiance for a Gaussian beam	Calculation for the central irradiance of a Gaussian beam
9.2 Focused Gaussian beam	Two-dimensional R-θ diffraction calculation for a focused Gaussian beam
9.3 Size of the incident beam	Two-dimensional R-θ diffraction calculation for the size of the incident beam
9.4 Truncated or untruncated Gaussian beam	Two-dimensional R-θ diffraction calculation for a truncated/untruncated Gaussian beam

(*continued overleaf*)

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9.7-1 Full-circle aperture	Two-dimensional X-Y diffraction calculation for a full-circle aperture
9.7-2 Ring aperture	Two-dimensional X-Y diffraction calculation for a ring aperture
9.8 Double diffraction	Two-dimensional R-θ diffraction calculation for double diffraction
9.9–9.10 Two-dimensional combination of one-dimensional diffractions	One-dimensional diffraction calculations and a two-dimensional combination
9.10 Rectangular aperture	Two-dimensional X-Y diffraction calculation for a rectangular aperture
9.11-1 Wave-front aberration	Two-dimensional VBA ray tracing and wave-front aberration
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